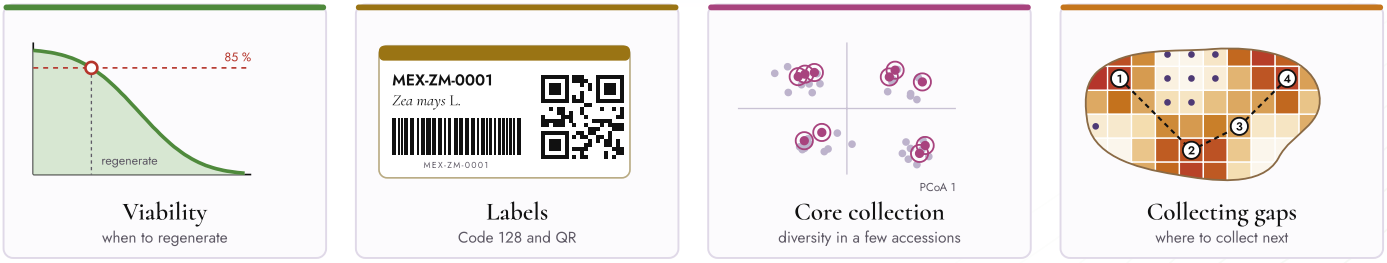
GermplasmPro

version 1.0.0

The genebank, accession by accession

El banco de germoplasma, accesión por accesión



Luis Ángel Barrera-Guzmán

2026 · LICENSED UNDER GPL-3.0 OR LATER

Ten blocks, from passport to report

GermplasmPro

User's Manual · The genebank, accession by accession

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PROGRAM	GermplasmPro, version 1.0.0 (1 October 2026)
MANUAL	First English edition, 2026, translated from the first Spanish edition. Describes version 1.0.0 of the program.
PROGRAM LICENSE	GNU General Public License, version 3 or later (GPL-3.0-or-later).
MANUAL RIGHTS	© 2026 Luis Ángel Barrera-Guzmán. All rights reserved.
INTERFACE LANGUAGE	Spanish or English, with the ES · EN switch on the top bar. The app opens in Spanish; in this manual the names of buttons, fields and menus are written as they appear on screen in English, for example Open the example collection.

HOW TO CITE THE PROGRAM

Barrera-Guzmán, L.Á. (2026). GermplasmPro: the genebank, accession by accession (version 1.0.0) [software]. https://github.com/luisangelbg/GermplasmPro

HOW TO CITE THIS MANUAL

Barrera-Guzmán, L.Á. (2026). *GermplasmPro: User's Manual* (1st English ed.). Distributed with GermplasmPro, version 1.0.0. https://github.com/luisangelbg/GermplasmPro

The citation of the program is the same one given by the How to cite button on the app's home page. The program can also be used online, without installing anything, at https://luisangelbg.github.io/GermplasmPro/.

> **PRIVACY**
>
> GermplasmPro runs entirely inside your browser. The passport data, characterization and management records you load are read on your computer and **are never sent to any server**; the collection is kept in that same browser's local storage so that it is not lost when you close the window. You can work with the coordinates of sensitive populations or with data from custodian families without anything leaving your machine, and without the internet: the maps use geometries included in the program.

The illustrations, diagrams and screenshots in this manual are original: they were generated with the program itself or with code written for the manual. The **example collection** is **fictitious**: the sites are real localities with approximate coordinates, but the accessions, people, stocks and dates are invented, and the example characterization is generated at random with a fixed seed. The viability constants included are those published for *Zea mays*; for other species they are an editable starting point. The map outlines are from **Natural Earth** (public domain).

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How to read this manual

The manual follows the same order as the app: ten numbered blocks that appear, from top to bottom, in the sidebar on the left. Each chapter explains one block and carries its number — Chapter 6 is Block 6, and its sections, figures and tables are numbered 6.1, 6.2…—. Each one begins with the idea from plant genetic resources management that the block needs, goes on to what the screen calculates and how to read it, and ends with what is worth checking before moving on. This introduction has no number: its sections run I.1, I.2…

A color for each block

The block's color appears on the opening page of its chapter, on the number of each section, on the tab at the edge of the page and in the table of contents, and it is the same as in the strip on the cover. That way you can tell at a glance which part of the app you are in.

Block	Name in the app	What it does
1	Home	Home page, the five kinds of genebank, seed behavior and the simulator "the life of an accession", which follows the material and its diversity for decades
2	Passport	Bring in the list of accessions (file, paste or template), recognize the columns as MCPD v2.1 descriptors, validate them, edit them, portray the collection and export it
3	Quality & duplicates	Coordinates that do not fall in the declared country, untidy names and catalogs, corrections proposed for your approval and the same collection entered twice
4	Collecting map	The accessions on the map, offline, colored by species, elevation, year or route; how many fall in each state and each country
5	Geographic diversity	Richness, diversity indices and Hill numbers per geographic unit, rarefaction, richness estimators, similarity between regions, Mantel test and elevation gradient
6	Collecting gaps	The grid of what is missing, a transparent priority index, the route for the next field trip, the risks to what you already have and the minimum set by complementarity
7	Characterization & core	Mixed morphological descriptors, Gower's distance, principal coordinates, clustering and a validated core collection
8	Genebank management	Viability lot by lot with the Ellis and Roberts equation, what has to be done and when on each route, how many plants and seeds to regenerate, and the workload for the next 25 years
9	Labels & register	Labels with Code 128 barcodes and QR codes, in formats measured in millimeters; sowing lists, delivery notes and the register book
10	Report	The collection status report, with the methods written out, to print or save as PDF; the .zip package and the passport in MCPD and in Darwin Core

Types of box

NOTE

Clarifications and practical tips that save time.

IMPORTANT

What can change your decisions if overlooked.

THEORY IN BRIEF

The idea behind a calculation, without unnecessary formulas.

EXAMPLE

A case worked step by step with the example collection.

DECISION RULE

If the result falls in this case…	**then** this is the recommended reading.

These are guides for interpretation, not laws: what you know about your species, your genebank and your sites always comes first.

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KEY FACT

A figure or a fact worth remembering; for example, the FAO/Bioversity multi-crop passport descriptors (MCPD v2.1) that the app recognizes, plus seven fields of its own for management and in situ conservation.

Writing conventions

Written like this	Means
Open the example collection	A button, checkbox or menu of the app, with its text as it appears on screen in English.
Block 3 › Is it the same collection twice?	A path: open the first item and look for the second inside it.
`ACCENUMB`, `GP_CONS`, `.csv`	A descriptor key, a file name or a value typed as it stands. Keys beginning with `GP_` are the GermplasmPro extension, not part of MCPD.
MEX-ZM-0039 · Tuxpeño de la costa	An accession from the example collection, with its number and its name as they appear in the table of Block 2.
25,000 m · 98% · 0.14	Thousands separated with a comma and decimals with a point. In its own text the manual writes the percent sign without a space (98%); where it quotes the screen, it writes it as the app in English shows it, with a space (98 %).

! THE FIGURES IN THIS MANUAL

The results in the text come from running the app with the **example collection** of Block 2 (46 accessions, with their planted errors) and, in Block 7, with the **example characterization** of 12 descriptors. The examples have a fixed seed and the calculations are deterministic: with those data you will see the same numbers. If a chapter starts from another state, such as the corrected collection, it says so first.

In

INTRODUCTION · BEFORE THE BLOCKS

Introduction to GermplasmPro

What the program is, which questions from the daily work of a genebank it answers, what data it accepts and how to go through it from start to finish. It includes the five ideas that hold up everything else —the accession and its passport, the route each species needs, the safety duplicate, the two clocks of conservation, and what you have versus what you are missing— and an honest list of what the program does not do.

I.1 What is GermplasmPro?

I.2 Questions it answers

I.3 Five ideas behind the manual

I.4 What data it accepts and what it delivers

I.5 What you need and how to open it

I.6 A tour of the interface

I.7 The example collection

I.8 How a piece of work proceeds

I.9 What the program does not do

I.1 What is GermplasmPro?

GermplasmPro is a platform for the daily work of a plant genetic resources collection **from the field notebook to the report**: it organizes the passport of each accession with the FAO/Bioversity multi-crop descriptors, finds errors and duplicates, draws the collecting map, measures geographic diversity, points out where collecting is still missing, builds a core collection from the characterization, projects the viability of each lot and warns what needs regenerating, prints labels with a barcode or QR code, and writes a report with the methods written out.

It works for **any species** and for **every kind of genebank**. Ex situ: the seed bank, the field genebank or living collection, slow-growth in vitro culture and cryopreservation; the DNA bank is recorded in the passport. In situ: the genetic reserve of crop wild relatives and conservation in the field of whoever sows it. A real collection almost always combines several of those routes, and the app handles them all at once. Maize and chayote turn up often on the screens and in this manual because they are good examples of two worlds —an orthodox seed that is kept in a jar and a crop that can only be conserved alive—, not because the program is meant for them.

Everything happens inside the browser. There is nothing to install, no programming is needed and the data do not leave the computer. It is written for genebank curators, field technicians, thesis students and those in charge of institutional or community collections: it knows that a passport almost always arrives from a spreadsheet with hand-typed headers, that the same collection is often there twice with the name spelled differently, that a longitude loses its sign when it passes from one program to another, that a lot that has gone years without a germination test is an invisible risk, and that an in situ plot nobody visits is also lost.

I.2 Questions it answers

Each block of the app answers a question that anyone who manages a collection really asks. These are the questions, in the order in which the blocks are worked through:

Block	The question	How it answers
1	Which kind of genebank suits my species? What happens to an accession over the years?	The five kinds of genebank, seed behavior and a simulator of viability and diversity over decades
2	What do I have, where did it come from and how is it conserved? Is my passport complete?	Reading the list, recognizing the columns, validation field by field and across fields, a portrait of the collection and completeness per descriptor
3	Are my data reliable? Do I have the same collection twice?	Coordinates against the country, names and catalogs, proposed corrections and pairs of possible duplicates with their score
4	Where did I collect? How many accessions are there per state and per country?	The offline collecting map, colored by species, elevation, year or route, and the tables by state and by country

Block	The question	How it answers
5	Where is the diversity of my collection? How much is still left to find?	Richness and indices per geographic unit, rarefaction, richness estimators, similarity between regions and elevation gradient
6	Where do I go on the next field trip? What could I lose of what I already have?	The grid of gaps, the priority index, the collecting route, the risks to the collection and the minimum set
7	Which accessions are alike? Which do I keep if I cannot evaluate them all?	Gower's distance with mixed data, principal coordinates, groups and a core collection with its validation
8	Which lot must be tested or regenerated, and when? How many plants do I sow?	Projected viability per lot, alerts and a work list per route, calculation of plants and seeds, and the workload over 25 years
9	How do I identify each envelope, jar, pot or cryovial? How do I record what goes out?	Labels with Code 128 and QR, sowing lists, delivery notes and the register book
10	How do I report on the state of the collection? How do I share the passport?	A report with written methods and references, a .zip package, the passport in MCPD and in Darwin Core

I.3 Five ideas behind the manual

You do not need to be a genebank curator to use the app, but there are five ideas that come back again and again. Understanding them turns the results from tables into decisions.

1. The accession is the unit, and its passport is its identity

An **accession** is a distinct sample of a species that the genebank conserves and manages as a unit: a maize collection from one community, a potato clone, a wild population in its reserve. What makes it useful is not the material but what is known about it: what it is, where it came from, who collected it and when, what type of material it is and how it is conserved. That is the **passport**, and the FAO/Bioversity multi-crop passport descriptors (MCPD v2.1) are the way of writing it that every genebank and information system understands. The app recognizes its 42 descriptors and adds seven fields of its own, those beginning with `GP_`, for the active conservation route, the in situ site and custodian, the stock, the last germination test and the last regeneration. Every block reads from that passport; that is why Block 2 is the way in.

2. Each species has its conservation route

A seed bank only works for **orthodox** seed, which tolerates drying and freezing. **Intermediate** seed does not tolerate deep cold, **recalcitrant** seed tolerates neither drying nor freezing, and in **vegetatively propagated** crops the seed does not reproduce the clone. For them there are the field, the in vitro jar and liquid nitrogen, and nothing replaces **in situ** conservation, where the population keeps evolving. The app handles the five routes and adapts alerts and deadlines to each one; the label format (envelope, jar, cryovial, stake…) is your choice in Block 9.

3. An accession without a safety duplicate depends on a single facility

The **safety duplicate** is a sample of the same accession kept at another institution, so that a fire, a long power cut or a pest does not take away the only specimen. The passport records it in `DUPLSITE`, and the app counts what share of the collection has one. The other duplicate, the **accidental** one —the same collection entered twice in the catalog, with the name spelled differently—, is the opposite: it protects nothing, inflates the collection and muddles the analyses. Telling one from the other, and finding the second before measuring diversity, is part of the work of Block 3.

4. An accession has two clocks

The first is the clock of the **material**: seed ages in the store at a rate that depends on its moisture content and the temperature, which the Ellis and Roberts viability equation can project; field plants die, jars get contaminated, frozen explants only partly recover. The second is the clock of **genetic integrity**: each regeneration, each subculture and each generation in the field is a bottleneck that redraws the allele frequencies, and the rare alleles are the first to be lost. Hence the genebank rule: regenerate rarely and with many individuals. The simulator in Block 1 shows the two clocks side by side; Block 8 applies them to every lot in your collection.

5. What you have and what you are missing

A collection is worth the diversity it represents, and that diversity has a geography: it is spread across regions, elevations and sites. Block 5 measures how much there is where you have already collected and how much you are still missing with the same effort; Block 6 draws the grid of what has never been collected and proposes where to go, with a priority index whose components are visible and can be moved. In both cases the analysis is **geographic and collection-based**: it counts accessions, species and sites, not climate or soil.

1.4 What data it accepts and what it delivers

WHAT GOES IN

- **The passport**: a table with one row per accession, which is opened from a `.csv`, `.tsv` or `.txt` file, pasted directly from a spreadsheet, typed into the guided template of the paste dialog or started blank. The headers may come in Spanish, in English or as the MCPD key: the app proposes which descriptor each column is and you confirm it.
- **The characterization**, in Block 7: another table, one row per accession and one column per morphological descriptor —numbers, categories and scales mixed, with empty cells—, which is joined to the passport by the accession number.
- **The management records**, in Block 8: the germination tests, regenerations and subcultures you carry out, and the viability constants of your species if you have them.
- **A project** saved earlier as `.json`, which brings back the collection exactly as it was, with its characterization and the settings of Blocks 3, 8 and 9.

WHAT COMES OUT

- **The clean passport** as `.csv`, in two versions: with the MCPD descriptors only, to send to any system, or complete, with the `GP_` extension, so as not to lose the internal management. And the project as `.json`.
- **The quality report** of Block 3 and the `.csv` tables of each block: the map points with their state and their cell, the diversity indices, the collecting route, the genebank work plan.
- **Figures** as SVG or as PNG at double resolution: the collecting map, the accumulation curve, the similarity heat map, the gap map.
- **Labels** ready to print, with barcodes and QR codes generated in the browser itself.
- **The report**: an `.html` file that opens in any browser and can be printed or saved as PDF, with one section per analysis, the methods written from what was actually run, and their references; and a `.zip` with everything, including the passport in MCPD and in Darwin Core.

I.5 What you need and how to open it

A computer with a recent browser —Edge, Chrome, Firefox or Safari from the last two or three years— and the program folder. There is no installation: you copy the folder and open index.html with a double click. It also works from a USB stick and on a computer without internet, and online at https://luisangelbg.github.io/GermplasmPro/.

1. Copy the whole GermplasmPro folder to your computer (or download it and unzip it). Inside there must be index.html and the folders js, css, data —which holds the map outlines— and tests.
2. Open index.html with a double click. If your institution blocks local pages, use Open GermplasmPro.bat on Windows. The first screen is Block 1, the home page.
3. Choose the language with ES · EN and the light or dark theme with the sun button. Both are remembered for next time.
4. If you want to check that the calculations come out right in your browser, open tests/index.html: these are the program's tests; all of them must come out green.

> **I** THE APP REMEMBERS THE COLLECTION
>
> The accessions are saved automatically in the browser of that computer (in its *local storage*); next to the counter in Block 2 the time of the last save is shown, and when you open the app again the collection is still there; if the browser does not allow saving (full storage or a private window), it says so in that same place. It is only a convenience: the real backup is the `.json` project downloaded with Save the project (.json), which can be opened on any other machine with Open a file (CSV, TSV or project). The Empty button of the table deletes what is saved, after asking for confirmation.

In

I.6 A tour of the interface

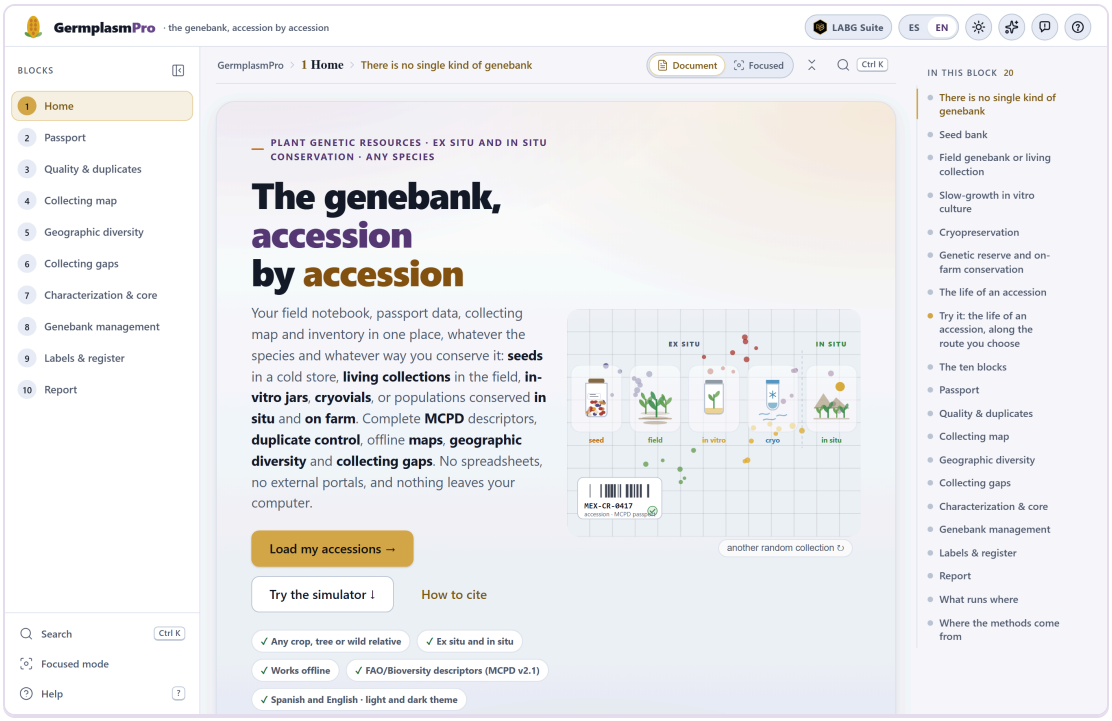


Figure I.1. The opening screen. At the top, the fixed bar with the brand, the link to the suite portal, the language, the theme, the animations, the problem report and the keyboard shortcuts. On the left, the sidebar with the ten blocks; in the center, the home page of Block 1; on the right, the In this block index, which follows your reading.

Element	What it is for
The block sidebar	On the left, the ten numbered blocks, always visible. The active one is highlighted in gold; the counter at the top (*1/10, 2/10…*) and the check mark next to each number say how many you have done. No block is locked: you can jump to any of them at any time. The sidebar collapses with its button or with the `[` key.
The path and the reading modes	Above the content, the path *GermplasmPro › block › section* tells you where you are. Document shows all the sections of the block one after another; Focused, one at a time, with Previous and Next (or Alt+← and Alt+→).
In this block	The index on the right: the sections of the block, with the current one marked. A click takes you to each one. In sections that have figures there is also the button to open them in the figure studio.
Search · **Ctrl K**	A palette that finds blocks, sections, controls and figures by name, with the recent ones first.
The cards	Each block is divided into numbered sections —"1 · Fixes the app can make for you", "2 · Does the point fall where it says?"…— that are read from top to bottom. The first ones are usually the data and assumptions; the following ones, the results.
The result tiles	A row of large figures with their label and a line of context below, like those of the portrait of the collection (Figure I.3). They are the summary of the section: what you would say in one sentence.

Element	What it is for
The export buttons	Next to each map, chart or table: Export SVG, Export PNG (2×), Table as CSV… What they download is what you see.
Previous · Next	At the foot of each block, the buttons that take you to the block before and the block after, with their number and name.
Help and shortcuts	The Help button in the sidebar and the question mark on the top bar show the keyboard shortcuts; Focused mode switches the reading mode from the same sidebar.

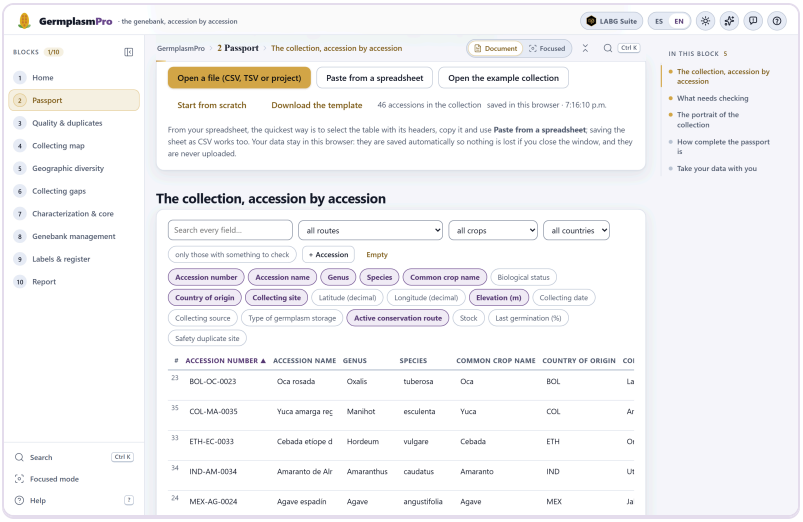


Figure I.2. A block in use: Block 2 with the example collection. The sidebar already marks *1/10*; below the buttons, the counter and the time of the last save; then the filters, the visible columns and the table.

I.7 The example collection

The Open the example collection button of Block 2 loads a small but realistic genebank, the same one that illustrates the whole of this manual. It is **fictitious**: the collecting sites are real localities with approximate coordinates, but the accessions, numbers, custodians, stocks and dates are invented. It serves to learn the app, to see what each block does before loading your collection and to reproduce the figures in the manual.

Figure I.3. The portrait of the example collection, in Block 2. The eight "countries" include the code `MX`, miswritten on purpose, and the six "routes" count the genetic reserve and the on-farm plot separately, since they are the two faces of in situ conservation.

What it contains	How much
Accessions	46, from 24 taxa and 18 genera: maize landraces and their wild relatives (18 accessions of *Zea mays*), beans, squashes, chili peppers, amaranths, husk tomato, potatoes, oca, cassava, agave, vanilla, avocado, cacao, coffee, chayote, oak, wheat and barley.
Countries of origin	7: Mexico (37 accessions), Guatemala, Peru, Bolivia, Colombia, Ethiopia and India; plus one accession with the country written `MX`.
Conservation routes	All five: seed bank (28), field genebank or living collection (8), in vitro culture (3), cryopreservation (1) and in situ, with 2 accessions in a genetic reserve and 3 in the fields of custodian families. One accession is left without a route, because its only storage code does not exist.
Coordinates	45 of the 46 accessions can be mapped (98%).
Safety duplicate	20 of the 46 have one at another institution (43%).
Characterization (Block 7)	With `Use the example characterization`: 12 descriptors for the 46 accessions —six quantitative, four qualitative and two ordinal—, with about 8% empty cells, as in a real field notebook.

THE PLANTED ERRORS

The collection deliberately contains the typical errors of a real file, so that Blocks 2 and 3 have something to find:

- a longitude that lost its minus sign, so that the point of a maize from Veracruz falls outside Mexico (MEX-ZM-0039, Tuxpeño de la costa);
- a 30 February as collecting date;
- a country written with two letters (`MX`) and an elevation of 25,000 m in the same accession;
- an invented storage code (15, which does not exist in MCPD);
- an accession without coordinates, with the site "described from memory";
- a maize with wild status collected in a granary, which also entered the genebank before it was collected;
- two pairs of accidental duplicates: the same collection entered twice, with the name in capitals or with extra spaces and the collecting number repeated.

With the collection just loaded, the validation in Block 2 reports **3 errors and 26 warnings in 21 of the 46 accessions**; Block 3 finds the point outside its country, proposes 4 corrections and brings together the two groups of duplicates.

1.8 How a piece of work proceeds

The order of the blocks is the order in which the work on a collection really happens: first the passport, then the cleaning, then the analysis of what you have and what you are missing, and finally the management of the living material and the paperwork. You do not have to go through all of it: an inventory ends at Block 3; a collecting plan needs Blocks 2 to 6; the work of the cold room, Blocks 2, 8 and 9. All the blocks read the same collection, so whatever is corrected in one is seen by the others.

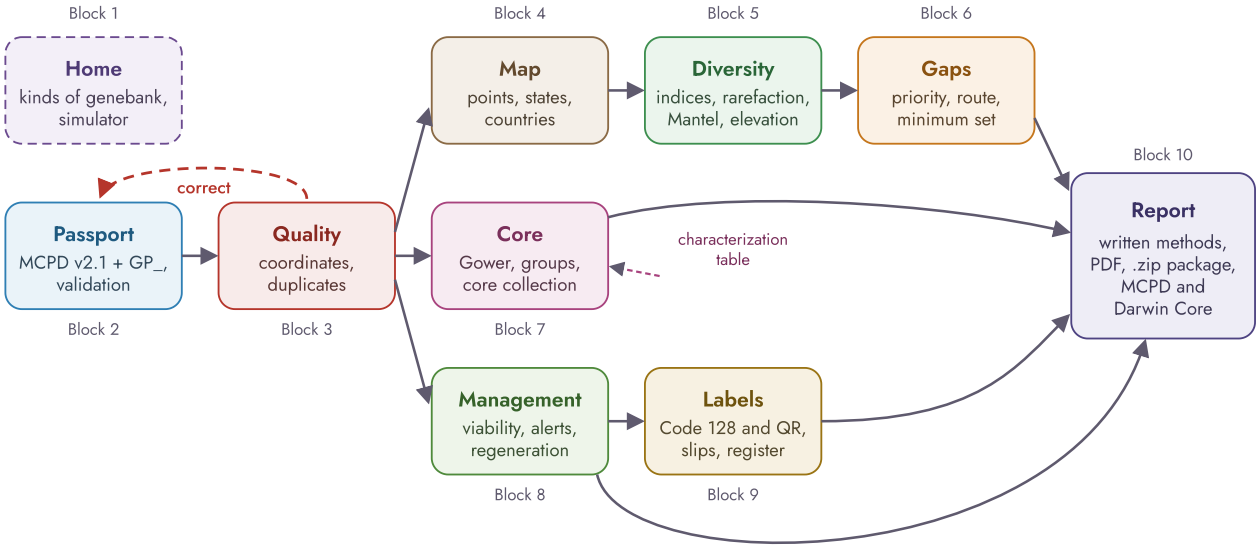


Figure I.4. The path of a piece of work through the blocks. The passport of Block 2 goes through the check of Block 3, and whatever is corrected there goes back to the passport. Three branches come out of the clean collection: the geographic one (map, diversity and gaps), the characterization branch, which also needs its own table, and the management of the living material with its labels. All of them flow into the report. Block 1 stands apart: it is for understanding the kinds of genebank before starting.

1. **Block 1.** If it is your first time, go through the kinds of genebank and try the simulator with your species: it tells you whether the route you use suits it and how long an accession stays alive with the management you give it.
2. **Block 2.** Bring in your list, confirm which descriptor each column is and review what the validation flags. Look at the portrait of the collection and how complete the passport is. Save the project.
3. **Block 3.** Run the check on the whole collection, apply the corrections that convince you, decide on each pair of possible duplicates and download the quality report. This step comes before any analysis: an unresolved duplicate counts twice, and a point in the sea falls in no region.
4. **Blocks 4, 5 and 6.** See where you collected, how much diversity there is in each region and how much is missing, and where it is worth going next time. Export the collecting route if you are going to the field.
5. **Block 7.** If you have characterization data, load them, check the type of each descriptor, read the groups and build the core collection; check with its validation that it really represents the rest.
6. **Blocks 8 and 9.** Review the assumptions of each route, deal with the alerts in the work list, record tests and regenerations, and prepare the labels, the sowing list or the delivery note for what leaves the genebank.
7. **Block 10.** Choose the sections of the report, review it as it comes out and download it, together with the package and the passport in whichever format you are asked for.

> **WHAT IS UPSTREAM RULES WHAT IS DOWNSTREAM**
>
> If you load another collection in Block 2, everything is redone and the check marks in the sidebar are cleared. If you apply a correction in Block 3, the map, the diversity and the gaps pick it up when you open them. The report of Block 10 is written with whatever exists at that moment: that is why it is best generated at the end, and generated again if something was changed.

1.9 What the program does not do

- **It is not an ecogeographic analysis.** Collecting gaps and diversity are calculated from geography and from the collection itself: accessions, species, cells, states and elevation. The app brings no climate or soil layers, so it does not say which environments are missing in an ecological sense, nor does it model the distribution of a species.
- **It neither downloads nor uploads data.** It does not query Genesys or any other portal, it does not look up names in online taxonomic catalogs and it does not georeference localities by name. It works with what you give it and delivers files that you decide where to send.
- **A viability projection is not a germination test.** The Ellis and Roberts equation estimates when the germination of a lot will drop, but it is only as good as its constants: the app brings those published for maize, and for any other species you have to enter your own before reading the years as a forecast. The decision to regenerate is taken with a real test, and the program says so.
- **It does not know your borders in detail.** Block 3 compares each coordinate with a rectangle enclosing the declared country: it catches gross errors, such as a lost sign, but it does not verify a municipality. The maps use simplified outlines, good for a collection, not for a boundary survey.
- **It does not replace your institution's information system.** It keeps the collection in the browser of one computer; it is not a database shared among several users, nor does it keep a log of who changed what. The backup is the `.json` project, and the exchange with other systems, the MCPD and Darwin Core files.
- **It does not read workbooks.** An `.xlsx` file is deliberately not opened: save it as `.csv` or copy and paste the block of cells. That is what lets the app open with a double click and work without internet.
- **It does not decide for you.** The corrections of Block 3 are proposed and not applied without your approval; duplicates are proposed with a score that you interpret; the collecting priorities and the core collection are reasoned suggestions. The app points out, explains and leaves the decision to whoever knows the collection.

1

CHAPTER ONE · BLOCK 1 · HOME

Home: kinds of genebank and the life of an accession

The first screen of the app needs no data. It presents the five kinds of genebank — four ex situ and one in situ—, says which seed behavior each one admits and brings a simulator that follows an accession for decades along the route you choose, with two clocks running: that of the material and that of its genetic diversity. It is the place to understand why a species is conserved the way it is before loading the collection.

1.1 What you see when you open the app

1.2 The kinds of genebank

1.3 Seed behavior and the species in the simulator

1.4 The simulator "the life of an accession"

1.5 The seed route and the viability equation

1.6 The other four routes

1.7 When the route does not suit the species

1.8 The rest of the home page and how to cite

1.1 What you see when you open the app

When you open index.html, Block 1 appears. At the top, in small capitals, the scope of the program —*Plant genetic resources · ex situ and in situ conservation · any species*— and, in large type, its motto: **"The genebank, accession by accession"**. The paragraph below sums up the promise: the field notebook, the passport data, the collecting map and the inventory of the collection in one place, whatever the species and whatever the way of conserving it, with no loose spreadsheets, no external portals and without the data leaving the computer.

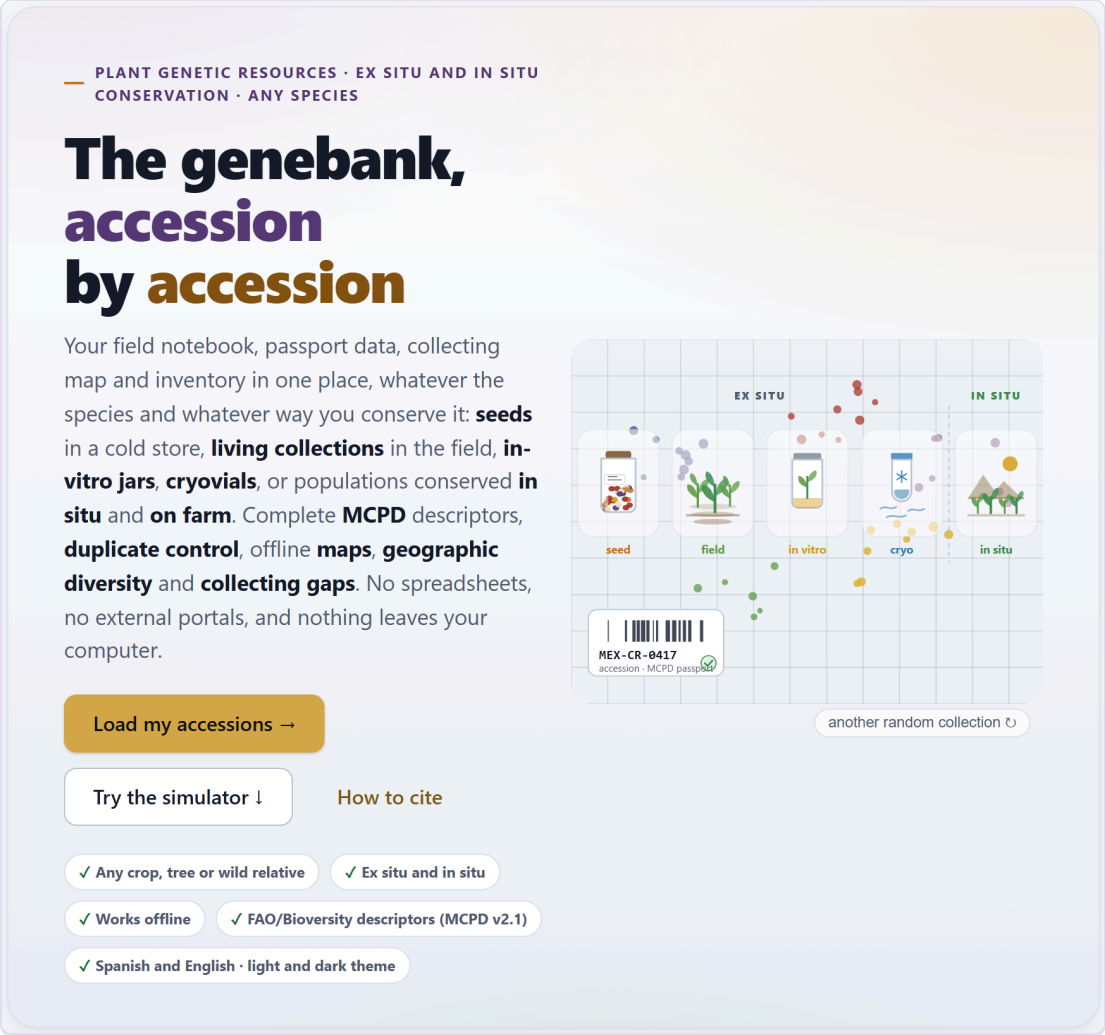


Figure 1.1. The home page. On the left, the motto, the introduction, the three buttons and the five badges; on the right, the illustration: the five conservation routes —seed, field, in vitro, cryo and in situ— over a grid with collecting points, and the label of an accession with its barcode.

Three buttons lead in. Load my accessions → takes you to Block 2, where the work with your collection begins; Try the simulator ↓ scrolls down to this block's simulator (Section 1.4), and How to cite opens the box with the citation of the program (Section 1.8). Below, five badges sum up what the program guarantees: *Any crop, tree or wild relative*; *Ex situ and in situ*; *Works offline*; *FAO/Bioversity descriptors (MCPD v2.1)*, and *Spanish and English · light and dark theme*.

The illustration is drawn by the program itself. The four ex situ routes are to the left of a dotted line and the in situ route, to the right; the collecting points cluster in five groups, as collections really cluster along roads and valleys. The label at the bottom (*MEX-CR-0417 · accession · MCPD passport*) is decorative: it does not correspond to any accession of the example collection. The another random collection ↻ button redraws the scene with other points.

> **BLOCK 1 DOES NOT READ YOUR COLLECTION**
>
> It is the only block that does not use the passport. Nothing you do here is saved, changes the collection or goes into the report of Block 10. That is why it is best used for thinking: change one control at a time and see what happens to the accession.

1.2 The kinds of genebank

The section There is no single kind of genebank starts with the essentials: a seed bank only works for species whose seed tolerates drying and freezing. Trees with recalcitrant seed and everything propagated by cutting, tuber or offset —potato, cassava, agave, vanilla, chayote— are conserved alive, in jars or in liquid nitrogen, and none of that replaces in situ conservation. Five cards present the routes; each one carries its *Ex situ* or *In situ* badge.

Card	What it is	Its weak point
Seed bank ex situ	Dry seed in sealed jars, in a cold room. Cheap, compact and safe.	It only lasts if someone watches germination and regenerates in time.
Field genebank or living collection ex situ	Living plants in a plot, an orchard or an arboretum. Essential for trees and recalcitrant species.	Expensive and exposed to the weather, to pests and to budget cuts.
Slow-growth in vitro culture ex situ	Plantlets or shoot tips in jars, with growth slowed down to subculture as rarely as possible. It saves space and cleans the material.	Every subculture costs labor and risks contamination and somaclonal variation.
Cryopreservation ex situ	Shoot tips, embryos or pollen in liquid nitrogen at −196 °C. The material stops aging and there are no more regeneration cycles.	All the risk lies at the entry (how much regrows after thawing) and in never running out of nitrogen.
Genetic reserve and on-farm conservation in situ	The population stays alive in its own site —a reserve of crop wild relatives or the field of whoever sows it each year— and keeps evolving.	Control is lost: there is selection, gene flow from neighboring populations and the risk that the site disappears.

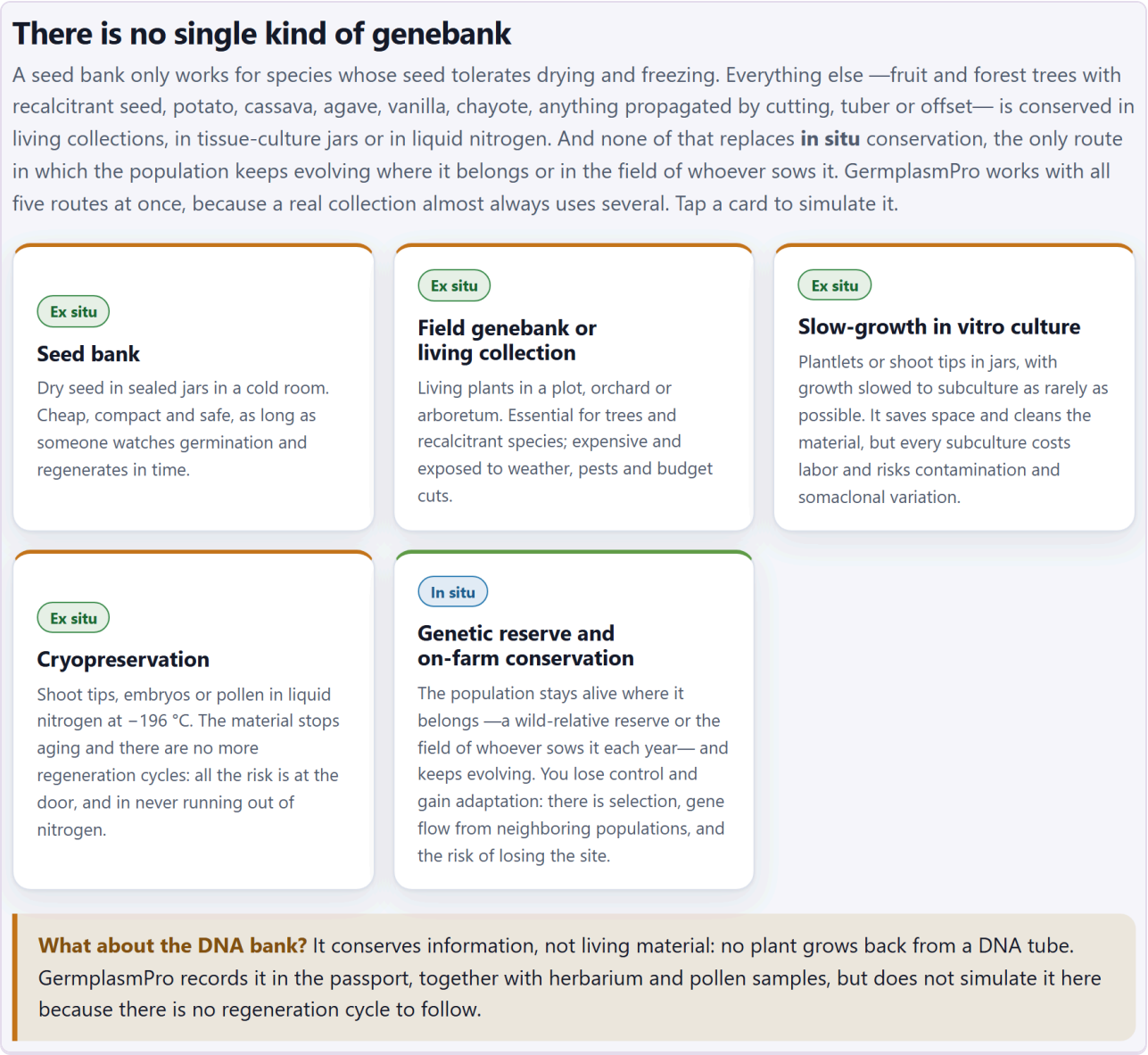


Figure 1.2. The five cards of the kinds of genebank and the box on the DNA bank. A click on a card opens the simulator with that route.

The two faces of in situ —the genetic reserve of crop wild relatives and on-farm conservation, in the fields of custodian families— share a card and a model in the simulator, because they are driven by the same dynamics: a population that reproduces on its own, with drift, gene flow and the risk of losing the site. In the passport, on the other hand, they are recorded separately (Chapter 2), and that is why the portrait of the example collection counts six routes (Figure I.3).

When you click a card, the app picks the first ready-made scenario for that route (Table 1.3), loads it into the simulator and scrolls down to it: *Seed bank* opens the FAO standard with maize; *Field genebank*, chayote; *In vitro culture* and *Cryopreservation*, potato, and *In situ*, the maize landrace in the farmer's field.

WHAT ABOUT THE DNA BANK?

Below the cards, a box makes it clear: the DNA bank conserves information, not living material —no plant grows from a tube of DNA—. GermplasmPro records it in the passport, together with herbarium and pollen samples, but does not simulate it, because there is no regeneration cycle to follow. That is why the introduction speaks of "five kinds of genebank": the sixth exists, but only as a record.

1.3 Seed behavior and the species in the simulator

Which route suits a species depends, first of all, on how its seed behaves when dried and cooled, and on whether the variety to be conserved reproduces by seed or is a clone. The app distinguishes four behaviors, with the same sentence it shows in the simulator's verdict:

Behavior	What the app says
Orthodox	Tolerates drying to 3–7% moisture and freezing: it is stored in jars at −18 °C.
Intermediate	Tolerates drying but not deep cold: it loses viability within a few years at −18 °C.
Recalcitrant	Tolerates neither desiccation nor freezing: a seed bank is not an option.
Vegetatively propagated	Its seed does not reproduce the genotype to be conserved, because the variety is a clone.

With these four behaviors the app decides, for each pair of species and route, whether the route **suits it**, whether it is **possible with caveats** or whether it **does not suit it**. Table 1.1 gathers those decisions; they are what produce the verdicts of Section 1.7.

Table 1.1. Which route each behavior admits, according to the app. ✓ suits it · ! possible, with caveats · ✗ does not suit it.

Behavior	Seed	Field	In vitro	Cryo	In situ
Orthodox	✓	✓	!	✓	✓
Intermediate	!	✓	✓	✓	✓
Recalcitrant	✗	✓	✓	✓	✓
Vegetatively propagated	✗	✓	✓	✓	✓

The two caveats have their reasons. Intermediate seed can be dried, but in a store at −18 °C its useful life is measured in years, not centuries, and it has to be checked very often. And for a species with orthodox seed the in vitro jar works, but it is an unnecessary expense: the seed bank does the same job for a fraction of the cost.

The species in the simulator

The Species selector has fifteen entries: fourteen species that illustrate the four behaviors and a generic entry for yours. It is not a closed catalog of what the app can handle —the passport accepts any taxon—, but real cases so that the simulator has something to work with. Each species carries a note that appears below the results.

Table 1.2. The species in the simulator, their behavior and what their note says.

Species	Seed	The app's note, in brief
Other species (choose its behavior)	whichever you choose	Declare how your species behaves and, if it goes to a seed bank, enter its viability constants before deciding anything with the years that come out.
Maize (*Zea mays*)	orthodox	Published constants. Outcrossing: regenerate with at least 100 plants and controlled pollination.
Common bean (*Phaseolus vulgaris*)	orthodox	Self-pollinating: fewer plants are enough, but it is best to harvest plant by plant.
Squash (*Cucurbita pepo*)	orthodox	Outcrossing, with specialist pollinators: isolation or hand pollination.
Chili pepper (*Capsicum annuum*)	orthodox	Self-pollinating with appreciable natural outcrossing: regenerate under insect-proof mesh.
Wheat (*Triticum aestivum*)	orthodox	The easiest case for a seed bank.
Coffee (*Coffea arabica*)	intermediate	Freezing kills it: it is conserved in the field, in vitro or cryopreserved.
Avocado (*Persea americana*)	recalcitrant	Living field collection; elite germplasm as grafted clones.
Cacao (*Theobroma cacao*)	recalcitrant	Field collections backed up in vitro and by cryopreserved shoot tips.
Chayote (*Sechium edule*)	recalcitrant	It germinates inside the fruit: field genebank or in vitro, with periodic replanting.
Oak (*Quercus* spp.)	recalcitrant	Conservation starts in situ, in seed stands and reserves.
Potato (*Solanum tuberosum*)	vegetative	The variety is a clone: in vitro and cryopreserved, with a field backup.
Cassava (*Manihot esculenta*)	vegetative	Propagated by stem cuttings: field and in vitro, with cryo as backup.
Agave (*Agave* spp.)	vegetative	Cultivars kept as offsets; many wild populations only in situ.
Vanilla (*Vanilla planifolia*)	vegetative	Narrow genetic base: in vitro, shaded field plots and in situ conservation of its relatives.

Only with Other species (choose its behavior) does the Its seed is selector appear, with the four options above; for the other species the behavior is given. Outcrossing and self-pollinating species are not distinguished in the calculation of drift —in all those that reproduce sexually two copies per plant are drawn—; the difference stays in the note, as management advice for regeneration.

1.4 The simulator "the life of an accession"

The section Try it: the life of an accession, along the route you choose follows an accession for decades with two clocks running, the same as in the fourth idea of the introduction (I.3): that of the **material** —the germination of the lot, the living plants of the plot, the jars in the laboratory, the explants that regrow after thawing or the size of the population in its site— and that of **genetic integrity**, because each bottleneck redraws the allele frequencies and takes the rare ones away.

Figure 1.3. The whole simulator, with the scenario Seed · community seed bank, no cold room (20 °C, 12 %). From top to bottom: the selectors, the verdict, the two charts, the route's sliders, the result tiles and the reading in words.

The controls

Control	What it is for
Scenario	Loads in one go one of the fourteen ready-made scenarios (Table 1.3): route, species and slider values.
Conservation method	The route, in two groups: *Ex situ* (the first four) and *In situ*. Changing it changes the sliders.
Species	One of the fifteen entries of Table 1.2.
Its seed is	Only with *Other species*: the behavior of your seed.
K_E and C_W	Only in the seed bank and with a species whose viability constants are editable (all seeded species except maize). See Section 1.5.
Another random run ↻	Repeats the simulation with another seed of the random number generator: the regenerations, the tests, the weather hits and the drift change; the parameters do not.

Below the charts are the **route's sliders**, different for each one (Sections 1.5 and 1.6); all of them share the last one, Simulation horizon (years), from 30 to 200 years. Each movement runs the simulation again instantly.

The ready-made scenarios

Table 1.3. The fourteen scenarios of the Scenario selector and what each one gives with the initial run (the one shown when the app opens, before pressing Another random run ↻).

Scenario	Alleles	H_e	F	And also
Seed · long-term genebank (FAO standard: −18 °C, 6 %)	100.0 %	100.0 %	0.0 %	p_{50} 883 years; no regeneration
Seed · medium-term cold room (5 °C, 8 %)	98.3 %	96.2 %	1.9 %	wheat; p_{50} 117 years; 3 regenerations in 200 years
Seed · community seed bank, no cold room (20 °C, 12 %)	69.4 %	85.0 %	16.8 %	p_{50} 11 years; 15 regenerations
Seed · the same community bank with no germination test for 15 years	59.4 %	79.2 %	20.9 %	6 regenerations with a mean N_e of 14
Seed · good cold room, but regenerated with 25 plants	80.0 %	91.8 %	7.6 %	3 regenerations with a mean N_e of 19
Field · chayote, recalcitrant, replanted every 3 years	53.3 %	73.9 %	33.6 %	33 replantings, 3 hits
Field · cacao tree collection, renewed every 25 years	46.7 %	74.3 %	28.7 %	4 replantings, 3 hits
In vitro · potato under slow growth, subcultured every 12 months	100.0 %	100.0 %	0.0 %	50 subcultures, 1,000 jars; fidelity 77.8%
In vitro · the same potato, but subcultured every 3 months	100.0 %	100.0 %	0.0 %	200 subcultures, 4,000 jars; fidelity 36.7%
Cryo · potato shoot tips in liquid nitrogen	100.0 %	100.0 %	0.0 %	100 of 200 explants recovered
Cryo · the same potato, but with only 30 explants frozen	100.0 %	100.0 %	0.0 %	12 of 30 recovered
In situ · maize landrace in the farmer's field (on-farm)	100.0 %	97.1 %	4.4 %	mean N_e 136; 3 hits
In situ · the same field, isolated and short of seed (no gene flow)	17.2 %	2.3 %	98.8 %	mean N_e 17; 2 hits
In situ · oaks in a crop-wild-relative reserve	100.0 %	99.7 %	0.3 %	mean N_e 777; 1 hit

The scenarios come in pairs on purpose: the same material with only one thing changed. Compare the community bank that tests every 3 years with the one that lets 15 go by; the good cold room that regenerates with 100 plants with the one that regenerates with 25; the potato subcultured every year with the one subcultured every three months; the 200 explants with the 30; the connected field with the isolated one. In each pair, what is lost is explained not by time but by management. In the potato pairs, since potato is a clone, no alleles are lost: what management changes is the fidelity, the labor and how much material is left so as not to lose the whole accession.

What the charts and the tiles show

- **The state of the material** (left). On the vertical axis, what is measured on each route: the germination of the lot (%), the living plants as a percentage of the target, genetic fidelity (% of lines without change), the recoverable material (%) or the size of the population as a percentage of the initial census. The chart title changes with the route and says what is in it. For seed, field and in situ, the dotted vertical lines are the cycles —regenerations, replantings, generations—; they are omitted when there are more than 60, as in the hundred in situ generations. In in vitro and cryo there are no lines: the title says only The state of the material. In the seed bank, in addition, the dots are the germination tests and the dashed red line, the regeneration threshold.
- **What is left of the original diversity** (right). Three curves as a percentage of the initial value: alleles retained (in in vitro, clonal variants retained), expected diversity (the expected heterozygosity, H_e/H_{e0}) and accumulated inbreeding (F), dashed.
- **The tiles.** Always the three for diversity — Alleles retained, Diversity retained and Inbreeding — plus those specific to each route, explained in Sections 1.5 and 1.6.
- **The reading in words.** A paragraph that puts the figures into a sentence and says what they mean; it adds warnings when the management calls for them and ends with the species note. If the accession was lost, it says so in red, with the year.

THEORY IN BRIEF · DRIFT IN EACH CYCLE

The simulated accession starts with 30 loci of six alleles each, with frequencies ranging from common to rare (about 0.45, 0.22, 0.15, 0.09, 0.06 and 0.03, with a small random variation between loci). Each time it passes through a bottleneck, the app redraws the frequencies with $\mathbf{2N_e}$ copies, two for each plant, jar or explant that passes on to the next cycle. In a **clonal** accession —potato, cassava, agave, vanilla or an *Other species* declared vegetatively propagated— there is nothing to draw: cuttings, jars and cryovials are copies of the same genotype, and in the field, in vitro and in cryo its alleles, its diversity and its F stay at 100, 100 and 0%. The only thing that puts a clone at risk is running out of material. An allele with a frequency of 0.03 has a good chance of not turning up in a small draw, and once lost it does not come back: that is why alleles are lost earlier and faster than expected diversity, which gives little weight to the rare ones. Inbreeding grows with each cycle:

$$F_t = 1 - (1 - F_{t-1})(1 - 1/(2N_e))$$

Hence the genebank rule: **few cycles, each with many individuals**. In situ the same happens in every generation, but gene flow from neighboring populations brings back part of what was lost (Section 1.6).

1

1.5 The seed route and the viability equation

In the seed bank, the clock of the material is the **Ellis and Roberts viability equation** (1980): from the store temperature and the seed moisture it gives how long the lot takes to lose germination. The simulator applies it year by year to a lot that enters with 96% germination and 2,500 seeds, and combines it with the genebank's germination tests and regenerations.

$$v = K_i - p/\sigma \cdot \log_{10} \sigma = K_E - C_W \log_{10} m - C_H t - C_Q t^2 \cdot \text{germination} = 100\ \Phi(v)$$

THEORY IN BRIEF · THE VIABILITY EQUATION

The germination of a lot does not fall in a straight line: the seeds die over time following a normal distribution, which is why the equation is written in **probits**, units of normal deviation. **v** is the viability in probits after **p** days; K_i, the initial viability of the lot in probits (96% is equivalent to 1.7507); **σ**, the days it takes viability to fall by one probit. The expected germination is Φ(v), the cumulative normal probability.

σ depends on the seed moisture **m** (% fresh weight) and on the temperature **t** (°C). K_E and C_W are specific to each species; C_H = 0.0329 and C_Q = 0.000478 describe the effect of temperature and hold for many orthodox species, so the app keeps them fixed. Since the moisture term is a logarithm, lowering moisture from 12 to 6% multiplies σ by $2^{2.115} \approx 4.3$; lowering the temperature from 20 to −18 °C multiplies it by about 19, and both together, by 84: from 6 to 505 years per probit.

The sliders and the genebank rules

Slider	Range	What it controls
Store temperature (°C)	−20 to 25	t in the equation.
Seed moisture content (%)	3 to 14	m in the equation, in steps of half a point.
Regeneration threshold (% germination)	50 to 95	If a test gives less, the lot is regenerated. It is the red line on the chart.
Years between checks	1 to 30	How often a germination test is done, with 100 seeds.
Plants used to regenerate	10 to 300	The seeds sown at each regeneration.

Each test uses 100 seeds and its result carries the sampling error of a real count (that is why the dots do not fall exactly on the curve). The lot is regenerated when a test falls below the threshold or when the stock drops to 400 seeds. At regeneration, the live fraction of what is sown germinates at random; 90% of the plants that emerge leave offspring —that is the N_e of the cycle—, and the new lot goes back to 96% germination with as many seeds as those plants yield. If fewer than two emerge, the lot is lost.

The seven tiles of this route are Half-viability period (the years to 50% germination, p50; above 999 it is written *>999*), Regenerations within the horizon, the three for diversity, Mean effective size of the regenerations and Seed spent on tests. The reading adds two warnings: if checks are every 10 years or more, and if regeneration uses 30 plants or fewer.

WORKED EXAMPLE · THE EQUATION BY HAND

The scenario of the community seed bank without a cold room: maize at 20 °C with 12% moisture, constants $K_E = 6.474$, $C_W = 2.115$.

$\log_{10} 12 = 1.07918$	$C_W \log_{10} m = 2.115 \times 1.07918 = 2.28247$
$C_H\ t = 0.0329 \times 20 = 0.65800$	$C_Q\ t^2 = 0.000478 \times 400 = 0.19120$
$\log_{10} \sigma = 6.474 - 2.28247 - 0.65800 - 0.19120 = 3.34233 \rightarrow \sigma = 10^{3.34233} = 2{,}199.5$ days = 6.02 years per probit	
$K_i = \Phi^{-1}(0.96) = 1.7507$ · at 50%, $v = 0 \rightarrow p_{50} = 1.7507 \times 6.02 = 10.54$ years	
at 85%, $v = \Phi^{-1}(0.85) = 1.0364 \rightarrow p = (1.7507 - 1.0364) \times 6.02 = 4.30$ years	
after 3 years: $v = 1.7507 - 3/6.02 = 1.2525 \rightarrow 89.5\%$ · after 6 years: $v = 0.7544 \rightarrow 77.5\%$	

The app writes *σ = 6 years per probit* and a half-viability period of **11** years (10.54 rounded). The lot crosses the 85% threshold at 4.3 years, but with tests every 3 years the year-3 test still gives about 89%, and it is the year-6 test that detects it, with the lot already at 77.5%: that is why the first of the 15 regenerations falls in year 6. With the same calculation, the FAO standard (−18 °C, 6%) gives $\log_{10} \sigma = 6.474 - 1.6458 + 0.5922 - 0.1549 = 5.2655$, σ = 504.6 years per probit and p50 = 1.7507 × 504.6 = **883** years, the figure in the tile; in 100 years the lot only drops to 94%.

The viability constants

Only **maize** comes with published constants ($K_E = 6.474$, $C_W = 2.115$), and with it the K_E and C_W boxes do not appear. For the other species that can go to a seed bank —common bean, squash, chili pepper, wheat, coffee and *Other species*— the boxes appear next to the species selector and start with the maize values, except coffee, which starts with $K_E = 5.2$ to reflect its short life. In other words: the wheat in the medium-term scenario ages like a maize until you give it its own constants.

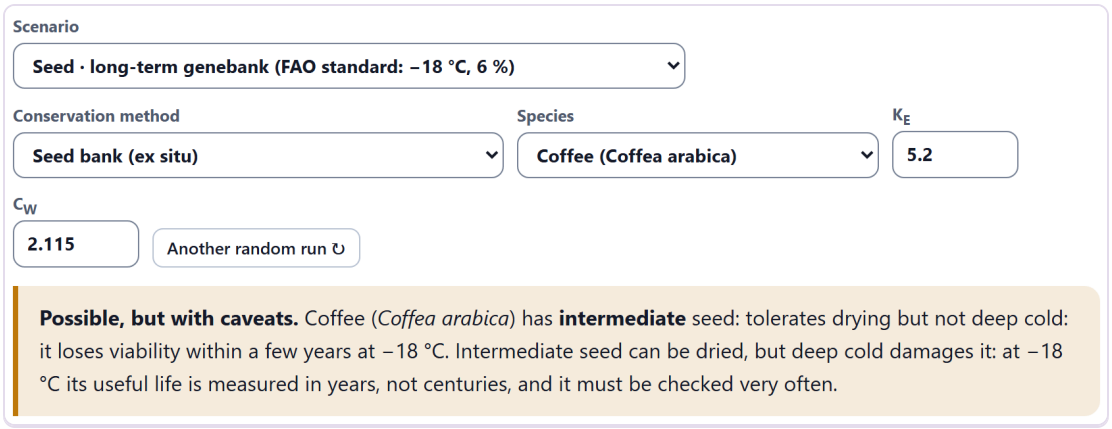


Figure 1.4. Coffee in the seed bank: the verdict Possible, but with caveats. and the K_E and C_W boxes, which start at 5.2 and 2.115. With those values, at −18 °C and 6% the lot has a half-viability period of 47 years.

1

! WITHOUT YOUR OWN CONSTANTS, THE YEARS ARE AN ILLUSTRATION

The What runs where section of the home page warns about it: for any species other than maize, K_E and C_W are a starting point. Take them from Kew's Seed Information Database or from the literature on your species before using the simulator's years to schedule a regeneration. What does hold without them is the shape of the responses: how much a lot gains by lowering moisture or temperature, and how much it costs to check late or to regenerate with few plants.

1.6 The other four routes

On the routes without stored seed there is no viability equation: the clock of the material is a different one on each. What does not change is the genetic clock, which runs with the same drift as in the box of Section 1.4.

Field genebank or living collection

A living collection does not age in a jar: it ages in the plot. Every year there is a probability of a hit —drought, pest or hurricane— that kills between 30 and 70% of the plants; every so many years the collection is renewed, and at that replanting 75% of the remaining plants leave offspring. Afterwards the plot returns to its number of plants. Replanting from seed redraws the genes of those plants; in a clone, which is replanted from cuttings, tubers or offsets, replanting copies the same genotype and takes no alleles away: only the hits that can leave the plot without plants count.

Slider	Range	What it controls
Years between replantings	1 to 30	How often the plot is renewed.
Plants kept in the collection	10 to 300	The target number of living plants.
Annual risk of drought, pest or hurricane (%)	0 to 15	The probability of a hit each year.

The tiles specific to this route are Hits to the collection, Replantings and Mean effective size. The left chart shows the living plants as a percentage of the target: each drop is a hit, and the line returns to 100 at the next replanting.

The reading puts it bluntly: here the enemy is not storage time, but how often the plot is renewed, how big it is and luck. The cacao scenario shows it from the other side: renewing every 25 years seems cheap, but three hits before the first renewal (in years 12, 18 and 21) leave only 2 of the 40 trees to renew it in year 24: in that single step the collection drops to 46.7% of its alleles, and the three following renewals, with 30 trees, no longer take any away. That is why the app insists that a field genebank always needs a backup: in vitro, cryopreservation or a second site.

Figure 1.5. Chayote in the field, replanted every 3 years with 60 plants. In 100 years, 33 replantings and 3 hits; the collection keeps 53.3% of its alleles and inbreeding reaches 33.6%.

Slow-growth in vitro culture

Here the clock of the material is **genetic fidelity**. Each subculture leaves behind part of the true-to-type material through somaclonal variation (the app assumes 0.5% per pass, a fixed value that has no slider) and loses jars to contamination or poor rooting; those that survive are multiplied until the number of lines is restored. With 50 passes, fidelity ends at 0.995^{50} = 77.8%; with 200, at 36.7%. In a clone such as potato, losing jars does not change the accession's alleles —all the lines are the same genotype—, so the diversity tiles stay at 100% and inbreeding at 0%. If the species reproduces sexually (a maize or a coffee in a jar), each pass does redraw the genes of the surviving lines, two copies per line, and diversity drops: maize with 20 lines and yearly subculture keeps 19.4% of its alleles in 100 years.

Slider	Range	What it controls
Months between subcultures	2 to 24	How often the material is moved to fresh medium.
Jars (lines) per accession	5 to 60	How many lines are maintained.
Jars lost per subculture (%)	0 to 20	The loss to contamination at each pass.

The tiles specific to this route are Subcultures, Genetic fidelity and Jars handled, which measures the laboratory's work: 1,000 jars in 50 years with yearly subculture and 20 lines, 4,000 if subcultured every three months. The reading recalls that lengthening the interval —with lower temperature, less light or osmotic media— is the only way to bring that cost down, and that the definitive backup is cryopreservation.

Cryopreservation

At −196 °C the material does not age, so the only bottleneck is the **entry**: of the explants that are frozen, only some regrow after thawing, and those are the accession that remains. If the material is a clone, that bottleneck takes no alleles away: the explants are copies of one genotype and it only decides whether there is material left to regrow. If they are embryos or pollen of a sexually reproducing species, on the other hand, the draw happens just once, in year 0, and inbreeding stays at 1/(2N), with N the explants recovered. Afterwards, the only thing that can happen is that the tank or the nitrogen supply fails, and then everything is lost.

Slider	Range	What it controls
Explants frozen	10 to 500	How many shoot tips or embryos go into the nitrogen.
Recovery after thawing (%)	5 to 95	What fraction regrows.
Tank failure risk (per thousand years)	0 to 20	The annual probability, per thousand, of losing the tank.

The tiles specific to this route are Explants recovered, Regeneration cycles (always 0) and Tank failures. The chart of the material is a flat line at the height of the recovery, which falls to zero if the tank fails. The two potato scenarios keep 100% of the alleles —it is a clone—, but they are not worth the same: with 200 explants and 50% recovery, 100 plants are left to regrow; with 30 and 40%, only 12. The app's reading puts it this way: all the risk is at the entry and in the nitrogen supply, so it pays to freeze enough explants and to work on the protocol until recovery goes up.

In situ: genetic reserve or on-farm conservation

The population lives in its site and reproduces on its own, one generation per year. In each generation there is drift with the effective size (a fraction of the census), and then gene flow arrives from neighboring populations, bringing back the frequencies of the regional gene pool. Every year there is also the risk of a hit to the site —drought, fire, land-use change, the variety no longer being sown— that takes between 40 and 100% of the population; if it survives, it recovers little by little, 15% per year, up to its usual census.

$$F_t = (1 - m)^2 \left[F_{t-1}(1 - 1/(2N_e)) + 1/(2N_e)\right]$$

With migration m, the two genes of an individual come from resident parents with probability $(1 - m)^2$, so inbreeding does not grow without limit: it settles at an equilibrium between drift and migration. Without gene flow (m = 0), the formula goes back to the one in the box of Section 1.4.

Figure 1.6. The isolated field: 80 plants, N_e of 25% of the census and no gene flow. In 100 generations it keeps 17.2% of its alleles and inbreeding reaches 98.8%. With 500 plants and 5% gene flow (the scenario of the connected field) it keeps 100% of its alleles and inbreeding stays at 4.4%.

Slider	Range	What it controls
Individuals in the population	20 to 3,000	The usual census.
Effective size (Ne as % of the census)	5 to 100	What part of the census really leaves offspring.
Gene flow from neighboring populations (%)	0 to 30	m, per generation.
Annual risk of losing the site (%)	0 to 10	The probability of a hit each year.

The tiles specific to this route are Generations, Hits to the site, Mean effective size and Final population. The reading changes depending on whether there is gene flow —conserving in situ means conserving a network of populations, not a single plot— and always closes with the same idea: it is the only route in which the accession keeps evolving and keeps its associated knowledge; what is gained in adaptation is lost in control, and that is why in situ and ex situ are complementary, not alternatives.

WHAT THE SIMULATOR SIMPLIFIES

It is a model for understanding, not for forecasting a particular accession: the initial diversity is that of 30 imaginary loci, the initial germination of the lot (96%) and its initial stock are fixed, and there is neither natural selection nor mutation. Its figures serve to compare management options with one another. Decisions about real lots are made in Block 8, with your tests and your regenerations.

1.7 When the route does not suit the species

Above the charts, a box always says what type of seed the chosen species has and whether the route suits it. It takes three forms, which come from Table 1.1:

- **Green**: *"… The chosen method suits it."* The simulation runs normally.
- **Orange,** Possible, but with caveats. The simulation runs, and the box explains the caveat: intermediate seed in a cold room (Figure 1.4) or an orthodox species in an in vitro jar.
- **Orange,** This route does not suit this species. The app **does not invent a simulation**: no viability curve comes out of a recalcitrant seed in a cold room. The left chart says No simulation: this material is not conserved this way., the sliders and tiles disappear, and the right chart offers, under Try instead:, one button for each route that does suit it. A click on any of them loads it and runs the simulation.

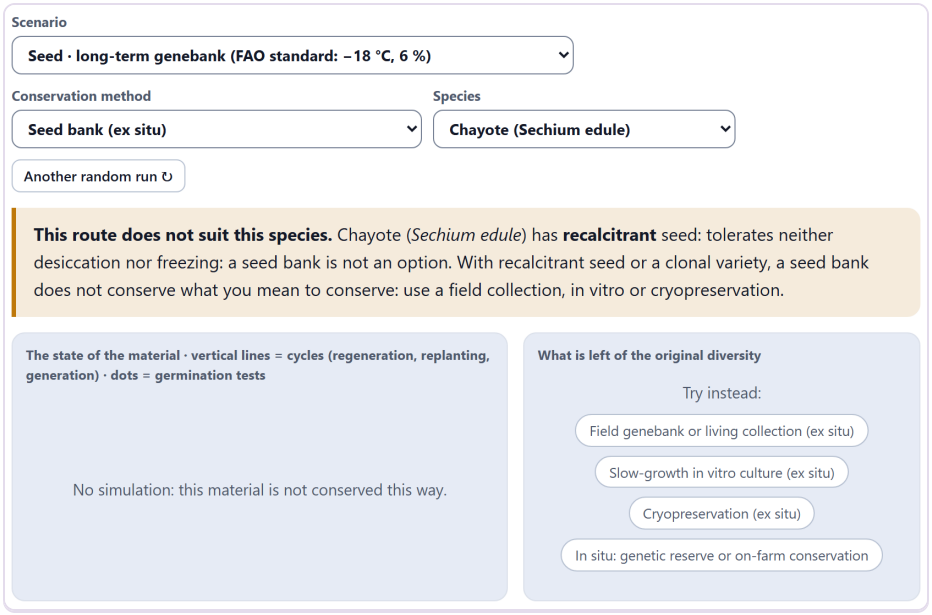


Figure 1.7. Chayote in the seed bank. The warning explains why —recalcitrant seed: a seed bank is not an option— and proposes the four routes that do suit it: field, in vitro, cryopreservation and in situ.

Only the seed bank can be blocked: with recalcitrant seed or with a clonal variety, the app replies that the seed bank does not conserve what you mean to conserve and proposes field, in vitro or cryopreservation. This happens with avocado, cacao, chayote and oak (recalcitrant), with potato, cassava, agave and vanilla (vegetative), and with *Other species* when you choose recalcitrant or vegetatively propagated in Its seed is.

1.8 The rest of the home page and how to cite

The life of an accession and the ten blocks

Between the cards and the simulator, the strip The life of an accession sums up in seven steps the order in which the work on a collection happens: *Collect* (site, elevation, informant), *Register* (MCPD passport), *Clean* (errors and duplicates), *Map* (maps and grids), *Analyze* (diversity and gaps), *Conserve* (whichever route each species needs) and *Deliver* (labels and report). They are Blocks 2 to 10 grouped by what they solve; Figure I.4 of the introduction draws the same path block by block.

After the simulator, the section The ten blocks recalls that all ten are ready —you can enter any of them from there or from the sidebar— and brings nine cards, from Block 2 to Block 10, with the name of the block and a line on what it does; a click takes you to the block. The same list, with the question each block answers, is in Section I.2.

What runs where, and where the methods come from

The What runs where card repeats what you already know from the introduction —everything is computed in the browser, nothing is uploaded, the maps work offline, and the app opens with a double click or with Open GermplasmPro.bat— and adds two boxes: the warning about the viability constants from Section 1.5, and an idea that sums up the program: the same accession written in three ways, a coordinate with the sign flipped, a lot that has gone twelve years without a germination test or an in situ plot nobody visits cannot be seen in a sheet with a thousand rows.

The list Where the methods come from gives the six sources of the block: the Ellis and Roberts equation (1980); FAO's genebank standards (2014) for seed, field, in vitro and cryopreservation; the MCPD v2.1 descriptors (Alercia, Diulgheroff and Mackay, 2015); the review by Engels and Ebert (2021) of the global ex situ conservation system; the book by Maxted, Dulloo and Ford-Lloyd (2016) on in situ and on-farm conservation, and Kew's Seed Information Database, for the viability constants and the behavior of each species. Appendix E gives them in full.

How to cite

Barrera-Guzmán, L.Á. (2026). GermplasmPro: the genebank, accession by accession (version 1.0.0) [software]. https://github.com/luisangelbg/GermplasmPro

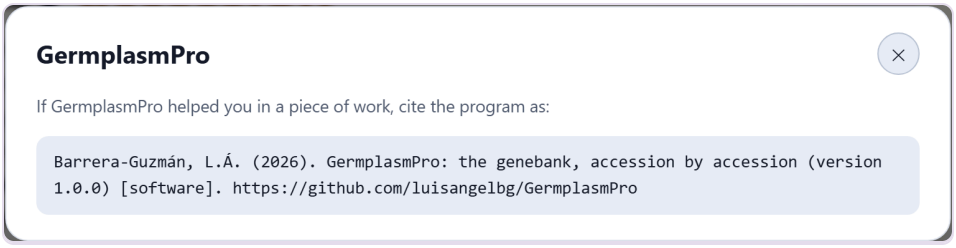


Figure 1.8. The box opened by the How to cite button on the home page.

It is the same citation as in the credits of this manual and in the CITATION.cff file in the program folder. The program has no digital object identifier: the citation carries the address of the repository. The box closes with its × button or with the Esc key.

When something goes wrong

What you see	Why it happens and what to do
No simulation: this material is not conserved this way.	The route does not suit the species (Section 1.7). Press one of the buttons under Try instead:.
The K_E and C_W boxes do not appear	They only appear in the seed bank and with species whose constants are editable; with maize they do not, because its constants are the published ones.
Its seed is does not appear	It only appears with Other species (choose its behavior).
The half-viability period says *>999*	The lot takes more than 999 years to reach 50%: the store is very cold or the seed very dry. It is a result, not an error.
The figures do not match those in this manual	Another random run ↻ was pressed or a slider was moved. Choose the scenario again or reload the page: the initial run is always the same.
The chart of the material has no vertical lines	In in vitro and in cryo the chart never has them. For seed, field and in situ, there are more than 60 cycles within the horizon (as in situ, one generation per year) and the app omits them so that the curve can be read.

DECISION RULE · WHAT TO DO IN BLOCK 1

If you do not know which route suits a species	**then** choose it in Species (or use *Other species* and declare its seed) and try the five routes: the verdict tells you which ones suit it.
If you run a seed bank without a cold room	**then** compare the two community seed bank scenarios: checking often matters as much as temperature.
If you are going to use the seed bank years for planning	**then** first enter the K_E and C_W constants of your species; only those of maize come published.
If a figure changes a lot with Another random run ↻	**then** the result depends on luck (hits, few individuals): run it several times and look at the range, not at one number.
If you already have your list of accessions	**then** Load my accessions → and go on to Chapter 2.

Nothing done in Block 1 affects the collection: the simulator keeps no state and does not go into the report.

2

CHAPTER TWO · BLOCK 2 · PASSPORT

The passport of the accessions

Block 2 is the gateway to the collection. Your list of accessions arrives here — from a file, from a spreadsheet paste or typed into a template—; the app recognizes which descriptor of the MCPD v2.1 standard each column is, checks every value on its own and against the others, and lets you correct it in a table or in the record of each accession. It ends with the portrait of the collection, the completeness of each descriptor and the files for taking the passport with you. Everything the following blocks do comes from here.

2.1 The MCPD v2.1 passport and the GP_ extension

2.2 Ways to bring in the data

2.3 Pasting from a spreadsheet

2.4 The guided template

2.5 What is each column?

2.6 Validation: errors and warnings

2.7 The table and the record of each accession

2.8 The portrait of the collection and completeness

2.9 Taking your data with you and automatic saving

2.1 The MCPD v2.1 passport and the GP_ extension

The first idea of the introduction (I.3) sums it up: the accession is the unit and its passport is its identity. GermplasmPro writes that passport with the **FAO/Bioversity Multi-Crop Passport Descriptors, version 2.1** (MCPD; Alercia, Diulgheroff and Mackay, 2015), the form that genebanks and information systems around the world understand. Each descriptor has a fixed upper-case key —`ACCENUMB`, `DECLATITUDE`, `SAMPSTAT`— and the app always shows it next to its full name, so that nobody has to memorize abbreviations.

The app recognizes the **42 descriptors** of the standard and sorts them into seven groups, the same ones that organize the record of each accession, the selector of the column mapping and the catalog of the template. Three are **mandatory** —the holding institute code, the accession number and the genus— and twelve more are **strongly recommended**: without them the accession exists, but it is worth less. To those 42 it adds seven fields of its own, the **GermplasmPro extension**, whose keys start with `GP_`.

Table 2.1. The groups of passport descriptors. Mandatory: `INSTCODE`, `ACCENUMB` and `GENUS`. Strongly recommended: the twelve in the last column.

Group	Descriptors	The strongly recommended ones in the group
Identification	6	— (two of the mandatory ones are here)
Taxonomy	6	`SPECIES`, `CROPNAME` (and the mandatory `GENUS`)
Collecting	7	`COLLDATE`, `COLLSRC`
Site and coordinates	10	`ORIGCTY`, `COLLSITE`, `DECLATITUDE`, `DECLONGITUDE`, `ELEVATION`
Origin and status	5	`SAMPSTAT`, `ACQDATE`
Donor and safety duplicates	5	—
Conservation and availability	3	`STORAGE`
GermplasmPro extension	7	— (outside the standard)

The 42 descriptors of the standard

Table 2.2 is a reference for deciding which descriptor each column of your sheet corresponds to. The name is the one shown in the app; the example is the one the app proposes in the record and in the template. The full descriptions are in the app itself: in the record, next to each box, and in the catalog of the template (Section 2.4). That is why this reference is here and not in an appendix: it is what you need at hand while you confirm the mapping.

Table 2.2. The MCPD v2.1 descriptors, by group. * mandatory · ° strongly recommended.

Key	Name in the app	What it holds · example
Identification		
`PUID`	Persistent unique identifier	An identifier that never changes and works outside your genebank · `doi:10.18730/ABC12`

Table 2.2. The MCPD v2.1 descriptors, by group. * mandatory · ° strongly recommended. *(continued)*

Key	Name in the app	What it holds · example
`INSTCODE` *	Holding institute code	WIEWS code: three country letters and three digits · `MEX006`
`ACCENUMB` *	Accession number	Unique in your genebank; never repeated or reused · `MEX-CR-0417`
`ACCENAME`	Accession name	Local or cultivar name; several, separated by semicolons · `Cónico norteño; Maíz azul`
`ACCEURL`	Accession URL	Its page in your institution's public catalog
`OTHERNUMB`	Other identification numbers	The numbers it has in other genebanks, separated by semicolons · `CIMMYTMA 2317; PI 484512`
Taxonomy		
`GENUS` *	Genus	Capitalized, not abbreviated · `Zea`
`SPECIES` °	Species	The epithet only, in lower case, or "sp." · `mays`
`SPAUTHOR`	Species authority	`L.`
`SUBTAXA`	Subtaxa	With its rank abbreviation · `subsp. mexicana`
`SUBTAUTHOR`	Subtaxa authority	`(Schrad.) H.H.Iltis`
`CROPNAME` °	Common crop name	`Maíz`
Collecting		
`COLLNUMB`	Collecting number	The number in the field notebook · `LAB-2019-033`
`COLLCODE`	Collecting institute code	WIEWS code · `MEX006`
`COLLNAME`	Collecting institute name	Its name, when there is no code
`COLLINSTADDRESS`	Collecting institute address	Postal address
`COLLMISSID`	Collecting mission identifier	Groups the accessions of the same trip · `SIERRA-2019`
`COLLDATE` °	Collecting date	YYYYMMDD, with hyphens where unknown · `20190812`, `2019----`
`COLLSRC` °	Collecting source	Code from the list: wild habitat, field, farm store, market… · `26`
Site and coordinates		
`ORIGCTY` °	Country of origin	ISO 3166-1 alpha-3 · `MEX`
`COLLSITE` °	Collecting site	From general to particular · `Oaxaca, Santiago Apoala, 3 km al norte`
`DECLATITUDE` °	Latitude (decimal)	Signed, negative to the south · `17.6423`
`DECLONGITUDE` °	Longitude (decimal)	Signed, negative to the west · `-97.3311`
`LATITUDE`	Latitude (degrees, minutes, seconds)	DDMMSSH format · `173832N`
`LONGITUDE`	Longitude (degrees, minutes, seconds)	DDDMMSSH format · `0971952W`

Table 2.2. The MCPD v2.1 descriptors, by group. * mandatory · ° strongly recommended. *(continued)*

Key	Name in the app	What it holds · example
`COORDUNCERT`	Coordinate uncertainty (m)	Radius in meters within which the real site lies · `250`
`COORDDATUM`	Coordinate datum	`WGS84`
`GEOREFMETH`	Georeferencing method	GPS, from a map, from a gazetteer or from the description · `GPS`
`ELEVATION` °	Elevation (m)	Meters above sea level · `2180`
Origin and status		
`SAMPSTAT` °	Biological status	Code from the list: wild, landrace, improved… · `300`
`ANCEST`	Ancestral data	Pedigree of breeding material
`BREDCODE`	Breeding institute code	WIEWS code
`BREDNAME`	Breeding institute name	Its name, when there is no code
`ACQDATE` °	Acquisition date	YYYYMMDD; it is not the collecting date · `20191104`
Donor and safety duplicates		
`DONORCODE`	Donor institute code	WIEWS code
`DONORNAME`	Donor institute name	Its name, when there is no code
`DONORNUMB`	Donor accession number	The number it had in the donor's collection
`DUPLSITE`	Safety duplicate site	WIEWS code of the genebank holding the backup; several, separated by semicolons · `NOR051`
`DUPLINSTNAME`	Safety duplicate institute name	Its name, when there is no code
Conservation and availability		
`STORAGE` °	Type of germplasm storage	One or several codes separated by semicolons: 10–13 seed, 20 field, 30 in vitro, 40 cryo, 50 DNA, 99 other · `13;40`
`MLSSTAT`	MLS status	0 not included, 1 included, 99 other · `1`
`REMARKS`	Remarks	Whatever does not fit in another field, prefixed with the key it refers to

Four descriptors take their value from a closed list of codes: `SAMPSTAT` (21 codes, from 100 wild to 999 other), `COLLSRC` (22, from 10 wild habitat to 99 other), `STORAGE` (9) and `MLSSTAT` (3); `GEOREFMETH` accepts four fixed texts. You do not need to learn them: in the table and in the record, typing in those boxes brings up the list with the meaning of each code, and under the value you can read its name (*Traditional cultivar/landrace* under *300*). The country list is another matter: the app bundles 119 ISO alpha-3 codes in common use in germplasm work, not the 249 entries of the standard, and that is why a well-formed code that is not in it only raises a warning.

The GermplasmPro extension

MCPD is a standard for **ex situ** collections: a population kept in its reserve or in the field of the farmer who sows it has no storage type and no acquisition date, but it has to be recorded all the same. And a genebank needs to know things the standard does not ask for, such as how much seed is left or when it was last tested. That is what the seven `GP_` fields are for. They are never confused with those of the standard: in the record they carry the tag `outside the standard`, in the completeness list they come last and the MCPD file does not include them (Section 2.9).

Table 2.3. The seven fields of the GermplasmPro extension and who uses them.

Key	Name in the app	What it covers
`GP_CONS`	Active conservation route	The route by which the material is managed today: `seed`, `field`, `invitro`, `cryo`, `dna`, `insitu` or `onfarm`. It includes the two in situ routes that `STORAGE` does not cover. If left blank, the app derives it from the first `STORAGE` code that has a route (13 → seed, 20 → field…); the other codes count as backups.
`GP_SITEID`	In-situ site identifier	The code of the reserve, holding or field, so that you can go back.
`GP_KEEPER`	Custodian farmer or community	Who keeps the population, with their consent.
`GP_STOCK`	Stock	How much there is today: seeds, plants, jars or cryovials, depending on the route.
`GP_GERMPCT`	Last germination (%)	The result of the last viability test, from 0 to 100.
`GP_GERMDATE`	Last test date	When that test was done, in YYYYMMDD.
`GP_REGENDATE`	Last regeneration date	When it was last regenerated, replanted or subcultured, in YYYYMMDD.

The route decides a great deal in the following blocks: which alerts Block 8 gives and, already in this one, which validations apply. The four stock and date fields are the starting point of genebank management in Block 8.

2.2 Ways to bring in the data

The first card of the block gathers five buttons. Next to them are the counter (*46 accessions in the collection*) and the time of the last save in the browser; Figure I.2 of the introduction shows that bar with the example collection loaded.

Button	What it does
`Open a file (CSV, TSV or project)`	Reads a .csv, .tsv or .txt file and moves on to the column mapping (Section 2.5). If it is a .json saved with `Save the project (.json)`, it opens it directly, without mapping.
`Paste from a spreadsheet`	Opens the paste assistant, with its two tabs: paste the table (Section 2.3) or build it column by column in the guided template (2.4).

Button	What it does
Open the example collection	Loads the 46 accessions used to illustrate this manual (Section I.7).
Start from scratch	Leaves the collection with a single empty accession and opens its record so you can fill it in (Section 2.7).
Download the template	Downloads passport-template.csv: the 49 columns, with their key as the header, and one row with the example of each descriptor.

When it reads a file, the app guesses the separator —tab, semicolon, comma or vertical bar— by choosing the one that splits every line into the same number of fields, not the most frequent one. It respects quotes, so a comma inside *"Veracruz, Huatusco"* does not split the cell, and it accepts line breaks inside a quoted cell. It removes the byte order mark that some spreadsheets put at the start and, if broken characters appear when reading it as UTF-8, reads it again as Windows-1252, the encoding spreadsheets usually use on Windows; that way accented letters arrive intact. An empty header is named *Column A*, *Column B*..., as in the sheet, and a repeated one is numbered (*Sitio (2)*) so that the column is not lost.

! BEFORE REPLACING THE COLLECTION, THE APP ASKS

Open the example collection, Start from scratch and opening a .json project replace the whole collection. If there are already accessions with any data, the app asks for confirmation and says how much will be lost: *The 46 accessions now in the collection will be lost. Continue?* If the collection is empty or has only blank rows, it does not ask. When importing a file or a paste, on the other hand, you choose in the mapping between Import and replace and Append to what is there. Empty and the × of each row also ask for confirmation. If the collection you are about to replace is yours, save the project first (Section 2.9).

i IF YOU USE THE DOWNLOADED TEMPLATE

The example row of passport-template.csv is only a format guide (`MEX-CR-0417` , `Zea` , `20190812` ...). Delete it before you bring the filled-in file back, or it will enter as one more accession. The columns you will not use can be left empty or removed: the app recognizes the remaining ones by their key.

2.3 Pasting from a spreadsheet

This is the quickest way to bring in a collection: in your spreadsheet you select the table with its headers, copy it, press Paste from a spreadsheet and paste with Ctrl + V. You do not have to aim at the box: the dialog accepts the paste anywhere. The Paste from the clipboard button does the same with one click, if the browser grants permission to read the clipboard; if it does not, the app asks you to paste with Ctrl + V. Text separated by commas, semicolons or tabs, typed or copied from anywhere else, works too.

The idea of the assistant is that nobody presses "read" blindly. As soon as the text comes in, the app reads it and shows what it understood, before importing anything:

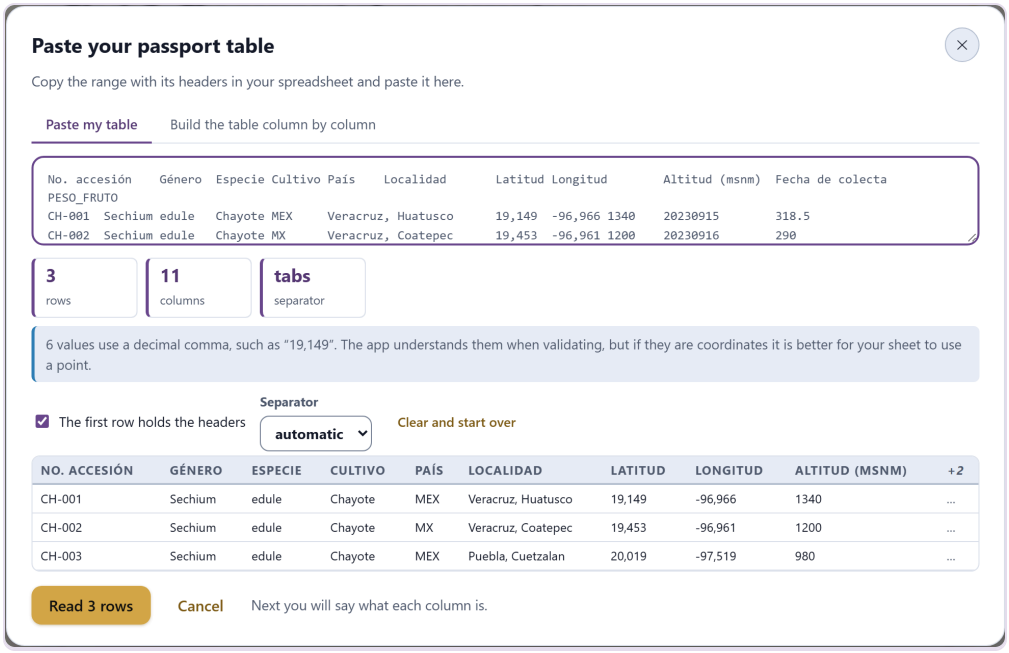


Figure 2.1. The assistant with a test paste: three chayotes copied from a spreadsheet, with the coordinates in decimal comma. At the top, the text as it arrived; then the three cards, the notice, the options and the preview already split into cells.

- **The cards.** How many data rows come out, how many columns and which separator split them.
- **The notices.** What usually goes wrong when copying from a sheet, explained in one sentence (Table 2.4).
- **The options.** The checkbox The first row holds the headers and the Separator selector (*automatic*, *tab*, *comma*, *semicolon* or *pipe*). Any change reads the text again instantly. Clear and start over empties the box.
- **The preview.** The first six rows, split into cells, with up to nine columns; the rest are summarized in a *+2* column, and below it says how many more rows there are.

The app ticks the headers checkbox the first time. To decide, it compares the first row with the ones that follow: a title does not have the shape of the data it heads —it is not a number over a column of numbers, nor a date over dates, nor a code like *MEX-001* over codes—. If the table holds only text, it checks whether the first row looks like a title: no numbers, no empty cells and no repeated texts. If it gets it wrong, simply untick or tick the box. Without headers, the columns are named *Column A*, *Column B*..., and in the mapping you choose by hand what each one is.

Table 2.4. The notices of the paste assistant. Only the last one prevents you from going on.

What the notice says	What happened and what to do
"1 row does not have the same number of cells as the rest (3)..."	Almost always a stray comma inside a text without quotes, or merged cells in the sheet. Check that row in the preview; if it is a comma, try another separator.
"Everything landed in a single column..."	If your table had more columns, the separator is not the right one: choose it in Separator. If you deliberately copied a single column, go ahead.

Table 2.4. The notices of the paste assistant. Only the last one prevents you from going on. *(continued)*

What the notice says	What happened and what to do
"6 values use a decimal comma, such as "19,149"..."	It is for information: the app turns the comma into a point when validating and importing numbers and coordinates. Even so, your sheet had better use a point.
"There is only one row and it is being taken as a header..."	You copied only the titles, or a single data row that looks like a title. Untick the headers box if it was data, or copy the whole table again. The button stays disabled until you fix it.

When everything is in order, the main button says how many rows it will read (Read 3 rows) and, when you press it, the dialog closes and the column mapping of Section 2.5 appears.

EXAMPLE · THE TEST PASTE OF THIS CHAPTER

Figures 2.1 and 2.3 use this paste, copied from a spreadsheet (the columns are separated by tabs; here they are trimmed):

```
No. accesión  Género   Especie  Cultivo  País  Localidad           Latitud  Longitud  …  Fech
CH-001        Sechium  edule    Chayote  MEX   Veracruz, Huatusco  19,149   -96,966   …  2023
CH-002        Sechium  edule    Chayote  MX    Veracruz, Coatepec  19,453   -96,961   …  2023
CH-003        Sechium  edule    Chayote  MEX   Puebla, Cuetzalan   20,019   -97,519   …  2023
```

The assistant reads **3 rows** and **11 columns** separated by **tabs**, ticks the headers box on its own and warns about the **6 values** with a decimal comma. It deliberately carries three problems that the mapping uncovers: a two-letter country, a 31 February and a characterization column. The headers are in Spanish, as they would be in a Mexican genebank's sheet; the app recognizes them all the same.

2.4 The guided template

The second tab of the assistant, Build the table column by column, is for those who do not have the table ready or do not know what to call the columns. It shows the full catalog of what the app understands, with the name of each column, its key, what it means and an example; you tick the ones you will use and a table ready to fill in appears below.

The catalog of columns

The catalog has **86 columns** in twelve groups that open and close: the eight of the passport (the 42 MCPD descriptors and the 7 of the extension) and four of **characterization**, with 37 descriptors that are not passport data: *Agronomic and plant traits* (11), *Fruit morphology* (10), *Chemical and quality traits* (12) and *Seed* (4). The latter carry their unit in brackets —*Fruit weight (g)*, *Soluble solids (°Brix)*, *Plant vigor (1–9)*— and a line saying how it is measured. Next to the name of each group, a counter says how many columns it has or, if you have already ticked some, how many of them (*2/6*). The three mandatory descriptors carry the tag mandatory.

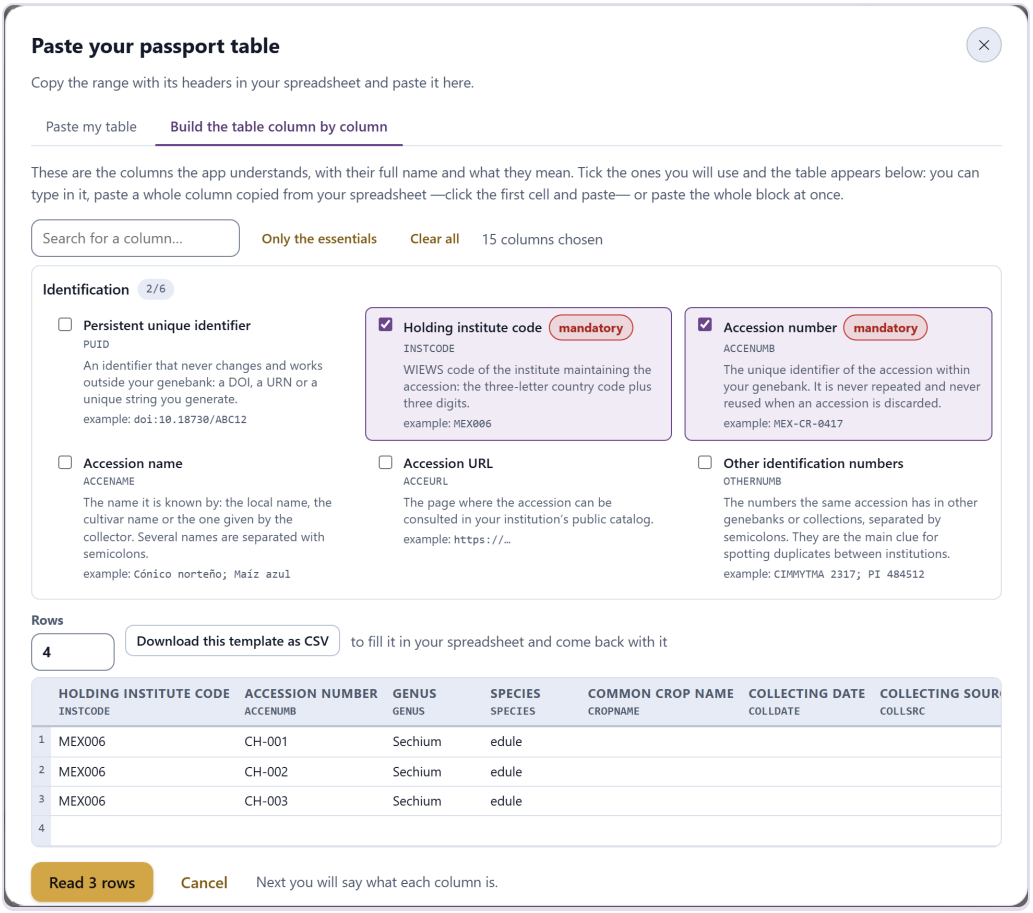


Figure 2.2. The guided template just opened: it already has the 15 essential columns ticked. At the top, the *Identification* group expanded; at the bottom, the grid with three rows typed in and the Read 3 rows button.

Control	What it is for
Search for a column...	Filters the catalog by name, by key or by what the description says, regardless of accents and case: "fruit" finds nine characterization columns. Groups with matches open by themselves.
Only the essentials	Leaves ticked only the 15 mandatory and strongly recommended columns (Table 2.1). It is the selection the tab opens with.
Clear all	Unticks everything, so you can start your own selection.
Rows	How many rows the grid has, from 1 to 500; it starts at 10.
Download this template as CSV	Downloads germplasmpro-template.csv with only the ticked columns, with the keys as headers and with whatever you have already typed into the grid. It is for filling it in in your spreadsheet and coming back with it.

The grid

Each ticked column appears in the grid with its full name and its key underneath, in the order in which you ticked it. You type in it as in a sheet. What makes it practical is pasting whole columns: copy a column of values in your spreadsheet —the accession numbers, for example—, click the first cell of the corresponding column of the grid and paste. The values are spread downwards, one per row, and if there are more of them than rows, the grid grows up to 500. If what you copied has several columns, they are also spread to the right, over the following columns. A single value, without tabs or line breaks, is pasted normally into the cell where the cursor is, just as in the search box.

Empty rows do not count: the main button says how many rows have something (`Read 3 rows`). When you press it, the grid follows the same path as a paste and reaches the column mapping, where all the keys are recognized at the first attempt.

2.5 What is each column?

Whether it comes from a file, a paste or the template, the table goes through the card `What is each column?` before entering the collection. At the top, a line says what was read (*Read 3 rows and 11 columns from pasted text. The app recognized 10 columns; check the rest.*); below it, one row per column of your file, with four items:

- `Column in your file`: the original header and, under it, its first three values, so you can tell what it is about.
- `Descriptor`: the selector with the 49 descriptors grouped as in Table 2.1, plus — *do not import* —. The app fills it with its proposal.
- `Key`: the chosen key, in green if it belongs to the standard and in another color if it belongs to the `GP_` extension.
- `How well it fits`: what would happen if you imported the column as that descriptor (explained in the box below).

How the app recognizes your columns

Real headers arrive in Spanish, in English, with accents, with full stops, with units or with the MCPD key. Before comparing them, the app normalizes them: it removes accents, case, spaces and punctuation, so that *No. accesión*, *no_accesion* and *NO ACCESIÓN* are the same thing. Then it searches in three passes, and each descriptor receives a single column:

1. **The exact key of the standard.** A column called `ACCENUMB`, `accenumb` or `GP_CONS` is recognized first.
2. **The synonyms.** The app bundles 224 common column names for the 49 descriptors, in Spanish and in English: *número de accesión*, *accession number*, *clave*; *país*, *country*, *iso3*; *altitud*, *msnm*, *elevación*; *custodio*, *agricultor*, *comunidad* for `GP_KEEPER`; *existencias*, *inventario* for `GP_STOCK`, and so on with the rest.

3 **Partial matches.** A header that contains a synonym of at least four letters is assigned to the longest match: *Altitud (msnm)* contains *altitudmsnm* and goes to `ELEVATION`; *Germinación %* goes to `GP_GERMPCT`.

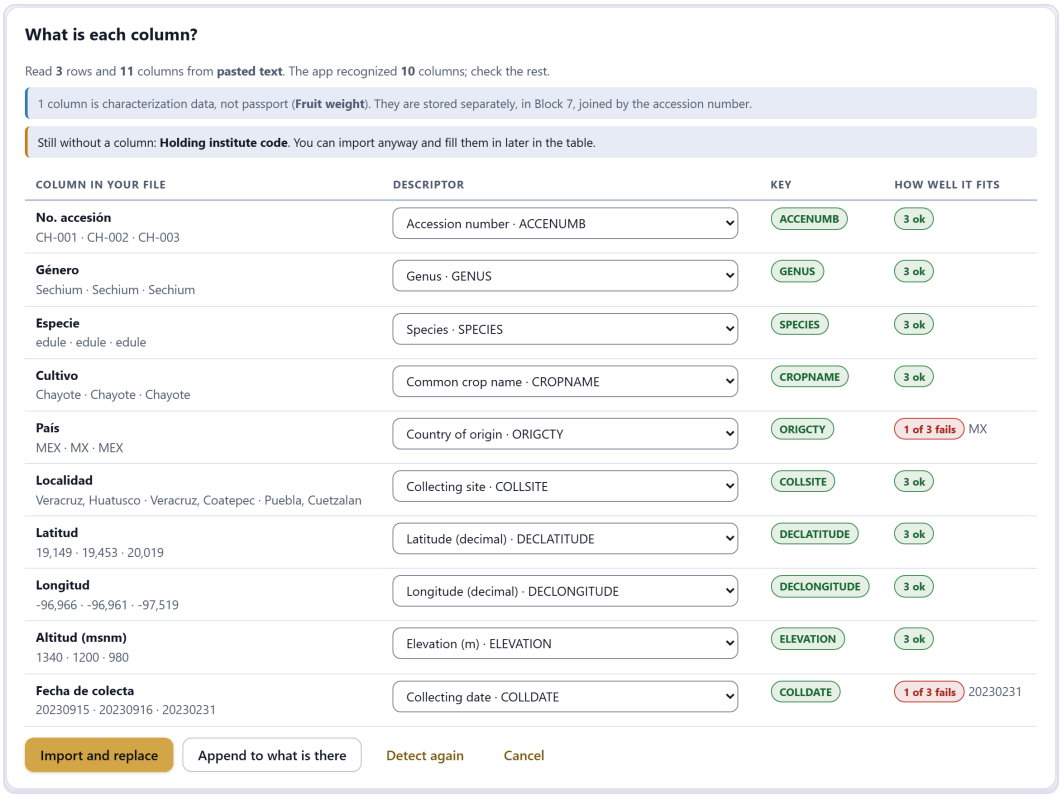


Figure 2.3. The mapping of the test paste. The ten passport columns were recognized; *PESO_FRUTO* was set aside as characterization. Under How well it fits, the country *MX* and the 31 February are visible before importing. (In the app, the list scrolls inside its frame when there are many columns.)

Coordinates take one more step, because the header is not enough: a *Latitud* column may hold *17.6423* or *173832N*. The app looks at up to ten values and, if most of them have the shape of degrees-minutes-seconds, sends it to `LATITUDE`; if they are decimals, to `DECLATITUDE` (and likewise for longitude), swapping the two columns if necessary. If only the degrees-minutes-seconds coordinates are present, the decimal ones, which the maps use, are computed on import.

Since each descriptor receives a single column, when two columns compete the first one wins: in a sheet with *Estado*, *Municipio* and *Localidad*, only one reaches `COLLSITE` and the others are left at — *do not import* —. If you choose by hand a descriptor that already had another column, the other one is unassigned automatically. Detect again undoes your changes and goes back to the app's proposal.

> **THEORY IN BRIEF · "HOW WELL IT FITS"**
>
> For each assigned column, the app validates all its non-empty values as if they already belonged to that descriptor, with the same rules as in Section 2.6. If none of them gives an error, it writes `3 ok` (with the number of values); if any fails, `1 of 3 fails` (or *2 of 3 fail*) and, next to it, the first value that fails. Only errors count, not warnings. It is the fastest way to discover a wrongly assigned column: if the elevation column ended up as collecting date, nearly all of them will fail. Better to see it here than later, with the data already inside.

What stays outside the passport

Below the summary line, two boxes may appear:

- **The characterization columns.** If a column is named exactly like a key of the characterization catalog (`PESO_FRUTO`, `BRIX`, `DIAS_FLORACION`…), the app does not look for an MCPD descriptor for it: it sets it aside, names it in the box (*1 column is characterization data, not passport (Fruit weight)*) and, on import, stores it in the characterization of Block 7, linked to its accession by the number. MCPD is an exchange standard and fields are not hung on it. A column with another name, such as *Peso del fruto*, is not recognized this way: either you rename it with the key, or you load it in Block 7.
- **Mandatory descriptors without a column.** *Still without a column: Holding institute code.* You can import anyway and fill it in later in the table; meanwhile, each accession will carry that error.

At the end, four buttons: `Import and replace` replaces the collection with what was read; `Append to what is there` adds it at the end of the current collection (without looking for repeats: Block 3 does that); `Detect again` and `Cancel`. On import, numeric values and coordinates go from decimal comma to decimal point, and countries to upper case. The imported collection is saved in the browser immediately, and the time of the save appears next to the counter (Section 2.9).

2.6 Validation: errors and warnings

Every time the collection changes, the app checks all the accessions, field by field and across fields, and gives each problem one of two levels. **Errors** are values the standard does not accept —a date that does not exist, a code that is not in the list, an empty mandatory field— and must be corrected. **Warnings** are values worth completing or confirming: an empty strongly recommended field, an odd elevation, two fields that disagree.

Table 2.5. The field-by-field validation rules.

Type of data	Error	Warning
Any empty descriptor	If it is mandatory: *A mandatory value is missing.*	If it is strongly recommended.
Dates (`COLLDATE`, `ACQDATE`, `GP_GERMDATE`, `GP_REGENDATE`)	Not in YYYYMMDD with hyphens where unknown; year before 1700 or in the future; month outside 01–12; day outside 01–31; a day that does not exist in that month (30 February).	—

Table 2.5. The field-by-field validation rules. *(continued)*

Type of data	Error	Warning
Decimal coordinates	Not a number; outside −90 to 90 or −180 to 180.	An exact zero, which is almost always a missing value written as zero.
Degrees, minutes and seconds	Does not follow DDMMSSH (*173832N*).	—
Numbers	Not a number; germination outside 0–100; negative uncertainty or stock.	Elevation outside −430 to 6,500 m.
Country (`ORIGCTY`)	Not three letters.	Three letters that are not in the list of 119 countries.
WIEWS codes (`INSTCODE`, `COLLCODE`, `BREDCODE`, `DONORCODE`, `DUPLSITE`)	—	Not three letters and three digits. `DUPLSITE` accepts several, separated by semicolons.
Code lists	A value that is not in the descriptor's list; in `STORAGE`, any code of the semicolon-separated list that does not exist.	—

Table 2.6. The rules that compare fields of the same accession.

What is compared	Level	What the app says
Only one of the two decimal coordinates	error	*The other coordinate is missing: one alone locates nothing.*
Decimal and degrees-minutes-seconds coordinates	warning	They do not match (they differ by more than 0.02°).
Collecting date later than acquisition date	warning	*The accession would have entered the genebank before being collected.*
Last test or regeneration earlier than acquisition	warning	The test (or the regeneration) predates the accession's arrival.
Wild status (100) and cultivated or store source (21, 22, 23, 26, 27)	warning	*It says wild but was collected in a cultivated site or a store.*
Landrace status (300) and wild habitat (10–15)	warning	*It says landrace but was collected in a wild habitat.*
In situ route with a storage type	warning	If you hold both, register two linked accessions.
In situ route without `GP_SITEID`	warning	*An in-situ record with no site identifier cannot be revisited.*
Ex situ route without `DUPLSITE`	warning	*No safety duplicate: the accession depends on a single facility.*
Ex situ route with `DUPLSITE` only at the holding institution itself	warning	*The safety duplicate is held by the same institution as the accession: it does not protect it if that institution loses its collection.*

The route changes what is required. An in situ record (genetic reserve or on-farm) is not asked for `STORAGE`, `ACQDATE`, `DUPLSITE` or `INSTCODE`: it has no storage, it never entered a genebank and it is not duplicated elsewhere. And the safety duplicate only counts if at least one of the sites in `DUPLSITE` is **an institution other** than the one in `INSTCODE`: a copy in the cold room next door does not protect against a fire in the same building. That same rule is the one used by the *With safety duplicate* tile of the portrait (Section 2.8). An accession without a recognizable route —neither `GP_CONS` nor a valid `STORAGE` code— does not receive the duplicate warnings.

What needs checking

The section What needs checking gathers the problems of the whole collection, grouped by descriptor and message, which is how they are fixed: errors first and, within each level, the most frequent ones. Above it, three figures: how many errors, how many warnings and in how many accessions. The show button of each row turns on the filter only those with something to check, clears the other filters, adds the problem's column to the table if it was not there and opens the record of the first affected accession.

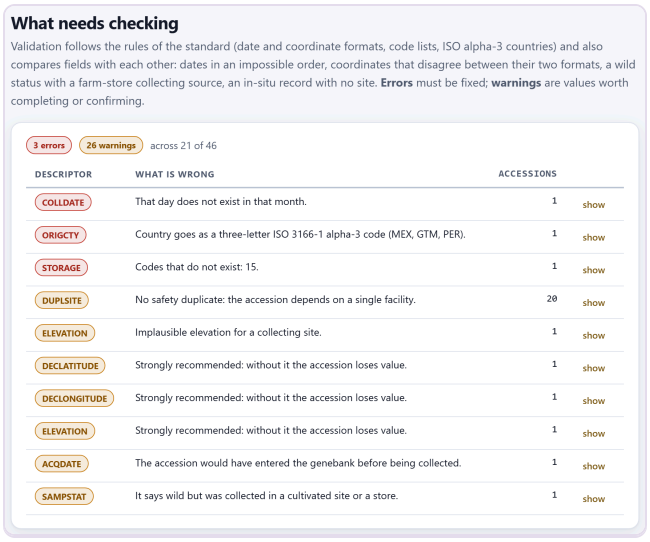


Figure 2.4. The validation of the example collection just loaded: 3 errors and 26 warnings in 21 of the 46 accessions, in ten groups.

EXAMPLE · THE 29 PROBLEMS OF THE EXAMPLE COLLECTION

The seeded errors (Section I.7) that Block 2 can see with a single accession in view are these:

MEX-CU-0040 · Calabaza pipiana	error in `COLLDATE`: *20190230*, a 30 February.
MEX-PH-0041 · Frijol tepari	error in `ORIGCTY` (*MX*, two letters) and warning in `ELEVATION` (*25000* m).
MEX-CH-0042 · Chile costeño	error in `STORAGE`: *Codes that do not exist: 15.* With no route, it is not asked for a duplicate either.
MEX-ZM-0043 · Maíz ancho de riego	three warnings: no latitude, no longitude and no elevation.
MEX-ZM-0044 · Maíz reventador	two cross-field warnings: wild status collected in a farm store, and an acquisition date (*20190105*) earlier than the collection.
20 ex situ accessions	warning in `DUPLSITE`: no safety duplicate.

They add up to 3 errors and 6 + 20 = 26 warnings, in 21 accessions (the five above, plus sixteen that only have the duplicate warning). The 20 accessions with a duplicate hold it at another institution (`NOR051`, `MEX008`, `BOL004` or `PER001`), so none of them receives the same-institution warning; Figure 2.6 triggers it on purpose. What Block 2 does **not** see: the unsigned longitude of MEX-ZM-0039 is a valid number (only Block 3 discovers that it falls outside Mexico), and the two pairs of accidental duplicates are, each one on its own, correct accessions.

2.7 The table and the record of each accession

The section The collection, accession by accession shows the collection in an editable table, 50 rows at a time, with the ← → arrows and the count (*showing 1–46 of 46*) at the foot. Above it are the search box, the filters and the buttons for adding and emptying.

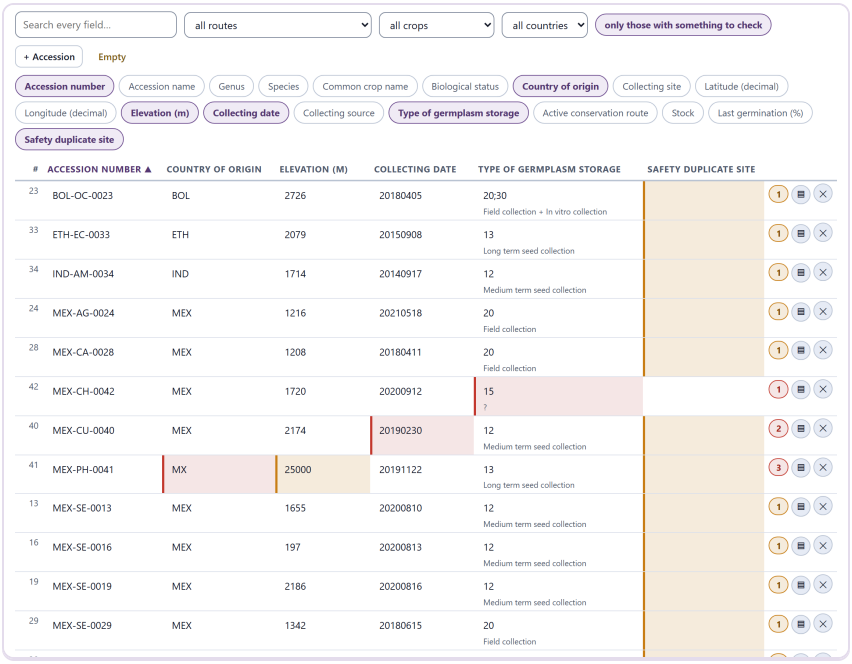


Figure 2.5. The table with only those with something to check (21 rows) and the columns chosen to see the problems: errors in red, warnings in ochre. The circle on the right counts the problems of each accession, in red if any of them is an error.

Control	What it is for
Search every field...	Searches the 49 fields at once, regardless of accents, case and hyphens: "chayote" leaves 3 rows.
all routes, all crops, all countries	Filter by conservation route (only those in use), by common crop name and by country of origin (with its name; a code that is not in the list, such as *MX*, appears as it is). With the example collection, *Field genebank or living collection* leaves 8 rows.
only those with something to check	Keeps only the accessions with some error or warning: 21 in the example collection.
+ Accession	Adds an empty accession at the end and opens its record.
Empty	Deletes the whole collection, including the copy stored in the browser, after asking for confirmation.
The 18 column tags	Choose which columns are shown; the highlighted ones are visible. At least two must remain. It opens with nine: accession number and name, genus, species, crop, country, site, elevation and route.

A click on a column title sorts by it (▲) and another reverses the order (▼); coordinates, elevation and the other numbers are sorted as numbers. The table opens sorted by accession number, and the gray number on the left is the position of the accession in the collection, which does not change when sorting or filtering. Each cell is edited by typing in it; the change is applied when you leave the cell and validation is redone at once. Code cells offer the list as you type and show the meaning underneath (a *?* if the code does not exist). Hovering over a flagged cell shows the problem's message. To the right of each row, ▤ opens the full record and ✕ deletes the accession, with confirmation.

The full record

The record shows the 49 fields of an accession, in the eight groups of Table 2.1. The title gives the accession number and the scientific name; on the right, its position (*44 / 46*) and the ← → arrows to move to the previous or next one without closing the dialog. Each line has the name of the descriptor with its key (and an asterisk if it is mandatory), the box and, on the right, the description of the descriptor or, if there is a problem, the message in bold, with the line tinted red or ochre. Empty text boxes show the descriptor's example as a guide.

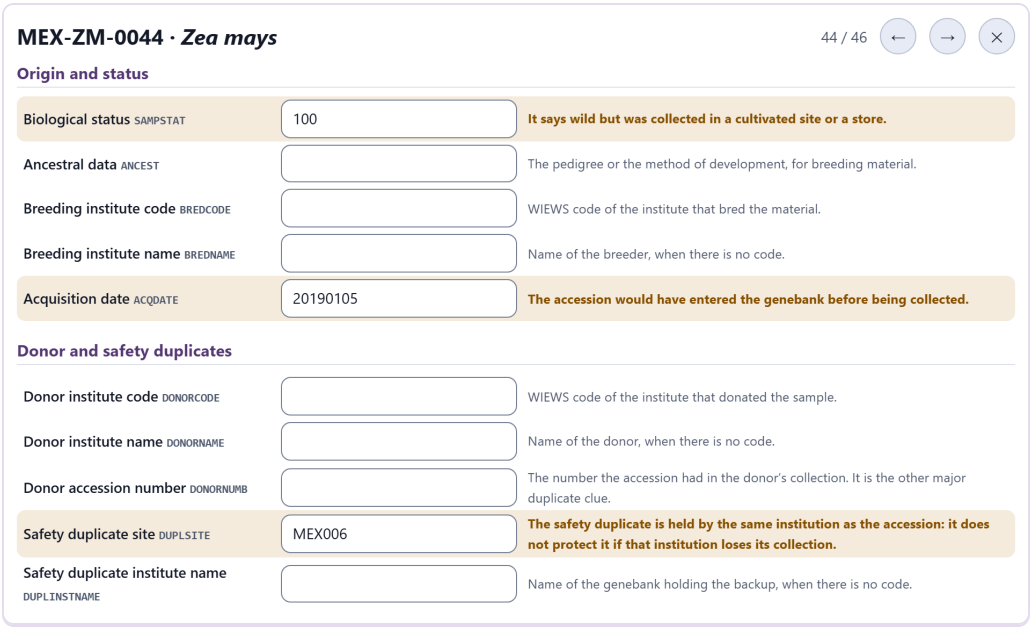


Figure 2.6. The record of MEX-ZM-0044 · Maíz reventador. Besides its two seeded warnings, *MEX006* was typed here as the duplicate site —the same code as the institute that holds it— and the warning changed from *No safety duplicate* to the same-institution one. With *NOR051* it disappears; with *MEX006;NOR051* too, because one of the two sites is another institution.

What you type in the record is applied when you leave the box, and everything is updated at once: the record, the table, the list of problems and the portrait. Start from scratch and + Accession open the record of an empty accession, which starts with 3 errors (the three mandatory fields) and 12 warnings (the strongly recommended ones); they go out as you fill it in. The record closes with ✕ or with Esc.

2.8 The portrait of the collection and completeness

The portrait of the collection begins with the five tiles already presented in the introduction (Figure I.3): accessions, with coordinates, with safety duplicate (in red while it does not reach 80%), countries and routes in use. Below them come six charts, which are redrawn with every change (Figure 2.7):

Chart	What it counts · with the example collection
Accessions by conservation route	Each route in use, declared or derived from `STORAGE`: 28 in seed, 8 in field, 3 in vitro, 3 on-farm, 2 in a genetic reserve and 1 cryopreserved. The accession without a route does not appear.
Accessions by species	The ten species with the most accessions (genus and epithet): *Zea mays* with 18, *Capsicum annuum* and *Sechium edule* with 3. All 24 are in the taxa tile.
Biological status	Up to eight statuses: 39 landraces, 4 wild, 2 natural and 1 breeder's line.
Country of origin	Up to eight countries: Mexico with 37, Guatemala and Peru with 2, and one each for Bolivia, Ethiopia, India, Colombia and *MX*.
Elevation of collecting sites (m)	Histogram in 12 classes of equal width between the minimum and the maximum: here, from 20 to 25,000 m. Empty elevations are left out; those of 0 m, at sea level, are included.
Collecting year	Histogram in 10 classes with the year of `COLLDATE`: from 2014 to 2022, with most between 2018 and 2020.

Figure 2.7. The charts of the portrait with the example collection. In the elevation chart, the accession at 25,000 m stretches the axis until nearly all the others are left in the first two bars: a data-entry error jumps out.

Long names are trimmed in the bars (*Traditional cultivar/landr…*). The charts count what is written, not what ought to be: the code *MX* appears as an eighth country until you correct it to *MEX*, and the elevation of 25,000 m distorts the histogram. That is why it pays to look at the portrait again after Block 3.

How complete the passport is

The last results section gives, for each descriptor, the percentage of accessions that carry it. The bar is green from 90%, ochre from 50 to 90% and red below 50%; an asterisk marks the mandatory ones, and hovering over a line shows the description of the descriptor. So as not to fill the list with zeros, optional descriptors that no accession carries are omitted; the mandatory ones, the strongly recommended ones and the seven of the extension always appear, the latter at the end.

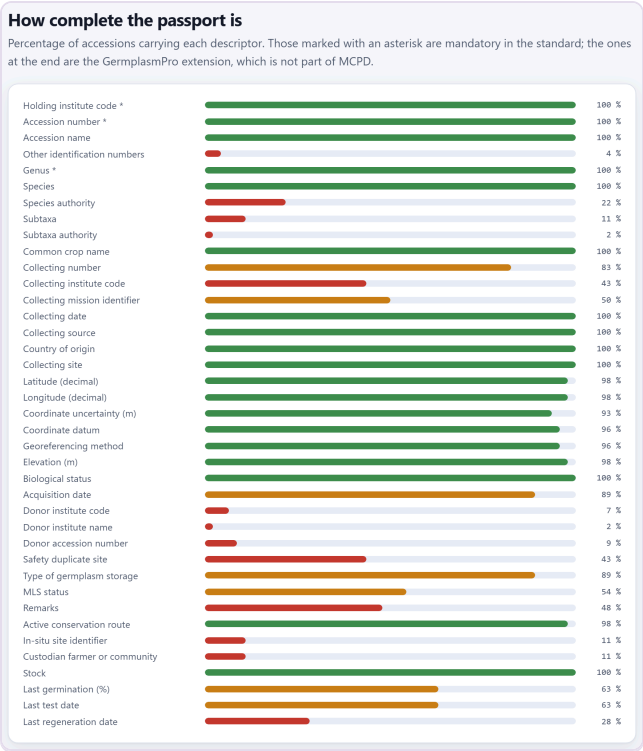


Figure 2.8. The completeness of the example collection: 39 lines; the 10 descriptors that no accession carries (such as `PUID` or `ANCEST`) are not shown.

It can be read in two ways. By descriptor, it says what is missing in the whole collection: here, the safety duplicate site (43%) is the most serious gap among those that matter for conservation, and the last regeneration date (28%) the one in the extension. And against what validation asks for: acquisition date and type of germplasm storage stand at 89% (41 of 46) and, even so, they produce no warnings, because the five accessions without them are in situ and must not have them. Completeness measures presence, not correctness: *MX* counts as a country present.

2.9 Taking your data with you and automatic saving

The section Take your data with you has three buttons, each with a different destination:

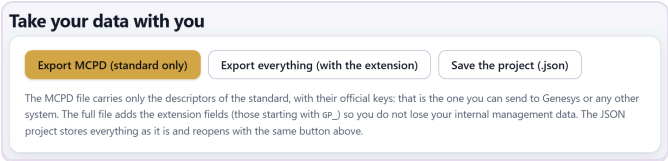


Figure 2.9. The three export buttons of Block 2.

Button	File	What it carries and what it is for
Export MCPD (standard only)	passport-mcpd.csv	The 42 descriptors of the standard, with their official keys as headers and in the order of Table 2.2. It is the one sent to Genesys, to SINAREFI or to any other system.
Export everything (with the extension)	passport-full.csv	The 49 fields: the standard and, at the end, the seven `GP_` ones, so that internal management data are not lost. It comes back in through Open a file (CSV, TSV or project) with all the columns recognized by their key.
Save the project (.json)	germplasmpro-collection.json	The accessions as they are, with all their fields; the characterization of Block 7; the threshold and the dismissed pairs of Block 3; the genebank assumptions, the custom viability constants and the calculator values of Block 8, and the label preferences of Block 9. It opens with the same button above, without going through the mapping, in any browser or computer.

The CSV files come out comma-separated and in UTF-8, and only cells that need it (those containing commas, quotes or line breaks) are quoted. The files are exported with the values as they are, errors included: fix first whatever validation flags. The project carries everything that was worked on and does not fit in the passport (version 2 of the format): when it is opened, each item is restored in its block. A project saved with an earlier version, which carried only the passport or the passport and the characterization, opens all the same and leaves as it was whatever it does not carry: the characterization, the Block 3 settings, those of the genebank and those of the labels.

Automatic saving

Every change to the collection —a cell, the record, an accession added or deleted, the example collection— is saved by itself in the browser's local storage a moment later, and *saved in this browser* · appears next to the counter with the time. When you reopen the app in that browser, the collection is still there. If the browser cannot save —full storage or a private window—, the time is replaced, in red, by *Could not save in this browser (storage full or private mode): download the .json project*: from then on your work lives only in that tab until you save the project. Even when it saves, it is not a backup: it lives in one browser on one computer and is lost if the browsing data are cleared. The backup is the .json project; save it at the end of every work session.

When something goes wrong

What you see	Why it happens and what to do
Accented letters arrive broken (*Ã©*, *�*)	The app already tries UTF-8 and Windows-1252; if they still come out wrong, save the sheet as "CSV UTF-8" or use the paste, which does not depend on the encoding.
Read 0 rows or the button disabled	You copied only the headers, or the headers box swallowed the only row (Table 2.4). In the template, no row has data.
In the mapping, a column was left at *— do not import —*	Its header does not resemble any synonym, or its descriptor was already taken by another column. Choose it by hand in the selector.
How well it fits says nearly all fail	The column is assigned to the wrong descriptor. Change the selector before importing.
The coordinates ended up as text or in another column	They came in degrees-minutes-seconds with another shape, or with a decimal comma and a comma separator. Check the separator in the assistant and the column's selector in the mapping.
Every accession has *A mandatory value is missing.* in `INSTCODE`	Your file did not carry the institute code. Add the column in your sheet and import again, or type it in the table.
A characterization column entered the passport	It is set aside only if its header is an exact key of the catalog (`PESO_FRUTO`). Set it to *— do not import —* and load it in Block 7.
The collection disappeared when the app was opened again	It was opened in another browser or with the browsing data cleared. Open the .json project.
Next to the counter it says *Could not save in this browser...*	The browser's storage is full or the window is private. Use Save the project (.json) now, and again before closing; in a normal window, with space, the notice goes back to *saved in this browser*.

DECISION RULE · WHAT TO DO IN BLOCK 2

If your collection is in a spreadsheet	**then** copy it with its headers and use Paste from a spreadsheet; check the cards and the notices before reading it.
If you do not know what to call your columns	**then** build the table in the guided template, starting with Only the essentials, or download it as CSV to fill it in in your sheet.
If in the mapping a column fails in How well it fits	**then** look at the first value that fails: if it is one, import it and correct it in the table; if it is nearly all of them, change the descriptor.
If validation gives errors	**then** correct them before going on; the safety duplicate warnings and the empty strongly recommended fields are pending work on the collection, not on data entry.
If an ex situ accession only has a copy at your own institution	**then** it has no safety duplicate: arrange one at another genebank and record it in `DUPLSITE`.
If you have finished loading and correcting	**then** save the .json project and move on to Block 3, before any analysis.

What you correct here is seen by every block; Block 3 then looks for what a single accession does not reveal: coordinates outside their country, untidy names and the same collection entered twice.

3

CHAPTER THREE · BLOCK 3 · QUALITY & DUPLICATES

Data quality and duplicates

Block 2 checks each value on its own; Block 3 checks what only shows when the whole collection is looked at: the coordinate that does not fall in the country the accession declares, the genus spelled two ways, the site with several spellings and, above all, the same collection entered twice. The app diagnoses, proposes the fix and explains why; applying it, merging two accessions or dismissing a pair is your decision.

3.1 What the block checks and how to read the summary

3.2 Does the point fall where it says? Coordinates against the country

3.3 Names and catalogs

3.4 Fixes, always with your approval

3.5 How duplicates are searched for

3.6 Groups, side-by-side comparison and the threshold

3.7 Merge, note or dismiss

3.8 The quality report

3.1 What the block checks and how to read the summary

Block 3 works on the collection you loaded in Block 2 (Chapter 2) and asks for nothing else: when you enter it, the check runs by itself, with the animated wait *Checking quality and duplicates*, and the result is laid out in five tiles and four sections. If there are no accessions yet, the block says so —Load your accessions in Block 2 first.— and offers the button Go to the passport.

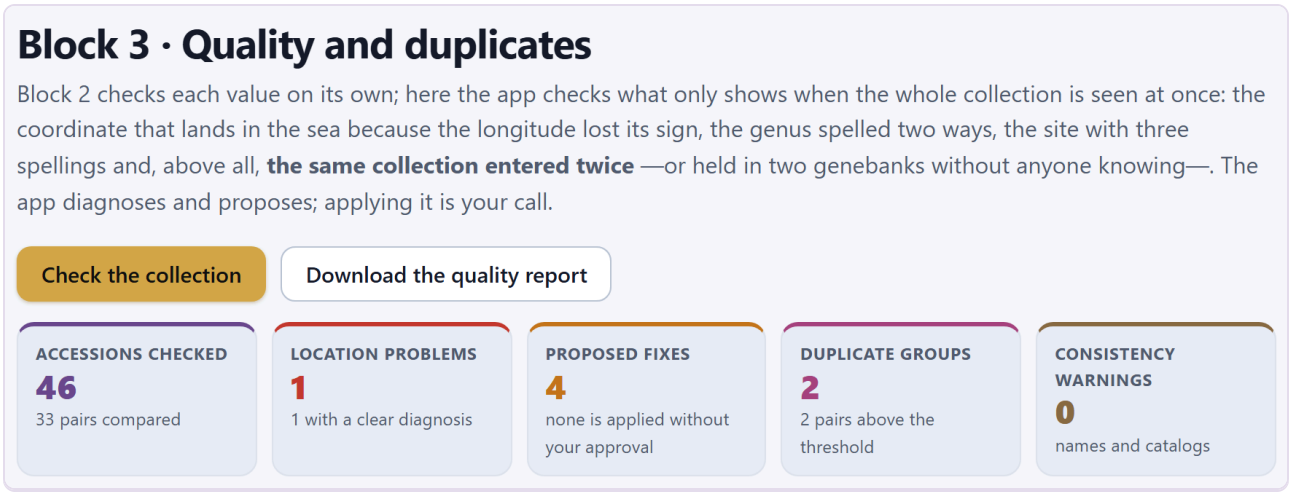


Figure 3.1. The header of the block with the example collection just loaded. The five tiles summarize the four sections below.

Two buttons govern the block. Check the collection runs the whole check again; it is needed when you changed something in Block 2 without leaving it, because on re-entering Block 3 the check is redone anyway. Download the quality report saves everything found in a `.csv` file (Section 3.8). Next to them, a green tag shows for a few seconds the result of the last thing you did: *Coordinate fixed.*, *Accessions merged.*, *Noted in both accessions.*

Table 3.1. The five summary tiles and what they show with the example collection.

Tile	Example	What it counts
Accessions checked	46	All those in the passport. Below, how many pairs were compared in the search for duplicates: 33 of the 1,035 possible (Section 3.5).
Location problems	1	The errors and warnings of section 2 (Section 3.2); below, how many of them come *with a clear diagnosis*, that is, are errors. Accessions that could not be compared with their country do not count.
Proposed fixes	4	The changes the app knows how to make and that await your approval (Section 3.4).
Duplicate groups	2	The groups of accessions that look like the same material; below, how many pairs are above the threshold (Sections 3.5 and 3.6).
Consistency warnings	0	The findings about names and catalogs (Section 3.3).

On screen, the sections come in the order in which it pays to work: `1 · Fixes the app can make for you`, `2 · Does the point fall where it says?`, `3 · Names and catalogs` and `4 · Is it the same collection twice?` First you clean up what has a safe fix, and at the end you decide about the duplicates, which is what calls for the most judgment. This chapter explains the checks first (3.2 and 3.3), because the fixes (3.4) come out of them.

> **WHICH SEEDED ERRORS EACH BLOCK CATCHES**
>
> Of the errors planted on purpose in the example collection (Section I.7), Block 3 finds the **unsigned longitude**, the **two-letter country** and the **two duplicates**. The rest —the 30 February, the elevation of 25,000 m, the invented storage code, the accession without coordinates and the dates in reverse order— are errors in an isolated value and are flagged by the validation of Block 2 (Chapter 2). They are not lost here: the quality report gathers them along with everything else.

3.2 Does the point fall where it says? Coordinates against the country

Each accession declares its country of origin in `ORIGCTY` and its site in `DECLATITUDE` and `DECLONGITUDE`. If the two do not agree, one of them is wrong, and almost always because of one of a few mechanical slips. The app compares each coordinate with a **bounding box** of the country: the rectangle, in degrees, running from its minimum to its maximum latitude and from its minimum to its maximum longitude. It bundles the boxes of **119 countries** —the whole of the Americas, Europe, a good part of Africa and Asia, Australia, New Zealand, Papua New Guinea and Fiji— and gives each one a margin of one degree per side, so as not to accuse sites on the coast or the border.

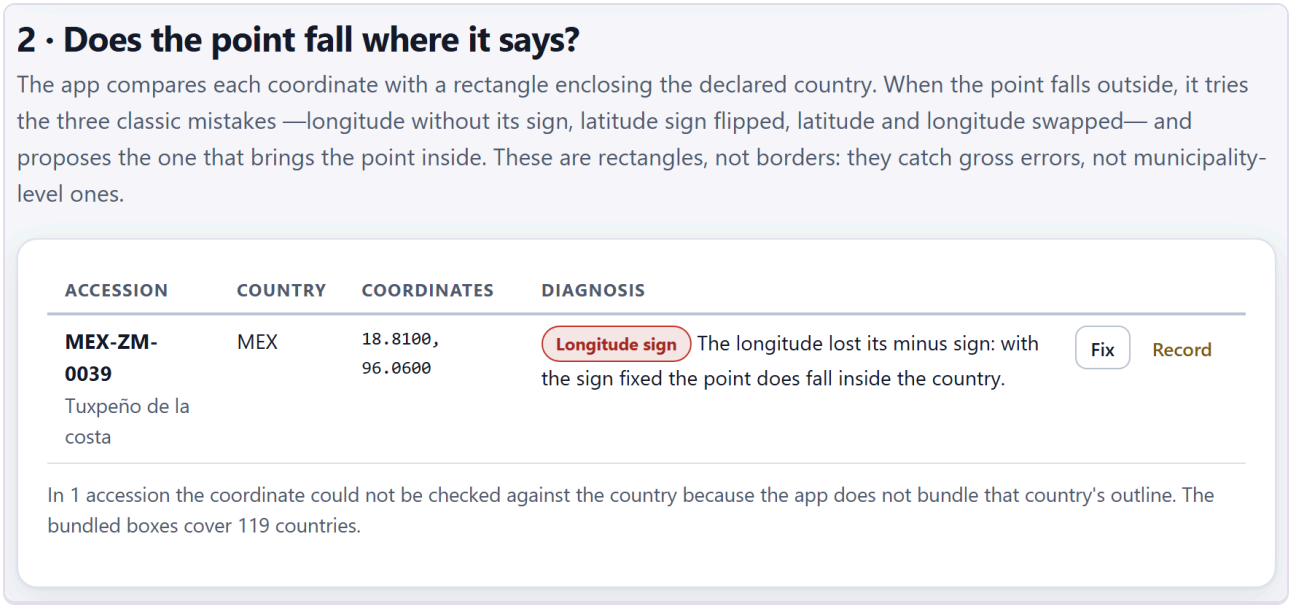


Figure 3.2. Section 2 with the example collection: the maize *Tuxpeño de la costa* declares Mexico with a positive longitude, which would place it in Myanmar. Below, the note about the accession that could not be compared.

The checks, in order

The app does not check an accession that lacks the latitude or the longitude, or that has them written as something that is not a number: Block 2 takes care of that. With both coordinates present, it checks in this order:

Tag	Level	What it detects and what the app says
Point (0, 0)	error	The point (0, 0): *is in the Gulf of Guinea: it is an empty field written as zero.* No fix is possible; the coordinate has to be looked up in the notebook.
Latitude = longitude	warning	Identical latitude and longitude: one was probably copied over the other.
Whole degrees	warning	Both are whole numbers: the site was rounded to degrees and carries more than 50 km of uncertainty.
Country without outline	note	The country is not among the 119: the coordinate is not compared. It does not count as a problem; it is summarized in a note under the table.
Longitude sign, Latitude sign, Swapped coordinates, Swapped and sign-flipped	error	The point falls outside the box and one of the classic slips brings it back inside (below). The app proposes that fix.
Outside the country	warning	The point falls outside and no simple fix repairs it: the coordinate or the country must be checked.

The classic slips and their diagnosis

When the point leaves the box, the app tries four repairs, one after another, and keeps the first one that brings it back into the country:

1. **The longitude lost its sign** (Longitude sign): the sign of the longitude is changed. It is the most common in the Americas, where all longitudes are negative and the sign gets lost when passing through a spreadsheet or when copying from a GPS that writes "W".
2. **The latitude has the wrong sign** (Latitude sign): the symmetric error, typical of countries in the southern hemisphere.
3. **Latitude and longitude swapped** (Swapped coordinates): each one ended up in the other's column.
4. **Swapped and with the sign changed** (Swapped and sign-flipped): both things at once.

The order only matters when two repairs would work, which is rare with small boxes. Each row of the table carries the number and name of the accession, the country, the coordinates as they stand and the diagnosis —a tag with the name of the problem and the explanation—, with two buttons: Fix applies the proposed fix on the spot (and the green tag says *Coordinate fixed.*), and Record opens the accession's record in Block 2 so that you can review it or correct it by hand.

WORKED EXAMPLE · THE MAIZE THAT LANDED ON THE OTHER SIDE OF THE WORLD

The box of Mexico runs from 14.3 to 32.8 in latitude and from −118.5 to −86.6 in longitude; with the one-degree margin, from 13.3 to 33.8 and from −119.5 to −85.6. *Tuxpeño de la costa* (MEX-ZM-0039) declares 18.8100, 96.0600: the latitude falls inside, but the longitude +96.06 is in the eastern hemisphere, in Myanmar. The first test, changing the sign of the longitude, gives −96.06, which does fall between −119.5 and −85.6: diagnosis Longitude sign, fix **96.0600 → −96.06**. The corrected point lies on the Gulf coastal plain, in Veracruz.

3

Below the table, the note In 1 accession the coordinate could not be checked against the country because the app does not bundle that country's o refers to the tepary bean MEX-PH-0041, which declares its country as *MX*. It is not that Mexico is missing: *MX* is the two-letter code, and the boxes use the three-letter one. As soon as you apply the fix *MX* → *MEX* (Section 3.4), the coordinate is compared, falls inside and the note disappears.

RECTANGLES, NOT BORDERS

A bounding box is generous: Mexico's includes part of Guatemala and of the Gulf, and Chile's a good part of Argentina. It serves the purpose it is used for here —catching a lost sign or two swapped columns, errors of hundreds or thousands of kilometers—, but it does not confirm that a point is in the right municipality or state. That fine check is done on the map of Block 4 (Chapter 4), which draws the 32 states of Mexico. For the same reason, a point that passes this check is not guaranteed to be right.

3.3 Names and catalogs

The section 3 · Names and catalogs looks for what is not a format error but still makes the collection untidy: things Block 2 cannot see because each value, on its own, is well written. Before comparing, the app **normalizes** the texts: it removes accents and punctuation, turns everything into lower case and collapses spaces. That is why "Maiz" and "Maíz" count as the same crop name, and in the example collection no warning comes up even though the duplicated accession writes "Maiz" without the accent.

Table 3.2. The checks on names and catalogs. All of them are warnings (ochre tag), except the site with several spellings, which is a note (blue tag).

Tag	What it looks for
Genus capitalization	A genus that is not written with an initial capital and the rest in lower case (*zea*, *ZEA*).
Epithet capitalization	An epithet with some capital letter (*Mays*).
Authority in the epithet	An epithet that carries something more: a space or an initial with a full stop, as in *mays L.* The authority belongs in `SPAUTHOR` and the infraspecific rank in `SUBTAXA`.
Misspelled genus	Two genera in the collection that are too similar to be different: at a **Levenshtein distance** of 1 or 2, with at least five letters in the longer one.

Table 3.2. The checks on names and catalogs. All of them are warnings (ochre tag), except the site with several spellings, which is a note (blue tag). *(continued)*

Tag	What it looks for
Crop names	The same taxon under several crop names. The taxon includes the subspecies or variety: teosinte (*Zea mays* subsp. *parviglumis*) does not count as another name for maize.
Site spelling	The same site written in several ways: identical after normalizing, different as they stand.
Repeated coordinates	Accessions with exactly the same coordinates (to four decimals) that declare different sites: almost always, a coordinate copied from the row above.

The Levenshtein distance is the minimum number of letters that must be inserted, deleted or changed to go from one word to another. From *Phaseolus* to *Phaseolous* there is one (the second *o* is extra); from *Capsicum* to *Capcisum*, two (two letters swapped cost two changes). The five-letter minimum prevents two short, different genera, which may differ by one or two letters, from accusing each other.

The example collection comes out clean: the section says Scientific names, crop names and sites are spelled consistently. and the warnings tile shows 0. For Figure 3.3, six values were altered by hand, as you could do in the record of Block 2, so that every type of warning shows up.

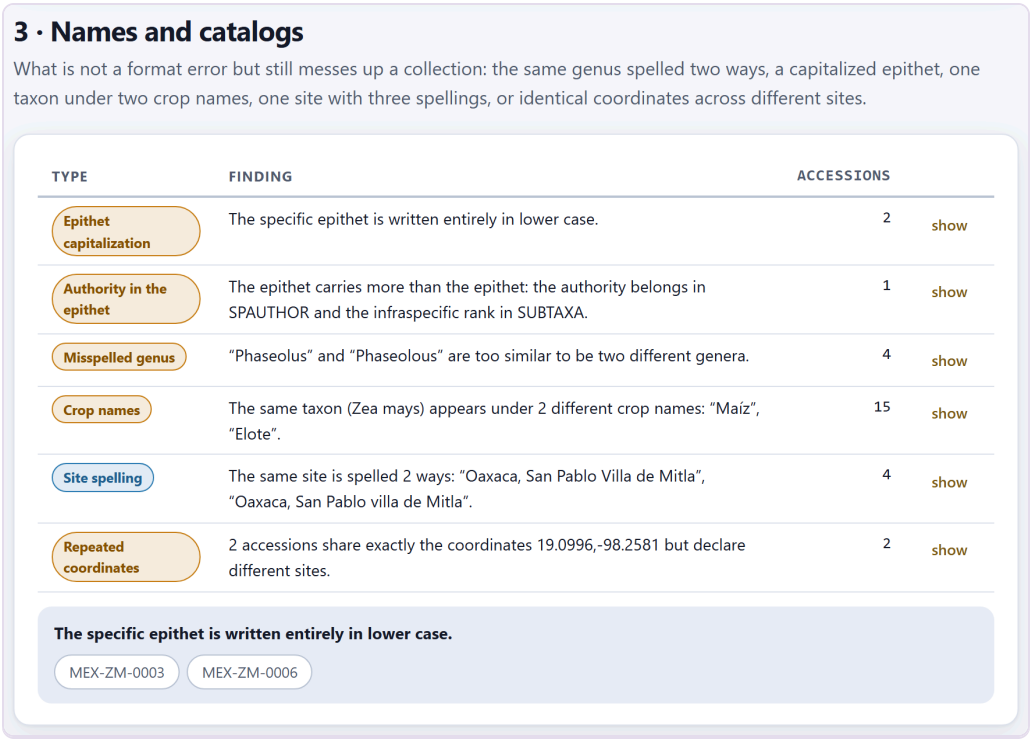


Figure 3.3. Section 3 with six values of the example collection altered for the figure: the genus of MEX-PH-0041 written *Phaseolous*; the epithet of MEX-ZM-0006 as *Mays* and that of MEX-ZM-0003 as *mays L.*; the crop of MEX-ZM-0004 as *Elote*; the site of MEX-ZM-0045 with *villa* in lower case, and the coordinate of MEX-SE-0019 copied into MEX-CH-0042. Below, the detail opened by show.

Each row gives the type, the finding and how many accessions it touches. The show button unfolds underneath the list of those accessions —up to 40— as buttons; each one opens its record in Block 2. Notice that *mays L.* comes up twice: as an epithet with a capital and as an epithet with an authority. In the crop warning, the taxon and the names appear as they are written in the passport, even though the app has already compared them in normalized form.

> **WARNING IS NOT FIXING**
>
> Of all these warnings, only the capitalization of the genus and the epithet has an automatic fix (Section 3.4). The others call for judgment: which of the two genera is the right one, whether "Elote" is an error or the local name you want to keep, which of the two accessions with the same coordinate has it right. The app takes you to the record; the decision, and the correction, are made in Block 2.

3

3.4 Fixes, always with your approval

The section 1 · Fixes the app can make for you gathers the repairs that have a single reasonable answer. None is applied by itself: each one is shown with the original value struck through and the proposed one in green, and it is applied when you say so.

1 · Fixes the app can make for you

Each one is proposed with the original and the new value in sight. They are applied only when you say so, by type or all at once.

DESCRIPTOR	WHAT IS PROPOSED	EXAMPLE	CASES	
Longitude (decimal)	**Longitude sign.** The longitude lost its minus sign: with the sign fixed the point does fall inside the country.	~~96.0600~~ → **-96.06**	1	Apply
Country of origin	**Two-letter country.** "MX" is the two-letter code; the standard asks for the three-letter one.	~~MX~~ → **MEX**	1	Apply
Accession name	**All caps.** All caps: proposed in sentence case (extra spaces removed too).	~~BOLITA AMARILLO~~ → **Bolita amarillo**	1	Apply
Accession name	**Extra spaces.** Extra spaces at the start, at the end or between words.	~~Chayote verde liso~~ → **Chayote verde liso**	1	Apply

Apply every fix (4) You can undo them by editing the accession in Block 2.

Figure 3.4. The four fixes the app proposes for the example collection, grouped by type and descriptor: on the left, the name of the descriptor; in the middle, the tag of the fix and its reason.

Table 3.3. The fixes the app knows how to propose: the tag it shows in the What is proposed column and the descriptors it touches.

Fix · descriptor	When it is proposed and what it does
The four of Section 3.2 (Longitude sign…) · decimal latitude and longitude	When the check of Section 3.2 found one of the classic slips: the sign change or the swap that brings the point back into the country.
Two-letter country · country of origin	When the country comes in two letters and the app knows it: it replaces it with the three-letter code the standard asks for. The app's table has 56 codes in common use (*MX*, *GT*, *PE*, *US*, *ES*…, and also *UK* for the United Kingdom).
Extra spaces · accession name, collecting site, common crop name	Extra spaces at the start, at the end or between words.
All caps · the same three	Only a text written **entirely** in capitals (with some word of four letters or more). The accession name and the crop name are turned into sentence case: *BOLITA AMARILLO* → *Bolita amarillo*. The site is turned into a place name, with a capital on every word except *de*, *la*, *del* and the other Spanish minor words: *SAN PABLO VILLA DE MITLA, OAXACA* → *San Pablo Villa de Mitla, Oaxaca*. It also removes the surplus spaces.
Genus capitalization, Epithet capitalization · genus, species	The genus with an initial capital; the epithet in lower case (only if it carries no authority, which has to be moved by hand).
Inferred route · active conservation route	The empty conservation route when it can be derived from the storage type (`STORAGE`).
Degrees, minutes, seconds · decimal latitude and longitude	The empty decimal coordinate when there is one in degrees, minutes and seconds (`LATITUDE`, `LONGITUDE`): it computes it, with six decimals.

Proposals are grouped by type and descriptor; each group shows its first case as an example and how many cases it has. The Apply button of a row applies that whole group (the green tag says, for example, *1 fix applied to Longitude (decimal)*.); Apply every fix (4), all of them at once. When fixes are applied, the passport is saved, the validation of Block 2 is redone and the check of Block 3 runs again. With the example collection, after applying the four, the tiles show 0 location problems and 0 fixes, and the note about the country without a box disappears; the two duplicate groups are still there, because correcting a name does not separate them.

> **BEFORE APPLYING EVERYTHING IN ONE GO**
>
> Sentence case leaves in lower case everything after the first word, proper names included: *TUXPEÑO DE VERACRUZ* would become *Tuxpeño de veracruz*. Whatever already mixes upper and lower case is left alone, so this only happens with names entered entirely in capitals. There is no undo button: the app itself reminds you, You can undo them by editing the accession in Block 2. With a large collection, apply by groups, check the examples and, if in doubt, export the passport first (Chapter 2).

3.5 How duplicates are searched for

An **accidental duplicate** is the same collection recorded twice in the catalog —when digitizing an old notebook, when receiving a safety duplicate back, when merging two inventories— or the same accession kept in two genebanks without anyone knowing (Section I.3). It is almost never written identically: a capital letter, a space, the coordinate or the accession number changes. That is why the app does not look for identical rows, but for pairs that resemble each other in several things at once, and each resemblance adds points.

Before anything else, the app looks for **repeated accession numbers**. That is not a duplicate candidate, it is an error: if there is one, a box at the start of section 4 (Repeated accession numbers) gives the number and the two rows, and recalls that an accession number is never repeated, not even when the material really is the same.

First, blocking

Comparing every pair grows very fast: 46 accessions give 1,035 pairs, and 5,000 would give more than twelve million. The app only compares the pairs that share at least one **blocking key**:

- the **collecting number** (`COLLNUMB`), normalized and without spaces;
- each of the **donor and other-collection numbers** (`DONORNUMB`, `OTHERNUMB`, separated by semicolons);
- the first five letters of the **accession name**, without spaces;
- the first four letters of the genus and the **half-degree cell** in which the coordinate falls;
- the **taxon and the year and month of collection**.

Keys shorter than three letters do not count, and a key shared by more than 400 accessions is ignored, because it distinguishes nothing. For example, *Bolita amarillo* (MEX-ZM-0002) has the keys `lab2019101` (as collecting number and as number), `bolit`, `zea` with the cell 17, −96.5 and `zea mays` with 2019-08. In the example collection, blocking leaves **33 pairs** out of the 1,035: that is the figure shown in the first tile.

Then, the score

Table 3.4. What adds and what subtracts in each pair. The total is clipped between 0 and 100.

Match	Points	Detail
The same collecting number	+40	Normalized: "LAB-2019-101" and "lab 2019 101" match.
They share a donor or other-collection number	+35	Not counting the collecting number, which already added.
Nearly identical names	+25	Similarity of 0.92 or more (below).
Similar names	+12	Similarity from 0.82 to 0.92.
Less than 1 km / 5 km / 25 km apart	+20 / +12 / +5	Haversine distance between the two points.
The same collecting date	+15	Full eight-digit date; if only the year matches, +5.
The same taxon	+10	Same genus and epithet.

Table 3.4. What adds and what subtracts in each pair. The total is clipped between 0 and 100. *(continued)*

Match	Points	Detail
The same collecting mission	+5	`COLLMISSID`.
Different genera	−35	Almost certainly not the same thing; it replaces the taxon bonus.
Different subtaxa	−15	Same genus, but both declare `SUBTAXA` and it does not match (subsp. *mexicana* versus subsp. *parviglumis*).

The weights say how strong each clue is. A collecting number is assigned by the collector, one per sample: a match is almost a signature. Being close, on the other hand, is weak: many different samples are collected in the same community. No clue on its own is enough to pass the threshold of 50 points; two or three together are needed.

Similarity between names

To compare names, the app normalizes them (without accents, case or punctuation: *Tuxpeño* and *TUXPENO* become equal) and computes two measures, keeping the larger one: the **Jaro-Winkler similarity** between the two strings, and a word-based measure, 0.55 + 0.45 × the proportion of shared words (Jaccard index), which only counts if they share at least one word. The second rescues names with extra or missing words: *Chayote verde liso* and *Chayote verde*.

THEORY IN BRIEF · THE JARO-WINKLER SIMILARITY

The Jaro similarity counts the letters that match between two strings **a** and **b** within a window —a letter of **a** is only paired with the same letter of **b** if it is at most ⌊max(|a|, |b|)/2⌋ − 1 positions away—, and the transpositions **t**: half the paired letters that end up in a different order. With **m** paired letters:

$$J = \left(\frac{m}{|a|} + \frac{m}{|b|} + \frac{m - t}{m} \right) / 3 \qquad JW = J + \ell \cdot p \cdot (1 - J)$$

Winkler added a bonus for a **common prefix**, because in names errors are seldom at the start: ℓ is the number of equal initial letters (up to 4) and p = 0.1. The app only applies the bonus if J is already 0.7 or more, so as not to bring together strings that do not resemble each other.

WORKED EXAMPLE · MARTHA AND MARHTA

```
window = ⌊6/2⌋ − 1 = 2 · all six letters are paired (T and H, one position apart) → m = 6
M A R T H A versus M A R H T A: two letters out of order → t = 2/2 = 1
J = (6/6 + 6/6 + 5/6) / 3 = 0.9444          common prefix M-A-R → ℓ = 3
JW = 0.9444 + 3 × 0.1 × (1 − 0.9444) = 0.9611, as in the app
```

The distance between the two sites

THEORY IN BRIEF · THE HAVERSINE DISTANCE

Between two points on the Earth, with latitudes φ_1, φ_2 and longitudes λ_1, λ_2 in radians, the distance over the surface of a sphere of radius R is:

$$a = \sin^2(\Delta\phi/2) + \cos\phi_1 \cos\phi_2 \sin^2(\Delta\lambda/2) \cdot d = 2R \arcsin\sqrt{a}$$

The app uses R = 6,371.0088 km, the mean radius of the Earth. Treating the Earth as a sphere produces errors of around 0.5%, irrelevant for deciding whether two collections are 1, 5 or 25 km apart. Unlike subtracting degrees, the formula takes into account that a degree of longitude is shorter away from the equator.

3

WORKED EXAMPLE · THE TWO BOLITA AMARILLO

MEX-ZM-0002 is at 16.9771, −96.4008 and MEX-ZM-0045 at 16.932, −96.3665.

$\Delta\phi = -0.0451° = -0.00078714$ rad	$\Delta\lambda = 0.0343° = 0.00059865$ rad
$\sin^2(\Delta\phi/2) = 1.5490 \times 10^{-7}$	$\cos\phi_1 \cos\phi_2 = 0.95642 \times 0.95665 = 0.91497$
$\sin^2(\Delta\lambda/2) = 8.9595 \times 10^{-8}$	$a = 1.5490 \times 10^{-7} + 0.91497 \times 8.9595 \times 10^{-8} = 2.3687 \times 10^{-7}$
$d = 2 \times 6{,}371.0088 \times \arcsin(4.8670 \times 10^{-4}) = 2 \times 6{,}371.0088 \times 4.8670 \times 10^{-4} = 6.2015$ km	

The app writes *6.2 km*, both in the header of the pair and in the reason, and adds 5 points, because the distance lies between 5 and 25 km.

3.6 Groups, side-by-side comparison and the threshold

The pairs that reach the threshold are gathered into **groups by transitivity**: if A resembles B and B resembles C, the three form a group even if A and C do not reach the threshold between them.

Each group is a card with the highest score of its pairs, the numbers of its accessions in passport order, the name of the first one and the button compare. With the example collection and the default threshold there are **2 groups and 2 pairs**, the two seeded duplicates, both with 100 points. The maize one adds up to exactly 100: the same taxon (+10), the same collecting number LAB-2019-101 (+40), nearly identical names (+25: after normalizing, *BOLITA AMARILLO* and *Bolita amarillo* are equal), 6.2 km (+5), the same date (+15) and the same mission (+5). The chayote one adds up to 102 —the same taxon, collecting number LAB-2018-102, name, 1.97 km (+12) and date— and is clipped to 100.

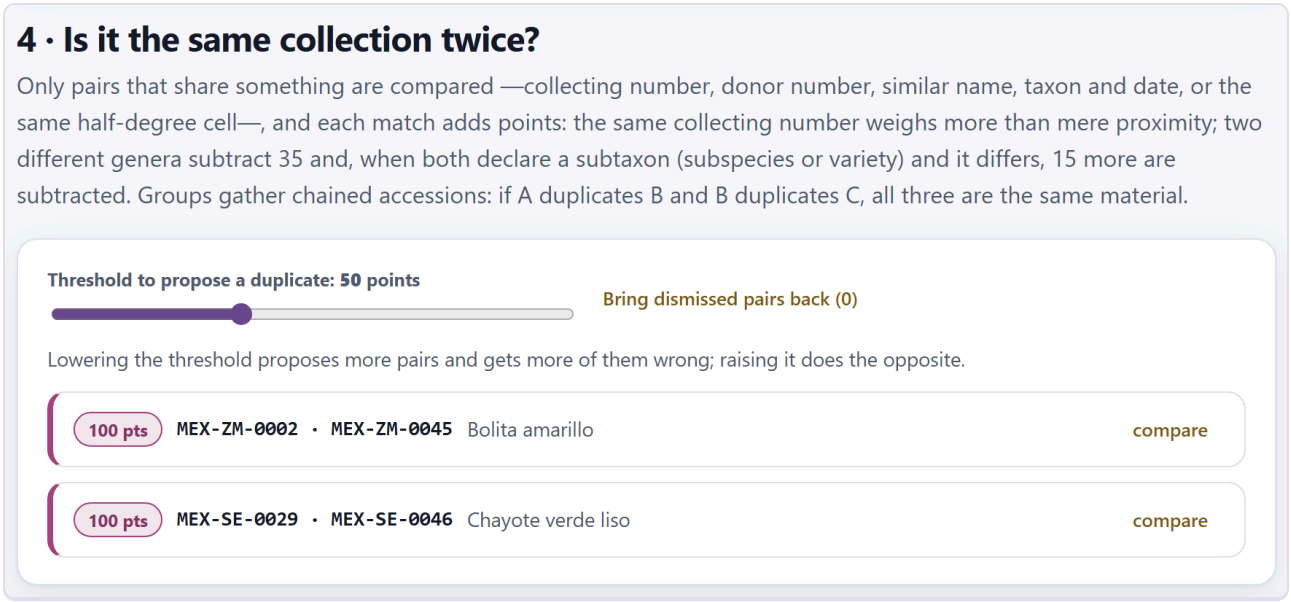


Figure 3.5. Section 4 with the example collection: the two seeded duplicates, each in its own group, with 100 points.

The side-by-side comparison

compare opens the card (and hide closes it). At the top, a table puts the accessions in columns, with the first one tagged as master, and up to fifteen descriptors in rows: accession number and name, genus, species, collecting number, date and site, latitude, longitude, elevation, holding institute, conservation route (with its name: *Seed bank*, *Field genebank or living collection*…), stock, other numbers and donor accession number. Rows that are empty in every accession are omitted; those that **differ** are highlighted in pink with the name in bold, so that you can see at a glance where they do not match.

Below come the **pairs** of the group, one per box: the two numbers, the score, the distance and the list of reasons, each with its points in brackets. Each pair carries its three decision buttons (Section 3.7). The distance is written the same way in the header and in the reason: in meters below 1 km, with one decimal up to 10 km (*6.2 km apart*) and in whole kilometers from there on.

The threshold

The slider Threshold to propose a duplicate runs from 25 to 95 points, in steps of 5, and starts at 50. When you release it, the check runs again; the value is saved in the browser for next time. The app itself sums up the trade-off: Lowering the threshold proposes more pairs and gets more of them wrong; raising it does the opposite.

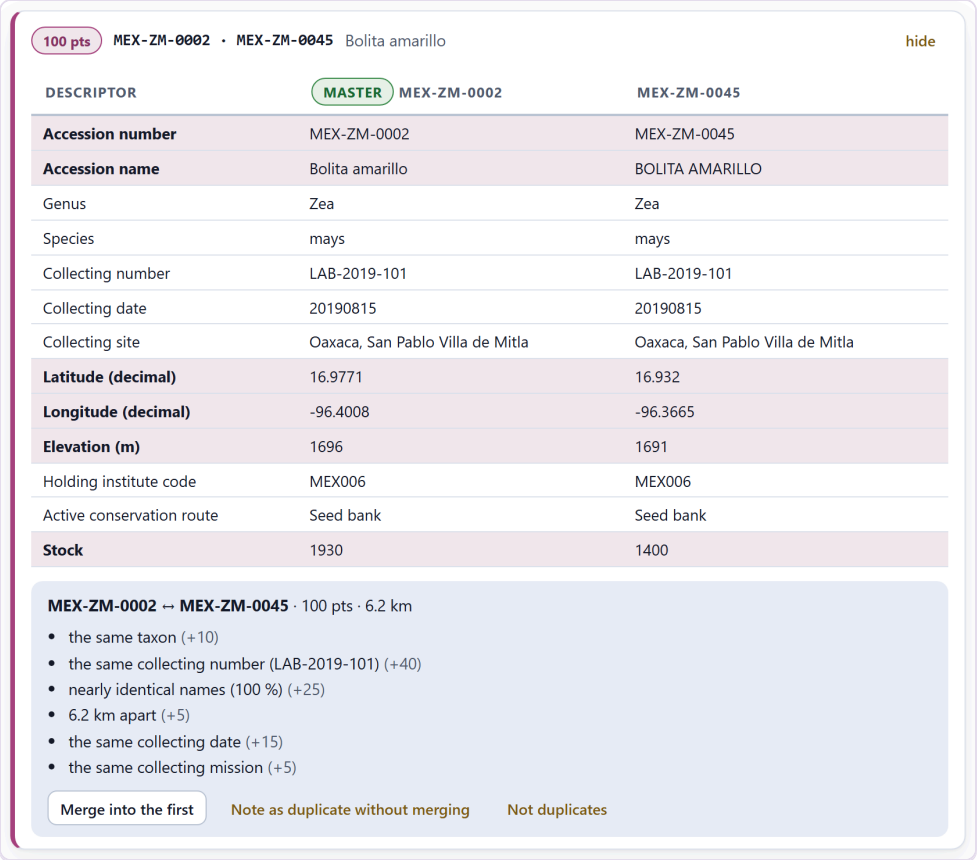


Figure 3.6. The comparison of the first group. The number, the form of the name, the coordinate, the elevation (1,696 and 1,691 m) and the stock differ; the collecting number and date, the site and the institute match.

Table 3.5. What the example collection proposes with different thresholds.

Threshold	Pairs	Groups	What comes in
40 to 95	2	2	Only the two seeded duplicates (100 points).
35	4	2	The spiny chayote MEX-SE-0030 (39 points with each one) joins the chayote group.
30	5	3	In addition, two maizes from the same mission with similar names (32).
25	8	5	Three more pairs, of 25 to 27 points (taxon, similar name, proximity).

Figure 3.7 shows the risk of transitivity with a low threshold. The spiny chayote adds up to 39 points with each smooth green chayote —the same taxon (+10), names 87% similar (+12), 3.4 and 2.3 km apart (+12) and the same year (+5)— and stays in the group, although it is another collection, from another day, and in its passport it is another variety (`SUBTAXA` *var. nigrum spinosum*).

The subtraction for different subtaxa does not stop it, because the smooth green chayotes declare no variety: the app only subtracts the 15 points when both sides declare a subtaxon and it does not match. That is why the decision is taken **pair by pair**: here the two 39-point pairs would be dismissed with Not duplicates and only the 100-point one would be resolved.

100 pts **MEX-SE-0029 · MEX-SE-0030 · MEX-SE-0046** Chayote verde liso hide

DESCRIPTOR	MASTER MEX-SE-0029	MEX-SE-0030	MEX-SE-0046
Accession number	MEX-SE-0029	MEX-SE-0030	MEX-SE-0046
Accession name	Chayote verde liso	Chayote erizo (cuspidatum)	Chayote verde liso
Genus	Sechium	Sechium	Sechium
Species	edule	edule	edule
Collecting number	LAB-2018-102	LAB-2018-103	LAB-2018-102
Collecting date	20180615	20180616	20180615

Figure 3.7. With the threshold at 35, the spiny chayote MEX-SE-0030 enters the group of the smooth green chayote by transitivity. The first rows of the comparison give away that it is another collection: another name, another number and another day.

3.7 Merge, note or dismiss

Each pair has three ways out. None is automatic, and all three change what the app proposes afterwards.

Merge into the first

Merge into the first turns the two accessions into one. Beforehand, the app asks for confirmation: *MEX-ZM-0045 will be merged into MEX-ZM-0002 and the second will disappear from the catalog. Continue?* The "first" is the one that comes earlier in the passport —usually the one that entered the catalog first— and it is the **master**. The merge follows three rules:

- **The master rules.** Every descriptor the master has filled in stays as it is; from the other one, only those that are empty in the master are taken. If the master has no donor accession number and the other does, it inherits it.
- **Stocks are added**, because they are two packets —or two groups of plants— of the same material.
- **The absorbed number is not lost**: it is added to the master's `OTHERNUMB`, so that any label, publication or distribution note that cites it can be traced.

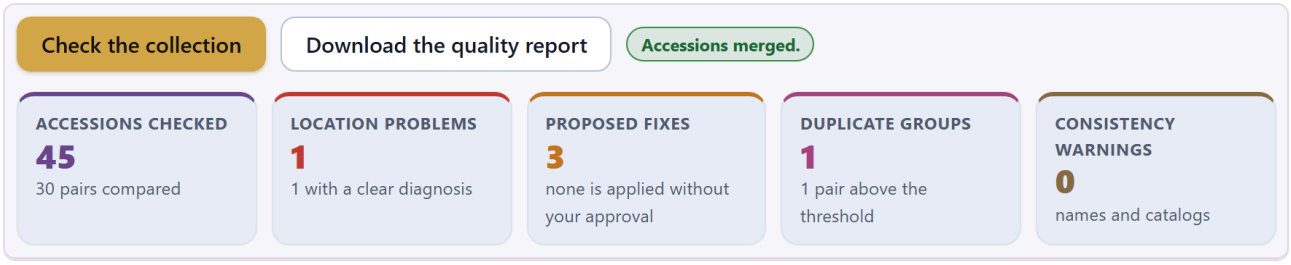


Figure 3.8. After merging MEX-ZM-0045 into MEX-ZM-0002: the catalog drops to 45 accessions and a single group remains. The fixes drop from 4 to 3 because the all-caps name belonged to the absorbed accession.

With the example collection, MEX-ZM-0002 ends up with 1,930 + 1,400 = **3,330** seeds, with *MEX-ZM-0045* among its other numbers, and keeps its name, its coordinate, its crop "Maíz" and its remark *Raza Bolita.*; the remark of the absorbed one, which explained how it had entered the catalog again, does not carry over, because the master already had its own. The merge has no undo button: if in doubt, export the passport first (Chapter 2).

> **WHAT THE MERGE DOES NOT DO FOR YOU**
>
> Adding stocks assumes the two lots are in the same condition. If one was tested a year ago with 89% germination and the other has another date, the master keeps only its own test: check in Block 8 (Chapter 8) whether the merged lot should be tested again. And if the two records are really the same material in two different genebanks, do not merge: note it.

Note as duplicate without merging

Note as duplicate without merging leaves both accessions in the catalog, but records the link in both: each one receives the other's number in `OTHERNUMB` and a sentence in `REMARKS` —in the example, *Declared duplicate of MEX-SE-0046.* and *Declared duplicate of MEX-SE-0029.*— (the sentence is written in the interface language). It is the way out for the same collection kept at two institutions, or for two lots you decide to keep apart. The pair is no longer proposed.

Not duplicates

Not duplicates dismisses the pair without touching the passport. The dismissal —like that of a noted pair— is saved in the browser and survives closing the app, so the pair does not come back to bother you. Each dismissal is remembered by the **identity** of the two accessions —the institute code and the accession number of each one—, not by their place in the table, so it remains valid even if you merge, re-sort or import more accessions. The button Bring dismissed pairs back shows in brackets how many dismissals correspond to a pair of the current collection and brings them all back at once. With the example collection, after merging the maize and noting the chayote, the groups tile shows 0 and the button says *(1)*.

3.8 The quality report

Download the quality report saves quality-report.csv, in UTF-8, with one row per finding and five columns: `type`, `level`, `accession`, `field` and `detail`. It gathers what Block 3 found and, at the end, what the validation of Block 2 found, so it is the complete list of pending work on the collection in a single file, to work on in a spreadsheet or to attach to a report. With the interface in Spanish, the file, its columns and its texts come out in Spanish (informe-calidad.csv).

Table 3.6. The five row types of the report and how many it carries with the example collection just loaded (37 in all).

type	Rows	level · accession · field
`geography`	2	`error`, `warn` or `info` · the accession · the descriptor checked (Section 3.2).
`consistency`	0	`warn` or `info` · all the accessions of the finding, separated by spaces · the name of the warning, as on screen (Section 3.3).
`correction`	4	`fix` · the accession · the descriptor; the detail carries "before → after" and the reason (Section 3.4).
`duplicate`	2	the score · the two accessions joined with "+" · the distance in km, with one decimal; the detail, the reasons (Section 3.5).
`passport`	29	`error` or `warn` · the accession · the descriptor: the 3 errors and 26 warnings of the Block 2 validation (Chapter 2).

Table 3.7. Some rows of the report for the example collection, exactly as they come out (the detail, abbreviated).

type	level	accession	field	detail
geography	error	MEX-ZM-0039	DECLONGITUDE	The longitude lost its minus sign…
geography	info	MEX-PH-0041	ORIGCTY	The app does not bundle the outline of MX…
correction	fix	MEX-PH-0041	ORIGCTY	MX → MEX · "MX" is the two-letter code…
duplicate	100	MEX-ZM-0002 + MEX-ZM-0045	6.2 km	the same taxon; the same collecting number (LAB-2019-101); …
duplicate	100	MEX-SE-0029 + MEX-SE-0046	2.0 km	the same taxon; the same collecting number (LAB-2018-102); …
passport	warn	MEX-ZM-0001	DUPLSITE	No safety duplicate: the accession depends on a single facility.

The report portrays the collection at the moment it is downloaded: dismissed or noted pairs do not appear, and neither does anything you have already corrected. If you are going to hand it in, download it at the end, after applying fixes and resolving duplicates, or download one before and another after to document the clean-up.

When something goes wrong

What you see	Why it happens and what to do
Load your accessions in Block 2 first.	There is no passport. Press Go to the passport and load your collection or the example one (Chapter 2).
An accession without coordinates does not appear in section 2	Without latitude or longitude there is nothing to compare with the country. Block 2 flags it.
The note says an accession could not be compared	Its country is not among the 119, or it is written in two letters. Apply the `ORIGCTY` fix if there is one; if the country is not included, check the point on the map of Block 4.
An obviously misplaced point does not show up as a problem	It falls inside the country's box, even if in the wrong state: the boxes are rectangles with a one-degree margin. Check it in Block 4.
A duplicate you know of does not appear	It shares no blocking key, does not reach the threshold or you dismissed it earlier. Lower the threshold or press Bring dismissed pairs back.

What you see	Why it happens and what to do
A group gathers accessions that are not the same	They came in by transitivity. Open the group and dismiss, pair by pair, those that are not.
A pair you had not resolved disappeared	You dismissed it or noted it earlier, perhaps in another session: the dismissal is saved with the numbers of the two accessions (Section 3.7). Press Bring dismissed pairs back.
You merged the wrong pair	There is no undo. Recover the accession from the passport you exported earlier, or enter it again in Block 2 with the number left in the master's `OTHERNUMB`.

DECISION RULE · WHAT TO DO IN BLOCK 3

If a point has a sign or swap diagnosis	**then** accept the fix if the corrected site agrees with `COLLSITE`; if not, open the record and look up the coordinate in the notebook.
If a fix is about form (two-letter country, spaces, capitals)	**then** apply it by group after looking at the example; watch out for proper names inside a name written entirely in capitals.
If a names or site warning appears	**then** decide which form is the right one and correct it in the record of Block 2.
If a pair matches in collecting number, date and taxon	**then** it is the same collection: merge it if it is in your genebank twice; note it if it is in two genebanks.
If a pair only resembles each other in name, proximity or year	**then** compare it calmly and, if the collecting number or date differ, mark it as Not duplicates.
If you are going to analyze or hand in the collection	**then** resolve all the groups first and download the quality report at the end.

Whatever is corrected, merged or noted in Block 3 stays in the passport: Block 2 shows it and Blocks 4 to 10 take it when they open.

4

CHAPTER FOUR · BLOCK 4 · COLLECTING MAP

The collecting map

Block 4 puts every accession at the site where it was collected. The map is drawn inside the app, with outlines that travel with it, so it works offline and no coordinate ever leaves your computer. It lets you see the collection at a glance — what there is, in which conservation route, at what elevation—, spot the point that landed where it should not, and count how many accessions fall in each state, each country or each cell of a grid, which is the gateway to Blocks 5 and 6.

4.1 A map that needs no internet

4.2 From the sphere to the screen: the projection

4.3 Which state and which country does each point fall in?

4.4 The extent: your data, Mexico, the world or a single state

4.5 Color, size and filters of the points

4.6 The choropleth and the grid

4.7 The counts by state and by country

4.8 Taking the map and the points with you

4.1 A map that needs no internet

Block 4 takes the accessions you loaded in Block 2 (Chapter 2), with whatever you corrected in Block 3 (Chapter 3), and draws one point for each accession that has decimal latitude and longitude. There is nothing to click when you open it: the map, its legend, the summary boxes and the tables are rebuilt every time you open the block. If there is no passport data yet, instead of the map you see Load your accessions in Block 2 first.

Figure 4.1. Section 1 · The accessions on the map with the example collection and the extent set to All of Mexico. Each point is an accession, colored by its conservation route.

The block has two sections. The first, 1 · The accessions on the map, gathers the controls, the map, the legend and the export buttons on one card (Figure 4.1). The second, 2 · How many fall in each state and each country, gives five summary boxes and two tables (Section 4.7). At the end, a box reminds you where the outlines come from.

Where the outlines come from

The outlines come from **Natural Earth**, a **public-domain** map data set, and are stored inside the app, in two files in the data/ folder:

File	Natural Earth data set	What it holds
mexico-states.js	admin-1 (states and provinces) at 1:10 million	The **32 federal entities** of Mexico, simplified with the Douglas–Peucker algorithm at a tolerance of 0.004° (about 400 m), with coordinates rounded to four decimals. Each state carries its name and its three-letter code (*OAX*, *PUE*, *CMX*...).
world.js	admin-0 (countries) at 1:110 million	The outlines of **174 countries**, simplified to two decimals and without Antarctica.

The difference in scale is deliberate. Mexico is drawn with the detail a state-level map calls for; the rest of the world, with the coarse line that is enough to place a collection in its country. Neither is fit to decide whether a point falls on one side or the other of a municipal boundary, or a hundred meters from the coast.

> **WHY IT MATTERS THAT THE MAP NEVER LEAVES YOUR COMPUTER**
>
> An online viewer asks a server for the tiles of the area you are looking at; here there is no request at all. That lets you work in the field without a signal, and it is no small matter when the coordinates belong to populations of crop wild relatives, to species prone to illegal collection or to the plots of custodian families (Section I.3) —data that many collections do not publish with all their decimals.

4.2 From the sphere to the screen: the projection

A coordinate is an angle on a sphere; the screen is flat. The app uses the simplest projection that gives a reasonable map of a country: the **equirectangular** projection (or *plate carrée*), in which meridians and parallels are straight lines at equal spacing, with one correction: the width of each degree of longitude is multiplied by the **cosine of the central latitude** of the extent.

> **THEORY IN BRIEF · THE EQUIRECTANGULAR PROJECTION WITH COSINE CORRECTION**
>
> A degree of latitude measures almost the same everywhere on Earth (about 111 km), but a degree of longitude shrinks toward the poles: it measures 111 km × cos φ at latitude φ. With an extent running from longitude λ_0 to λ_1 and from latitude φ_0 to φ_1, the central latitude is $\varphi_m = (\varphi_0 + \varphi_1)/2$, and each point is taken to the screen with:
>
> $$k = \cos \phi_m \cdot x = o_x + (\lambda - \lambda_0) \cdot k \cdot s \cdot y = o_y + (\phi_1 - \phi) \cdot s$$
>
> where s is the number of pixels per degree of latitude, the largest value at which the extent fits in the 900 × 560 canvas leaving a 10 px margin, and o_x, o_y center the drawing. The **y** coordinate grows downwards, which is why the latitude is subtracted from the maximum: north stays at the top. The shape is faithful near φ_m; for a country or a region, that is enough.

WORKED EXAMPLE · THE BLUE MAIZE FROM APOALA ON THE MAP OF MEXICO

The All of Mexico extent runs from −118.5 to −86.6 in longitude and from 14.3 to 32.8 in latitude. MEX-ZM-0001 lies at 17.6871, −97.2788.

$\phi_m = (14.3 + 32.8)/2 = 23.55°$	$k = \cos 23.55° = 0.91671$
width = 31.9° × 0.91671 = 29.243	height = 32.8 − 14.3 = 18.5°
s = min(880/29.243, 540/18.5) = min(30.09, 29.189) = 29.189 px per degree	
$o_x = 10 + (880 - 29.243 \times 29.189)/2 = 23.21$ · $o_y = 10$	
x = 23.21 + (−97.2788 + 118.5) × 0.91671 × 29.189 = 23.21 + 567.84 = 591.0	
y = 10 + (32.8 − 17.6871) × 29.189 = 10 + 441.13 = 451.1, the center of the circle in the app's SVG	

Without the correction, the width would be 31.9° and Mexico would come out 9% wider than it really is at that latitude.

4

The scale bar, the north arrow and the graticule

In the lower right corner, the **scale bar** shows a round distance —1, 2, 5, 10, 25, 50, 100, 200, 250, 500, 1,000, 2,000 or 5,000 km— that fits in 140 pixels of the canvas. It is measured at the central latitude: on the map of Mexico, 1° of longitude at 23.55° of latitude measures 101.93 km by the haversine formula (Section 3.5) and takes up 26.76 px, so each pixel is worth 3.81 km and the bar chosen is the **500 km** one, 131 px long. That measure is exact only at the central latitude: at latitude φ, a horizontal pixel measures cos φ / cos φ_0 times what the bar says. That is why the app works out how much the horizontal scale changes within the extent and decides which bar is honest: if it changes by less than 10% (Mexico, which runs from 14° to 33° and stays under 9%, or any single state), the bar stands alone; if it changes by up to 35%, the bar says at which latitude it holds, such as *2,000 km at 7° N* in the automatic extent of Figure 4.2; beyond that, as with the whole world, no bar is drawn and the map says No scale bar: it varies with latitude instead (Figure 4.6). In the lower left corner is the **north arrow**, which always points up. Behind the land, a graticule of meridians and parallels is drawn every 20° if the extent spans more than 60° of longitude, every 5° if it spans more than 14°, every 2° if it spans more than 4°, and every degree in smaller extents.

THE SCALE IS EXACT ONLY AT THE CENTER

On a map of a single state or of Mexico, the bar holds for the whole map with a small error. In a tall extent it no longer does: the bar that says "at 7° N" is true at that latitude and, away from it, east–west distances are exaggerated, more and more toward the poles; over the whole world the error would exceed 80%, which is why there is no bar there. To measure the distance between two collecting sites, do not use the bar; use the distance column of the duplicates (Chapter 3) or the calculations of Blocks 5 and 6, which work on the sphere.

4.3 Which state and which country does each point fall in?

The app does not trust the passport text to tell where an accession is. It does not read the state written in `COLLSITE` or the country in `ORIGCTY`: it looks at **where the point falls** and searches for the state or the country whose outline contains it. This is called a **point-in-polygon** test.

> **THEORY IN BRIEF · THE EVEN–ODD RULE**
>
> A ray is cast eastwards from the point, and the number of times it crosses the polygon's boundary is counted. If it crosses an odd number of times, the point is inside; if even, outside. The rule respects holes: a point inside a hole crosses the outer boundary and the hole's boundary, twice, and ends up outside; and it works equally for states with islands, which are several rings. To avoid testing all 32 full outlines against every point, the app first discards the states whose bounding rectangle does not contain the point, and runs the fine test only on those that remain. The same goes for countries.

With the example collection, the test assigns a state to the 36 accessions that fall in Mexico, and for all of them it matches the state written at the start of their collecting site. Those that do not match their passport are two, and both are seeded errors (Section I.7):

Accession	What the passport says	Where the point falls	What the app does
MEX-ZM-0039 *Tuxpeño de la costa*	`ORIGCTY` = MX; site "Veracruz, Tlalixcoyan"; 18.8100, 96.0600	Myanmar (Burma), no state	It counts it in Myanmar, not in Mexico or in Veracruz. It is the longitude without its sign that Block 3 diagnoses (Section 3.2).
MEX-PH-0041 *Frijol tepari*	`ORIGCTY` = MX; site "Sonora, Álamos"	Mexico, Sonora	It counts it in Mexico and in Sonora: the point is right, even though the country code is misspelled.

The accession without coordinates, MEX-ZM-0043 (*Maíz ancho de riego*, site "Por el rumbo de Guerrero"), does not appear on the map or in the counts, even though its site names a state: the Without coordinates box shows 1. Points at (0, 0) and out-of-range coordinates (latitude beyond 90° or longitude beyond 180°) are not drawn either.

The map is the fastest way to see a coordinate error: a maize from the Gulf coast shows up on the other side of the world (Figure 4.2). Correct it in Block 3 — Fix in section 2— and return to Block 4: the point lands in Veracruz, at 18.81, −96.06, the country table loses Myanmar and Mexico rises from 36 to 37 accessions, Veracruz rises from 6 to 7 accessions and from 4 to 5 species (*Zea mays* is added), and the Countries box drops from 8 to 7. What does not change is the automatic extent, which still spans from the Americas to India, because those accessions really are from Ethiopia and India.

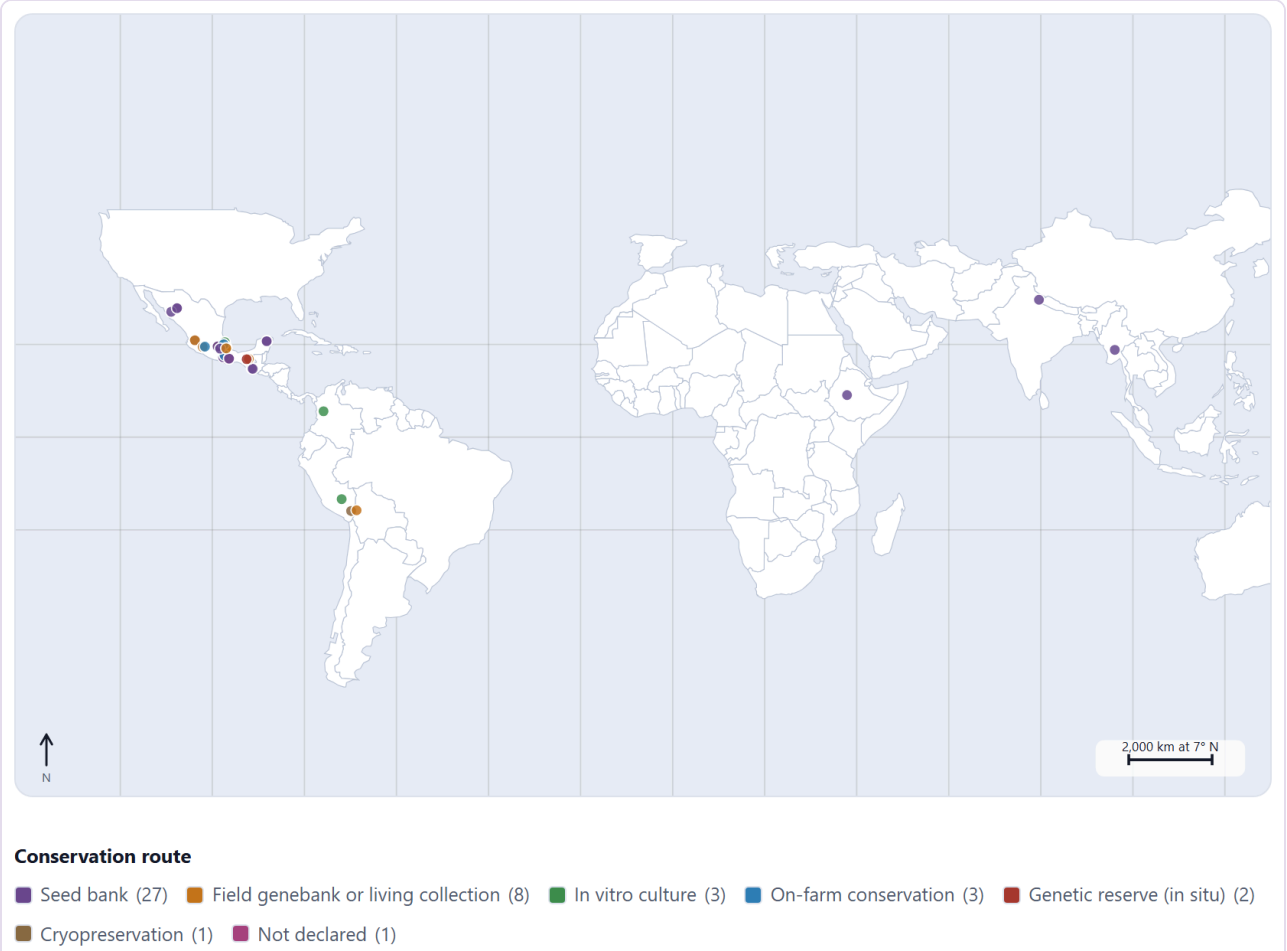


Figure 4.2. The automatic extent with the freshly loaded collection. The point on the right, in Myanmar, is the maize from Veracruz with the unsigned longitude: it stretches the extent all the way to Southeast Asia.

> **TWO CHECKS THAT COMPLEMENT EACH OTHER**
>
> Block 3 compares the point with the rectangle of the declared country, with a one-degree margin; it catches errors of hundreds or thousands of kilometers (Section 3.2). Block 4 assigns the point to the outline of a state, and that is where the fine error shows: a point of a maize "from Puebla" that falls in Tlaxcala, or one "from Oaxaca" that falls in the sea, passes the Block 3 check and gives itself away here, in the state table or in the point's record.

4.4 The extent: your data, Mexico, the world or a single state

The Extent menu decides which part of the Earth fills the canvas. Points outside the extent are not drawn, but they still count in the summary boxes, the tables and the points file (Sections 4.7 and 4.8).

Option	What it shows
Fit to my data	The default. The rectangle that contains all the points, with a 15% margin on each side (at least 0.4°), and never less than 1.5° wide or high, so that a collection from a single municipality does not end up as a close-up with no landmarks. With the example collection it runs from −139.7 to 126.8 in longitude and from −22.7 to 36.4 in latitude (Figure 4.2).
All of Mexico	The rectangle of Mexico, from −118.5 to −86.6 and from 14.3 to 32.8: it draws 38 of the 45 points (Figure 4.1).
The world	From −180 to 180 in longitude and from −60 to 84 in latitude (Figure 4.6).
A single state (the 32, by name)	The rectangle of the state, with an 8% margin. The neighboring states and their points are drawn too; the chosen one gets the thick outline, in purple (Figure 4.3).

You can also reach a state without the menu: **tap the state on the map** to zoom into it, and tap it again to go back to All of Mexico. The buttons with the name of each state in the state table do the same (Section 4.7).

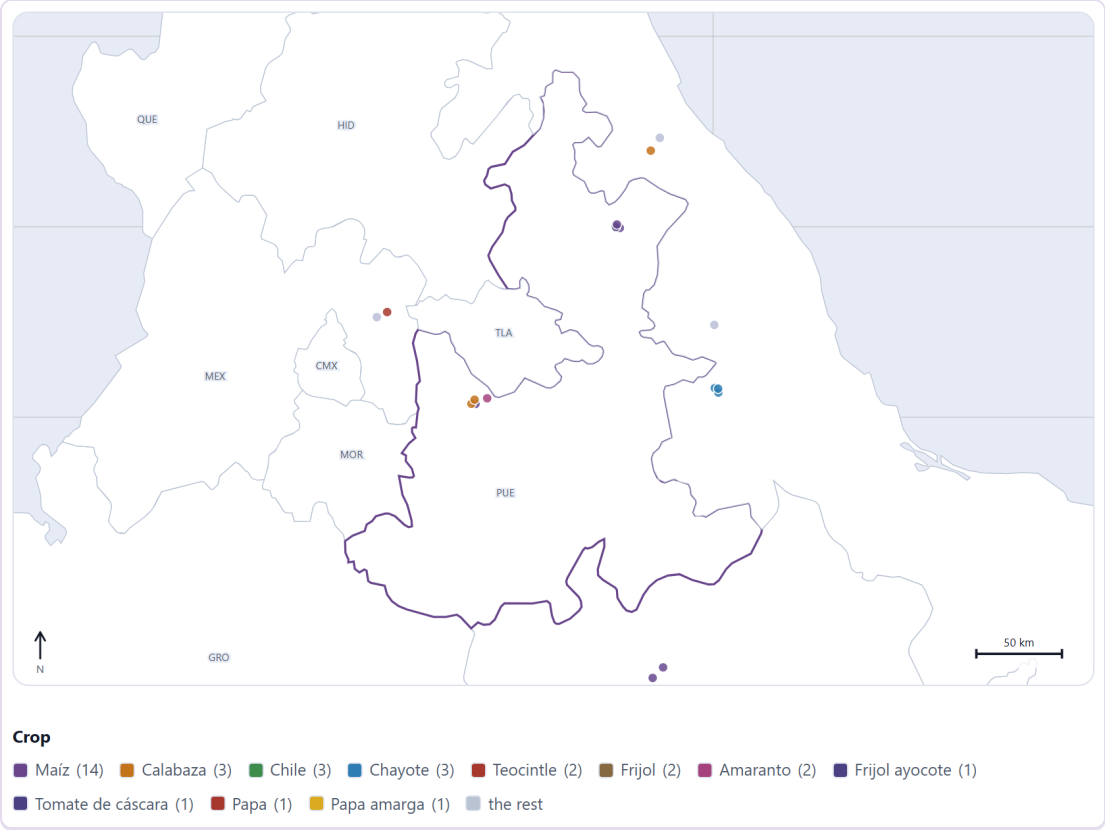


Figure 4.3. The extent set to Puebla, with the points colored by crop. The bar drops to 50 km. The points of Veracruz, Tlaxcala and Oaxaca that fall inside the rectangle are drawn too.

When the states are drawn

The 32 states are drawn only when the extent spans **45° of longitude or less** and touches the rectangle of Mexico. In a wider extent, Mexico is drawn as one more country, with the 1:110 million outline, and the state codes disappear. That is why, with the example collection, the automatic extent shows no states: the accessions from Ethiopia and India make it as wide as the world. If your collection comes from Mexico and other countries, choose All of Mexico to see the states.

The state codes button, on by default, labels each state with its three-letter code at the point of its interior farthest from the edge (in its largest part, if the state has islands, so that the code of Colima does not land in the sea). If two codes overlap, the app shifts one of them by a few pixels; a code without a count that fits nowhere is left out. The codes carry a halo in the background color so that they can be read over points and lines; with a state choropleth, the code also carries the count in parentheses (Figure 4.5). Turn it off when the map is going into a publication and the labels get in the way. Watch out for two codes: *MEX* is the State of Mexico and *CMX* is Mexico City, not the country.

Hovering over and tapping a point

When you hover over a point, the point grows and, below the map, a line summarizes the accession: number, name, taxon, site, the state where the point falls, the elevation and the route. For MEX-ZM-0001 it reads *MEX-ZM-0001 · Maíz azul de Apoala · Zea mays · Oaxaca, Santiago Apoala · Oaxaca · 1993 m · Seed bank*. **Tapping** the point opens its record in Block 2, to review or correct it. The foot of the card reminds you: Tap a point to open its record; tap a state to zoom into it.

4.5 Color, size and filters of the points

The Point color menu chooses which passport variable colors the points. There are eight, plus A single color:

Table 4.1. The variables that color the points and their legend with the example collection (45 points).

Option · descriptor	Type	Legend with the example collection
Conservation route · active route	categorical	The default. Seed bank (27), field genebank or living collection (8), in vitro culture (3), on-farm conservation (3), genetic reserve (2), cryopreservation (1) and *not declared* (1). The route is the one declared in `GP_CONS` or, if there is none, the one deduced from `STORAGE`, as in Chapter 2.
Crop · `CROPNAME`	categorical	Maíz (14), Calabaza, Chile and Chayote (3 each)…, and *the rest*.
Species · genus and epithet	categorical	*Zea mays* (17), *Capsicum annuum* and *Sechium edule* (3)…, and *the rest*.
Biological status · `SAMPSTAT`	categorical	Traditional cultivar/landrace (38), wild (4), natural (2), breeder's line (1).
Country of origin · `ORIGCTY`	categorical	Mexico (36), Guatemala and Peru (2), Bolivia, Ethiopia, India, Colombia and *MX* (1). Here the passport text does rule, not where the point lies.
Collecting source · `COLLSRC`	categorical	Field (23), backyard, kitchen or home garden (6), farm store (5)… ten sources in all.

Table 4.1. The variables that color the points and their legend with the example collection (45 points). *(continued)*

Option · descriptor	Type	Legend with the example collection
Elevation · ELEVATION	continuous	From 20 to 3,797 m; the 25,000 m one stays gray, off the scale (Figure 4.4).
Collecting year · year of COLLDATE	continuous	From 2014 to 2022.

With a **categorical** variable, the app counts how many points there are of each value and gives a color from its palette to the **eleven most frequent values**; the rest go in gray, under the label *the rest*. The legend lists each value with its number of points in parentheses. Crop and Species have more than eleven values in the example collection (23 crop names as written and 24 taxa), so in those two the gray lumps several together. Note that the crop counts what is written: *Maíz* (14) and *Maiz* without the accent, the one of the seeded duplicate, are two different values until you resolve the duplicate in Block 3.

With a **continuous** variable, the color comes from a ramp that runs from pale gold to gold, ochre and purple, linear between the minimum and the maximum of the points. The pale gold at the low end stands apart from the grayish blue of the sea and the white of the land. Points with no value go in gray and off the scale. For elevation, an empty field means *no data*, not 0 m, and an elevation that is impossible on land —below −500 m or above 7,000 m— also goes in gray, so that a data-entry error does not stretch the scale. The legend shows the ramp in six steps with the minimum and the maximum below its ends (years without a thousands separator: *2014*, *2022*) and, if there are gray points, how many: for elevation, no data or outside −500 to 7,000 m (1).

Figure 4.4 is the counterpart of the elevation histogram in Block 2 (Figure 2.7), where the same absurd value stretches the axis until almost all the bars sit in the first two classes. Here the impossible value does not enter the scale: the 20 to 3,797 m range is shared among the remaining 44 points, and the gray point in Sonora stays in view as a pending task. Tap it, correct the elevation in its Block 2 record and, when you come back, the point takes its color and the legend stops counting it in gray.

Size by stock

The Size menu has two options. With fixed size, the default, every point has a 4 px radius. With size by stock, the radius grows with the square root of the stock (GP_STOCK): r = 2.5 + 5.5 √(stock / maximum). With the square root, the **area** of the circle, which is what the eye compares, grows in proportion to the stock. In the example collection the maximum is 3,200 seeds, and the radii range from 2.8 to 8 px. It is the way to see at a glance where the scarce material is, the material that should be regenerated (Chapter 8).

The crop and route filters

The Crop and Route menus leave on the map a single crop (from those written in CROPNAME, in alphabetical order) or a single route (from those in use). Unlike the extent, the filter does change everything else: the summary boxes, the tables, the choropleth, the grid and the points file count only what is filtered. With the crop *Maíz*, 14 points remain; with the route *Field genebank or living collection*, 8. To go back to everything, choose all crops and all routes.

Figure 4.4. The points colored by elevation with the freshly loaded collection. The tepary bean from Sonora, with its elevation of 25,000 m (a seeded error), stays gray, off the scale, which runs from 20 to 3,797 m. There are no purple points in Mexico: the maximum is in the Andes, outside the extent, because the scale is computed with all the points and not only with those in view.

4.6 The choropleth and the grid

The points say where each accession is; when there are many, they overlap and no longer say how many there are in each place. That is what the choropleth, which paints each unit by its count, and the grid, which counts in equal cells, are for.

The choropleth

The Choropleth menu has three options, besides no choropleth:

- accessions per state: each Mexican state painted by how many points fall in it;
- species per state: by how many distinct species (genus and epithet) there are among those points;
- accessions per country: each country, by how many points fall in it. If the map shows the states, they are all painted with the value of Mexico.

The color comes from the same ramp as the points, but starts at 15% so that a unit with a single accession stands apart from empty land: a unit with n accessions is painted with the shade 0.15 + 0.85 × n / maximum. Units without accessions stay white. The legend only says *few* and *many*; the figures are in the codes on the map and in the tables (Section 4.7). Over a choropleth or a grid, each point carries a thick white border and a thin dark ring, so that it shows even when it has the same color as the unit that contains it.

Figure 4.5. The accessions per state choropleth, with A single color for the points. Ten states have collections; Oaxaca and Puebla, with 7 each, carry the purple of the maximum.

With the example collection, the accessions per state choropleth paints ten states: Oaxaca and Puebla with 7, Veracruz with 6, Chiapas with 4, Michoacán with 3, Chihuahua, Yucatán, the State of Mexico and Jalisco with 2, and Sonora with 1 (Figure 4.5). The species choropleth changes the order: Puebla comes out on top with 5 species, Veracruz and Chiapas with 4, and Oaxaca drops to 3, because four of its seven accessions are maize. The two readings answer different questions: where most has been collected, and where the most collected diversity is —the diversity that Block 5 measures properly.

> **A COUNT CHOROPLETH FAVORS LARGE UNITS**
>
> Chihuahua, with two accessions from the same site, is painted as strongly as Jalisco with two of two species, and its huge area draws more attention than that of Tlaxcala. A choropleth of absolute counts corrects neither for the size of the unit nor for collecting effort. To compare regions, use the grid, with equal cells, or the rarefied indices of Block 5 (Chapter 5).

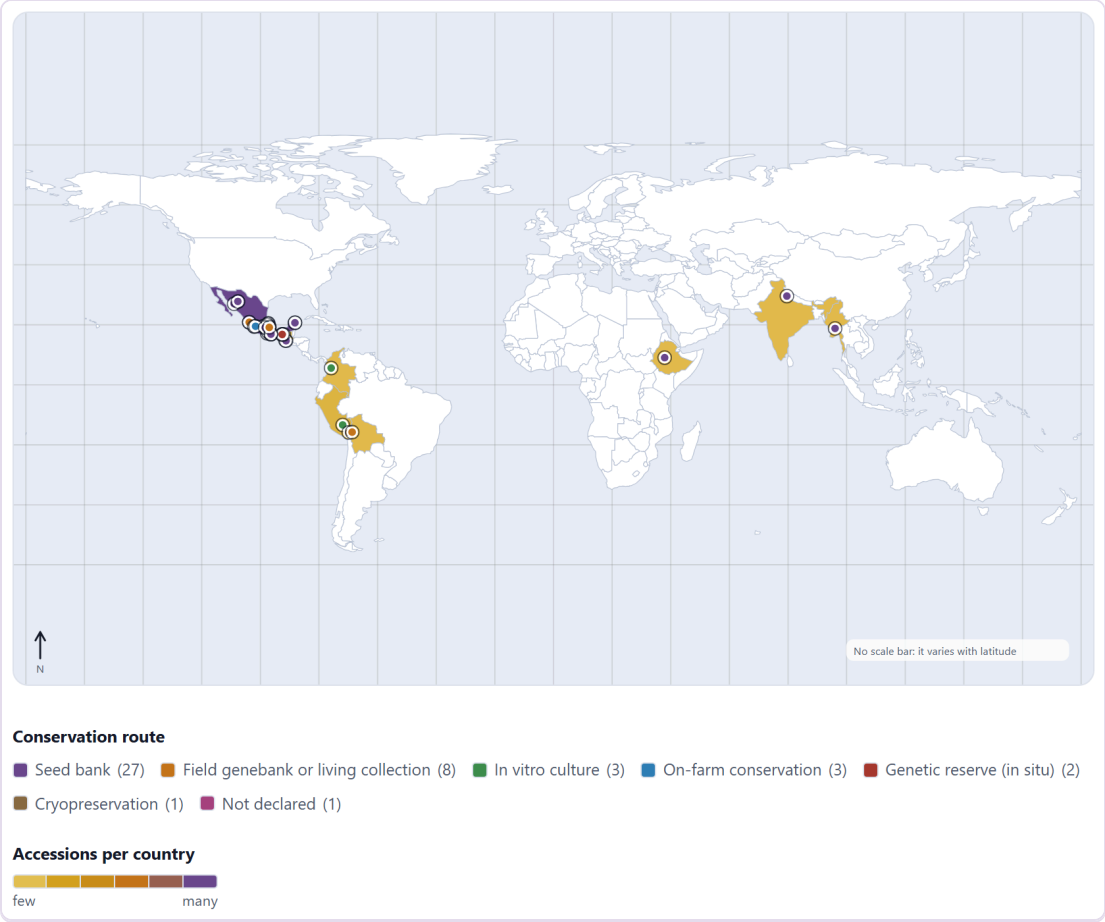


Figure 4.6. The extent The world with the accessions per country choropleth and the freshly loaded collection. Myanmar is painted because of the maize from Veracruz with the unsigned longitude.

The country choropleth makes the error of Figure 4.2 even more visible: a whole country painted by a single misplaced point (Figure 4.6). With 36 accessions against 2 at most in the others, Mexico takes the purple and every other country the lightest shade; in a collection like this one, the country choropleth only tells "Mexico" from "the rest".

The grid

The Grid menu lays square cells of 0.25°, 0.5°, 1° or 2° over the map, aligned with whole degrees: the 1° cell containing the point (λ, φ) runs from $\lfloor\lambda\rfloor$ to $\lfloor\lambda\rfloor + 1$ and from $\lfloor\varphi\rfloor$ to $\lfloor\varphi\rfloor + 1$. Only cells with at least one point are drawn, semi-transparent, in the shade of the ramp that matches their count (from 25% of the ramp, so that a single-point cell shows). When you hover over a cell, the app shows its number of accessions.

The cell size changes the answer (Table 4.2): with small cells nearly every cell holds a single point; with large cells, different sites are lumped together. Correcting the maize from Veracruz does not change the number of 1° cells: its point leaves a cell in Myanmar and occupies another in Veracruz. A 1° cell measures about 111 km from north to south and, at the latitude of central Mexico, about 105 km from east to west; its area in square kilometers changes with latitude, which is worth remembering when comparing cells in Chihuahua and Chiapas. The grid in this block is for looking; the cell-based analysis is done in Blocks 5 and 6 (Chapters 5 and 6), with their own cell size.

Figure 4.7. The 1° grid over the map of Mexico. The purple cell, from −99° to −98° and from 19° to 20°, holds the 6 accessions from Cholula and Texcoco.

Table 4.2. The occupied cells at each size, with the 45 points of the freshly loaded example collection.

Cell	Occupied	With a single point	Maximum	The most populated cell
0.25°	29	18	4	A tie: San Pablo Villa de Mitla, Oaxaca, and San Andrés Cholula, Puebla.
0.5°	26	14	4	A tie between Mitla, Cholula and the cell of Coatepec and Huatusco, Veracruz.
1°	21	9	6	−99° to −98°, 19° to 20°: Cholula and Texcoco.
2°	18	7	7	−98° to −96°, 16° to 18°: the seven points of Oaxaca.

4.7 The counts by state and by country

Section 2 · How many fall in each state and each country opens with five summary boxes and goes on with two tables. Everything is counted with the points left by the crop and route filters, regardless of the extent, and by the place where each point falls (Section 4.3).

Table 4.3. The five summary boxes, before and after correcting the longitude of MEX-ZM-0039 in Block 3.

Box	Before	After	What it counts
On the map	45	45	The accessions with usable coordinates, *of 46 accessions*.
Without coordinates	1	1	Those that cannot be mapped (MEX-ZM-0043). The box turns ochre while there is any.
Mexican states	10	10	The states with at least one point. The corrected maize falls in Veracruz, which already had collections.
Countries	8	7	The countries *according to where the point falls*: Myanmar drops out after the correction.
Mapped species	24	24	The taxa (genus and epithet) with at least one point.

Figure 4.8. The summary boxes and the tables with the freshly loaded collection. The state table scrolls to show the remaining six states.

The Mexico, by state table gives, for each state with collections, from the one with the most to the one with the fewest, the number of accessions, of species and up to three crops. The state name is a button that zooms the map into it. With the example collection, Oaxaca and Puebla lead with 7 accessions; after the correction, Veracruz catches up with 7 and 5 species (Table 4.4). If no accession falls in Mexico, the table says so: No accession falls in Mexico.

Table 4.4. The state table of the example collection, after correcting MEX-ZM-0039.

State	Accessions	Species	Crops (up to three)
Oaxaca	7	3	Maíz, Frijol, Chile
Puebla	7	5	Maíz, Calabaza, Amaranto
Veracruz	7 (was 6)	5 (was 4)	Calabaza, Vainilla, Café
Chiapas	4	4	Maíz, Frijol ayocote, Cacao
Michoacán	3	3	Teocintle, Aguacate, Frijol
Chihuahua · Yucatán · México · Jalisco	2 each	1 · 2 · 2 · 2	Maíz · Maíz, Chile · Teocintle, Trigo · Chile, Agave
Sonora	1	1	Frijol tepari

The table The world, by country gives accessions and species for each country, with the country name in the interface language. Before the correction: Mexico 36 (18 species), Guatemala 2, Peru 2, and Bolivia, Ethiopia, India, Colombia and Myanmar with 1; afterwards, Mexico 37 and Myanmar disappears. The tepary bean counts in Mexico even though its passport says *MX*, because the point rules. If a point does not fall inside any country on the map —in the sea, for example—, the app counts it in the country declared in its passport; if no country is declared either, it is left out of the table and a note below says how many there are: *usually points at sea, that is, coordinates with an error*.

> **WHAT THESE COUNTS ARE FOR**
>
> They are the first geographic portrait of the collection: where collecting effort is concentrated and which states or countries are missing. The state table is the starting point for Block 5, which computes the diversity of each state, and for Block 6, which looks for uncollected areas. If a state shows fewer accessions than you expected, check the records of its points: one of them probably falls in the neighboring state.

4.8 Taking the map and the points with you

The Map title (optional) field writes a title in bold in the upper left corner of the map; it shows on screen and goes into the exports. Below the map, three buttons save what you see:

Button · file	What it saves
Export SVG collecting-map.svg	The vector map, 900 × 560 units, with everything in view: extent, colors, choropleth, grid, codes, title, scale and north arrow. It enlarges without losing sharpness and can be edited in any vector drawing program. The legend is not included: it sits on the page, outside the drawing.
Export PNG (2×) collecting-map.png	The same map as an image, at twice the resolution: 1,800 × 1,120 pixels, with a background. If the browser fails to convert it, the app warns *Could not convert to PNG; download the SVG instead.*
Points as CSV (with state and cell) map-points.csv	One row per point, with the crop and route filters applied but regardless of the extent: with the example collection, 45 rows plus the header.

The SVG and the PNG are redrawn for the file with the **fixed colors of the light theme**, even if you are working in the dark theme: on screen the map uses the theme colors, which change with it, and a file has to look the same everywhere, on white paper or on a slide. That is why the exported SVG carries no color variables, only literal values.

Table 4.5. The twelve columns of map-points.csv and three rows of the freshly loaded example collection (the accession name omitt

ACCENUMB	TAXON	CROPNAME	DECLATITUDE	DECLONGITUDE	ELEVATION	STATE	POINT_COU
MEX-ZM-0001	Zea mays	Maíz	17.6871	-97.2788	1993	Oaxaca	MEX
MEX-ZM-0039	Zea mays	Maíz	18.81	96.06	20		MMR
MEX-PH-0041	Phaseolus acutifolius	Frijol tepari	27.028	-108.937	25000	Sonora	MEX

The columns the block adds are the valuable ones: `STATE`, the state where the point falls (empty outside Mexico); `POINT_COUNTRY`, the country where it falls, in three letters, next to `PASSPORT_COUNTRY`, the one declared in `ORIGCTY`; `CELL_1DEG`, the 1° cell as "longitude|latitude" of its south-west corner (*-98|17* is the cell from −98° to −97° and from 17° to 18°); and `ROUTE`, the conservation route with its internal code (*seed*, *field*, *invitro*, *cryo*, *insitu*, *onfarm*). Sorting the file by the two country columns, or filtering the rows where they differ, is the fastest way to find, in a large collection, the points that do not fall where the passport says: in the example, exactly the two accessions of Section 4.3 stand out.

The figure studio

When you hover over the map, the figure studio button appears in its corner, the same one as for the Block 2 charts (Section I.6); the In this block index also offers it next to section 1. The studio opens the map in split screen to adjust the text size and the line width, change the palette and export to PNG, SVG, PDF or TIFF in millimeters or inches, at 300, 600 or 1,200 dpi, which is what journals usually ask for. Since the on-screen map uses the theme colors, the studio can export it with those of the light theme even if you work in the dark one.

When something goes wrong

What you see	Why it happens and what to do
Load your accessions in Block 2 first.	There is no passport data. Load your collection or the example one (Chapter 2).
An accession does not appear on the map	It lacks the decimal latitude or longitude, they are (0, 0) or out of range; or a crop or route filter left it out. If it only has degrees, minutes and seconds, Block 3 offers to convert them (Section 3.4).
The automatic map shows half the world	Some point lies far from the rest: another continent or a lost sign. Look at where the isolated point falls; if it is an error, correct it in Block 3. If it is real, choose All of Mexico or a single state.
The states and their codes are not shown	The extent spans more than 45° of longitude or does not touch Mexico. Choose All of Mexico or a single state; also check that state codes is on.
An accession counts in another state or another country	The point falls there, whatever the passport says. Tap the point, check the coordinate in the record and compare it with the collecting site.

What you see	Why it happens and what to do
With elevation, a point comes out gray	It has no elevation, or the one it has is impossible (outside –500 to 7,000 m), as in Figure 4.4. Tap it and correct the value in its record.
A value comes out gray as *the rest*	Only the eleven most frequent values get their own color. Filter by crop or by route.

DECISION RULE · WHAT TO DO IN BLOCK 4

If you have just loaded or corrected the collection	**then** look at the automatic extent first: an isolated point on the other side of the map is almost always a lost sign.
If the point falls in a state or country other than the passport's	**then** tap it, check the coordinate in its record and decide whether the error is in the coordinate or in the text.
If you want to see where collecting effort is concentrated	**then** use the grid, with equal cells, rather than the accessions choropleth.
If you are looking for scarce material	**then** use `size by stock` and filter by the route you are interested in.
If the map is going into a publication	**then** give it a title, decide whether to keep the state codes and export it as SVG or from the figure studio; save the points CSV as well.

The map does not modify the passport; Blocks 5 and 6 use the same points and states.

5

CHAPTER FIVE · BLOCK 5 · GEOGRAPHIC DIVERSITY

Geographic diversity

A collection is not worth how many envelopes it has, but how many distinct things it holds and where. Block 5 counts that diversity by region, by grid cell or by elevation belt; it compares it at equal effort, so that the state with the most field trips does not look the most diverse for that reason alone; it estimates how much is still left to collect; and it asks whether nearby regions hold the same things. Everything comes from the passport: there is nothing else to enter.

5.1 Units, types and the block summary

5.2 The diversity indices, unit by unit

5.3 Rarefaction: comparing at the same effort

5.4 How much is missing? Accumulation, estimators and coverage

5.5 Exclusive types and the index map

5.6 Do the regions resemble each other? Similarity and clustering

5.7 The Mantel test and turnover between units

5.8 The elevation gradient

5.9 Exporting, and how everything changes with the unit

5.1 Units, types and the block summary

Block 5 works on the passport accessions that have coordinates (Chapter 2): when you open it, or when you change any selector, it computes everything behind the animated wait *Computing diversity and the Mantel test*. With the example collection, **45 of the 46 accessions** come in; the one without coordinates (MEX-ZM-0043) stays out of the whole block. If there are no accessions, the block says `Load your accessions in Block 2 first.`

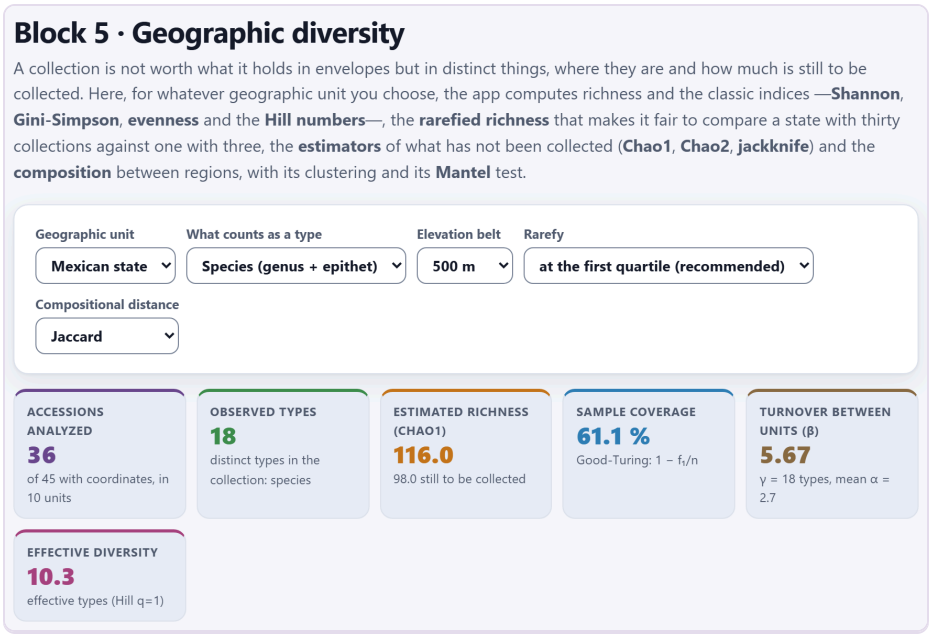


Figure 5.1. The block header with the example collection and the default values. The `Cell size` selector appears only when the unit is `Grid cell`.

Two decisions govern everything that follows: **how the territory is divided** (the unit) and **what counts as distinct** (the type). The other selectors fine-tune details.

Table 5.1. The six selectors of the block and their options (the default in bold).

Selector	Options and what they do
`Geographic unit`	`Mexican state`: each point goes to the state whose outline contains it, and points outside Mexico form no unit. `Country`: the country where the point falls (if it falls in none, the one in `ORIGCTY`). `Grid cell`: squares of 0.25°, 0.5°, 1° or 2°. `Elevation belt`: elevation belts according to `ELEVATION`; accessions without elevation do not come in.
`What counts as a type`	`Species (genus + epithet)`, `Crop` (`CROPNAME`), `Genus` or `Accession name` (`ACCENAME`). The species does not distinguish subspecies: teosinte counts as *Zea mays*. An empty value counts as the type "—".
`Cell size`	`0.25°`, `0.5°`, `1°` or `2°`; only with the unit set to grid cells. The cells are the same ones the Block 4 map draws (Chapter 4).

Table 5.1. The six selectors of the block and their options (the default in bold). *(continued)*

Selector	Options and what they do
Elevation belt	100 m, 250 m, 500 m or 1000 m. It is used for the belt unit and, always, for the gradient section (Section 5.8).
Rarefy	The common effort at which richness is compared (Section 5.3): at the first quartile (recommended), at the median, at the smallest unit or at 5, 10 or 20 accessions.
Compositional distance	Jaccard, Sørensen or Bray-Curtis (Section 5.6).

The six summary boxes summarize all the units taken together, not a single one. All the totals come from the same accessions: those that fall in some unit. With the unit set to states, the example collection distributes 36 of its 45 accessions with coordinates among 10 states. The other nine —the two maizes from Guatemala, the two potatoes, the oca, the barley, the amaranth from India, the cassava and the maize whose longitude lost its sign— enter no calculation, and a note at the start of section 1 says so: *9 of the 45 with coordinates are left out because they fall outside Mexico.* With another unit the note gives another reason (belts, for example, leave out the accessions without a valid elevation).

Table 5.2. The summary boxes with the example collection.

Box	Example	What it says
Accessions analyzed	36	Those that fall in some unit; below, out of how many with coordinates (45) and in how many units (10).
Observed types	18	The distinct types among the accessions analyzed, the γ richness: here, 18 species.
Estimated richness (Chao1)	116.0	How many types there would be in total according to Chao1, and how many would still be left to collect (98.0). With this collection, unreliable (Section 5.4).
Sample coverage	61.1 %	The Good–Turing coverage (Section 5.4).
Turnover between units (β)	5.67	Whittaker's beta; below, γ = 18 types and the mean richness per unit, α = 2.7 (Section 5.7).
Effective diversity	10.3	The Hill number of order 1 of the accessions analyzed (Section 5.2).

! CLEAN BEFORE YOU MEASURE

Block 5 counts what is in the passport, errors included. In the uncorrected example collection, the maize *Tuxpeño de la costa*, with its unsigned longitude, does not fall in Veracruz (it falls in Myanmar, which appears as a "country" if you choose that unit) and stays out of the states; the tepary bean, at 25,000 m, stays out of the belts and the gradient, because the app only accepts elevations from −500 to 7,000 m; and the two duplicates count twice. Sort out Blocks 2 and 3 first (Chapters 2 and 3). The figures in this chapter are those of the freshly loaded collection, so that you can reproduce them; where cleaning changes something important, it is said.

5.2 The diversity indices, unit by unit

Section 2 · Where is it? is a table with one row per unit and ten columns. A click on a header sorts by that column —from highest to lowest the first time, except for Unit, which goes in alphabetical order— and another click reverses the order; the arrow marks the active column, and cells with "—" always go last, in both directions. By default, the units run from the one with the most accessions to the one with the fewest (belts, from the lowest to the highest).

2 · Where is it?

UNIT	ACCESSIONS ▼	RICHNESS S	RAREFIED TO 2	H'	1 − D	J'	HILL q1	EXCLUSIVE	MEAN ELEVATION
Oaxaca	7	3	1.52	0.80	0.449	0.72	2.22	0	1,836 m
Puebla	7	5	1.86	1.48	0.735	0.92	4.37	4	1,661 m
Veracruz	6	4	1.80	1.24	0.667	0.90	3.46	4	939 m
Chiapas	4	4	2.00	1.39	0.750	1.00	4.00	3	1,834 m
Michoacán	3	3	2.00	1.10	0.667	1.00	3.00	1	1,780 m
Chihuahua	2	1	1.00	0.00	0.000	–	1.00	0	2,311 m
Yucatán	2	2	2.00	0.69	0.500	1.00	2.00	0	59 m
México	2	2	2.00	0.69	0.500	1.00	2.00	1	2,248 m
Jalisco	2	2	2.00	0.69	0.500	1.00	2.00	1	1,228 m
Sonora	1	1	–	0.00	0.000	–	1.00	1	–

The rarefied column compares every unit at the same effort (2 accessions); units with fewer are left blank because rarefying upwards would be making things up. "Exclusive" counts the types —species— found in no other unit: if that unit is lost, they go with it. J' shows "—" when the unit holds a single type: with one there is no spread to measure.

Figure 5.2. The table of indices by state with the example collection. Sonora has no mean elevation: its only accession, the tepary bean, carries the impossible elevation of 25,000 m.

Table 5.3. The columns of the table.

Column	What it is
Accessions	*N*: how many accessions fall in the unit.
Richness S	*S*: how many distinct types it holds.
Rarefied to 2	The types that would be expected with the same number of accessions in every unit; the number in the header is that effort (Section 5.3).
H', 1 − D, J', Hill q1	Shannon, Gini–Simpson, Pielou's evenness and effective diversity (below); J' shows "—" with a single type. Richness S is the Hill number q0; q2 has no column, but it equals 1 / (1 − (1 − D)).
Exclusive	The types found in no other unit (Section 5.5).
Mean elevation	The mean of `ELEVATION` over the unit's accessions. Empty or impossible elevations (outside −500 to 7,000 m) do not count; if none is left, "—".

THEORY IN BRIEF · SHANNON, GINI–SIMPSON AND PIELOU

In a unit with N accessions spread over S types, let $p_i = N_i/N$ be the proportion of type i. The classic indices are:

$$H' = -\Sigma\ p_i \ln p_i \cdot 1 - D = 1 - \Sigma\ p_i^2 \cdot J' = H' / \ln S$$

Shannon (H') grows with the number of types and with how even their distribution is; it uses the natural logarithm. **Gini–Simpson** (1 – D) is the probability that two accessions drawn at random, with replacement, are of different types. **Pielou** (J') runs from 0 to 1 and measures evenness alone: 1 means that all types have the same number of accessions. With a single type, J' = 0/0 is undefined: the app writes "—" (and leaves the cell empty in the CSV).

THEORY IN BRIEF · HILL NUMBERS

Shannon, Gini–Simpson and Pielou are on different scales and cannot be compared with one another. **Hill numbers** translate them to a single scale, that of **effective types**: how many equally abundant types would give the same diversity.

$$q0 = S \cdot q1 = \exp(H') \cdot q2 = 1 / \Sigma\ p_i^2$$

q0 counts all types equally; q1 weights them by their proportion; q2 gives even more weight to the abundant ones. Always q0 ≥ q1 ≥ q2, and the gap between them tells how dominated the unit is by a few types.

WORKED EXAMPLE · THE SEVEN ENVELOPES FROM OAXACA

Oaxaca has 7 accessions: 5 of *Zea mays*, 1 of *Phaseolus vulgaris* and 1 of *Capsicum annuum*, so p = 5/7, 1/7, 1/7.

$H' = -(5/7 \cdot \ln 5/7 + 2 \cdot 1/7 \cdot \ln 1/7) = 0.24034 + 0.55597 = 0.7963 \rightarrow$ the app writes 0.80

$\Sigma\ p^2 = (25 + 1 + 1)/49 = 27/49 = 0.5510$	$1 - D = 22/49 = 0.449$
$J' = 0.7963 / \ln 3 = 0.7963 / 1.0986 = 0.72$	$q1 = e^{0.7963} = 2.22$
$q0 = 3 \cdot q2 = 49/27 = 1.81 \rightarrow$ 3 types, but "worth" 2.2 even ones, or 1.8 if the abundant ones weigh more	

How to read it for a decision: Oaxaca and Puebla have the same 7 accessions, but Puebla holds 5 species with J' = 0.92 and Oaxaca 3 with J' = 0.72: in Oaxaca, five of seven envelopes are maize. Oaxaca is well represented for maize and poorly for everything else. At the other extreme, Chiapas has 4 accessions and 4 species (J' = 1, q1 = 4): every collection brought something new, a sign that the collection has not started repeating itself there and that more effort would keep yielding new types.

THE WHOLE COLLECTION HAS INDICES TOO

The same formulas, applied to the 36 accessions analyzed, give H' = 2.336, 1 − D = 0.821, J' = 0.808, q1 = 10.34 and q2 = 5.59. The Effective diversity box shows q1 = 10.3: there are 18 species, but the weight of maize (14 of the 36 accessions) makes them equivalent to about 10 equally represented species.

5.3 Rarefaction: comparing at the same effort

Richness grows with the number of accessions examined: a state with thirty collections will almost always have more types than one with three, even if its territory is not more diverse. The app itself reminds you of this at the foot of the panel: Why rarefaction is not a whim. **Hurlbert's rarefaction** (1971) answers how many types you would expect in each unit if you had taken the same number *n* of accessions from all of them.

THEORY IN BRIEF · HURLBERT'S RAREFACTION AND ITS VARIANCE

If *n* of *N* accessions are taken without replacement, the probability of not seeing type *i*, which has N_i, is $\pi_i = C(N - N_i, n) / C(N, n)$, where C is the number of combinations. The expected richness is the sum of the probabilities of seeing it:

$$E[S_n] = \Sigma\,(1 - \pi_i) \cdot \mathrm{Var} = \Sigma\,\pi_i(1 - \pi_i) + 2\,\Sigma_{i<j}\,(\pi_{ij} - \pi_i\pi_j)$$

The variance is that of Heck, van Belle and Simberloff (1975); $\pi_{ij} = C(N - N_i - N_j, n) / C(N, n)$ is the probability of seeing neither of the two. The app computes the combinations with the logarithm of the gamma function, so that they do not overflow with large collections. With $n = N$, rarefaction gives *S* with zero variance.

WORKED EXAMPLE · OAXACA RAREFIED TO 2 ACCESSIONS

$C(7, 2) = 21 \cdot \pi_{maize} = C(2, 2)/21 = 1/21 \cdot \pi_{bean} = \pi_{chili} = C(6, 2)/21 = 15/21$

$E[S_2] = (1 - 1/21) + 2 \times (1 - 15/21) = 0.9524 + 0.5714 = 1.5238 \rightarrow 1.52$, as in the table

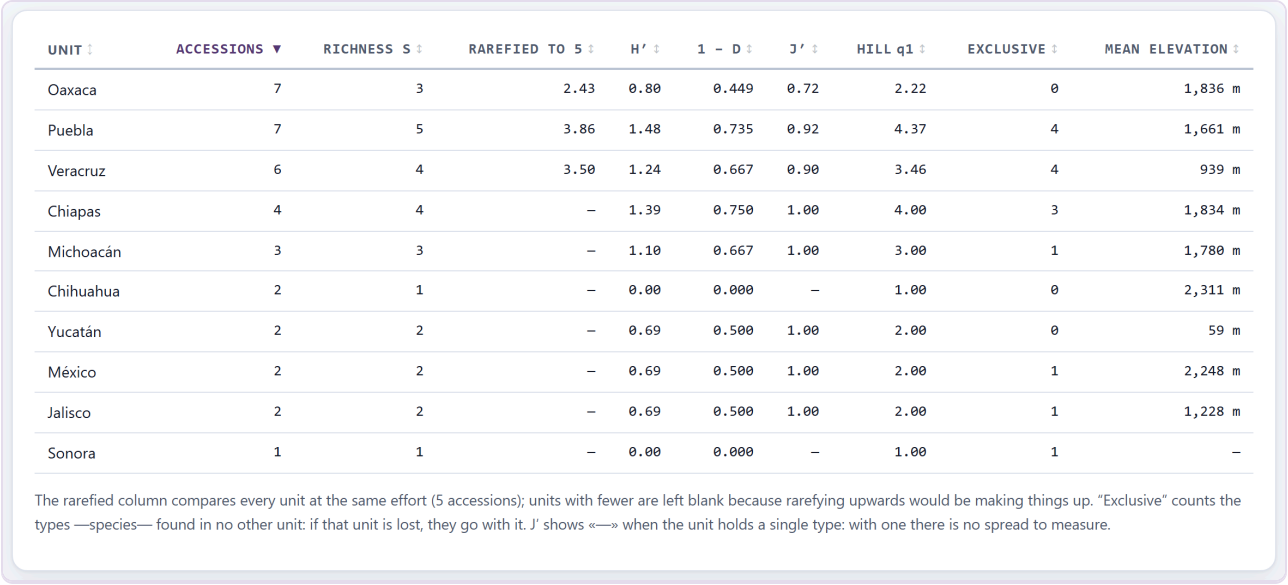
$\mathrm{Var} = 0.4535 - 0.2041 = 0.2494 \cdot$ standard deviation $= 0.50$

It can be checked by hand: with two envelopes there is only 1 type if both are maize, C(5, 2)/21 = 10/21; in the other 11 cases out of 21 there are 2 types. P(S = 2) = 0.5238, and the variance of such a variable is 0.5238 × 0.4762 = 0.2494.

The Rarefy selector chooses *n*. Units with fewer than *n* accessions are left with "—", because, as the note under the table says, rarefying upwards would be making things up. The relative options never go below 2.

Table 5.4. The common effort with the example collection (10 states, from 1 to 7 accessions).

Option	n	What remains
at the first quartile (recommended)	2	Nine states; only Sonora, with one accession, is left blank. It is a value that three out of four units reach.
at the median	3	Five: Chiapas and Michoacán 3.00, Puebla 2.60, Veracruz 2.45, Oaxaca 1.86.
at the smallest unit	2	The same as the quartile: the smallest has 1, and the app does not go below 2.
at 5 accessions	5	Three: Puebla 3.86, Veracruz 3.50, Oaxaca 2.43 (Figure 5.3).
at 10 or at 20 accessions	10, 20	None: the whole column shows "—".



UNIT	ACCESSIONS ▼	RICHNESS S	RAREFIED TO 5	H'	1 − D	J'	HILL q1	EXCLUSIVE	MEAN ELEVATION
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Puebla	7	5	3.86	1.48	0.735	0.92	4.37	4	1,661 m
Veracruz	6	4	3.50	1.24	0.667	0.90	3.46	4	939 m
Chiapas	4	4	–	1.39	0.750	1.00	4.00	3	1,834 m
Michoacán	3	3	–	1.10	0.667	1.00	3.00	1	1,780 m
Chihuahua	2	1	–	0.00	0.000	–	1.00	0	2,311 m
Yucatán	2	2	–	0.69	0.500	1.00	2.00	0	59 m
México	2	2	–	0.69	0.500	1.00	2.00	1	2,248 m
Jalisco	2	2	–	0.69	0.500	1.00	2.00	1	1,228 m
Sonora	1	1	–	0.00	0.000	–	1.00	1	–

The rarefied column compares every unit at the same effort (5 accessions); units with fewer are left blank because rarefying upwards would be making things up. "Exclusive" counts the types —species— found in no other unit: if that unit is lost, they go with it. J' shows «—» when the unit holds a single type: with one there is no spread to measure.

Figure 5.3. The table with Rarefy set to at 5 accessions: only the three states with 5 or more accessions can be compared at that effort.

The trade-off is clear: a small n lets you compare almost every unit, but discriminates little (with $n = 2$ the largest possible value is 2); a large one compares better, but fewer units. Look at two or three efforts and check whether the order of the units holds. In the example it does: at 2, 3 and 5 accessions, Oaxaca always stays below Puebla and Veracruz, even though it has as many accessions or more. That is the robust conclusion.

5.4 How much is missing? Accumulation, estimators and coverage

Section `1 · How much is there and how much is missing?` looks at all the units together. On the left is the **accumulation curve**: the rarefaction of the accessions analyzed for n = 1, 2..., 36, with a band of ±1.96 standard deviations (Heck's variance from the previous section), and four dashed lines with the estimators of total richness. On the right, `Richness against effort, unit by unit`: each unit as a point (accessions against types), with its least-squares line and Pearson's coefficient.

Figure 5.4. The note on the accessions analyzed, the warning about Chao1 and section 1 with the example collection. The curve shows no flattening at all: each new accession still brings, on average, almost 0.4 new species.

How to read the curve. If it flattens at the end, collecting more of the same hardly brings new types any more; if it ends still rising, as here, the collection is far from saturated. The example one goes from 17.22 expected types with 34 accessions to 18 with 36. The scatter on the right answers another question: with r = 0.83, the richness of each state is explained mostly by how many accessions it has, and that is why the table carries the rarefied column.

THEORY IN BRIEF · THE ESTIMATORS OF TOTAL RICHNESS

They all use the rare types, because those are the trace of what you have not seen yet (Chao, 1984, 1987; Burnham and Overton, 1978; Good, 1953). With **abundances** (accessions per type, over all units together), f_1 is the number of types with a single accession (*singletons*) and f_2 those with two (*doubletons*). With **incidences** (in how many units each type occurs, out of m units), Q_1 and Q_2 are the types present in one and in two units.

$$\text{Chao1} = S + f_1^2 / (2 f_2) \quad (\text{if } f_2 = 0\text{: } S + f_1(f_1 - 1)/2)$$

$$\text{Chao2} = S + ((m - 1)/m) \cdot Q_1^2 / (2 Q_2)$$

$$\text{Jackknife 1} = S + Q_1(m - 1)/m \cdot \text{Jackknife 2} = S + Q_1(2m - 3)/m - Q_2(m - 2)^2 / (m(m - 1))$$

$$\text{Good–Turing coverage} = 1 - f_1 / N$$

WORKED EXAMPLE · THE ESTIMATORS OF THE EXAMPLE COLLECTION

Abundances: $N = 36$, $S = 18$, $f_1 = 14$ (14 species with one accession), $f_2 = 1$ (*Phaseolus vulgaris*)

$\text{Chao1} = 18 + 14^2 / (2 \times 1) = 116$	coverage $= 1 - 14/36 = 0.611 = 61.1\%$

Incidences in the $m = 10$ states: $S = 18$, $Q_1 = 15$, $Q_2 = 1$ (*P. vulgaris*, in Oaxaca and Michoacán)

$\text{Chao2} = 18 + 0.9 \times 225/2 = 119.25$	$\text{Jackknife 1} = 18 + 15 \times 0.9 = 31.5$

$$\text{Jackknife 2} = 18 + 15 \times 17/10 - 1 \times 64/90 = 18 + 25.5 - 0.71 = 42.79$$

On the curve, the labels are rounded: Chao2 119, Chao1 116, Jackknife 2 43, Jackknife 1 32. **Coverage** is the probability that the next accession you collect belongs to a type you already have: here, 61%; with a probability of 39%, the next one would be a new species.

With a single doubleton, the Chao1 fraction explodes. When there are **fewer than three doubletons**, the app puts the warning Careful with Chao1 in this collection. at the top of the section, but the summary box still shows 116.0: do not quote it without the warning. Jackknife 1 is the most stable; Chao2, with a single type in two units, explodes just like Chao1.

WHAT THE TOTALS MEASURE DEPENDS ON THE UNIT

The curve, the coverage and the four estimators come from the same 36 accessions, so they can be compared with one another; but they speak only of what was collected in Mexico. If you want them to cover the whole collection, change the unit to Country or to Grid cell: with those, all 45 accessions with coordinates and the 24 species come in (Section 5.9).

5.5 Exclusive types and the index map

A type is **exclusive** to a unit when it appears in no other. The Exclusive column counts them, and their sum is the Q_1 of the previous section. In the example collection there are 15, distributed as follows: Puebla 4 (*Cucurbita pepo*, *Amaranthus hypochondriacus*, *Physalis philadelphica*, *Cucurbita argyrosperma*), Veracruz 4 (*Cucurbita moschata*, *Vanilla planifolia*, *Coffea arabica*, *Sechium edule*), Chiapas 3 (*Phaseolus coccineus*, *Theobroma cacao*, *Quercus crassifolia*), and one each in Michoacán (avocado), the State of Mexico (wheat), Jalisco (agave) and Sonora (tepary bean). Oaxaca, with 7 accessions, has none.

For the genebank, an exclusive type is a risk and a priority at the same time: if the accessions of that unit are lost, the type disappears from the collection. They are good candidates for the safety duplicate and for recollecting in the same region; Block 6 (Chapter 6) also reviews the types with a single accession, which are not the same thing: a type can be exclusive to a unit with several accessions.

Below the table, with the unit set to states, countries or cells, comes the Index map: a choropleth map that colors each unit by the index you choose in its selector — Accessions, Observed richness (S) (the default), Rarefied richness, Shannon (H′), Gini-Simpson (1 − D), Evenness (J′), Effective types (Hill q=1) or Exclusive types —, with a sentence beside it that explains what it measures. Units without a value stay white; the ramp runs from the minimum to the maximum of the units, and both are written at its ends. The map uses the outlines and the projection of Block 4, explained in Chapter 4. With the unit set to elevation belts there is no map.

Figure 5.5. The index map with Exclusive types. The three lightest states have accessions, but no type of their own; the white ones have no accessions.

Figure 5.6. With Geographic unit set to Grid cell of 1°, the map paints the cells over the states. The richest, with 5 species in 6 accessions, is the one at 19.50, −98.50.

With cells, the label of each unit is the coordinate of its center. The 1° grid distributes the 45 accessions among 21 cells, nine of which have a single accession. It is the most honest unit when the states differ greatly in size, but the finer it is, the more cells with one or two accessions and the less you can rarefy.

5.6 Do the regions resemble each other? Similarity and clustering

Section 3 · Do the regions resemble each other? compares the **composition** of the units: not how much they hold, but whether they hold the same things. Only units with at least two accessions come in, and at least three such units are needed; in the example, 9 states come in (Sonora stays out).

THEORY IN BRIEF · JACCARD, SØRENSEN AND BRAY–CURTIS

For two units A and B, with a and b types and c types in common, and with x_k, y_k the accessions of type k in each, the dissimilarities of Jaccard (1912), Sørensen (1948) and Bray and Curtis (1957) are:

$$\text{Jaccard} = 1 - c / (a + b - c) \cdot \text{Sørensen} = 1 - 2c / (a + b) \cdot \text{Bray–Curtis} = \Sigma\, |x_k - y_k| \,/\, \Sigma\, (x_k + y_k)$$

All three run from 0 (identical composition) to 1 (nothing in common). Jaccard and Sørensen look only at presences; Sørensen gives double weight to what is shared and never exceeds Jaccard. Bray–Curtis also weighs how many accessions there are of each type.

WORKED EXAMPLE · OAXACA AND YUCATÁN

Oaxaca: maize 5, bean 1, chili 1. Yucatán: maize 1, chili 1. They share $c = 2$ of $a = 3$ and $b = 2$ types.

$$\text{Jaccard} = 1 - 2/3 = 0.333 \qquad \text{Sørensen} = 1 - 4/5 = 0.200$$

$$\text{Bray–Curtis} = (|5 - 1| + |1 - 0| + |1 - 1|) \,/\, (7 + 2) = 5/9 = 0.556$$

With Jaccard it is the most similar pair of the nine states, and that is why it is the first pair the tree joins.

With the dissimilarity matrix, the app clusters the units by **UPGMA** (unweighted pair-group method with arithmetic means): it joins the closest pair, replaces the two by a group whose distance to each unit is the size-weighted mean of its members' distances, and repeats until a single group remains. Each join is drawn at a height equal to half its distance; Oaxaca and Yucatán join at 0.167 (half of 0.333).

How to read it. In the heat map each cell is the similarity (1 – dissimilarity) between two units: the darker, the more they share, according to the 0, 50 and 100% legend; the diagonal is each unit against itself. Hover over a cell to see the pair and its percentage. In the example, Veracruz splits off from all the others at the maximum height, 0.5: its four species are exclusive, and it shares none with anyone. This is not an oddity of the territory but an effect of the data: its maize is the one that lost the sign of its longitude. With the Block 3 corrections applied, Veracruz recovers its maize (it ends up with 7 accessions and 5 species) and is no longer isolated. Chihuahua and México (0.25) and the Michoacán–Oaxaca–Yucatán group are the maize blocks with one companion.

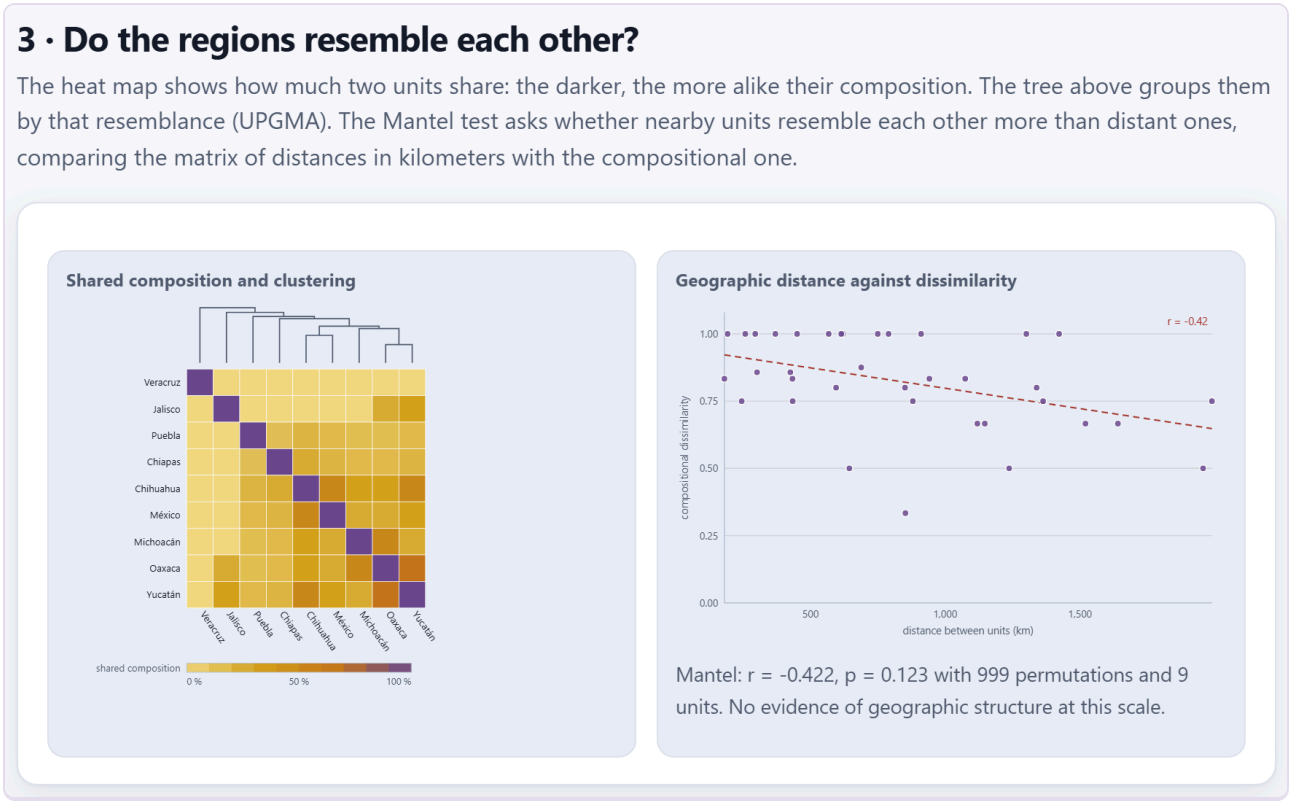


Figure 5.7. Section 3 with Jaccard. The rows and columns of the heat map follow the order of the dendrogram's leaves, so the groups appear as blocks of color along the diagonal.

Changing the distance changes the tree: with Sørensen the order is almost the same; with Bray–Curtis, which weighs abundance, Oaxaca and Yucatán go from 0.333 to 0.556, because the five maizes of Oaxaca carry weight, and the closest pairs are now states with two similar accessions (Chihuahua, México and Yucatán with one another, at 0.500).

5.7 The Mantel test and turnover between units

Do nearby units hold the same things, and distant ones different things? The **Mantel test** asks this by comparing two matrices: the distances in kilometers between the units (haversine, Section 3.5) and the compositional dissimilarity. It is the right-hand chart of Figure 5.7: one point for each pair of units —36 with nine states— and the least-squares line as a visual aid.

THEORY IN BRIEF · THE MANTEL TEST

In the Mantel test (1967), the statistic is Pearson's correlation *r* between the elements of the two matrices below the diagonal. Since those elements are not independent (each unit enters many pairs), their significance is not taken from a table: the rows and columns of one of the matrices are permuted at random, 999 times, *r* is recomputed each time, and

$$p = (\text{number of permutations with } |r_{perm}| \geq |r| + 1) / (999 + 1)$$

The test is two-tailed. The permutations use a fixed seed, so the same analysis always gives the same *p*. A positive and significant *r* says that "what is close is alike".

With the example collection, the note below the chart says *Mantel: r = -0.422, p = 0.123 with 999 permutations and 9 units. No evidence of geographic structure at this scale.* The negative sign —distant states would be slightly more alike— means nothing with $p = 0.123$ and nine units; what it does say is that composition is not ordered by geography but by what was collected on each trip. With $p < 0.05$ the sentence changes to *There is geographic structure: what is close is alike.*

The position of each state or country is the center of its bounding rectangle, and that of each cell, its center: with large states, the distance between centers is a rough approximation. With the unit set to elevation belts, the test does not use kilometers but the **elevation difference between the centers of the belts**, in meters, and the axis, the title and the note say so: with the example, *Mantel (against elevation difference): r = 0.405, p = 0.176 with 999 permutations and 6 units. No evidence of elevational structure with these belts.*

At least four units with two or more accessions are needed. If the test cannot be computed, the note gives the reason: that there are too few units with a composition to compare (by country, only 3 of the 8 have two or more accessions, and the app suggests using states, grid cells or elevation belts), or that all the dissimilarities are equal and there is no variation to correlate (by accession name, no state shares names with another).

Turnover between units

The Turnover between units (β) box holds **Whittaker's beta**, the simplest measure of how much composition changes from one unit to another:

$$\beta = \gamma / \bar{\alpha} - 1 \quad (\gamma\text{: types of all units together; } \bar{\alpha}\text{: mean richness per unit})$$

With the example, $\gamma = 18$, $\bar{\alpha} = (3 + 5 + 4 + 4 + 3 + 1 + 2 + 2 + 2 + 1)/10 = 2.7$ and $\beta = 18/2.7 - 1 =$ **5.67**. $\beta = 0$ would mean that every unit has all the types; a large β, that each unit holds a small and distinct part of the collection, as here. For the genebank, a high β means that no region substitutes for another: losing a state, or ceasing to collect there, leaves gaps that the others do not fill. Since γ and $\bar{\alpha}$ come from the same units, β can be compared between analyses with the same unit; between different units, it cannot: more and smaller units always give a larger β (Table 5.5).

5.8 The elevation gradient

Section 4 · The elevation gradient sorts into elevation belts the accessions with coordinates and a valid elevation (from −500 to 7,000 m), whatever unit is chosen (with the unit set to belts it does not appear, because the table already is that). It draws every belt from the lowest to the highest with data, the empty ones included. For each belt with accessions, two bars: accessions above, distinct types below, with both figures alongside; an empty belt carries 0 · 0 — no collections in this belt and a dotted line. The highest goes at the top. The belt width is that of the Elevation belt selector, and a footnote says how many accessions are included, how many belts are empty and how many accessions were left out for having no elevation or an impossible one.

Figure 5.8. The gradient with 500 m belts and, below it, the block's three export buttons. The belt from 3,000 to 3,500 m is empty; the 25,000 m tepary bean is left out and the note says so.

In the example, more than half the collection lies between 1,500 and 2,500 m (26 of the 44 included, with 6 and 9 species), the elevation of the central valleys where the maizes are concentrated. In the 2,000 to 2,500 m belt, 15 accessions give only 9 species; in the 0 to 500 m belt, 5 give 4. The lowlands are under-collected, but the little there is is varied: that is where a field trip would pay off most.

The empty belts are the most useful reading of the chart: from 3,000 to 3,500 m there is no collection at all, between the bitter potato at 3,797 m and everything else, so that elevation is not represented. And an impossible elevation does not vanish silently: the note counts it and says where to fix it: *1 accession has an impossible elevation and is left out (accepted range −500 to 7,000 m; fix it in the accession record, in Block 2)* (Chapter 2).

5.9 Exporting, and how everything changes with the unit

At the end of the panel there are three buttons (Figure 5.8):

- Table as CSV saves geographic-diversity.csv, in UTF-8, with one row per unit and the columns `unit`, `key` (the code of the state, of the country, of the cell or the lower limit of the belt), `accessions`, `richness`, `rarefied`, `shannon`, `gini_simpson`, `pielou` (empty with a single type), `hill_q1`, `exclusive`, `mean_elevation`, `lat` and `lon` (the position used by the Mantel test; empty for belts). The indices carry three decimals, not rounded to two as on screen.
- Curve as SVG saves accumulation-curve.svg, the curve with its estimators.
- Heat map as SVG saves composition-similarity.svg, the heat map with its dendrogram.

The SVG files carry the light theme colors, so that they look the same in a document. As in the other blocks, the figures can also be opened in the figure studio (Section I.6). The CSV does not store the settings: write down which unit, type and effort you used.

The same collection, other units

All the results depend on the unit and the type. With the example collection:

Table 5.5. What changes with the unit and the type (uncorrected example collection; the other selectors at their defaults).

Unit · type	Units · accessions	γ	β	Mantel and what is worth noting
State · species	10 · 36	18	5.67	r = −0.422, p = 0.123, 9 units. The figures of this chapter.
Country · species	8 · 45	24	6.38	Mexico holds 36; only 3 of the 8 countries have two or more: a heat map, but no Mantel test. Myanmar comes from the unsigned longitude.
1° cell · species	21 · 45	24	12.62	r = −0.408, p = 0.055, 12 cells. Chao1 = 224, Chao2 = 234, jackknife 1 = 44.0.
0.5° cell · species	26 · 45	24	15.00	r = −0.399, p = 0.077, 12 cells; 14 of the 26 cells have a single accession.
500 m belt · species	7 · 44	23	4.37	Against elevation difference: r = 0.405, p = 0.176, 6 belts. No index map; the 25,000 m tepary is left out.
State · genus	10 · 36	14	4.38	Chao1 drops to 50: there are more genera with two or more accessions.
State · crop	10 · 36	18	5.67	Oaxaca rises to 4 types: the crop separates what the species lumps together.
State · accession name	10 · 36	35	9.00	Almost every accession is a type: no state shares names, so Mantel has no variation to correlate.

The species is the natural type for the interspecific diversity of a multi-crop genebank; in a single-crop genebank, the accession name (the landrace or the race) is what matters, provided it is written consistently (Section 3.3). The state serves to report to those who manage the territory; the cell, to compare fairly and plan trips; the belt, to see which environments are missing.

When something goes wrong

What you see	Why it happens and what to do
Load your accessions in Block 2 first.	There is no passport data. Load your collection or the example one (Chapter 2).
Accessions analyzed is smaller than the accessions with coordinates	Some fall in no unit (outside Mexico, or with no valid elevation for belts); the note in section 1 says how many and why.
The rarefied column is full of "—"	The effort is larger than the accessions of almost every unit. Choose at the first quartile (recommended) or a larger unit.
Careful with Chao1 in this collection. appears	There are fewer than three types with two accessions. Read jackknife 1 and the curve, not Chao1.
at least three units with two accessions each are needed	There is nothing to compare composition with. Use a larger unit (2° cells, 1,000 m belts).
The Mantel note says it cannot be computed	Read the reason: too few units with two or more accessions (use larger units) or all dissimilarities equal (change the type).
The gradient note says an accession has an impossible elevation	A mistyped elevation (outside –500 to 7,000 m). It is left out of the belts, the gradient and the mean elevation; correct it in its Block 2 record.
The Index map is not shown	The unit is the elevation belt, which has no geography.

DECISION RULE · WHAT TO DO IN BLOCK 5

If you are going to measure diversity	**then** first correct coordinates, elevations and duplicates (Blocks 2 and 3): each error changes the units, the tree and the estimators.
If you want to know which unit is best represented	**then** compare the rarefied column, at two or three efforts, and not the raw richness; keep the order that holds.
If a unit has S equal to N, or J′ = 1 with few accessions	**then** sampling is lacking there: each collection brought something new. It is a candidate for the next field trip (Block 6).
If a unit has many accessions and a low q1	**then** it is saturated: collect other species there.
If the accumulation curve does not flatten or the coverage is low	**then** it is far from saturated; report jackknife 1 as a floor for what is missing, and Chao1 only without the warning.
If a unit has exclusive types	**then** secure their safety duplicate and consider recollecting there.
If the Mantel test gives $p < 0.05$ with a positive r	**then** diversity has geographic structure: spread your collecting over the territory instead of concentrating it.
If a belt is missing from the gradient	**then** that elevation is not represented: note it as a gap for Block 6.

Block 5 does not change the passport: it only measures it. What turns out poor or absent here is what Block 6 turns into priority cells and a collecting route.

6

CHAPTER SIX · BLOCK 6 · COLLECTING GAPS

Collecting gaps

Block 5 measured what the collection holds; Block 6 looks at what it lacks. It throws a grid over the territory, marks the cells without any collection and ranks them with a priority index whose four components you can see and weight as you like, so that the next field trip goes where it pays off most. Separately, it reviews the gaps a map does not show —types with a single accession, without a safety duplicate, only in a field genebank, only in situ or only as DNA— and looks for the smallest set of localities that brings together all the diversity. Everything comes from the passport and from geography: it is a geographic and collection gap analysis, not an ecogeographic one.

6.1 The territory, the grid and the block summary

6.2 The map of what is missing

6.3 The priority index, component by component

6.4 Moving the weights: what you are after on the next trip

6.5 The collecting route for the GPS

6.6 What you already have and could lose

6.7 The indispensable minimum: complementarity

6.8 How everything changes with the cell, the territory and the type

6.1 The territory, the grid and the block summary

A **collecting gap** is a part of the territory, or of a crop's diversity, that the collection does not hold. Block 6 looks for them with what is already in the passport (Chapter 2): the coordinates, the species, the conservation route and the safety duplicate. When you open it, or when you change any selector, it recomputes everything behind the animated wait *Searching for collecting gaps*. If there are no accessions, the block says `Load your accessions in Block 2 first.`

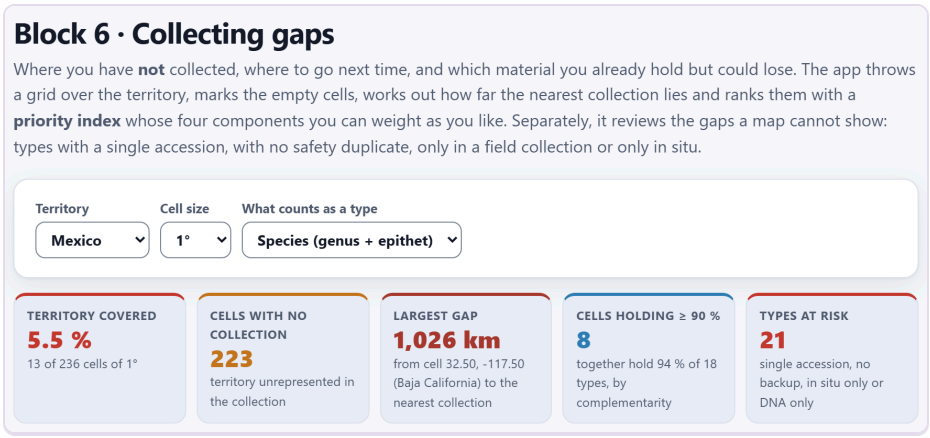


Figure 6.1. The block header with the example collection and the default values.

Three selectors govern the analysis; the other controls are inside each section.

Table 6.1. The three header selectors (the default option in bold).

Selector	Options and what they do
`Territory`	`Mexico`: the grid keeps the cells that touch the land of some state. `The world`: those that touch the land of some country, between 60° south and 84° north; the deficit in the index is then computed by country (Section 6.8).
`Cell size`	`0.5°`, `1°` or `2°`. The cells are laid out as in Blocks 4 and 5: squares aligned with the meridians and parallels that are multiples of the size.
`What counts as a type`	`Species (genus + epithet)`, `Crop`, `Genus` or `Accession name`, with the same criterion as Block 5 (Table 5.1). It decides what "richness" is, what "exclusive" is and which types are reviewed in section 3. An empty value does not count as a type in this block.

The five summary boxes sum up the result. With the example collection:

Table 6.2. The summary boxes with the example collection.

Box	Example	What it says
`Territory covered`	5.5 %	The proportion of cells with at least one collection: 13 of 236 cells of 1°. Below 25% the figure turns red.
`Cells with no collection`	223	The empty cells: the territory the collection does not represent (Section 6.2).

Table 6.2. The summary boxes with the example collection. *(continued)*

Box	Example	What it says
Largest gap	1,026 km	The greatest distance from an empty cell to the nearest collection; below, which cell it is: 32.50, −117.50, in Baja California.
Cells holding ≥ 90 %	8	How many cells, chosen by complementarity, are enough to bring together at least 90% of the types; below, how much they really hold (94%) and the total number of types within the territory (18; Section 6.7).
Types at risk	21	The types whose risk adds up to 3 points or more: a single accession, only in situ, only as DNA, or no duplicate together with another signal (Section 6.6).

Next to the map's Paint selector, the app reports how many accessions fall outside the analyzed area. With the territory set to Mexico there are **9**: the two maizes from Guatemala, the two potatoes from Peru, the oca from Bolivia, the barley from Ethiopia, the amaranth from India, the cassava from Colombia and the maize *Tuxpeño de la costa*, whose unsigned longitude sends it to Myanmar. The other 36 of the 45 accessions with coordinates come in; the one without coordinates (MEX-ZM-0043) enters no geographic calculation, though it does enter section 3.

> **A GEOGRAPHIC AND COLLECTION ANALYSIS, NOT AN ECOGEOGRAPHIC ONE**
>
> An ecogeographic analysis, such as that of Ramírez-Villegas et al. (2010), asks which combinations of climate, soil and vegetation the collection is missing; that requires environmental layers, and GermplasmPro neither bundles nor downloads them, because nothing it does leaves your computer. Block 6 answers with what the passport and geography provide: where there are no collections, how far away the nearest one is, what lies around it and which types depend on a single accession. The note at the foot of the panel says so (Figure 6.2). If you need the environmental step, export the route of Section 6.5 and take it to a modeling tool; the elevation belt (Section 6.6) is the only environmental variable the block looks at.

How far this analysis goes. This is a **geographic and collection** gap analysis, not an ecogeographic one: knowing which combinations of climate, soil and vegetation your collection is missing requires environmental layers, and GermplasmPro neither bundles nor downloads them —nothing this app does leaves your computer—. If you need that step, export the coordinates here and take them to an environmental modeling tool. What this block does do is everything the passport and the geography allow, which is most of the work of deciding where to go.

Figure 6.2. The note that closes the Block 6 panel.

6.2 The map of what is missing

Section 1 · The map of what is missing draws the grid over the outlines of Block 4 (Chapter 4). Each square is a cell that touches the land of the territory or that holds some accession. Cells with collections are green, with a dot at their center; empty ones are painted according to the Paint selector, and ten white circles with a dark border and number mark the top ten priorities of the index (Section 6.3). If two circles would overlap, one is moved aside and a line joins it to its cell.

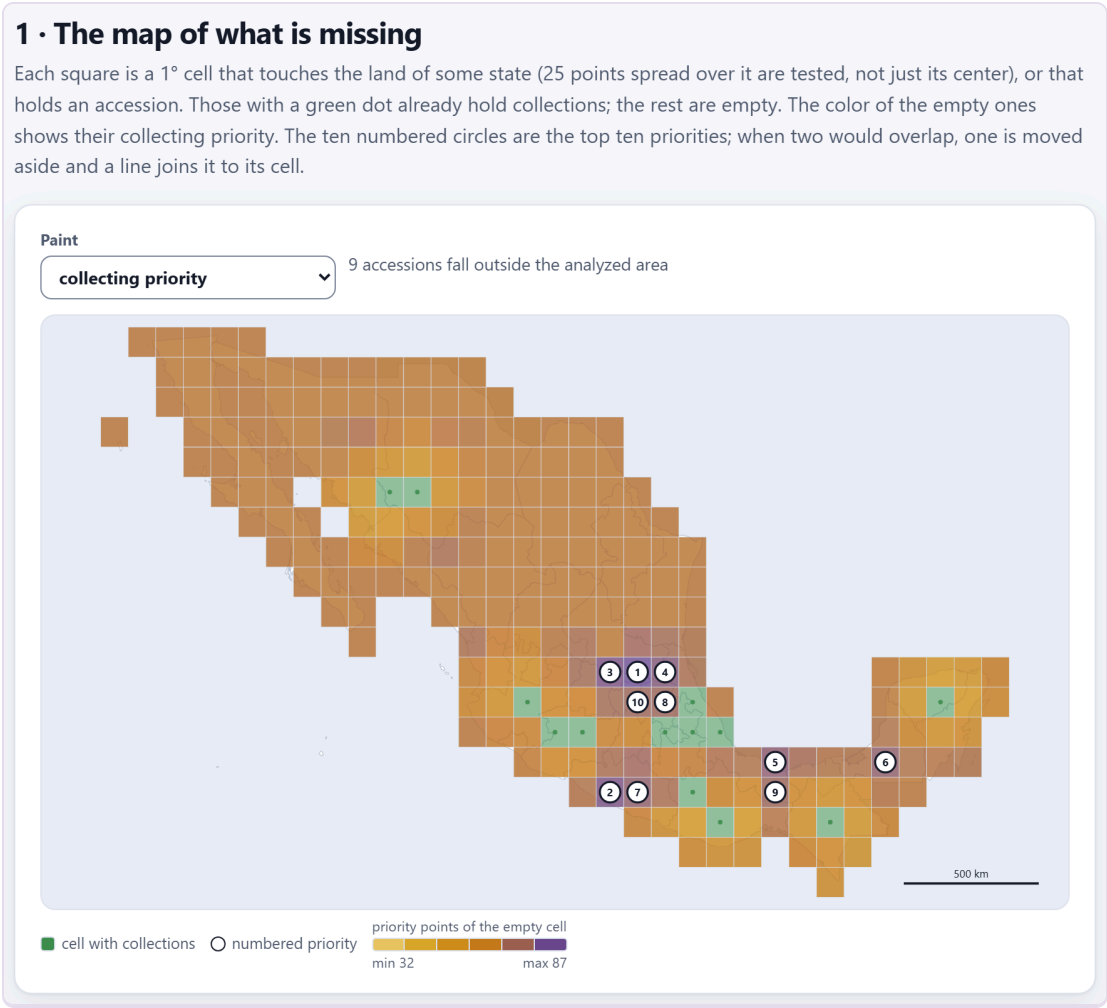


Figure 6.3. The gap map painted by priority, with the example collection. The darkest cells surround the area where the collections are concentrated, but without touching it.

At the lower right there is a scale bar in kilometers, with the same criterion as in Block 4 (Section 4.2): on its own for Mexico, with the latitude at which it holds if the extent is tall and, for the whole world, replaced by the notice No scale bar: it varies with latitude. When you hover over a cell, a label gives its center and, if it is empty, the distance to the nearest collection and its priority points (for example, *29.50, -118.50 · empty · 976 km · 60 pts*, the cell of Guadalupe Island); if it is occupied, how many accessions and types it holds. The legend changes with the mode: it says what the ramp measures and writes its minimum and maximum.

Table 6.3. The three ways of painting the map, with their ramp in the example collection.

Paint	What the ramp colors
collecting priority (default)	priority points of the empty cell, from 32 to 87; the darkest is the one with the most points. Occupied cells in green.
distance to nearest collection	km from the empty cell to the nearest collection, from 59 to 1,026: light next to the collections, purple at the extremes (Figure 6.4).
richness of occupied cells	types per occupied cell, from 1 to 5; empty cells stay uncolored and the legend says so (Figure 6.5).

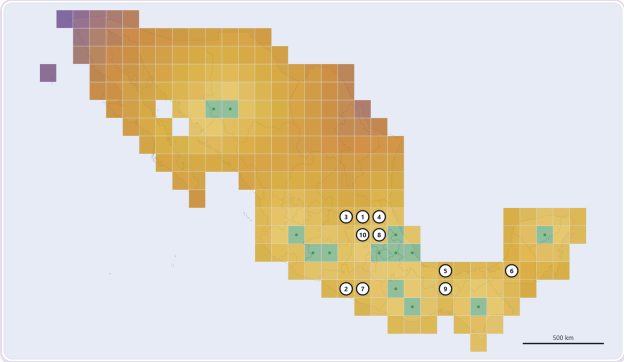


Figure 6.4. Painted by distance. The purple in the north-west is the largest gap, 1,026 km from the nearest collection.

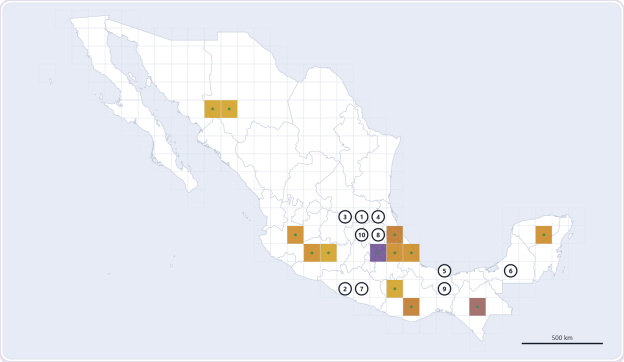


Figure 6.5. Painted by richness. The purple cell, 19.50, −98.50, holds five species.

The map is exported with the Map as SVG button (Section 6.5) and, like the other maps, it can be opened in the figure studio (Section I.6). For screen readers, the map and the curve of section 4 carry a description with their main figures.

THEORY IN BRIEF · THE GRID, THE EDGE CELLS AND THE DISTANCE

With cells of size t degrees, an accession with longitude λ and latitude φ falls in the cell with indices $i = \lfloor \lambda / t \rfloor$ and $j = \lfloor \varphi / t \rfloor$, whose center is $((i + ½)t, (j + ½)t)$. To know whether the cell belongs to the territory, the app does not look only at its center: it tests a 5 × 5 = 25 point lattice spread over it (with The world, 9 points with 1° and 2°, and 4 with 0.5°) and keeps the cells where some point falls on land. The fraction of points that falls in each state is its **land share** of the cell. This is how coastal cells whose center lies in the sea exist, and a state smaller than the cell still gets its own cell: if no sample point touches it, the app anchors it to the cell of an interior point of the state. All 32 states have at least one cell at any size.

The distance from an empty cell to the nearest collection is the smallest great-circle distance (haversine, Section 3.5) between its center and all the accessions with coordinates, including those that fall outside the area.

$$d = 2R \cdot \arcsin \sqrt{\left[\sin^2(\Delta\phi/2) + \cos \phi_1 \cos \phi_2 \sin^2(\Delta\lambda/2) \right]} \text{ with } R = 6{,}371.0088 \text{ km}$$

Each accession counts in the state where **its own point** falls, not in that of its cell's center. In the example, the cell 19.50, −98.50 is shared by the State of Mexico (32% of its land), Hidalgo, Puebla (24% each) and Tlaxcala (20%); its six accessions count four for Puebla and two for the State of Mexico, which is where they were collected. An accession that falls in the sea next to the coast counts in its cell, which exists even though its center also falls in the sea. With the territory set to Mexico, only an accession that falls in another country is left out.

6.3 The priority index, component by component

Not all empty cells are worth the same for the next field trip. Section 2 · Where to go next time gives them a score that adds up four components, each scaled between 0 and 1 and multiplied by its weight. The default weights add up to 100, so the score runs from 0 to 100.

THEORY IN BRIEF · THE PRIORITY INDEX OF AN EMPTY CELL

$$P = w_d \cdot \min(d, 250) / \min(d_{max}, 250) + w_r \cdot R_v / R_{max} + w_e \cdot \delta_{cell} + w_x \cdot X_v / X_{max}$$

Distance ($w_d = 40$): the distance d to the nearest collection, capped at 250 km —beyond that, the territory is equally unknown— and divided by the greatest distance among the empty cells (also capped). **Neighborhood richness** ($w_r = 25$): R_v, the distinct types in the cell's neighborhood (with 1°, the 5 × 5 block of cells centered on it), divided by the largest R_v in the grid. **Exclusive types nearby** ($w_x = 15$): X_v, the types in the neighborhood that occur in only one cell of the whole grid, divided by the largest X_v.

State deficit ($w_e = 20$). The collecting density of a state is ρ_e = accessions whose point falls in it ÷ its area in km²; its deficit, $\delta_e = 1 - \rho_e / \rho_{max}$: the best covered state is worth 0 and those with nothing are worth 1. A cell shared by several states takes the mean of their deficits weighted by the land share of each:

$$\delta_{cell} = \Sigma_e\, a_e\, \delta_e / \Sigma_e\, a_e \quad (a_e\text{: land share of state } e \text{ in the cell})$$

The neighborhood is the cells within about 220 km on each side, whatever the cell size: 2 cells with 1°, 4 with 0.5° and 1 with 2°. The section says so: *within about 220 km (2 cells of 1° each way)*. On equal scores, the cell with more richness around goes first; if still tied, the farthest one, and then the northernmost and the westernmost.

The table lists the 25 cells with the most points. Its columns are the priority number, the Cell center (latitude, longitude, with three decimals), the State with the most land in the cell —followed by "+*n*" if it touches others, which show when you hover (*Also touches: Querétaro, Hidalgo, Guanajuato*)—, the distance To nearest collection, the Richness nearby (R_v), the Score rounded to a whole number and, under Why, a bar that splits the score among the four components with their colors: purple for distance, green for richness, ochre for the deficit and magenta for exclusive types. The bar also carries the breakdown as text, for screen readers, and the table can be navigated with the keyboard.

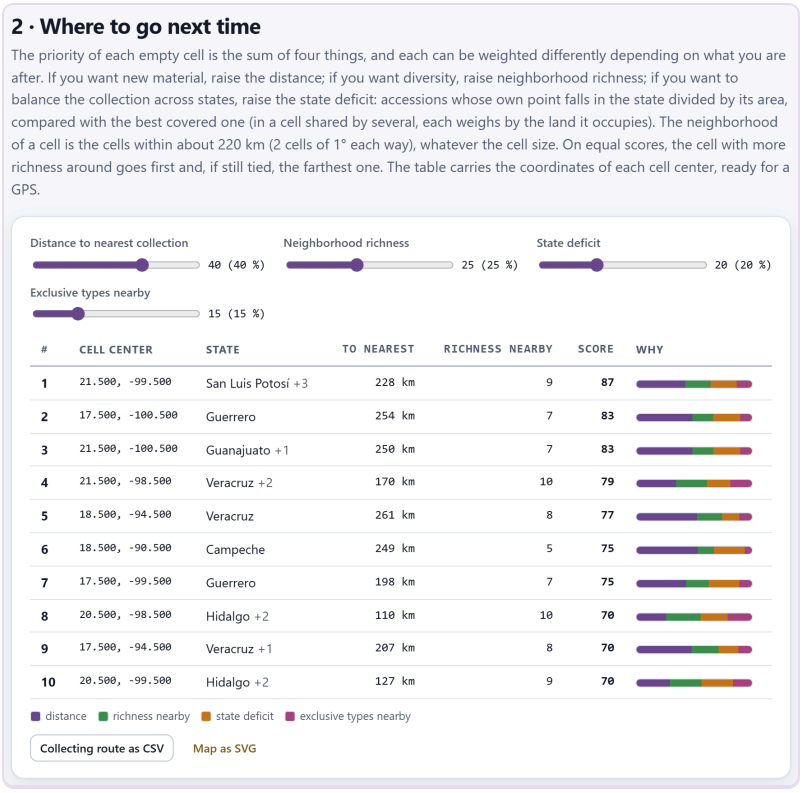


Figure 6.6. The weights and the first ten rows of the priority table (the table holds 25 and scrolls).

6

WORKED EXAMPLE · WHY THE CELL 21.50, −99.50 COMES FIRST

It is a cell of San Luis Potosí (56% of its land) that also touches Querétaro, Hidalgo and Guanajuato. Its 5 × 5 cell neighborhood (from −101.5° to −97.5° in longitude and from 19.5° to 23.5° in latitude) contains four occupied cells: the one at 19.50, −98.50 with 6 accessions and 5 species, one in Puebla with 2 (maize and husk tomato), one in Veracruz with 3 (squash, vanilla and maize) and one in Michoacán with 1 bean.

Distance: the nearest accession is MEX-ZP-0012, at 19.5523, −98.8155: $\Delta\phi = 1.948° \approx 216.6$ km, $\Delta\lambda \cdot \cos\phi \approx 71.3$ km → $d = 228.0$ km

$d_{max} = 1{,}026$ km > 250 → $40 \times 228.0 / 250 = 40 \times 0.912 = 36.48$

Richness: $R_v = 9$ species (12 accessions); $R_{max} = 12$ → $25 \times 9/12 = 18.75$

Deficit: the four states it touches have no accessions → $\delta = 1$ → $20 \times 1 = 20$

Exclusive: $X_v = 4 + 1 + 2 = 7$; $X_{max} = 9$ → $15 \times 7/9 = 11.67$

$P = 36.48 + 18.75 + 20 + 11.67 = 86.90$ → the table writes 87

The seven exclusive types are the four species of the cell 19.50, −98.50 found nowhere else (*Cucurbita pepo*, *Amaranthus hypochondriacus*, *Triticum aestivum* and *Cucurbita argyrosperma*), the husk tomato from Puebla, and the squash (*C. moschata*) and the vanilla from Veracruz. The cells that reach $R_{max} = 12$ and $X_{max} = 9$ are three empty ones in the band at 18.5° of latitude, between Puebla and Oaxaca, which have almost the whole collection of central Mexico in their neighborhood.

How to read it. The first three cells lie some 230 to 250 km from any collection and in states without a single accession, but with a rich neighborhood: unknown territory next to a diverse area, the typical case in which a trip is worth it. The eighth, 20.50, −98.50, in Hidalgo, is only 110 km from a collection, but surrounded by 10 species and 9 exclusive types: it rises because of richness, not distance, and its bar shows it with more green and magenta than purple. The 223 empty cells range from 86.90 points down to 31.83, the cell at 16.50, −97.50, in Oaxaca, at the bottom of the list.

> **THE STATE DEFICIT, WITH THIS COLLECTION**
>
> Puebla is the best covered: 7 accessions in 34,689 km², 2.02 per 10,000 km², so its deficit is 0. The others are measured against it: State of Mexico 0.548 (2 accessions), Veracruz 0.582 (6), Oaxaca 0.626 (7), Chiapas 0.732 (4), Yucatán 0.743 (2), Michoacán 0.749 (3), Jalisco 0.877 (2), Chihuahua 0.960 (2) and Sonora 0.973 (1); the remaining 22 states, Tlaxcala included, have no accessions and are worth 1. The cell 21.50, −98.50, fourth on the list, is 48% Veracruz, 32% San Luis Potosí and 20% Hidalgo: $\delta = 0.48 \times 0.582 + 0.32 + 0.20 = 0.799$, and its deficit contributes $20 \times 0.799 = 15.99$ points.

6.4 Moving the weights: what you are after on the next trip

The four sliders run from 0 to 60 in steps of 5. Each has its name above it and, to its right, the weight and, in parentheses, the share it represents of the total: with the default weights, 40 (40 %), 25 (25 %), 20 (20 %) and 15 (15 %). The calculation is redone when you release the slider. The weights do not need to add up to 100: if you raise one to 60, the points can exceed 100 and the percentages are redistributed.

Table 6.4. The first five cells when a single weight is raised to 60 (the other three at their defaults).

Weight raised to 60	First five cells (points)
None (defaults)	San Luis Potosí 86.9 · Guerrero 82.9 · Guanajuato 82.9 · Veracruz (21.50, −98.50) 79.0 · Veracruz (18.50, −94.50) 76.6
Distance (60, 50%)	San Luis Potosí 105.1 · Guerrero 102.9 · Guanajuato 102.9 · Veracruz (18.50, −94.50) 96.6 · Campeche 95.1
Neighborhood richness (44%)	San Luis Potosí 113.1 · Veracruz (21.50, −98.50) 108.1 · Guerrero 103.3 · Guanajuato 103.3 · Veracruz (18.50, −94.50) 100.0
State deficit (43%)	San Luis Potosí 126.9 · Guerrero 122.9 · Guanajuato 122.9 · Campeche 115.2 · Guerrero (17.50, −99.50) 114.7
Exclusive types nearby (41%)	Veracruz (21.50, −98.50) 124.0 · San Luis Potosí 121.9 · Hidalgo (20.50, −98.50) 115.2 · Veracruz (21.50, −97.50) 111.2 · Veracruz (18.50, −96.50) 109.7

The San Luis Potosí cell appears in every case: it is remote, next to a rich area and to states without accessions, so no reasonable weight removes it. That is the robust conclusion. Raising the exclusive types pulls the list toward central Mexico, to empty cells between diverse collections, where finding more of the same good material is likely; raising the distance or the deficit pushes it toward unexplored territory, such as Campeche or the Isthmus.

> **THE EXTREMES: A SINGLE WEIGHT, OR NONE**
>
> If you keep only the distance (the other three at 0), the 127 empty cells at 250 km or more tie at 60 points, because the cap levels them; they are ordered by the tie-break —more richness around, then farther away—, which puts 18.50, −94.50, in southern Veracruz, first. Always combine at least two components. If you set all four to 0, the app does not invent an order: instead of the table it writes *All the weights are zero. With nothing to weigh there is no priority index, and ordering the cells that way would invent a ranking: raise at least one of the four sliders.*, the percentages stay at 0% and the CSV button is disabled.

6.5 The collecting route for the GPS

Below the priority table there are two buttons (Figure 6.6):

- Collecting route as CSV saves collecting-priorities.csv, in UTF-8 with a byte-order mark (so that a spreadsheet reads the accents correctly), with up to 200 empty cells ordered by priority; with the example collection, the first 200 of the 223. The center coordinates carry four decimals, ready to load into a GPS or a route sheet.
- Map as SVG saves collecting-gaps.svg, the map in the painting mode you have chosen, with the light theme colors.

Table 6.5. The header and the first three rows of the CSV with the example collection.

rank	latitude	longitude	state	country	km_to_nearest	neighborhood_richness	neighborhood_acces
1	21.5	-99.5	San Luis Potosí	MEX	228	9	12
2	17.5	-100.5	Guerrero	MEX	254	7	9
3	21.5	-100.5	Guanajuato	MEX	250	7	9

The CSV has three columns that the screen does not show: `country` (the ISO alpha-3 code), `neighborhood_accessions` and `exclusive_nearby` (X_v); the score carries one decimal and the distance (`km_to_nearest`) whole kilometers. The `state` column is the state with the most land in the cell. The file does not store the weights, and the cell size only in `cell_degrees`: write down which weights you used.

The center of a cell is a reference point, not a collecting site: with 1° cells, the square is about 110 km on a side and its center may fall on a hill, in a reservoir, in an urban area or, in edge cells, in the sea. Use it to plan the route and choose in the field, within the square, the sites where the crop or the wild species you are looking for actually grows.

6.6 What you already have and could lose

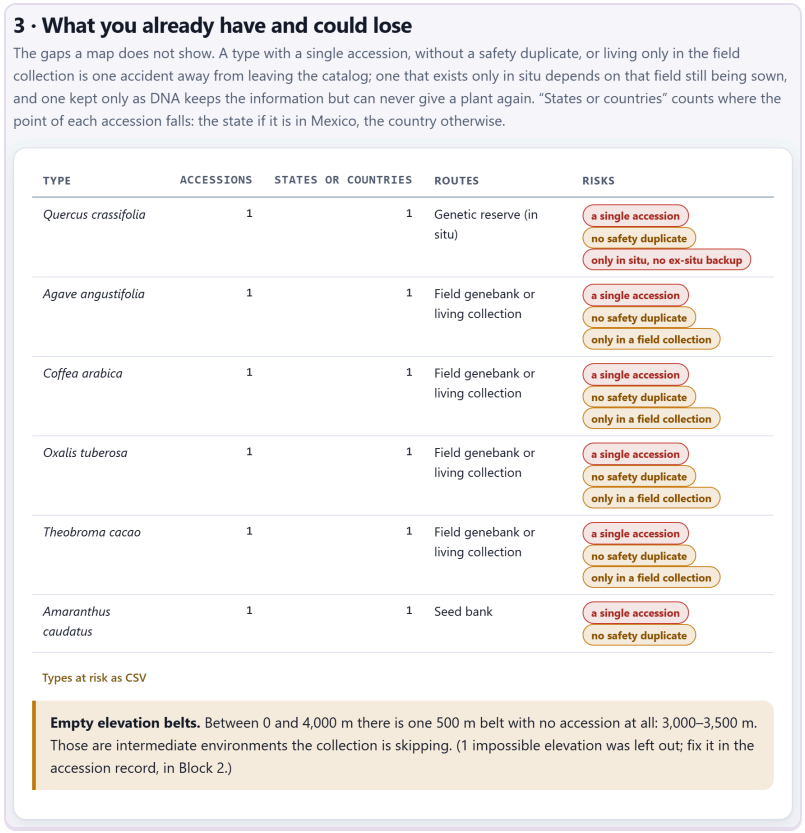
Section 3 · What you already have and could lose reviews the gaps a map does not show. It does not look at the territory but at each type in the collection —**all 46 accessions**, with or without coordinates— and gives it risk labels according to how many accessions it has, through which routes it is conserved and whether it has a safety duplicate.

Table 6.6. The seven risk signals and the weight of each.

Label	Weight	When it appears
a single accession	3	The type has a single accession in the whole passport.
only in situ, no ex-situ backup	3	All its accessions are in a genetic reserve or in on-farm conservation.
only as DNA: cannot regrow a plant	3	It has a DNA bank, but no route from which a plant can be regenerated (seeds, field, in vitro or cryopreservation), with or without in situ records.
no safety duplicate	2	None of its accessions has in `DUPLSITE` an institution other than the one that conserves it (the same rule as the Block 2 validation, Section 2.6).
only in a field collection	2	Of the routes that regenerate a plant, the only one it has is the field genebank or living collection.
no coordinates	1	None of its accessions has coordinates.
all from one state or country	1	It has several accessions, but they all fall in the same Mexican state or in the same country.

The route of each accession is the one in `GP_CONS` or, if missing, the one deduced from `STORAGE` (Chapter 2). The States or countries column counts where the point of each accession falls: the state if it falls in Mexico, the country if it falls elsewhere (or the one in `ORIGCTY` if it falls in the sea). The table orders the types from the highest to the lowest total risk, at equal risk from fewer to more accessions, and then alphabetically. The weight is not shown in the table: you can tell it from the order and the colors, red for the serious labels and ochre for the rest. If more than 40 types have some label, the table shows the 40 most serious and says so (*Showing 40 of 45 types with some risk…*). The type name is in italics only when the type is a species or a genus. The Types at risk box counts those that add up to 3 or more.

How to read it for a decision. Each label suggests a different action: *a single accession* calls for recollecting or requesting another sample; *no safety duplicate*, for sending a copy to another institution (Block 8, Chapter 8); *only in a field collection*, for an in vitro or cryopreserved backup, because a pest or a frost can wipe out the whole living collection; *only in situ*, for an ex situ backup, because the population depends on the reserve or the plot continuing to exist; *only as DNA*, for obtaining living material, because a plant cannot be regenerated from DNA.



3 · What you already have and could lose

The gaps a map does not show. A type with a single accession, without a safety duplicate, or living only in the field collection is one accident away from leaving the catalog; one that exists only in situ depends on that field still being sown, and one kept only as DNA keeps the information but can never give a plant again. "States or countries" counts where the point of each accession falls: the state if it is in Mexico, the country otherwise.

TYPE	ACCESSIONS	STATES OR COUNTRIES	ROUTES	RISKS
Quercus crassifolia	1	1	Genetic reserve (in situ)	a single accession; no safety duplicate; only in situ, no ex-situ backup
Agave angustifolia	1	1	Field genebank or living collection	a single accession; no safety duplicate; only in a field collection
Coffea arabica	1	1	Field genebank or living collection	a single accession; no safety duplicate; only in a field collection
Oxalis tuberosa	1	1	Field genebank or living collection	a single accession; no safety duplicate; only in a field collection
Theobroma cacao	1	1	Field genebank or living collection	a single accession; no safety duplicate; only in a field collection
Amaranthus caudatus	1	1	Seed bank	a single accession; no safety duplicate

Types at risk as CSV

Empty elevation belts. Between 0 and 4,000 m there is one 500 m belt with no accession at all: 3,000–3,500 m. Those are intermediate environments the collection is skipping. (1 impossible elevation was left out; fix it in the accession record, in Block 2.)

Figure 6.7. The first six rows of the risk table (it has 22), the CSV button and the note on the empty elevation belts.

The Types at risk as CSV button saves types-at-risk.csv with every type that has some label —22 in the example, even when the table is cut at 40—, with the columns `type`, `accessions`, `states_or_countries`, `routes` (by name, as in the table), `duplicates`, `with_coordinates`, `risks` and `weight`. It is the work list for Block 8.

▸ EXAMPLE · THE OAK, THE CHAYOTE AND THE BEAN

Quercus crassifolia: 1 accession, only in a genetic reserve, no duplicate → 3 + 3 + 2 = 8, first on the list

Theobroma cacao: 1 accession, only in a field genebank, no duplicate → 3 + 2 + 2 = 7

Sechium edule: 3 accessions, only in a field genebank, no duplicate, all in Veracruz ("all from one state or country") → 2 + 2 + 1 = 5

Phaseolus vulgaris: 2 accessions, in seeds and on farm, no duplicate → 2: it appears in the table, but does not count as "at risk"

Of the 24 species, 22 have at least one label and 21 add up to 3 or more. Only maize (18 accessions, 10 with a duplicate, in 9 states or countries) and chili (3 accessions, 2 with a duplicate) have none. The oak is the most fragile case: a recalcitrant acorn, conserved only in a reserve, with no copy in another institution.

The empty elevation belts

Below the table, when needed, there is a note with the intermediate elevation belts that have no accession at all. The app sorts the elevations of the mapped accessions into 500 m belts —always 500, whatever the Block 5 selector says—, from the belt of the lowest to that of the highest, and lists those left empty. It leaves out impossible elevations, below −500 m or above 7,000 m (the same range as in Block 5), so that a single mistyped elevation does not invent dozens of empty belts; at least three valid elevations are needed.

With the example collection, 44 elevations come in, from 20 to 3,797 m (belts from 0 to 4,000 m), and the note says that there is **one 500 m belt with no accession at all: 3,000–3,500 m**; it adds *1 impossible elevation was left out; fix it in the accession record, in Block 2*: the 25,000 m of the tepary bean. It is the same belt that was missing from the Block 5 gradient (Section 5.8). The empty intermediate belt is a coarse environmental gap, the only one the block can see without climate layers: no accession lies between 3,000 and 3,500 m, between the potato at 2,997 m and the bitter potato at 3,797 m, both from Peru. It is the zone of the potato and the Andean tubers, and the collection skips it.

6.7 The indispensable minimum: complementarity

Section 4 · The indispensable minimum answers a different question: if you could only conserve —or revisit— a few localities, which ones would bring together the most diversity? On the left, the curve Types captured by number of cells kept; on the right, the table The indispensable cells, in order, with the cell, what it Adds and the Cumulative total in types and as a percentage (up to 20 rows).

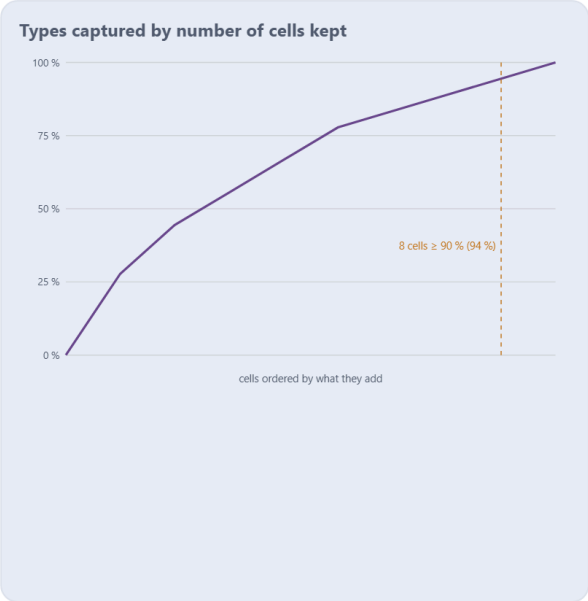
> **THEORY IN BRIEF · COMPLEMENTARITY BY THE GREEDY METHOD**
>
> It starts from all the occupied cells and the set of types they hold together. At each step it chooses the cell that adds the most types **not yet covered** —not the richest, but the one that best complements those already chosen—; if two tie, the one with more types in total wins and, if they are still tied, the one that comes first in the grid. Its types are marked as covered and the step is repeated until all are covered. Cells that add nothing do not enter the list. The greedy method does not guarantee the exact minimum, but it usually stays within one cell of it and is the method used to choose reserves.

The total is **18** and not 24 because, with the territory set to Mexico, only the cells of Mexico count: the six species found only outside it —the two potatoes, the oca, the barley, the Andean amaranth and the cassava— do not come in. With the territory set to The world, all 24 come in and 15 cells are needed to cover them; 13 bring together 92% (Section 6.8).

How to read it. A curve that rises slowly and almost in a straight line, like this one, says that each locality holds something of its own: there is no redundancy, and closing any of the nine cells takes at least one species away from the collection. If the curve reaches 90% early, a few localities are enough and the rest are backup. Complementarity uses only presences, not how many accessions there are: a cell with a single envelope of an exclusive type is worth as much as one with twenty.

4 · The indispensable minimum

By complementarity: if you could only keep a few localities, which ones would capture the most diversity? The curve adds cells one at a time, always starting with the one that contributes most. It helps to prioritize in-situ sites and to see how much would be lost by closing a locality.



The indispensable cells, in order

#	CELL	ADDS	CUMULATIVE
1	19.50, -98.50	+5	5 (28 %)
2	16.50, -92.50	+3	8 (44 %)
3	20.50, -97.50	+2	10 (56 %)
4	16.50, -96.50	+2	12 (67 %)
5	19.50, -96.50	+2	14 (78 %)
6	20.50, -103.50	+1	15 (83 %)
7	19.50, -102.50	+1	16 (89 %)
8	19.50, -97.50	+1	17 (94 %)
9	27.50, -108.50	+1	18 (100 %)

Figure 6.8. The complementarity curve and the nine indispensable cells. The dashed line marks the first cell with which 90% is reached, and its label says how much is actually brought together.

EXAMPLE · THE NINE CELLS OF THE EXAMPLE COLLECTION

1. 19.50, −98.50 (Puebla and State of Mexico): *Zea mays*, *Cucurbita pepo*, amaranth, wheat, *C. argyrosperma* → +5 = 5 of 18

2. 16.50, −92.50 (Chiapas): runner bean, cacao, oak (maize is already in) → +3 = 8

3–5. 20.50, −97.50: *C. moschata* and vanilla · 16.50, −96.50: bean and chili · 19.50, −96.50: coffee and chayote → +2 each = 14

6–9. agave · avocado · husk tomato · tepary bean → +1 each = 18 (100%)

Cells 3 and 4 tie on everything (they add 2 and have 3 species); the first in the grid wins. The remaining four of the 13 occupied cells —the one with the maizes of Chihuahua, the one in Oaxaca with 3 maizes, the one in Yucatán and the one with the bean from Michoacán— add nothing the others do not have. With 7 cells, 16 of 18 types are reached (89%); with 8, 17 (94%): that is why the box says 8 and "together hold 94 %".

6.8 How everything changes with the cell, the territory and the type

Everything above depends on the three header selectors. With the example collection:

Table 6.7. What changes with the territory and the cell size (type: species; default weights).

Territory · cell	Cells	Occupied	Coverage	Largest gap	≥ 90%	Outside	First priority
Mexico · 0.5°	831	18	2.2%	1,143 km	10 of 18	9	21.25, −98.75 (Hidalgo), 83.6
Mexico · 1°	236	13	5.5%	1,026 km	8 of 18	9	21.50, −99.50 (San Luis Potosí), 86.9
Mexico · 2°	69	11	15.9%	1,021 km	7 of 18	9	17, −101 (Guerrero), 82.9
The world · 1°	17,012	21	0.1%	10,581 km	13 of 24	0	−13.50, −69.50 (Peru), 69.6
The world · 2°	4,549	18	0.4%	10,640 km	11 of 24	0	−13, −69 (Peru), 69.8

A fine cell spreads the collection over more squares, leaves less territory "covered" and separates collections that a coarse one lumps together; the coarse cell gives a more optimistic coverage. The nine accessions outside the area are the same at any size: those from other countries. Since the neighborhood is measured in kilometers (about 220 km on each side), the index gives almost the same result at any size: the first cells fall in the same zone, between the Huasteca, northern Guerrero and the Bajío. With 0.5°, the first five are Hidalgo (21.25, −98.75) 83.6 · Guerrero (17.75, −100.25) 82.0 · Guanajuato (21.25, −100.25) 81.2 · Querétaro (21.25, −99.25) 79.4 · Hidalgo (20.75, −98.75) 77.2, and the ramp runs from 25 to 84 points; with 2°, Guerrero (17, −101) 82.9 · Hidalgo (21, −99) 82.5 · Veracruz (19, −95) 81.1 · Guanajuato (21, −101) 76.8 · Campeche (19, −91) 75.4, with the ramp from 42 to 83. The fine cell pinpoints the place; the coarse one gives fewer cells that are easier to cover.

With The world, the deficit is computed by country, with the same density per area: Guatemala, with 2 accessions in a small territory, is the best covered (0) and Mexico ends up almost the same (0.004); Peru is worth 0.917, Myanmar 0.921, Bolivia 0.950, Ethiopia 0.952, Colombia 0.953 and India 0.983, and the countries without accessions, 1. That is why the first priority moves to southern Peru, near the potatoes: −13.50, −69.50 with 69.6 points with 1° cells, and −13, −69 with 69.8 with 2° cells. The table gives the Country instead of the state, and the slider is called Country deficit. Coverage and the largest gap stop being useful —the largest gap, at 10,581 km, is in New Zealand, and the card says so—; this territory serves for the complementarity of an international collection, where all 24 species come in, and for planning trips outside Mexico by filtering the CSV route on the `country` column.

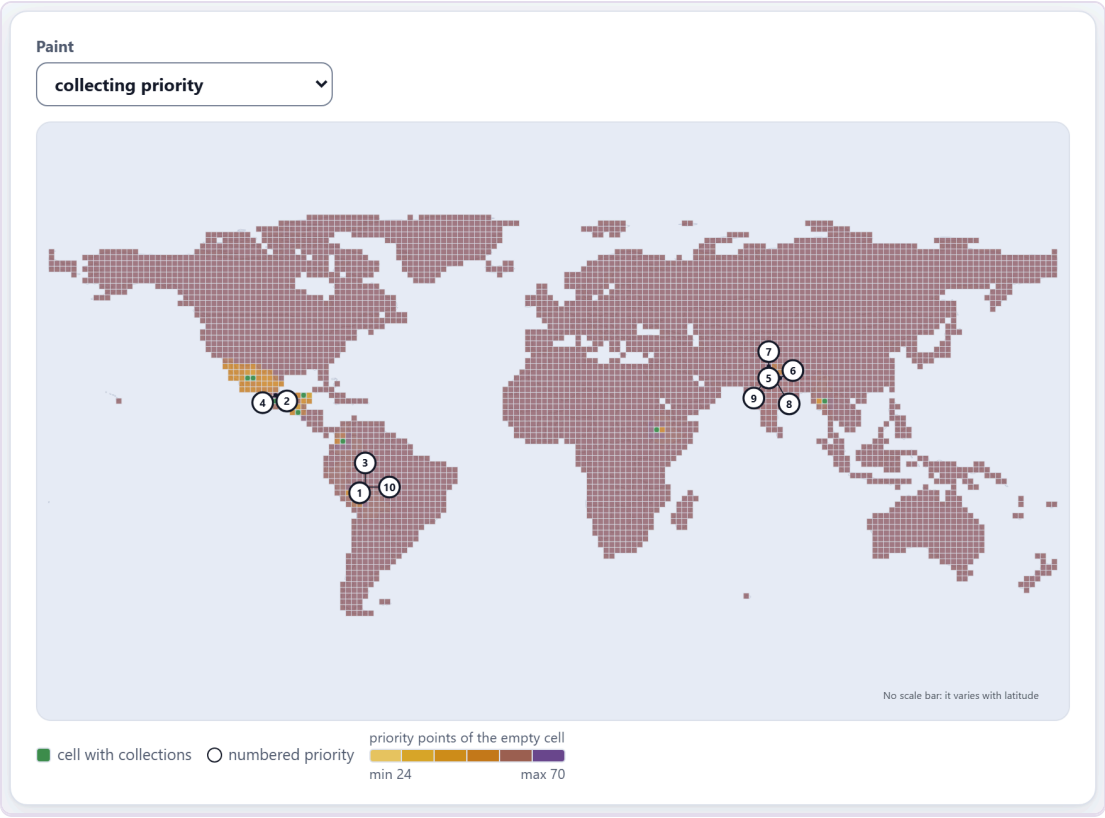


Figure 6.9. The territory set to The world with 2° cells: the priorities are split among the Andes, Mexico and India. Priorities that would overlap are moved aside and a line joins them to their cell.

The type changes the richness, the exclusive types, the complementarity and section 3, but not the grid or the coverage. With Genus there are 14 types in Mexico, 6 cells bring together at least 90% and 12 types are at risk; with Crop, 18 types, 8 cells and 17 at risk; with Accession name, 35 names, 11 cells and 45 at risk, because almost every accession is a type with a single accession. In that mode the risk table warns that it shows 40 of 45, and the names are not in italics.

When something goes wrong

What you see	Why it happens and what to do
Load your accessions in Block 2 first.	There is no passport data. Load your collection or the example one (Chapter 2).
N accessions fall outside the analyzed area	With the territory set to Mexico, their points fall in another country, through an error or because they come from abroad. Check the coordinates in Block 3 or change the territory to The world.
"+2" appears next to the state	The cell touches two other states; hover to see which. The cell is labeled with the one with the most land, and its deficit blends those of all of them.
Many cells with the same points	All the weight is on a single component: with distance alone, cells at 250 km or more tie, and with the deficit alone, those of states without accessions. Combine at least two weights (Section 6.4).

What you see	Why it happens and what to do
All the weights are zero...	The four sliders are at 0. Raise at least one: the table and the CSV come back.
more occupied cells are needed in section 4	Fewer than two cells contribute types: there is no curve to draw. Use a finer cell or the The world territory.
A type appears "at risk" although you know it has a copy	`DUPLSITE` is missing, or it names the same institution as `INSTCODE`. Complete it in Block 2.
Showing 40 of N types with some risk	More than 40 types have labels; the table shows the most serious. Save the CSV to see them all.
The belt note says it left out impossible elevations	There are elevations below –500 m or above 7,000 m. Correct them in the accession record, in Block 2.

DECISION RULE · WHAT TO DO IN BLOCK 6

If you are going to plan a field trip	**then** first correct the coordinates (Block 3): a lost sign moves an accession out of the territory and opens a gap that does not exist.
If a cell stays at the top with the default weights and with each weight raised	**then** it is a robust priority: put it on the route.
If you are after new, little-known material	**then** raise the distance and the state deficit; if you are after diversity close to what you already know, raise richness and exclusive types.
If you want to pinpoint where to go within a priority zone	**then** go down to 0.5° cells: the neighborhood is still about 220 km and the index points to the same zone, but in squares half as wide.
If a type has *a single accession*, is *only in situ* or *only as DNA*	**then** put it at the top of the work list: collect another sample or obtain living material for an ex situ backup.
If a type has *no safety duplicate* or is *only in a field collection*	**then** send a duplicate to another institution or make an in vitro or cryopreserved backup (Block 8).
If there is an empty intermediate elevation belt	**then** include cells at that elevation in the route, even if they are not among the top of the index.
If the complementarity curve rises almost in a straight line	**then** no locality is expendable: protect all the cells in the table and keep the one of each exclusive type.

Block 6 does not change the passport: it only says where and what is missing. The CSV route is the bridge to the field; whatever comes back from the trip enters again through Block 2.

7

CHAPTER SEVEN · BLOCK 7 · CHARACTERIZATION & CORE

Characterization and core collection

The passport says who each accession is and where it comes from; characterization says what it is like. Block 7 takes what you measured in the field and in the laboratory —numbers, scales, colors and empty cells, all together—, describes the variation trait by trait, measures how alike the accessions are with Gower's distance, lays them out on a map of resemblance and in a tree and, with all that, builds a core collection: the small subset you can characterize in depth or evaluate in the field without moving the whole genebank. Finally, it checks with the criteria of Hu *et al.* that the core truly represents the collection.

7.1 Bringing in the characterization and joining it to the collection

7.2 The traits: type, weight and variation

7.3 Gower's distance

7.4 The map of resemblance: principal coordinates

7.5 Clustering and the silhouette

7.6 The core collection: strata, allocation and choice

7.7 Does it represent well? The criteria of Hu *et al.*

7.8 Exporting, and how the core changes with the options

7.1 Bringing in the characterization and joining it to the collection

Block 7 works with two tables: the passport from Block 2 (Chapter 2) and a characterization table with **one row per accession and one column per descriptor**. The two are joined through the accession number, `ACCENUMB`, which is the key of the whole app. If there is no passport, the block says `Load your accessions in Block 2 first.`; if there is a passport but no characterization, `No characterization data yet: load yours or open the example ones with the button above.`

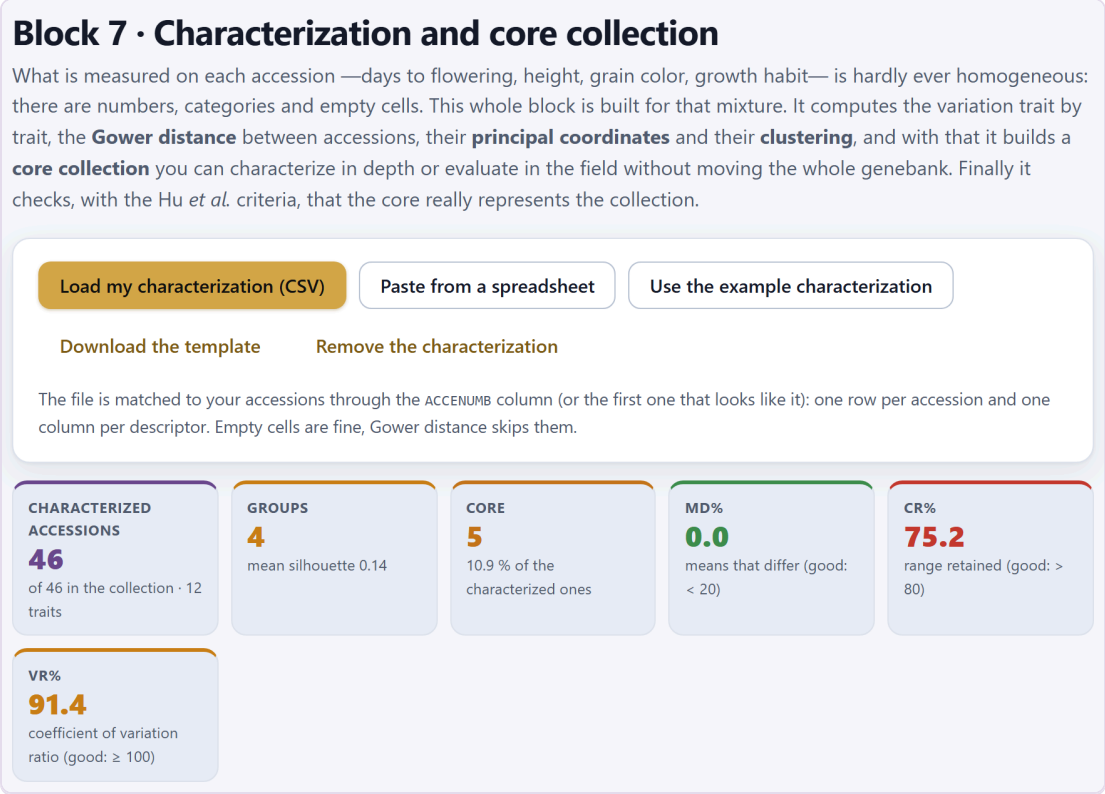


Figure 7.1. The header of the block with the example collection, the example characterization and the default values. The color of the number in each tile is its traffic light: green if the figure is in the good range, ocher or red if it is worth checking.

Table 7.1. The five buttons of the header.

Button	What it does
`Load my characterization (CSV)`	Reads a delimited .csv, .tsv or .txt file and opens the confirmation card (Figure 7.2).
`Paste from a spreadsheet`	Opens the same paste window as Block 2 (Section 2.3), titled `Paste your characterization`. Copy the block of cells with its header row and paste it; it also ends in the confirmation card.
`Use the example characterization`	Generates the example characterization for the loaded accessions and computes the whole block at once, with the animated wait *Computing characterization and core*.

Table 7.1. The five buttons of the header. *(continued)*

Button	What it does
Download the template	Saves characterization-template.csv: the `ACCENUMB` column already filled with all your accessions and six empty columns with the keys of the descriptor catalog: `DIAS_FLORACION`, `ALTURA_PLANTA`, `LARGO_FRUTO`, `COLOR_EXTERNO`, `FORMA_FRUTO` and `HABITO`. These are the same keys Block 2 recognizes as characterization on import (Section 2.5). Remove the ones you do not measure and add your own.
Remove the characterization	Asks first and, if you confirm, deletes the saved characterization and leaves the block empty. The Block 2 accessions are not touched; if you loaded it from a file, just load it again.

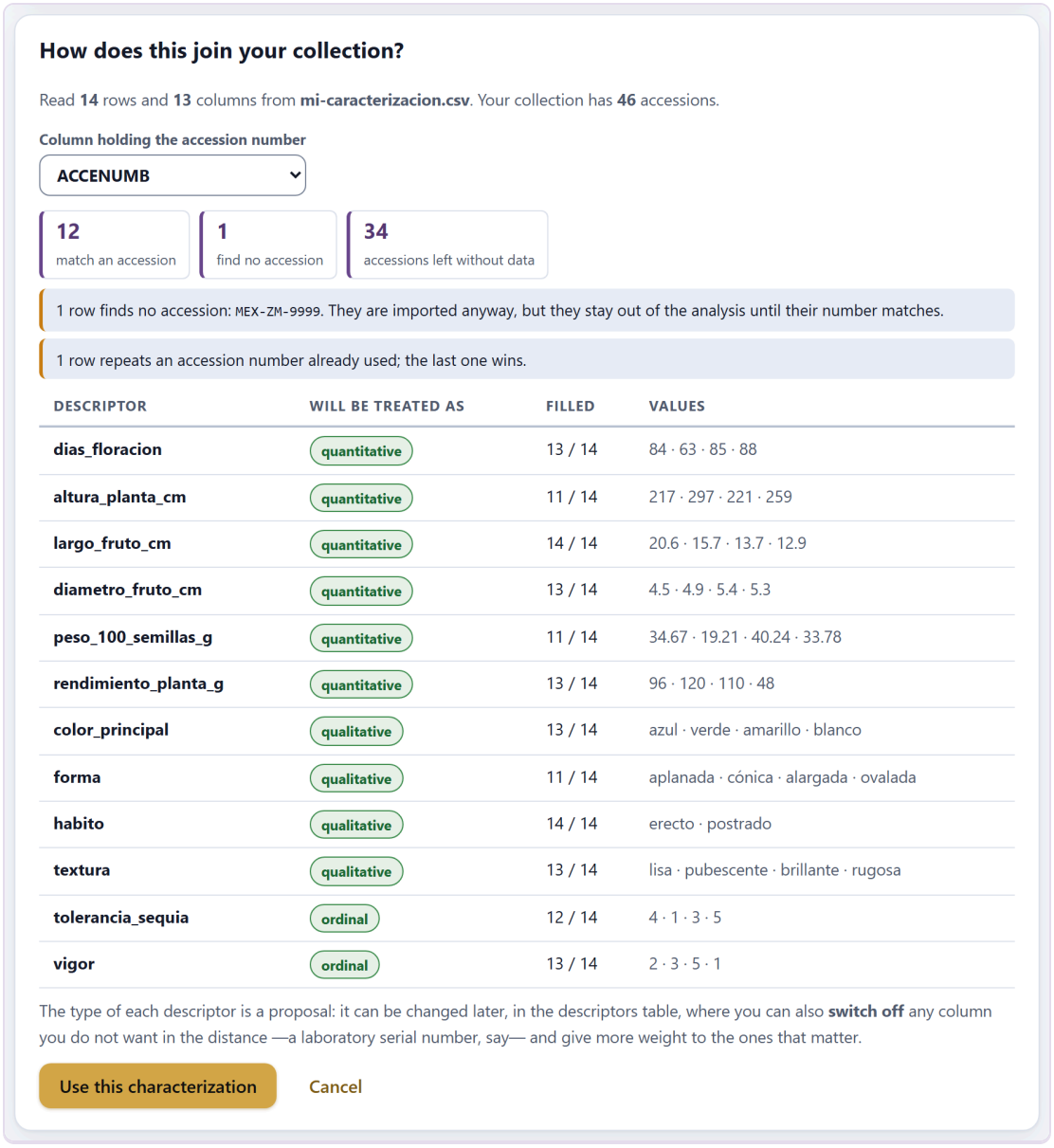
The characterization is saved in the browser's local storage, next to the passport: if you close the app and open it again, it is still there. It never leaves your computer.

> **THE EXAMPLE CHARACTERIZATION**
>
> Twelve descriptors for the 46 accessions of the example collection, **generated by the app with synthetic data**: six quantitative (`dias_floracion`, `altura_planta_cm`, `largo_fruto_cm`, `diametro_fruto_cm`, `peso_100_semillas_g`, `rendimiento_planta_g`), four qualitative (`color_principal`, `forma`, `habito`, `textura`) and two 1-to-5 scales (`tolerancia_sequia`, `vigor`). The quantitative ones come from a normal distribution with a mean specific to each genus —an agave takes years to flower and an oak is meters tall—; growth habit follows the genus (trees are arboreal, vanilla and chayote are climbers), and color, shape, texture and the two scales are drawn at random. About 8% of the cells are left empty on purpose. The same collection always yields the same values, so you can reproduce every figure in this chapter. The names of the columns and of the classes (*azul*, *erecto*…) are data and stay as the app writes them.

The "How does this join your collection?" card

When you load or paste a table, the app does not accept it yet: first it shows the only thing that really matters, how many of its rows find their accession. The key column is chosen automatically —the first one whose header resembles `ACCENUMB`, *accesión* or *accession*; if none does, the first column— and it is changed in the Column holding the accession number selector; the count is redone instantly.

The three tiles say how many rows match an accession, how many find no accession and how many accessions of the collection are accessions left without data. If no row matches, the red warning says that the key column is almost certainly the wrong one. Orphan rows are imported anyway, but they stay out of the analysis until their number matches; if a number is repeated, the last row wins (in the figure, 1 row repeats an accession number already used; the last one wins.). The table below proposes the type of each descriptor (Section 7.2) and shows how many cells are filled and up to four distinct sample values. Use this characterization accepts it and computes the block; Cancel discards it.



How does this join your collection?

Read **14** rows and **13** columns from **mi-caracterizacion.csv**. Your collection has **46** accessions.

Column holding the accession number

ACCENUMB

12	1	34
match an accession	find no accession	accessions left without data

1 row finds no accession: `MEX-ZM-9999`. They are imported anyway, but they stay out of the analysis until their number matches.

1 row repeats an accession number already used; the last one wins.

DESCRIPTOR	WILL BE TREATED AS	FILLED	VALUES
dias_floracion	quantitative	13 / 14	84 · 63 · 85 · 88
altura_planta_cm	quantitative	11 / 14	217 · 297 · 221 · 259
largo_fruto_cm	quantitative	14 / 14	20.6 · 15.7 · 13.7 · 12.9
diametro_fruto_cm	quantitative	13 / 14	4.5 · 4.9 · 5.4 · 5.3
peso_100_semillas_g	quantitative	11 / 14	34.67 · 19.21 · 40.24 · 33.78
rendimiento_planta_g	quantitative	13 / 14	96 · 120 · 110 · 48
color_principal	qualitative	13 / 14	azul · verde · amarillo · blanco
forma	qualitative	11 / 14	aplanada · cónica · alargada · ovalada
habito	qualitative	14 / 14	erecto · postrado
textura	qualitative	13 / 14	lisa · pubescente · brillante · rugosa
tolerancia_sequia	ordinal	12 / 14	4 · 1 · 3 · 5
vigor	ordinal	13 / 14	2 · 3 · 5 · 1

The type of each descriptor is a proposal: it can be changed later, in the descriptors table, where you can also **switch off** any column you do not want in the distance —a laboratory serial number, say— and give more weight to the ones that matter.

Use this characterization Cancel

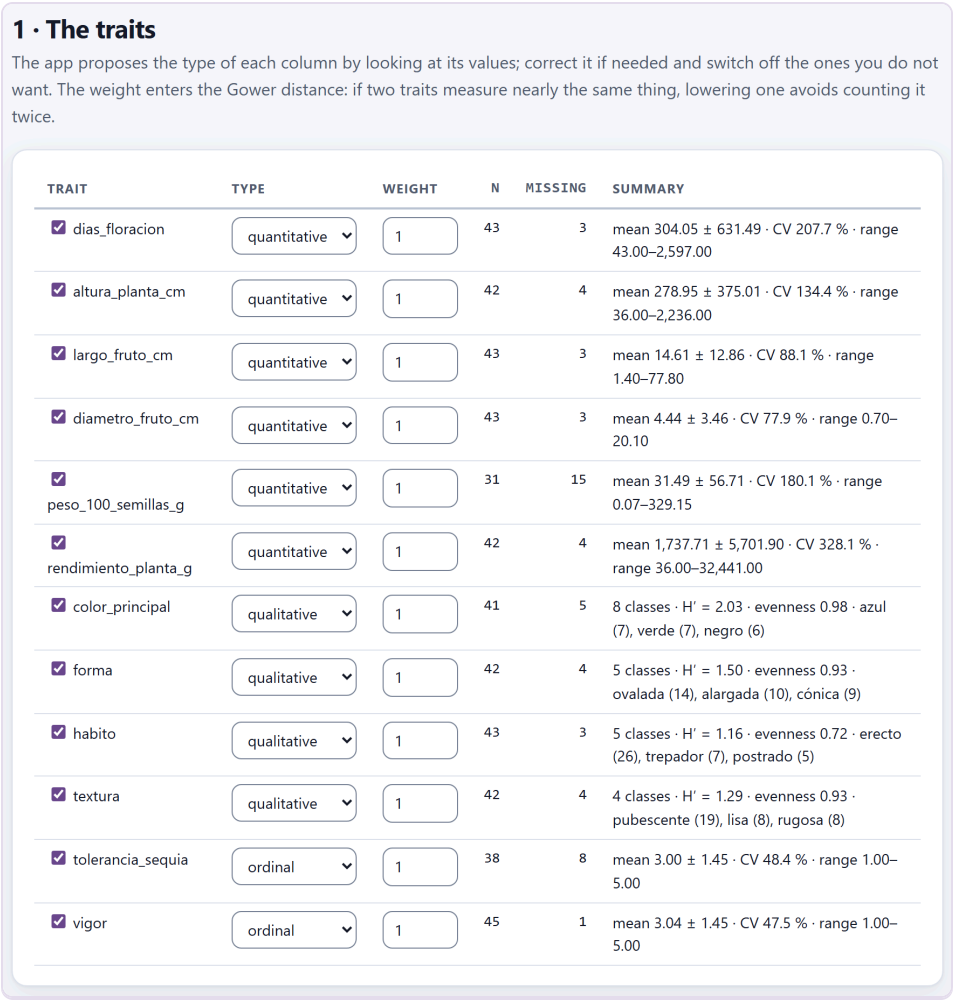
Figure 7.2. The confirmation card with a test file: the first 12 rows of the example characterization, a row with a number that does not exist (MEX-ZM-9999) and a repeated row. The descriptor table is shown expanded; in the app it scrolls inside its box.

AN UNCHARACTERIZED ACCESSION IS NO OBSTACLE

Only the accessions that have a row in the characterization enter Block 7: the Characterized accessions tile says how many there are, "of 46 in the collection". The distance, the tree, the core and its validation are computed on them. If you characterized half the genebank, the core represents that half, not the whole genebank.

7.2 The traits: type, weight and variation

The 1 · The traits section has one row per column of the characterization. In each one you decide three things: whether the trait enters (the checkbox next to its name), what type it is and how much it weighs.



1 · The traits

The app proposes the type of each column by looking at its values; correct it if needed and switch off the ones you do not want. The weight enters the Gower distance: if two traits measure nearly the same thing, lowering one avoids counting it twice.

TRAIT	TYPE	WEIGHT	N	MISSING	SUMMARY
dias_floracion	quantitative	1	43	3	mean 304.05 ± 631.49 · CV 207.7 % · range 43.00–2,597.00
altura_planta_cm	quantitative	1	42	4	mean 278.95 ± 375.01 · CV 134.4 % · range 36.00–2,236.00
largo_fruto_cm	quantitative	1	43	3	mean 14.61 ± 12.86 · CV 88.1 % · range 1.40–77.80
diametro_fruto_cm	quantitative	1	43	3	mean 4.44 ± 3.46 · CV 77.9 % · range 0.70–20.10
peso_100_semillas_g	quantitative	1	31	15	mean 31.49 ± 56.71 · CV 180.1 % · range 0.07–329.15
rendimiento_planta_g	quantitative	1	42	4	mean 1,737.71 ± 5,701.90 · CV 328.1 % · range 36.00–32,441.00
color_principal	qualitative	1	41	5	8 classes · H′ = 2.03 · evenness 0.98 · azul (7), verde (7), negro (6)
forma	qualitative	1	42	4	5 classes · H′ = 1.50 · evenness 0.93 · ovalada (14), alargada (10), cónica (9)
habito	qualitative	1	43	3	5 classes · H′ = 1.16 · evenness 0.72 · erecto (26), trepador (7), postrado (5)
textura	qualitative	1	42	4	4 classes · H′ = 1.29 · evenness 0.93 · pubescente (19), lisa (8), rugosa (8)
tolerancia_sequia	ordinal	1	38	8	mean 3.00 ± 1.45 · CV 48.4 % · range 1.00–5.00
vigor	ordinal	1	45	1	mean 3.04 ± 1.45 · CV 47.5 % · range 1.00–5.00

Figure 7.3. The table of traits with the example characterization. Thousands carry a comma, as the app writes them: *2,597.00* is 2,597 days.

The type. The app proposes it by looking at the values in the column: if more than 90% of the non-empty values are numbers (with a decimal point or comma) and there are more than six distinct values, it is quantitative; if they are numbers but six or fewer distinct ones, ordinal; in any other case, qualitative. Correct it when the proposal does not fit: a 1-to-9 scale with all its values present comes out quantitative, and a numeric color code (1 = white, 2 = yellow…) comes out ordinal or quantitative when it is really qualitative. A column with no data at all appears with the label empty instead of the selector, with the checkbox and the weight disabled: it stays out of the analysis.

The checkbox. Clearing it takes the trait out of the whole block: out of the distance, the core and its validation. Use it for columns that are not descriptors, such as a laboratory serial number or a sampling date.

The weight. It goes from 0 to 5 in steps of one half, and is 1 by default. It only enters Gower's distance (Section 7.3): a trait with weight 2 counts as two. If two traits measure nearly the same thing —fruit length and fruit weight, for example—, lowering the weight of one avoids counting the same difference twice. With weight 0 the trait no longer influences the distance, but it stays in the validation of the core.

The summary. The `n` and `Missing` columns count filled and empty cells. For quantitative and ordinal traits, the summary gives the mean ± the standard deviation, the coefficient of variation and the range; for qualitative ones, the number of classes, the Shannon diversity H', the evenness and the three most frequent classes with their accessions.

THEORY IN BRIEF · THE VARIATION OF A DESCRIPTOR

For a quantitative trait with n values, the standard deviation is the sample one (dividing by $n - 1$) and the coefficient of variation expresses it as a percentage of the mean:

$$s = \sqrt{\left[\ \Sigma\ (x_i - \bar{x})^2\ /\ (n - 1)\ \right]} \cdot CV = 100 \cdot s\ /\ |\bar{x}|$$

For a qualitative trait with k classes, let p_j be the proportion of accessions in class j. Phenotypic diversity is measured, as in Block 5, with the Shannon index in natural logarithm, and evenness divides it by its maximum possible value:

$$H' = -\Sigma\ p_j\ \ln\ p_j \cdot evenness = H'\ /\ \ln\ k$$

Evenness is 1 when all classes have the same number of accessions; with a single class it is undefined and the app writes "—", like the J' of Block 5. A descriptor with a high H' and evenness close to 1 discriminates well; one with a class that takes almost everything contributes little to the distance.

WORKED EXAMPLE · WHAT THE EXAMPLE TABLE SAYS

The 46 example accessions belong to 18 genera, and it shows: days to flowering go from 43 to 2,597 (CV 207.7%) because next to a bean there are an agave, an avocado and an oak; yield per plant, from 36 to 32,441 g (CV 328.1%). The weight of 100 seeds is the most incomplete, with 15 empty cells, because the example gives no seed weight to eight genera (potato, oca, cassava, agave, vanilla, avocado, cacao and chayote). Among the qualitative traits, `color_principal` spreads 41 accessions over 8 nearly even classes: ln 8 = 2.079 and H' = 2.03, so evenness is 2.03/2.079 = 0.98. `habito` has 5 classes, but 26 of 43 accessions are *erecto*: H' = 1.16 and evenness 1.16/ln 5 = 1.16/1.609 = 0.72.

The six tiles of the header summarize the result of the whole block; each one is explained in its section.

Table 7.2. The summary tiles with the example and the default values.

Tile	Example	What it says
Characterized accessions	46	The accessions with a row in the characterization; below, out of how many in the collection and how many traits enter (12).
Groups	4	The groups from cutting the tree; below, the mean silhouette (0.14). Green if above 0.25 (Section 7.5).
Core	5	The accessions in the core and what percentage of the characterized ones they are (10.9%).
MD%	0.0	Means that differ between core and collection; good below 20 (Section 7.7).
CR%	75.2	Range retained; good above 80.
VR%	91.4	Coefficient of variation ratio: the relative variation retained; good from 100, the same threshold in the label and in the traffic light.

7.3 Gower's distance

To say that two accessions are alike you have to combine into a single number differences in different things: days, centimeters, colors and scales. Euclidean distance does not work —it only accepts numbers, and the trait with the largest figures would decide everything— and hardly any distance accepts empty cells. **Gower's distance** (1971) solves both problems: each trait contributes a difference between 0 and 1, and the average is taken only over the traits that both accessions have measured.

7

THEORY IN BRIEF · GOWER'S DISTANCE

Between accessions i and j, each trait k contributes a partial difference d_{ijk}:

$$\text{quantitative or ordinal: } d_{ijk} = |x_{ik} - x_{jk}| \,/\, R_k$$
$$\text{qualitative: } d_{ijk} = 0 \text{ if same class, } 1 \text{ otherwise}$$

where R_k is the range of the trait (maximum – minimum) over all the characterized accessions; if the range is 0, 1 is used. Classes are compared ignoring case. The distance is the average weighted by the weights w_k, taken only over the traits both have measured ($\delta_{ijk} = 1$) and with a weight greater than 0:

$$d_{ij} = \Sigma_k \, w_k \, \delta_{ijk} \, d_{ijk} \,/\, \Sigma_k \, w_k \, \delta_{ijk}$$

It is 0 if the two accessions are identical in everything they share and 1 if they are opposite in everything. Ordinal traits are treated as quantitative: from 1 to 3 there is twice the distance as from 1 to 2.

The app does not show the matrix on screen —with 46 accessions there are 1,035 pairs—, but it exports it in full with the Distance matrix button (Section 7.8). In the example, distances go from 0.0086 to 0.8752, with a mean of 0.393. If two accessions share no measured trait, their distance is left empty and, in the principal coordinates, counts as 0.

WORKED EXAMPLE · TWO MAIZES OF THE EXAMPLE, MEX-ZM-0001 AND MEX-ZM-0002

MEX-ZM-0001 is missing its height; MEX-ZM-0002, its 100-seed weight and its drought tolerance. Those three traits do not enter; nine remain. The ranges are those of the 46 accessions.

dias_floracion 84 and 63	21 / 2,554 = 0.00822
largo_fruto_cm 20.6 and 15.7	4.9 / 76.4 = 0.06414
diametro_fruto_cm 4.5 and 4.9	0.4 / 19.4 = 0.02062
rendimiento_planta_g 96 and 120	24 / 32,405 = 0.00074
vigor 2 and 3	1 / 4 = 0.25000
color · forma · textura	azul/verde, aplanada/cónica, lisa/pubescente: 1 + 1 + 1
habito	erecto/erecto: 0
d = (0.00822 + 0.06414 + 0.02062 + 0.00074 + 0.25 + 3 + 0) / 9 = 3.34372 / 9 = 0.3715 → the exported matrix says 0.3715	

Look at the size of each term: a 21-day difference in flowering weighs 0.008, because the range is stretched by the oak, with 2,597 days; three different colors, shapes and textures weigh 3. In this collection, the distance is decided by the qualitative traits. That is why MEX-ZM-0002 is closer to a bean (MEX-SE-0013, d = 0.3674) than to another maize (0.3715), and the agave MEX-AG-0024 is 0.7502 away from MEX-ZM-0001.

ONE EXTREME VALUE FLATTENS ALL THE OTHERS

Since each quantitative trait is divided by its range, a single very different accession —or a data-entry error, such as a height of 25,000 cm— squeezes the differences of all the others toward zero and leaves the distance in the hands of the qualitative traits. If your collection mixes very different species, characterize and build a separate core for each species or crop, or transform very skewed traits, such as yield, to logarithms in your spreadsheet first. Always check the range in the summary before reading the tree.

7.4 The map of resemblance: principal coordinates

The 2 · The map of resemblance section puts the distance matrix into two plots: on the left, the **principal coordinates** (PCoA), a plane where accessions that are alike lie close together; on the right, the **tree** of the clustering (Section 7.5). Three selectors control them.

Table 7.3. The selectors of section 2 (the default in bold).

Selector	Options and what they do
Tree	Ward.D2 or UPGMA (Section 7.5). Changes the tree, the groups and, if the core is stratified by groups, the core.
Groups	From 2 to 8; by default, **4**. Where the tree is cut.

Table 7.3. The selectors of section 2 (the default in bold). *(continued)*

Selector	Options and what they do
Color by	group, crop, species or core. Only changes the plot on the left; the tree is always by group. Each category gets a color and a shape (circle, square, triangle or diamond): eight colors times four shapes distinguish up to 32 categories; beyond that, the least frequent are pooled into others, with their number in parentheses in the legend.

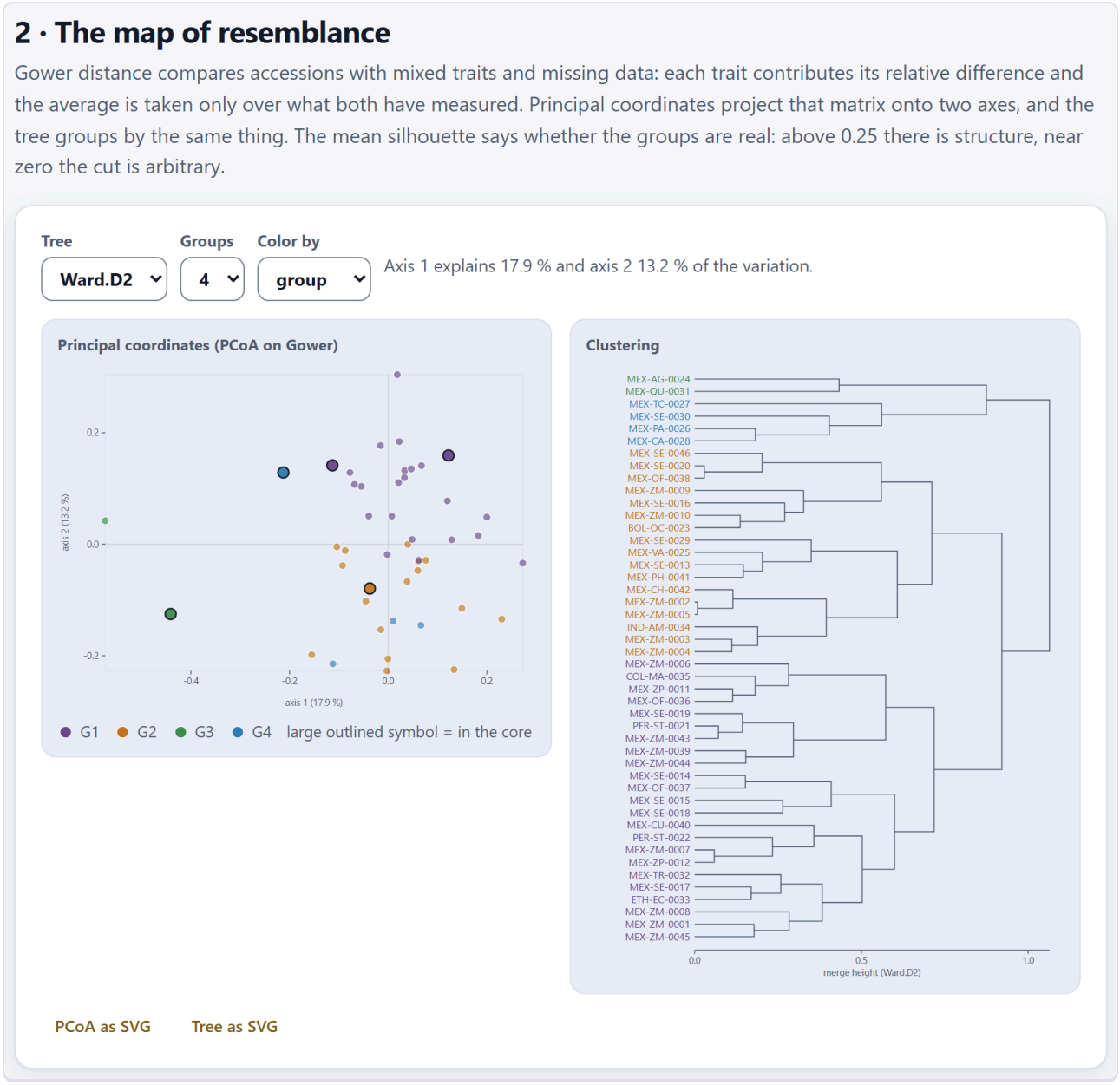


Figure 7.4. The map of resemblance with the default values. The tree shows all its leaves, each next to its name, with the merge-height scale at the bottom.

THEORY IN BRIEF · PRINCIPAL COORDINATES ANALYSIS

Principal coordinates analysis (Gower, 1966) looks for points on a plane whose distances resemble those of the matrix as closely as possible. With D the distance matrix of n accessions, A is formed with $a_{ij} = -\frac{1}{2}\, d_{ij}^2$ and is centered by rows and by columns:

$$b_{ij} = a_{ij} - \bar{a}_{i\cdot} - \bar{a}_{\cdot j} + \bar{a}_{\cdot\cdot}$$
$$B = \Sigma\ \lambda_m\ u_m\ u_m^{\top}$$
$$\text{coordinate}_{im} = u_{im}\ \sqrt{\lambda_m}$$

The eigenvalues λ_m of B (the app obtains them with the Jacobi method) measure how much variation each axis captures. The percentage the app writes on each axis is λ_m divided by the sum of the **positive** eigenvalues. Gower with missing data is not a Euclidean distance, so some eigenvalues come out negative; the app leaves them out of the sum and applies no correction.

In the example, the sum of the positive eigenvalues is 5.518 and the first one is 0.987, so axis 1 explains 0.987/5.518 = 17.9%; the second, 0.726/5.518 = 13.2%. Together, the plane captures 31% of the variation: it is an approximate portrait. Of the 46 eigenvalues, 24 are negative. With mixed traits and many species this is normal; if the first two axes gathered more than half, the plane would be a faithful portrait.

How to read it. Each point is an accession; hover over it to see its number and its category. The tick marks on the axes are unitless coordinates: what matters is the distance between points, not their absolute position. The large points with a dark outline are the core (Section 7.6). In Figure 7.4, the two green points on the left are the oak and the agave (G3), and the large blue one is the cacao (G4); groups G1 and G2, the most numerous, mix in the center, which is what the low silhouette announces.

7.5 Clustering and the silhouette

The tree on the right clusters the accessions with the same Gower matrix. Each leaf is an accession, written in the color of its group and with its branch attached to its name; branches join toward the right, and the scale at the bottom gives the merge height: the further to the right two branches join, the more different they are. With Ward the height is in Ward.D2 units; with UPGMA, in Gower's distance. The app cuts the tree into the chosen number of groups and calls the resulting groups G1, G2… Below the two plots, PCoA as SVG saves pcoa-gower.svg and Tree as SVG, gower-tree.svg, both with the colors of the light theme.

THEORY IN BRIEF · WARD.D2 AND UPGMA

UPGMA joins at each step the two groups with the smallest mean distance between their members. **Ward.D2** (Murtagh and Legendre, 2014) joins the pair whose merger least increases the within-group sum of squares; it works with squared distances and updates with the Lance-Williams formula, the same criterion as R's `hclust(…, method = "ward.D2")`. Ward tends to give groups of similar size; UPGMA respects the distances better, but leaves the rare accessions isolated and lumps the rest into one huge group. The cut into k groups splits, again and again, the highest remaining node.

THEORY IN BRIEF · THE SILHOUETTE

The **silhouette** (Rousseeuw, 1987) says whether each accession sits well in its group. With $a(i)$ the mean distance from i to the others in its group and $b(i)$ the mean distance to the nearest neighboring group:

```
s(i) = [ b(i) − a(i) ] / max{ a(i), b(i) }
mean silhouette = Σ s(i) / n
```

s goes from −1 to 1: near 1, the accession is much closer to its group than to any other; near 0, it is on the boundary; negative, it would be better in another group. An accession alone in its group scores 0. Above 0.25 there is structure; below it, the cut is almost arbitrary.

With the example, Ward in 4 groups gives G1 with 23 accessions (twelve of *Zea mays*, two of them teosinte, the two potatoes, the wheat, the barley, the cassava…), G2 with 17 (other maizes, beans, the chayote, the vanilla, the oca…), G3 with 2 (the agave and the oak) and G4 with 4 (avocado, cacao, coffee and a chayote). The mean silhouette is 0.14: G3 has 0.26 and the other three between 0.13 and 0.14, and two accessions have a negative silhouette. The groups are not species: since the colors and textures of the example were drawn at random, they mix species. Table 7.4 shows what happens with other cuts.

Table 7.4. Mean silhouette and group sizes by tree and number of groups, with the example characterization.

Groups	Ward.D2	Sizes	UPGMA	Sizes
2	**0.298**	6 · 40	**0.394**	2 · 44
3	0.116	6 · 17 · 23	0.272	2 · 1 · 43
4	0.140	23 · 17 · 2 · 4	0.161	2 · 1 · 1 · 42
5	0.121	17 · 4 · 2 · 9 · 14	0.148	2 · 1 · 1 · 3 · 39
6	0.133	14 · 9 · 4 · 2 · 7 · 10	0.137	39 · 3 · 1 · 1 · 1 · 1
8	0.142	9 · 4 · 7 · 2 · 6 · 4 · 4 · 10	0.020	3 · 1 · 1 · 1 · 1 · 1 · 1 · 37

How to read it to decide. The highest silhouette in the example is not the most useful partition: UPGMA with 2 groups gives 0.394 because it separates the agave and the oak from the other 44 accessions, and that organizes nothing. Ward with 2 groups (0.298) separates six perennials from the rest —the agave, the oak, the avocado, the cacao, the coffee and a chayote—: it is the only clear structure in this collection. To build the core by groups, a cut with groups of reasonable size is better, even if the silhouette is modest, than one with single-accession groups. Look for the number of groups where the silhouette has a local peak and the groups make biological sense.

7.6 The core collection: strata, allocation and choice

A **core collection** (Frankel and Brown, 1984) is a small subset that keeps, with a minimum of repetition, most of the diversity of the whole collection. It serves to characterize in depth, to evaluate in the field or to serve requests without touching the whole genebank. The 3 · The core collection section builds it in three decisions: how many accessions, among which strata they are allocated and which ones are chosen within each stratum.

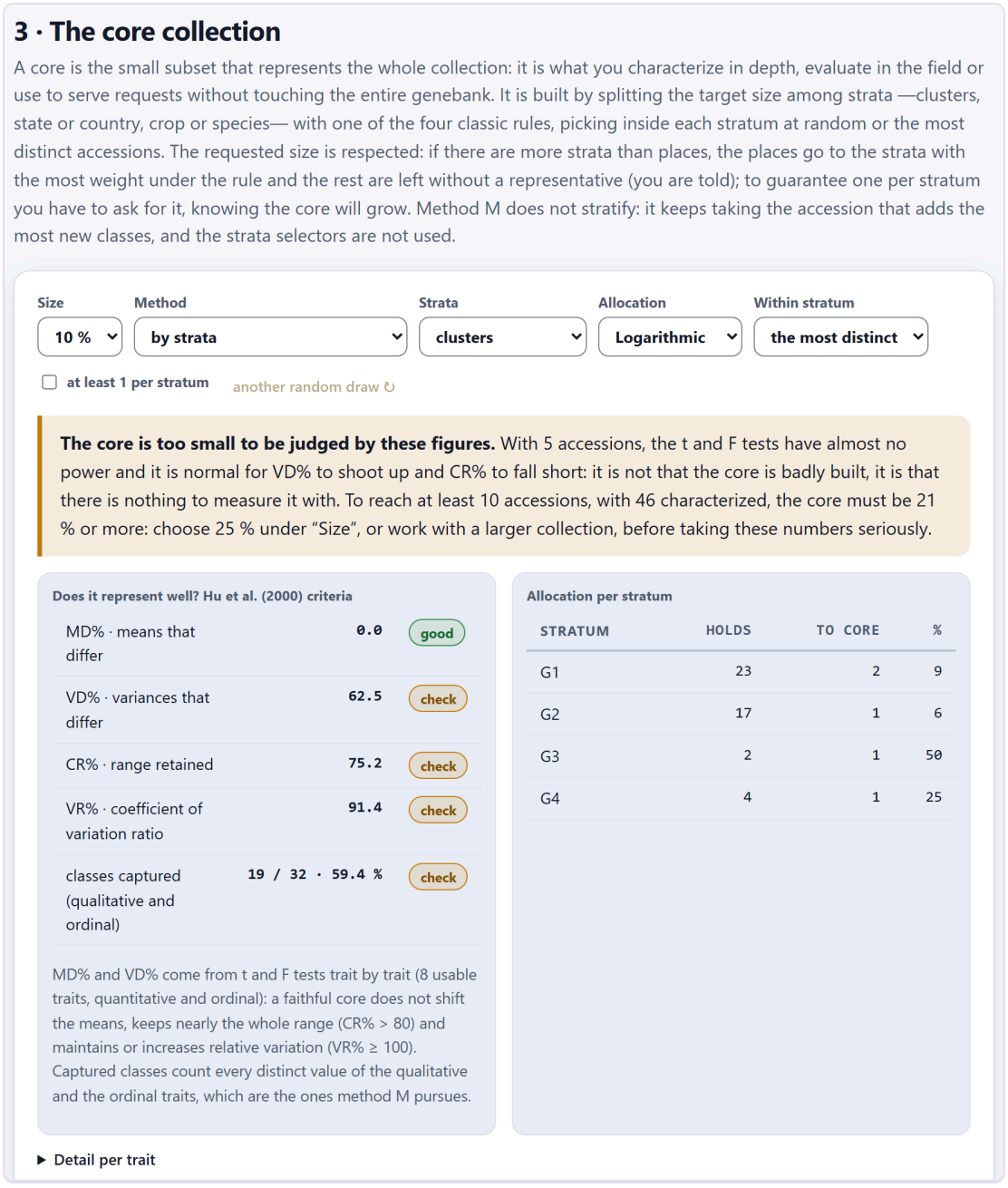


Figure 7.5. The default core: 10%, stratified by the Ward groups, logarithmic allocation and the most distinct accessions within each stratum. With 5 accessions, the app warns that the core is too small to be judged.

The Allocation per stratum table says, for each stratum, how many accessions it Holds, how many go To core and what percentage that is; strata left without a representative appear in gray, with 0. With method M, a sentence takes its place saying how many accessions were chosen and how many classes were left out. Below, the list of the core (Figure 7.9) gives the number, name, species, group and stratum of each chosen accession.

Table 7.5. The core controls (the default in bold).

Control	Options and what they do
Size	5, **10**, 15, 20, 25 or 30% of the characterized accessions, rounded to the nearest integer (10% of 46 = 4.6 → 5). The app respects it, unless you tick the checkbox below.
Method	by strata: allocates and chooses as explained in this section. method M (maximum classes): does not stratify; it keeps taking the accession that adds the most new classes. With method M the strata, allocation and choice selectors and the checkbox, which it does not use, are disabled.
Strata	clusters (the G groups of section 2), state or country (the Mexican state where the point falls or, if it falls in none, the country), crop (CROPNAME), species (genus + epithet) or no strata (a single stratum, whole collection).
Allocation	How many accessions each stratum gets: Constant, Proportional, Logarithmic or Square root.
Within stratum	the most distinct or at random. The another random draw ↻ button changes the seed of the draw; it is only active with at random.
at least 1 per stratum	Cleared by default. When ticked, every stratum contributes at least one accession even if that exceeds the requested size; the app warns what the actual size is.

THEORY IN BRIEF · THE FOUR ALLOCATION RULES AND THE CHOICE BY MAXIMUM DISTANCE

With a target size T and strata of sizes $n_1, \ldots, n_g$, each stratum is entitled to the quota $q_h = T \cdot w_h / \Sigma w$, where the weight w_h depends on the rule (Brown, 1989):

```
Constant: w = 1 · Proportional: w = n_h
Logarithmic: w = ln(n_h + 1) · Square root: w = √n_h
```

The constant rule gives the same to every stratum and favors the small ones; the proportional one reproduces the imbalance of the collection; the logarithmic and square-root rules lie in between, and the logarithmic one is recommended when strata are very unequal. Quotas are not integers, and the app distributes them **by largest remainder**: if there are at least as many places as strata, each stratum first gets one; then, place by place, the next one goes to the stratum furthest from its quota (q_h minus what it already has; on a tie, to the largest), never exceeding the accessions the stratum holds. If there are **more strata than places**, no one gets one up front: the places go to the strata with the largest quota and the others are left without a representative. That way the sum is always T. Only with at least 1 per stratum does each stratum get one even if the sum exceeds T.

The most distinct: within the stratum, the app starts with the pair furthest apart in Gower's distance and adds, one by one, the accession whose distance to the nearest of those already chosen is largest (maximin criterion). If the stratum gets only one, it keeps one of the two from the furthest pair. **At random**: sampling without replacement with a fixed seed, so that it can be repeated.

Method M (Schoen and Brown, 1993): all classes of the qualitative and ordinal traits are counted ("habito = trepador", "vigor = 4"…); at each step the accession that adds the most classes not yet covered enters. When no new class remains, it fills the size with the accessions that have the most classes.

MORE STRATA THAN PLACES: EITHER STRATA ARE MISSING OR THE CORE GROWS

With the example and the size at 20% (target: 9), stratifying by species leaves 24 strata for 9 places. The app respects the size and warns `15 of 24 strata are left without a representative.`: the places go to *Zea mays*, to two of the four beans, to two of the three squashes, to the chili, the amaranth, the husk tomato and the chayote, and the potato, the agave, the cacao and 12 other species do not enter. By state or country, 9 of 18 are left without a representative, and by crop, 14 of 23. If you tick `at least 1 per stratum`, the core by species rises to **24** accessions and the warning changes: `The core has 24 accessions, not the 9 requested (20 %).`, 52.2% of the characterized ones. With fine strata you have to choose between a core of the requested size that leaves strata out and one that covers them all but is no longer small; use coarser strata (clusters or no strata) if neither suits you.

WORKED EXAMPLE · THE LOGARITHMIC ALLOCATION OF THE DEFAULT CORE

T = 10% of 46 = 4.6 ⟶ 5. The Ward groups have 23, 17, 2 and 4 accessions.

```
w = ln 24, ln 18, ln 3, ln 5 = 3.178, 2.890, 1.099, 1.609 · Σ w = 8.776
quotas q = 5 · w / Σ w = 1.81, 1.65, 0.63, 0.92
4 strata ≤ 5 places → one to each stratum: 1, 1, 1, 1; 1 place left
distance to the quota: 0.81, 0.65, −0.37, −0.08 → the place goes to G1
final allocation: G1 2, G2 1, G3 1, G4 1 = 5 → as in the Allocation per stratum table
```

From G1 the maize MEX-ZM-0043 and the squash MEX-CU-0040 enter; from G2, the chili MEX-CH-0042; from G3, the agave MEX-AG-0024, and from G4, the cacao MEX-TC-0027. The two-accession group contributes half of its own; the 23-accession group, fewer than one in ten. That is exactly what is sought: what is rare weighs more than what is abundant.

STRATA BY STATE AND POINTS OUTSIDE MEXICO

With `state or country`, each accession goes to the Mexican state where its point falls; if it falls in none, to the country where it falls or, without coordinates, to the one in its passport. The table writes them with their name, and a state is never confused with a country: in the example, `México` is the State of Mexico (the teosinte and the wheat) and `Mexico (no state)` is the maize without coordinates, MEX-ZM-0043. The maize with the unsigned longitude, MEX-ZM-0039, forms the `Myanmar` stratum, because that is where its point falls: 18 strata in all. Correct the coordinates first in Blocks 2 and 3.

7.7 Does it represent well? The criteria of Hu *et al.*

A core is judged by comparing it with the collection it came from. The `Does it represent well? Hu et al. (2000) criteria` box gives five figures, each with its `good` or `check` traffic light. The first four use the quantitative and ordinal traits with at least three values in the collection and two in the core (the *usable traits*, eight in the example); the fifth, the classes of the qualitative and ordinal traits.

THEORY IN BRIEF · MD% AND VD%

For each usable trait the core is compared with the whole collection (of which the core is part). With m usable traits:

```
MD% = 100 · (traits with different means) / m
VD% = 100 · (traits with different variances) / m
```

Means are compared with Welch's t test (Welch-Satterthwaite degrees of freedom) and variances with the two-tailed F test, both at 5%:

$$t = (\bar{x}_N - \bar{x}_C) \,/\, \sqrt{s_N^2/n_N + s_C^2/n_C}$$

$$F = s^2_{larger} \,/\, s^2_{smaller}$$

THEORY IN BRIEF · CR%, VR% AND THE CLASSES CAPTURED

The range and the coefficient of variation are averaged over the m traits (N = core, C = collection; R = maximum – minimum):

$$CR\% = \text{mean of } 100 \cdot R_N / R_C$$

$$VR\% = \text{mean of } 100 \cdot CV_N / CV_C$$

```
classes captured = 100 · Σ classes in the core / Σ classes in the collection
(qualitative and ordinal: each distinct value is a class)
```

Hu *et al.* (2000) only set two thresholds: a good core has MD% < 20 and CR% > 80. Of VR% they say the larger, the better; the app asks for VR% ≥ 100, that the core does not lose relative variation compared with the collection, and uses that same threshold in the tile, in the traffic light and in the note about VD%. It also shows in green VD% < 20 and classes captured above 80%.

The yellow warning in Figure 7.5, The core is too small to be judged by these figures., appears whenever the core has fewer than 10 accessions, because with so few the t and F tests have almost no power. The warning computes how much to raise it: with 46 characterized, the core must be 21 % or more: choose 25 % under "Size". With 25% (12 accessions) the warning disappears, CR% rises to 99.2 and VR% to 99.3, still a hair short of the 100 threshold; with 30% (14 accessions) the core passes all three criteria (Figure 7.6).

With 14 accessions, the core keeps 99.3% of the range, 100.7% of the relative variation and 31 of the 32 classes (only the color "pinto" is missing), without changing any mean. VD% stays at 50: four of eight traits have a variance different from the collection's. As the note under the criteria table explains, that is not a flaw: by picking the accessions most distinct from each other, the core ends up more variable than the collection and the F test detects it. A high VD% is only worrying when it comes with a low VR%. The note appears when VD% is above 20 with VR% ≥ 100 and CR% > 80.

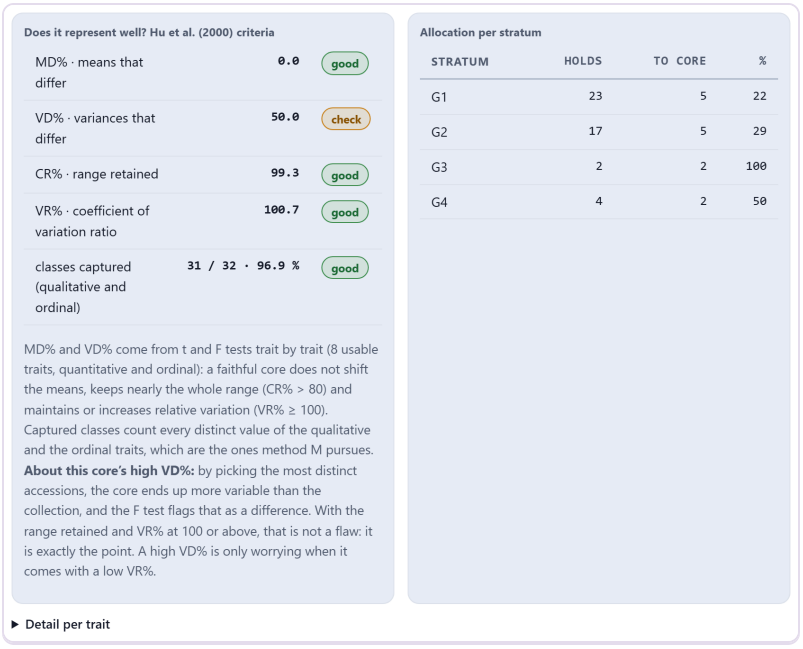


Figure 7.6. The validation and the allocation with the size at 30% (14 accessions). The small-core warning no longer appears, and the note explaining the high VD% does. Below the two boxes, the Detail per trait drop-down remains closed.

▸ WORKED EXAMPLE · VIGOR AND CLASSES IN THE DEFAULT CORE

The five accessions of the core have vigor 3, 1, 1, 4 and 5; the 45 in the collection with a value, mean 3.04 and standard deviation 1.45.

```
core: x̄ = 14/5 = 2.8 · s = √[(0.04 + 3.24 + 3.24 + 1.44 + 4.84)/4] = √3.2 = 1.789 · CV = 63.9%
collection: CV = 100 · 1.445/3.044 = 47.5% → VR of vigor = 100 · 63.9/47.5 = 134.6
range: core 1–5, collection 1–5 → CR of vigor = 100 · t = −0.30, p = 0.78: the mean does not differ
classes: color 4/8, shape 3/5, habit 3/5, texture 2/4, tolerance 3/5, vigor 4/5 → 19/32 = 59.4%
```

The app averages the same over the eight usable traits. The range comes out badly in two: height (CR = 9.5) and the weight of 100 seeds (2.4), because none of the five accessions has an extreme value of those traits — the oak with 329 g per 100 seeds, for example, did not enter—; that is why CR% stays at 75.2. All these figures, trait by trait, are in the Detail per trait drop-down (Figure 7.7).

To see which trait fails, open Detail per trait, below the criteria and the allocation (Figure 7.7): for each usable trait, the means, the *p* of the *t* and *F* tests (in red if below 0.05), the CR% and the VR%; for each qualitative or ordinal trait, the classes, the H' and the missing classes.

▼ Detail per trait

TRAIT	COLLECTION MEAN	CORE MEAN	P (T)	P (F)	CR%	VR%
dias_floracion	304.05	711.58	0.208	0.027	99.4	68.6
altura_planta_cm	278.95	512.17	0.215	0.036	100.0	85.5
largo_fruto_cm	14.61	19.48	0.425	0.056	98.0	112.9
diametro_fruto_cm	4.44	5.71	0.367	0.121	99.5	106.8
peso_100_semillas_g	31.49	72.33	0.469	0.004	97.9	96.9
rendimiento_planta_g	1,737.71	3,966.82	0.473	0.020	99.8	73.2
tolerancia_sequia	3.00	2.73	0.622	0.596	100.0	122.6
vigor	3.04	2.50	0.281	0.491	100.0	139.2

TRAIT	TYPE	CLASSES	IN THE CORE	H' COLLECTION	H' CORE	MISSING
color_principal	qualitative	8	7	2.03	1.87	pinto
forma	qualitative	5	5	1.50	1.52	—
habito	qualitative	5	5	1.16	1.47	—
textura	qualitative	4	4	1.29	1.17	—
tolerancia_sequia	ordinal	5	5	1.61	1.52	—
vigor	ordinal	5	5	1.59	1.44	—

Figure 7.7. The Detail per trait drop-down open, with the 30% core: the two tables take the full width of the card, with all their columns in view.

In the example, the four different variances are those of flowering, height, 100-seed weight and yield: the core brings in the agave, the oak, the cacao and the vanilla, which stretch those traits. Their means rise a lot — flowering, from 304.05 to 711.58 days— but no *p* of the *t* test falls below 0.05, because the variation within the core is also enormous.

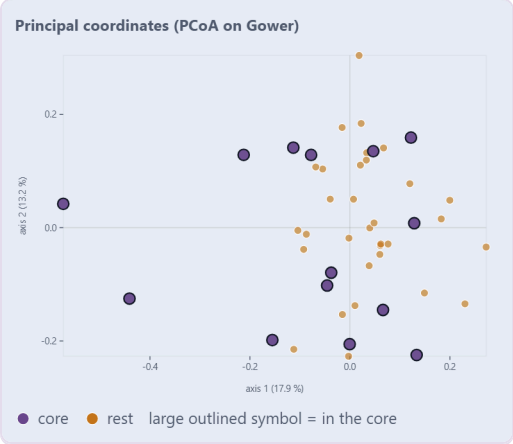


Figure 7.8. The principal coordinates with Color by on core and the 30% core: the fourteen chosen accessions cover the edges of the plane.

Figure 7.8 is the best visual test: a well-built core covers the whole plane and its edges, without crowding in the center. If the large points concentrate in one area, the core is missing part of the diversity even if the figures come out well.

In the example, the two isolated points on the left —the oak and the agave— are in the core, as are the furthest ones at the top, bottom and right.

7.8 Exporting, and how the core changes with the options

ACCESSION	NAME	SPECIES	GROUP	STRATUM
MEX-ZM-0001	Maíz azul de Apoala	*Zea mays*	G1	G1
MEX-ZM-0006	Cacahuacintle de Cholula	*Zea mays*	G1	G1
MEX-ZM-0009	Maíz salpor	*Zea mays*	G2	G2
MEX-AG-0024	Agave espadín	*Agave angustifolia*	G3	G3
MEX-VA-0025	Vainilla de Papantla	*Vanilla planifolia*	G2	G2
MEX-TC-0027	Cacao criollo de Ocosingo	*Theobroma cacao*	G4	G4
MEX-SE-0029	Chayote verde liso	*Sechium edule*	G2	G2
MEX-SE-0030	Chayote erizo (cuspidatum)	*Sechium edule*	G4	G4
MEX-QU-0031	Encino rojo, rodal semillero	*Quercus crassifolia*	G3	G3
MEX-OF-0037	Maíz arrocillo, parcela comun…	*Zea mays*	G1	G1
MEX-CU-0040	Calabaza pipiana	*Cucurbita argyrosperma*	G1	G1
MEX-CH-0042	Chile costeño	*Capsicum annuum*	G2	G2
MEX-ZM-0043	Maíz ancho de riego	*Zea mays*	G1	G1
MEX-SE-0046	Chayote verde liso	*Sechium edule*	G2	G2

Export the core as CSV | Characterization + group + core | Distance matrix

Figure 7.9. The list of the 30% core and the three export buttons. The list is shown expanded; in the app it scrolls inside its box. A name longer than 30 characters is cut with "…" (*Maíz arrocillo, parcela comun…*); the full name appears on hover.

Look at the list carefully: the 30% core includes *Chayote verde liso* twice, MEX-SE-0029 and MEX-SE-0046. They are one of the pairs of duplicates seeded in the example collection (Sections 3.5 to 3.7): the app does not tell them apart because the characterization treats them as distinct accessions. Merge duplicates in Block 3 before building the core, or you will spend a place on repeating an accession.

At the end of the section there are three buttons (the two SVG plots are saved from section 2, Section 7.5):

- **Export the core as CSV** saves core-collection.csv, one row per accession in the core with the columns `ACCENUMB`, `ACCENAME`, `TAXON`, `CROPNAME`, `group`, `stratum` (with its readable name, as in the table; empty with method M), `core` and its two principal coordinates, `pcoa1` and `pcoa2`, with four decimals.
- **Characterization + group + core** saves characterization-full.csv: the same columns for **all** the characterized accessions plus the values of the traits that enter the analysis.
- **Distance matrix** saves gower-distances.csv, the square Gower matrix with four decimals: the accession numbers go in the first column, headed `accession`, and in the header row.

The `core` column says *yes* or *no*. The CSV files are in UTF-8 with a byte order mark, so that Excel respects accented characters. None of them stores the settings: write down the options you used.

The same example, other options

The core does not always improve with size, and no allocation rule always wins. With the example characterization, stratified by the Ward groups and with the most distinct accessions:

Table 7.6. Validation by size (logarithmic allocation, four Ward groups, the most distinct). Classes: out of the 32 of the qualitative and ordinal traits.

Size	Core	MD%	VD%	CR%	VR%	Classes
5%	2	16.7	33.3	46.1	111.1	34.4%
10%	5	0.0	62.5	75.2	91.4	59.4%
15%	7	0.0	50.0	78.8	92.5	78.1%
20%	9	0.0	37.5	78.8	86.8	84.4%
25%	12	0.0	62.5	**99.2**	99.3	90.6%
30%	14	0.0	50.0	**99.3**	**100.7**	96.9%

With 5% there are 2 places for 4 groups and the app warns `2 of 4 strata are left without a representative.` 30% is the first size that passes MD%, CR% and VR% at once: from 25% on, the group of the agave and the oak contributes both its accessions and the range is almost complete.

Table 7.7. Other options with the size at 20% (target: 9 accessions); the rest, default.

Option	Core	MD%	VD%	CR%	VR%	Classes
Constant allocation (3 · 2 · 2 · 2)	9	0.0	75.0	99.2	99.9	84.4%
Proportional or square root (4 · 3 · 1 · 1)	9	0.0	37.5	78.8	91.4	87.5%
Logarithmic (3 · 3 · 1 · 2)	9	0.0	37.5	78.8	86.8	84.4%
Logarithmic, at random	9	0.0	37.5	73.0	87.2	68.8%
No strata, the most distinct	9	0.0	75.0	98.6	96.6	84.4%
Method M	9	0.0	62.5	91.4	96.6	100%
By state or country (9 of 18 without a representative)	9	0.0	12.5	56.9	72.3	71.9%
By crop (14 of 23 without a representative)	9	12.5	37.5	40.2	69.9	75.0%
By species (15 of 24 without a representative)	9	12.5	37.5	46.1	73.5	81.3%
By species, at least 1 per stratum	24	0.0	0.0	100	107.7	100%

Three lessons. The most distinct beat random choice (CR% 78.8 versus 73.0). Method M captures all the classes and keeps range and variation. And with few places, fine strata leave the rare ones out; with `at least 1 per stratum` the figures improve, but only because the core is now 52% of the collection. With few places, stratify by clusters or do not stratify.

When something goes wrong

What you see	Why it happens and what to do
Load your accessions in Block 2 first.	There is no passport. Load your collection or the example one (Chapter 2).
No characterization data yet...	There is a passport, but no characterization, or none of its rows matches. Load it, paste it or use the example one.
No row finds its accession.	The key column is not the accession-number column, or the numbers are not written as in the passport (spaces, zeros, hyphens). Change the column in the card or correct the numbers in your spreadsheet.
The mean silhouette is very low with any number of groups	The collection has no clear groups with these traits, or an extreme value flattens the differences (Section 7.3). Check the ranges, lower the weight of randomly assigned or unreliable traits, or work by species.
The core is too small to be judged by these figures.	The core has fewer than 10 accessions. Raise the size to the one the warning itself suggests (25% with 46 characterized).
... strata are left without a representative.	There are more strata than places. Use coarser strata, raise the size or tick at least 1 per stratum, knowing that the core will grow.
The core has ... accessions, not the ... requested	at least 1 per stratum is ticked and there are more strata than places. Clear it if you need the requested size.
The core includes the same accession twice under different numbers	Unmerged duplicates. Resolve them in Block 3 (Chapter 3).

✓ DECISION RULE · WHAT TO DO IN BLOCK 7

If you are going to load your characterization	**then** use the same accession numbers as the passport and check in the card that almost all rows match; merge the Block 3 duplicates first.
If two traits measure nearly the same thing	**then** lower the weight of one to 0.5, or clear the checkbox of one.
If the range of a trait is stretched by a single accession	**then** check that it is not an error; if it is real, work by species or transform the trait before loading it.
If the mean silhouette is above 0.25	**then** the groups are real: stratify the core by clusters; if it is low, do not stratify, use method M or build a core per crop.
If the small-core warning appears	**then** raise the size to 25 or 30% before reading the figures.
If MD% < 20, CR% > 80 and VR% ≥ 100	**then** the core represents the collection well; a high VD% with those figures is a sign that it picked the extremes well.
If CR% falls short	**then** the extremes are missing: use the most distinct, the constant allocation or a larger size, and look in the PCoA for the edge left without large points.
If rare classes are missing (see Detail per trait)	**then** try method M, which pursues them one by one.

Block 7 does not change the passport or the characterization: it only measures them and makes proposals. The core you export is a working list; the accessions all remain in the genebank, and Block 8 takes care of keeping them alive.

8

CHAPTER EIGHT · BLOCK 8 · GENEBANK MANAGEMENT

Running the genebank

The Block 1 simulator followed an imaginary accession; Block 8 does the same with your lots. From the last germination test, the stock, the regeneration dates and the conditions of your store, it projects with the Ellis and Roberts equation where each seed lot stands today and how many years it has left before the threshold; it checks each conservation route —seeds, field, in vitro, cryopreservation, reserve and on-farm plot— and raises alerts sorted by urgency; it records the tests, regenerations, replantings, subcultures, checks and visits you carry out; it computes how many plants and how many seeds are needed to regenerate without losing rare alleles, and it draws the workload for the next 25 years.

8.1 The genebank dashboard

8.2 The assumptions: the store and the intervals

8.3 Today's viability, lot by lot

8.4 The alerts, route by route

8.5 The work list

8.6 Recording what is done, and the plan as CSV

8.7 How many plants and how many seeds

8.8 The workload for the next 25 years

8.9 The genebank by route

8.1 The genebank dashboard

Block 8 works with four passport fields that Chapter 2 introduced (Section 2.1): `GP_STOCK` (stock), `GP_GERMPCT` (last germination), `GP_GERMDATE` (date of that test) and `GP_REGENDATE` (last regeneration, replanting, subculture, check or visit), plus the conservation route (`GP_CONS`, or the one inferred from `STORAGE`) and the safety duplicate (`DUPLSITE`). An empty cell is missing data, not a zero. Everything is recomputed on entering the block, with no wait: the calculations are instantaneous. If there are no accessions, the block says Load your accessions in Block 2 first.

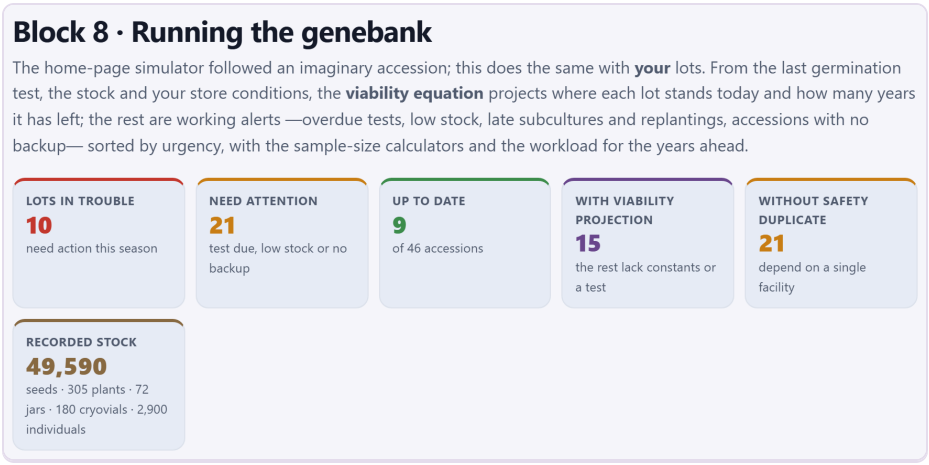


Figure 8.1. The header of the block with the example collection and the default assumptions, on October 31, 2026.

> **THE NUMBERS DEPEND ON THE DATE**
>
> Everything the block computes is measured from **today**, your computer's date: the years since the last test, the projected viability, the overdue checks and visits. The figures in this chapter were taken on **October 31, 2026** with the example collection; if you open it on another day, the dates change and, with them, some decimals and perhaps an alert. Incomplete dates are completed at the midpoint: a year without a month counts from June 15, and a month without a day, from the 15th.

The six tiles summarize the state of the genebank. Each accession receives the **level** of its most serious alert (Section 8.4): *in trouble*, *needs attention*, *a datum is missing* or *up to date*.

Table 8.1. The dashboard tiles with the example collection.

Tile	Example	What it counts
Lots in trouble	10	Accessions with at least one critical (red) alert. In red if there is any.
Need attention	21	Accessions whose most serious alert is a warning (ocher).

Table 8.1. The dashboard tiles with the example collection. *(continued)*

Tile	Example	What it counts
Up to date	9	Accessions with no alert at all, "of 46 accessions". The remaining 6 only have a missing-data notice, which has no tile of its own: 10 + 21 + 9 + 6 = 46.
With viability projection	15	Seed lots with viability constants for their species and a dated test: the 15 maizes and teosintes in the seed bank.
Without safety duplicate	21	Ex situ accessions with no copy in another institution (Section 8.4).
Recorded stock	49,590	Seeds, in large type, and below them the other units, each on its own: seeds · 305 plants · 72 jars · 180 cryovials · 2,900 individuals. The seeds add up those in the seed bank (49,070) and those of the lot with no declared route (520); the individuals, those in the reserves and in the on-farm plots.

Stock is not added across units: a seed, a field plant and a jar are not worth the same. By route they appear in section 4 (Section 8.9). Below the four sections of the block, a note reminds you that A projection is not a substitute for a test. (Figure 8.6).

8.2 The assumptions: the store and the intervals

The 1 · The assumptions behind every figure section gathers fifteen sliders. They are the conditions of your store and the working intervals and minimums of your genebank, route by route; everything the block says further down follows from them. The number on the right changes as you drag, and the calculation is redone when you release. The values are saved in the browser and kept when you close the app.

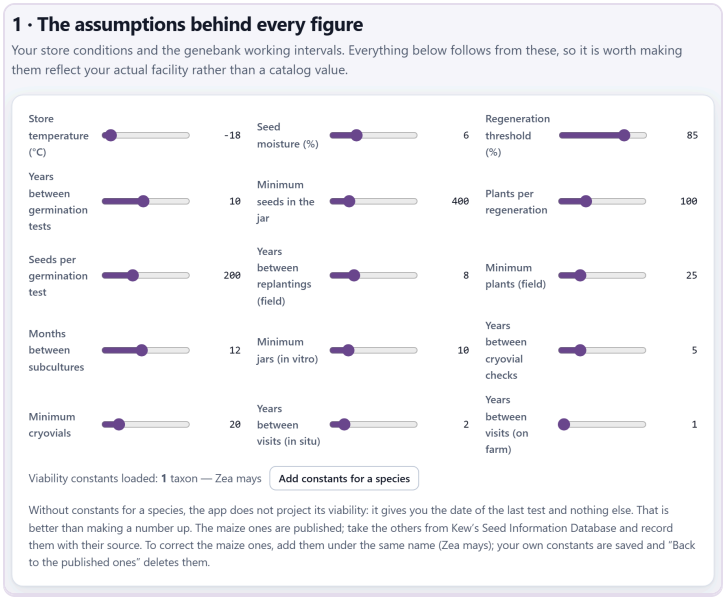


Figure 8.2. The default assumptions. The store is the one in the FAO standard for long-term conservation: –18 °C and 6% moisture.

Table 8.2. The fifteen sliders (default in bold).

Slider	Range · step	What it governs
Store temperature (°C)	−20 to 25 · 1 · **−18**	The *t* of the viability equation (Section 8.3).
Seed moisture (%)	3 to 14 · 0.5 · **6**	The *m* of the equation, in % of fresh weight.
Regeneration threshold (%)	50 to 95 · 5 · **85**	Below this germination, measured or projected, the lot is in trouble and must be regenerated.
Years between germination tests	1 to 20 · 1 · **10**	When a seed lot's test falls due and how often it is repeated in the workload.
Minimum seeds in the jar	50 to 2,000 · 50 · **400**	Below it, stock in trouble; below 1.5 times the minimum (600), a warning that it is getting close.
Plants per regeneration	20 to 300 · 10 · **100**	The plants you regenerate with: they give the seeds to sow in the alerts, the third line of the calculator (Section 8.7), the plants in the workload and the sowing list of Block 9.
Seeds per germination test	50 to 500 · 25 · **200**	What it proposes to deduct when you record a test (Section 8.6).
Years between replantings (field)	1 to 30 · 1 · **8**	The cycle of the field genebank, counted from `GP_REGENDATE`.
Minimum plants (field)	5 to 100 · 5 · **25**	Below it, the field accession is in trouble.
Months between subcultures	2 to 24 · 1 · **12**	The interval of slow-growth in vitro culture.
Minimum jars (in vitro)	2 to 50 · 1 · **10**	Below it, a warning of few jars.
Years between cryovial checks	1 to 20 · 1 · **5**	How often a cryovial is thawed to check that the material regrows.
Minimum cryovials	5 to 100 · 5 · **20**	Below it, a warning of few cryovials.
Years between visits (in situ)	1 to 10 · 1 · **2**	How often a genetic reserve is visited.
Years between visits (on farm)	1 to 10 · 1 · **1**	How often an on-farm conservation plot is visited.

The viability constants

Below the sliders, the line Viability constants loaded says how many taxa have constants and names the first four. By default there is only one, 1 taxon — Zea mays, with the constants published by Ellis and Roberts: K_E = 6.474, C_W = 2.115, C_H = 0.0329 and C_Q = 0.000478.

The Add constants for a species button opens four browser prompts in a row: the taxon (it suggests those in your collection), K_E, C_W and the source. The two constants propose the maize values; C_H and C_Q are not asked and take the values above, which describe the effect of temperature in many orthodox species (Section 1.5). Write the taxon exactly as the **genus and species** of the passport, separated by a space: *Phaseolus vulgaris*, not *P. vulgaris* or *bean*. If you add a taxon that already has constants, the new ones replace them; that is also how the maize ones are corrected, under the name *Zea mays*.

Your own constants are saved in the browser, and the line says so: with some for the common bean it shows 2 taxa — Zea mays, Phaseolus vulgaris (1 your own) and a second button, Back to the published ones, which, after asking you, deletes all your own constants and leaves only the published maize ones.

ONLY MAIZE COMES WITH PUBLISHED CONSTANTS

For the other twelve species with seed lots in the example collection the app projects nothing: it says the constants are missing and works only with the value and date of the last test. That is better than making up a number. Take K_E and C_W for your species from Kew's Seed Information Database or from the literature, and record the source in the fourth prompt. Accepting the proposed values for another species amounts to assuming that it ages like a maize.

8.3 Today's viability, lot by lot

For every lot in the seed bank —or with no declared route, which the block treats the same way— with constants for its species, a germination value and a test date, the app projects today's germination and the years it has left before crossing the threshold. It is the same equation as the Block 1 simulator (Section 1.5), applied from the lot's last real test.

THEORY IN BRIEF · THE ELLIS AND ROBERTS VIABILITY EQUATION

$$v = K_i - p / \sigma \cdot \log_{10} \sigma = K_E - C_W \log_{10} m - C_H t - C_Q t^2 \cdot \text{germination} = 100\ \Phi(v)$$

$\mathbf{K_i}$ is the germination of the last test expressed in probits: $K_i = \Phi^{-1}(g_0/100)$; **p**, the days elapsed since that test; **σ**, the days it takes viability to fall by one probit, and **Φ**, the cumulative normal. With the threshold *u*, the years the lot has left **from today** are

$$\text{years to threshold} = \sigma \cdot [\Phi^{-1}(g_0/100) - \Phi^{-1}(u/100)] / 365.25 - \text{years since the test}$$

If the result is negative, the lot has already crossed the threshold. Germination is bounded between 0.01 and 99.99% before converting it to probits, because Φ^{-1} of 0 or of 100 is infinite.

WORKED EXAMPLE · THE MAÍZ AZUL DE APOALA (MEX-ZM-0001)

The last test gave 88% on March 10, 2024; on October 31, 2026, 2.64 years (965.6 days) have passed. The store is at –18 °C with 6% moisture.

$$\log_{10} \sigma = 6.474 - 2.115 \times 0.77815 - 0.0329 \times (-18) - 0.000478 \times 324 = 6.474 - 1.6458 + 0.5922 - 0.1549 = 5.2655$$

$$\sigma = 10^{5.2655} = 184{,}305 \text{ days} = 504.6 \text{ years per probit}$$

$$K_i = \Phi^{-1}(0.88) = 1.17499 \qquad v = 1.17499 - 965.6 / 184{,}305 = 1.16975$$

$$\text{today's germination} = 100 \cdot \Phi(1.16975) = 87.89\% \rightarrow \text{the table writes } 88\ \% \rightarrow 88\ \%$$

$$\text{years to threshold} = 504.6 \times (1.17499 - 1.03643) - 2.64 = 69.91 - 2.64 = 67.27 \text{ years}$$

At –18 °C the lot loses a tenth of a point in two and a half years: the projection has nothing to warn about, and the lot's only alert is that it has no safety duplicate.

The Teocintle de Balsas (MEX-ZP-0011) is another case: its last test gave 81% on September 14, 2023, already below the 85% threshold; the projection puts it today at 80.83%, which is also written 81%, and its "years to threshold" are −83.1, because under those conditions it would take 83 years to fall from 85 to 81%.

The projected viability appears in the `Germination` column of the work list, after an arrow: `88 % → 88 %` is the last test and today's value. Below it goes the time since the test, in years with one decimal: `2.6 y ago`, or `no test recorded` if there is no date.

What temperature and moisture weigh

At −18 °C and 6% moisture the difference between the test and the projection is invisible. The conditions of the store are what change the picture: move the first two sliders and watch how the collection responds.

Table 8.3. Four stores with the example collection (85% threshold; the other assumptions, default).

Store	σ (years per probit)	MEX-ZM-0001 today	Years to threshold	In trouble	Up to date
−18 °C, 6% (default)	504.6	87.89%	67.27	10	9
5 °C, 6%	122.8	87.56%	14.37	10	9
5 °C, 8%	66.8	87.19%	6.61	10	9
20 °C, 12% (no store)	6.0	76.91%	−1.81	19	0

In a cold room at 5 °C the blue maize goes from having 67 years ahead of it to having 14, and with seed at 8% moisture, fewer than 7; the dashboard does not change yet, but the workload does: at 5 °C and 6% regenerations appear in years 14 and 20 (Section 8.8). In a community seed bank with no store, at 20 °C and 12%, nine more maizes cross the threshold and the lots in trouble go from 10 to 19; the other four —those at 93 to 95%— are left with fewer than 3 years and need attention.

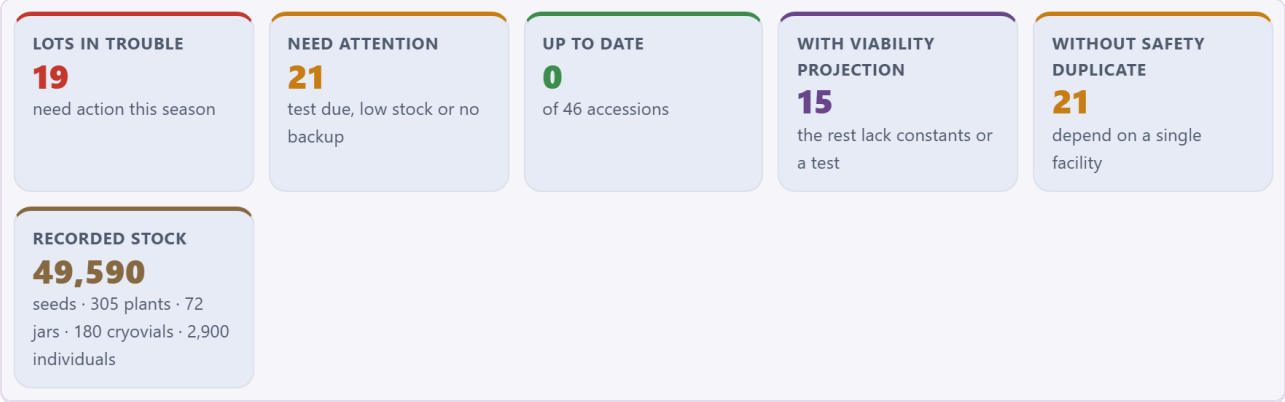


Figure 8.3. The dashboard with the store of a community seed bank: 20 °C and 12% moisture. No lot is up to date.

8.4 The alerts, route by route

Each conservation route has its own clock, and the block applies to each accession the checks of its route. Each alert has a level: **critical** (in red, 3 points), **warning** (in ocher, 2 points) or **missing datum** (in blue, 1 point). Table 8.4 gathers them all, with the default minimums and intervals.

With the example collection, the 10 accessions in trouble are four seed lots below the threshold —the teosinte (81%), the irrigated ancho maize (72%), the vaquita bean (84%) and the tepary bean (79%)—, three field accessions with few plants —the cacao, the avocado and a chayote— and the three in vitro cultures, with 32 to 34 months without subculture against an interval of 12. The 6 that only have a missing datum are seed lots without constants and with everything else in order: the runner bean, the squash, two chilies, the husk tomato and the wheat.

Table 8.4. The Block 8 alerts and how many accessions of the example collection have them.

Route	When it appears and what it says	Level	Ex.
Seed bank (or no declared route)	The last test already gave less than the threshold: `The last test gave N %, below the 85 % threshold. Regenerating with 100 plants means sowing about S seeds.`	critical	4
	The test is above the threshold, but today's projection is not: `The projection puts it at N % germination...`, with the same sentence about the seeds.	critical	0
	It has fewer than 3 years left before crossing the threshold: `About N years left...`	warning	0
	It has a germination value, but there are no constants for its species: `No viability constants for...`	datum	14
	It has no test date: `It has never had a germination test.`	critical	0
	More years than the interval have passed since the last test: `N years since the last test...`	warning	0
	Stock below the minimum (`N seeds left...`, critical) or below 1.5 times the minimum (`Stock (N) is getting close to the minimum.`, warning).	critical · warning	0 · 2
	The stock cell is empty: `No stock recorded.`	datum	0
Field genebank	No date of the last replanting: `No date for the last replanting.`	warning	2
	More years than the cycle since `GP_REGENDATE`: `N years since the last replanting...`	critical	2
	Fewer plants than the minimum: `Only N plants left (the minimum is 25): drift at the next replanting will be severe.`	critical	3
In vitro culture	No date of the last subculture: `No date for the last subculture.`	warning	0
	More months than the interval since `GP_REGENDATE`: `N months since the last subculture...`	critical	3
	Fewer jars than the minimum: `N jars left (the minimum is 10).`	warning	0

Table 8.4. The Block 8 alerts and how many accessions of the example collection have them. *(continued)*

Route	When it appears and what it says	Level	Ex.
Cryopreservation	More years than the interval since the last check (`GP_REGENDATE`) or, if there is none, since it entered the genebank (`ACQDATE`): `N years since it entered the genebank, with no check recorded...` ; with neither date, `No date for the last check of the cryovials.`	warning	1
	Fewer cryovials than the minimum: `N cryovials left (the minimum is 20).`	warning	0
Reserve and on-farm plot	More years than the interval since the last visit, which is `GP_REGENDATE` or, if missing, the collecting date: `N years since the last recorded visit to the site...`	warning	5
Every ex situ route	No site in `DUPLSITE` is an institution other than `INSTCODE` : `No safety duplicate in another institution.`	warning	21

Below the threshold there is a single alert, the most serious one: if the measured test is already below, the app states it and mentions the projection only when it shows after rounding (*...below the 85 % threshold; the projection puts it at N % today*). The teosinte gets no such mention, because 80.83% is also 81%. The closing sentence gives the seeds to sow to regenerate with the plants in the assumptions and today's germination (the projected one, if there is one): S = ⌈ plants / (g · 0.9) ⌉, as in the calculator (Section 8.7). If the stock is not enough, it adds: *...and only N are left.*

The safety duplicate follows the same rule as the Block 2 validation (Section 2.6): `DUPLSITE` may hold several codes separated by semicolons, and it is enough for one to differ from the code of the institution that holds the accession. In situ routes do not require it; all the others do, including cryopreservation and the DNA bank. The only cryopreserved accession in the example, the bitter potato (PER-ST-0022), has no check recorded and entered the genebank in 2018; its alert says *8.2 years since it entered the genebank, with no check recorded (the plan is every 5).*

8.5 The work list

The `2 · What needs doing` section is the list for the season: one row per accession, sorted by urgency, as the paragraph under the title explains. First comes the **level**: the accessions in trouble, then those needing attention, those with a missing datum and, last, those up to date. Within each level, the **score**, the sum of the points of its alerts (3, 2 or 1), from highest to lowest; on equal score, the accession number, in alphabetical order. It shows up to 200 rows.

EXAMPLE · THE ORDER OF THE FIRST ROWS

```
MEX-PH-0041, tepary bean: 79% < 85 → 3 + no constants 1 + 430 < 600 → 2 + no duplicate 2 = 8
MEX-TC-0027, cacao: replanting overdue (8.7 > 8 years) 3 + 20 plants < 25 → 3 + no duplicate 2 = 8
MEX-SE-0046, chayote: no replanting date 2 + 15 plants → 3 + no duplicate 2 = 7
MEX-PA-0026, avocado: 3 + 3 = 6; MEX-SE-0013, vaquita bean: 84% → 3 + no constants 1 + no duplicate 2 = 6
MEX-ZM-0043, irrigated ancho maize: 72% → 3 + no duplicate 2 = 5
COL-MA-0035, MEX-VA-0025, MEX-ZP-0011 and PER-ST-0021: a single critical alert, 3 each
needs attention: MEX-CH-0042, costeño chili: 1 + 2 + 2 = 5; MEX-CA-0028 and PER-ST-0022: 2 + 2 = 4; …
```

The 10 accessions in trouble come first, even though the teosinte only adds up to 3 points and the costeño chili, which only needs attention, adds up to 5. The first two tie at 8 points, and the tepary bean comes before the cacao by its accession number (MEX-PH before MEX-TC); the same goes for the avocado and the vaquita bean, with 6, and for the four 3-point lots. After the 21 needing attention come the 6 with a missing datum and, last, the 9 up to date.

Table 8.5. The columns of the work list.

Column	What it shows
Accession	The accession number and, below it, the first 24 characters of the name.
Route	The conservation route, in a label in the color of the accession's level: red in trouble, ocher if it needs attention, blue if a datum is missing, green if it is up to date.
Stock	`GP_STOCK` and, below it, its unit: seeds, plants, jars, cryovials or individuals; "—" if the cell is empty.
Germination	Seed only: the last test, today's projection after the arrow if there is one, and the time since the test. In the other routes, "—".
What is going on and what to record	The alerts, each with a bar in the color of its level (or the up to date label), and below them the buttons to record the work done (Section 8.6).

Two selectors filter the list. Show leaves everything, only the accessions in trouble (10), those that need attention (21), those with a missing datum, missing data (6), or those up to date (9). Route leaves a single route; each option says how many accessions it holds, for example Seed bank (28) or Not declared (1), which leaves only the costeño chili (MEX-CH-0042).

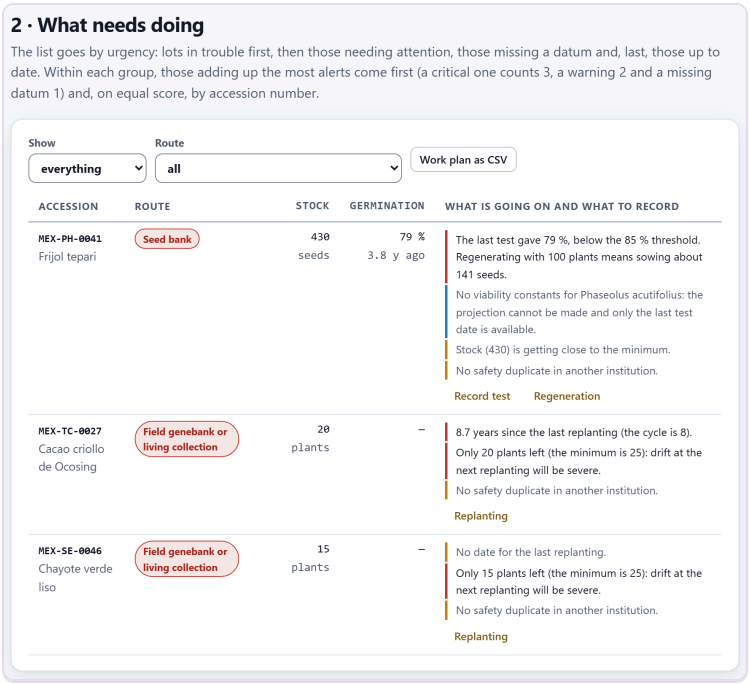


Figure 8.4. The paragraph that explains the order and the first three rows of the work list, in trouble with 8, 8 and 7 points.

8.6 Recording what is done, and the plan as CSV

The buttons on each row write into the passport what you did today. Each route has its own, and they work with the browser's prompt windows; if you cancel any of the prompts, nothing is saved. To record a regeneration not yet tested, leave the germination empty and accept. When you finish, the passport is saved in the browser, the Block 2 table is updated and Block 8 is recomputed.

Table 8.6. The buttons for each route (the date they write is today's, as YYYYMMDD).

Button · route	Prompt	Changes in the passport
Record test · seed and no route	The result (%, from 0 to 100; accepts a decimal comma) and how many seeds were used, proposing the number in the assumptions (200; 0 for no deduction).	`GP_GERMPCT` (with one decimal), `GP_GERMDATE`, and subtracts the seeds from `GP_STOCK` without going below 0; if no stock is recorded, it deducts nothing.
Regeneration · seed and no route	The stock after regenerating and the germination of the new lot, with 96% proposed; both may be left blank.	`GP_REGENDATE`; `GP_STOCK` if you gave a stock; `GP_GERMPCT` and `GP_GERMDATE` if you gave a germination.
Replanting · field; Subculture · in vitro; Check · cryopreservation	The stock after the work, in its unit (plants, jars or cryovials); blank if unchanged.	`GP_REGENDATE` and, if you gave a stock, `GP_STOCK`.
Visit · reserve and on-farm plot	Only confirms: Record a visit to the site of … today?	`GP_REGENDATE`.

EXAMPLE · A TEST THAT GOES BADLY AND THE REGENERATION THAT FIXES IT

On the Maíz azul de Apoala (MEX-ZM-0001), Record test with 80% and 200 seeds leaves `GP_GERMPCT` = 80, `GP_GERMDATE` = 20261031 and `GP_STOCK` = 1,600 (1,800 − 200). The lot goes from needing attention to being in trouble, with 5 points and the alert *The last test gave 80 %, below the 85 % threshold. Regenerating with 100 plants means sowing about 139 seeds.* (100 / (0.80 × 0.9) = 138.9). Then, Regeneration with 3,000 seeds and 97% leaves `GP_STOCK` = 3,000, `GP_GERMPCT` = 97 and both dates at 20261031; the lot is left with only the alert that it has no safety duplicate. In the same way, Check on the cryopreserved potato with 170 cryovials removes its overdue-check alert.

The changes enter the passport like any other edit: they show in the Block 2 record and travel in its exports. There is no history of tests; each record replaces the previous one. If you need to keep the series, export the passport before each season.

The work plan as CSV

The Work plan as CSV button saves work-plan.csv, in UTF-8, with one row per accession with any alert —37 with the example collection— in passport order and without applying the filters. If none has alerts, the app says Nothing pending: the whole collection is up to date.

Table 8.7. Seven of the twelve columns of the work plan, with two rows of the example collection.

ACCENUMB	route	urgency	stock	unit	last_germination	projected_germination
MEX-ZM-0001	Seed bank	needs attention	1800	seeds	88	87.9
MEX-ZP-0011	Seed bank	in trouble	900	seeds	81	80.8

The other five columns are `ACCENAME`, `TAXON` (genus and species), `years_since_test`, `years_to_threshold` (67.3 and −83.1 in these two rows) and `actions`, the accession's alerts separated by a vertical bar. The route, the urgency, the unit and the actions come out in the language of the interface; the years, with one decimal. Cells with no data are left empty: in the routes that are not seed, the five germination and year cells. Sort it by `urgency` in your spreadsheet and share it out among those who do each task.

8.7 How many plants and how many seeds

The 3 · How many plants and how many seeds section has two panels. The one on the left, Plants needed, answers the question of every regeneration: with how many plants rare alleles are not lost. It has three sliders: Frequency of the allele to keep (1 to 20%, default 5%), Desired certainty (80 to 99%, default 95%) and Lot germination (%) (20 to 100% in steps of 5, default 85%). The result changes as you drag.

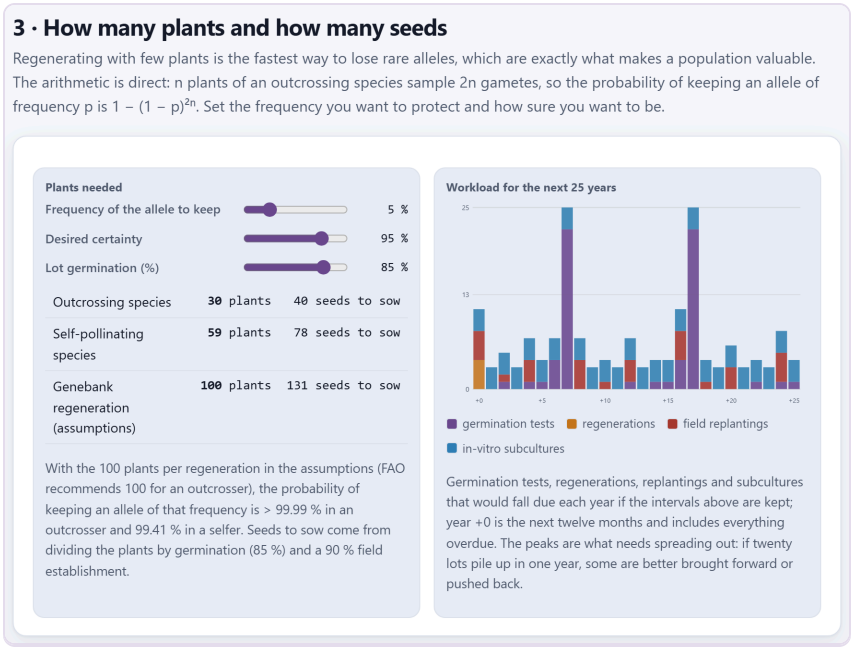


Figure 8.5. The sample-size calculator with its default values and the workload of the example collection.

THEORY IN BRIEF · THE PROBABILITY OF KEEPING A RARE ALLELE

A regeneration with n plants is a sample of gametes. In an **outcrossing** species, each plant contributes two independent gametes, and the probability that an allele of frequency p is left out of the $2n$ gametes is $(1 - p)^{2n}$. In a **self-pollinating** species, nearly homozygous, each plant contributes in practice a single effective gamete, and the exponent is n. To keep the allele with probability P:

```
P(keep) = 1 − (1 − p)^(k·n)  ·  n = ⌈ ln(1 − P) / (k · ln(1 − p)) ⌉  ·  k = 2 in outcrossers, 1 in selfers
```

The seeds to sow come from dividing the plants by the germination g and a fixed field establishment of 90%: seeds = ⌈ n / (g · 0.9) ⌉.

The first two lines say how many plants are needed in an outcrossing and in a self-pollinating species, and how many seeds to sow to obtain them. The third, `Genebank regeneration (assumptions)`, takes the plants per regeneration from section 1 and gives their seeds; the note below gives, with them, the probability of keeping the allele in the two kinds of species and, if they are not enough, how many would be needed. Probabilities within a hundredth of 100% are written `> 99.99 %`.

WORKED EXAMPLE · THE DEFAULT VALUES AND A 1% ALLELE

p = 0.05, P = 0.95: ln 0.05 = −2.99573; ln 0.95 = −0.05129	
outcrossing: 2.99573 / (2 × 0.05129) = 29.20 → 30 plants	selfing: 2.99573 / 0.05129 = 58.40 → 59 plants
seeds: 30 / (0.85 × 0.9) = 39.2 → 40	seeds: 59 / 0.765 = 77.1 → 78
100 plants: seeds 100 / 0.765 = 130.7 → 131; $1 - 0.95^{200}$ = 99.9965%, written > 99.99%; $1 - 0.95^{100}$ = 99.41%	
p = 0.01: outcrossing 2.99573 / (2 × 0.01005) = 149.04 → 150 plants and 197 seeds; selfing 299 and 391	
100 plants and p = 0.01: $1 - 0.99^{200}$ = 86.60% (outcrossing); $1 - 0.99^{100}$ = 63.40% (selfing)	

With p = 1% the note adds "100 plants do not reach the requested certainty in an outcrosser: 150 would be needed." A 5% allele is kept with 30 plants of an outcrosser; a 1% one requires 150. That is why it pays to set the lowest frequency you care about, not the most convenient one, and to carry that number over to the Plants per regeneration slider.

FAO recommends 100 plants to regenerate an outcrosser, and that is the slider's default value. The calculation assumes a random sample and unrestricted crossing: if only part of the plants flower and cross, the effective number is smaller than the number you sow, and you have to sow more.

8.8 The workload for the next 25 years

The right-hand panel of section 3 is a stacked bar chart: for each year, from +0 to +25, how many **germination tests** (purple), **regenerations** (orange), **field replantings** (red) and **in-vitro subcultures** (blue) would fall due if the intervals of section 1 are kept. Year +k is the span from k to k + 1 years from today: +0 is the next twelve months, and it includes **everything overdue**. This is what decides a genebank's budget: how many sowings, how much greenhouse space and how many hands are needed each year.

Table 8.8. How each task is scheduled.

Task	When it falls
Germination test	In seed lots (or with no declared route): interval – years since the last test, and every interval from then on. If it is overdue or the lot was never tested, the first one falls in year +0.
Regeneration	In year +0 if the last test is already below the threshold, whether or not the species has constants; in lots with a projection, the year they cross the threshold. Each one counts the plants per regeneration from the assumptions.
Field replanting	Cycle – years since the last replanting, and every cycle from then on; in year +0 if it is overdue or there is no date.
In vitro subculture	Interval – time since the last subculture, and every interval from then on, event by event: with 24 months, one year it falls due and the next it does not.

EXAMPLE · THE WORKLOAD OF THE EXAMPLE COLLECTION

```
year +0: 4 regenerations (teosinte, vaquita bean, tepary bean, irrigated ancho maize; ≈
400 plants) + 4 replantings (avocado and cacao, overdue; coffee and a chayote, no date)
+ 3 subcultures = 11
```

```
years +7 and +17: 22 tests + 3 subcultures = 25, the top of the scale
```

```
in 26 years: 61 tests, 4 regenerations, 28 replantings and 78 subcultures (3 jars × 26)
```

The two peaks are the 22 lots tested between March and April 2024: their tests fall due together in 7.4 years and repeat ten years later. This is the case the panel's note points out: it pays to bring some of those tests forward to spread them over the neighboring years. Another four lots, tested in 2023, fall in year +6.

The chart has an alternative text with the summary —totals, year +0 and the busiest year— for screen readers. When you hover over a segment, a label gives the year, how many tasks there are and, for regenerations, how many plants they add up to: `year +0: 4 regenerations (≈ 400 plants)`. The vertical scale has three marks: 0, the midpoint and the maximum (with the example collection, 0, 13 and 25). The chart also goes into the Block 10 report (Chapter 10).

The workload changes with the assumptions. With tests every 5 years, the peaks of 22 tests repeat in years 2, 7, 12, 17 and 22. With 24 months between subcultures, each jar is subcultured 13 times in 26 years, not 26. With the store at 5 °C and 8% moisture, other regenerations are added to the 4 in year +0 in years 6, 10, 13, 17 and 21; at 20 °C and 12%, 16 regenerations fall in year +0.

8.9 The genebank by route

The `4 · The genebank by route` section summarizes the collection by conservation route, from the one with the most accessions to the one with the fewest: how many accessions, how many in trouble and how many need attention (in colored labels when they are not 0) and the sum of their stock with its unit.

How to read it. The seed bank concentrates the accessions, but its problems are few and almost all about backup: 4 of 28 lots in trouble. The field and in vitro routes are the opposite: 3 of 8 and 3 of 3 in trouble, with 305 plants and 72 jars in total. These are the routes lost through neglect —a replanting that was not done, a subculture that slipped— and the ones that need the most hands per accession. The `Not declared` row is the costeño chili, which has no `GP_CONS` and whose `STORAGE` holds a code that does not exist in MCPD (15, one of the seeded errors flagged by the Block 2 validation): the block checks it as seed, but you should declare its route in Block 2.

The footnote, `A projection is not a substitute for a test.`, sums up the limit of the block: the equation says what *should* happen to a lot under those conditions, not what did happen to it. Incomplete drying, a badly sealed jar or a power cut appear in no formula. Use the projection to decide which lots to test first, not to skip the test.

4 · The genebank by route

ROUTE	ACCESSIONS	IN TROUBLE	NEED ATTENTION	STOCK
Seed bank	28	4	9	49,070 seeds
Field genebank or living collection	8	3	5	305 plants
In vitro culture	3	3	0	72 jars
On-farm conservation	3	0	3	1,350 individuals
Genetic reserve (in situ)	2	0	2	1,550 individuals
Cryopreservation	1	0	1	180 cryovials
Not declared	1	0	1	520 seeds

A projection is not a substitute for a test. The viability equation says what *should* happen to a lot under those conditions, not what did happen: an incomplete drying, a badly sealed jar or a three-week power cut appear in no formula. Use it to decide which lots to test first, not to skip the test. And if the app has no constants for your species, it projects nothing: it gives you the date of the last test and stops there.

Figure 8.6. The example genebank by route and the note that closes the panel.

When something goes wrong

What you see	Why it happens and what to do
Load your accessions in Block 2 first.	There is no passport. Load your collection or the example one (Chapter 2).
No viability constants for… in almost every lot	Only maize comes with constants. Add those of your species with Add constants for a species (Section 8.2).
I added constants and the lot still has no projection	The taxon does not match letter for letter the genus and species of the passport, or the lot has no `GP_GERMPCT` or `GP_GERMDATE`.
It has never had a germination test. or No stock recorded.	The `GP_GERMDATE` or `GP_STOCK` cells are empty. Fill them in the Block 2 record, or do the test and record it.
The Germination column says "—"	The route is not seed, or the lot has no recorded test.
The alert says there are not enough seeds to regenerate	The stock is smaller than the seeds to sow. Regenerate with what there is and request material from the safety duplicate.
Year +0 of the workload is very heavy	Everything overdue falls there. Spread what you can over the season and record what you do.
I recorded something by mistake	There is no undo. Correct `GP_GERMPCT`, `GP_GERMDATE`, `GP_REGENDATE` or `GP_STOCK` in the Block 2 record.

DECISION RULE · WHAT TO DO IN BLOCK 8

If you are reading the block for the first time	**then** first set the real temperature and moisture of your store, and the intervals and minimums of your genebank, not those of the standard.
If your species are not maize	**then** add their constants with their source; without them, work with the tests and their dates.
If the last test or the projection is below the threshold	**then** regenerate the lot this season, with the plants the calculator asks for the rarest allele you care about, and sow the seeds the alert gives.
If it has fewer than 3 years left before the threshold	**then** bring its germination test forward: the projection decides what to test first, the test decides whether to regenerate.
If a field accession has few plants or an overdue replanting	**then** replant it first and make an in vitro or cryopreserved backup.
If an in vitro culture is past its subculture interval	**then** subculture it before any seed lot: the jar does not wait.
If the cryovials have no check recorded	**then** thaw one, check that it regrows and record the check.
If an ex situ accession has no safety duplicate	**then** send a copy to another institution and note it in `DUPLSITE`.
If the workload has a peak	**then** bring some of those tests forward to spread them over the neighboring years.

Block 8 writes to the passport only when you record a test, a regeneration, a replanting, a subculture, a check or a visit; the rest is arithmetic. Export the passport every season to keep the history of your lots, which the app does not store.

9

CHAPTER NINE · BLOCK 9 · LABELS & REGISTER

Labels and register

Everything that enters or leaves the genebank carries paper: the label on the envelope, the jar, the stake or the cryovial, and the sheet that travels with the material. Block 9 prints those labels at their real size, in millimeters, with a Code 128 barcode or a QR code that the app itself generates, with no outside libraries; and it prepares the genebank's three worksheets: the sowing list for a regeneration, the delivery note and the collection register book. It computes nothing new: it takes the passport from Block 2, the core collection from Block 7 or the lots in trouble from Block 8, and puts them on paper.

9.1 What gets printed

9.2 The five formats and how many fit on a sheet

9.3 What each label says

9.4 The Code 128 barcode

9.5 The QR code and the chayote in the middle

9.6 Printing and checking

9.7 The sowing list

9.8 The delivery note

9.9 The register book

9.1 What gets printed

The block has two sections: 1 · What gets printed and in which format, with the controls, and 2 · The sheet, exactly as it will print, with the preview. What you see in the second section is exactly what the Print button sends to the printer: when printing, the app hides the title, the controls and the warnings. Below the controls, when needed, ochre warnings appear about what is worth checking before printing (Sections 9.3 to 9.5). If there are no accessions, the block says Load your accessions in Block 2 first.

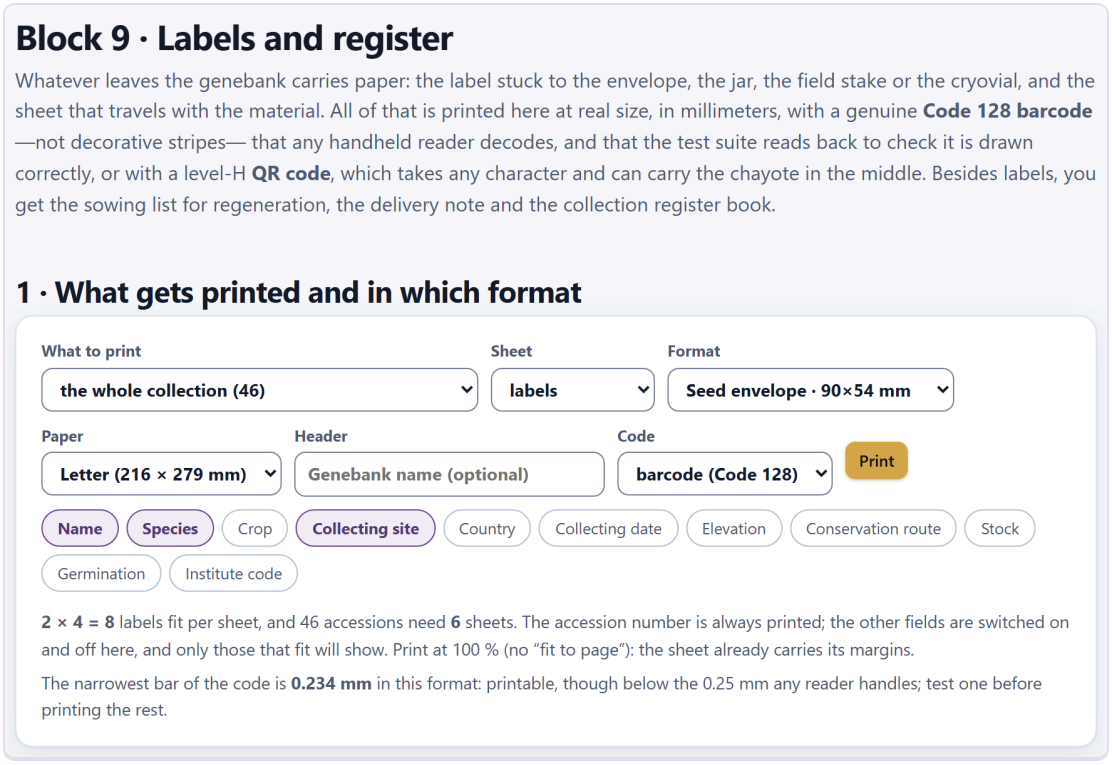


Figure 9.1. Section 1 with the default values and the example collection.

The What to print selector decides which accessions go in, and it applies to the labels and to the three worksheets alike:

Table 9.1. The four options of What to print, with the example collection.

Option	Ex.	Which accessions it takes
the whole collection (46)	46	The whole passport, in its order. This is the default option.
the core collection from Block 7 (5)	5	The accessions of the core you built in Block 7 (Chapter 7). With the example characterization and the default 10% core, they are MEX-AG-0024, MEX-TC-0027, MEX-CU-0040, MEX-CH-0042 and MEX-ZM-0043. While there is no core, the option says (not built yet) and is disabled.

Table 9.1. The four options of What to print, with the example collection. *(continued)*

Option	Ex.	Which accessions it takes
the lots in trouble from Block 8 (10)	10	The accessions with at least one critical alert in Block 8 (Section 8.4): the same 10 as in the Lots in trouble box. The app runs the Block 8 computations even if you have not opened it, so the number is there from the first visit.
my selection (0)	—	The ones you tick by hand in the list that appears below (Figure 9.2).

If the core is cleared after you choose it —for example, when you load another characterization—, the option stays selected with no accessions: the preview says No accessions in this selection., the Print button is disabled and a warning asks There is no core yet: build it in Block 7.

The Sheet selector chooses what gets printed: labels, sowing list, delivery note or register book. The format, paper, code and field controls only appear with labels. The Header is there with any sheet: on labels it goes at the top of each one, and on the three worksheets, in the top right corner, above the date.

Picking accessions by hand

With my selection, a list with tick boxes appears below section 1, one per accession, with its number, the first 28 characters of its name and its species. The Search box filters by number, name, genus, species or collecting site, ignoring case; the tick the N listed button ticks every accession the filter leaves, and untick all clears the selection. The list shows up to 400 accessions; if there are more, narrow the search.

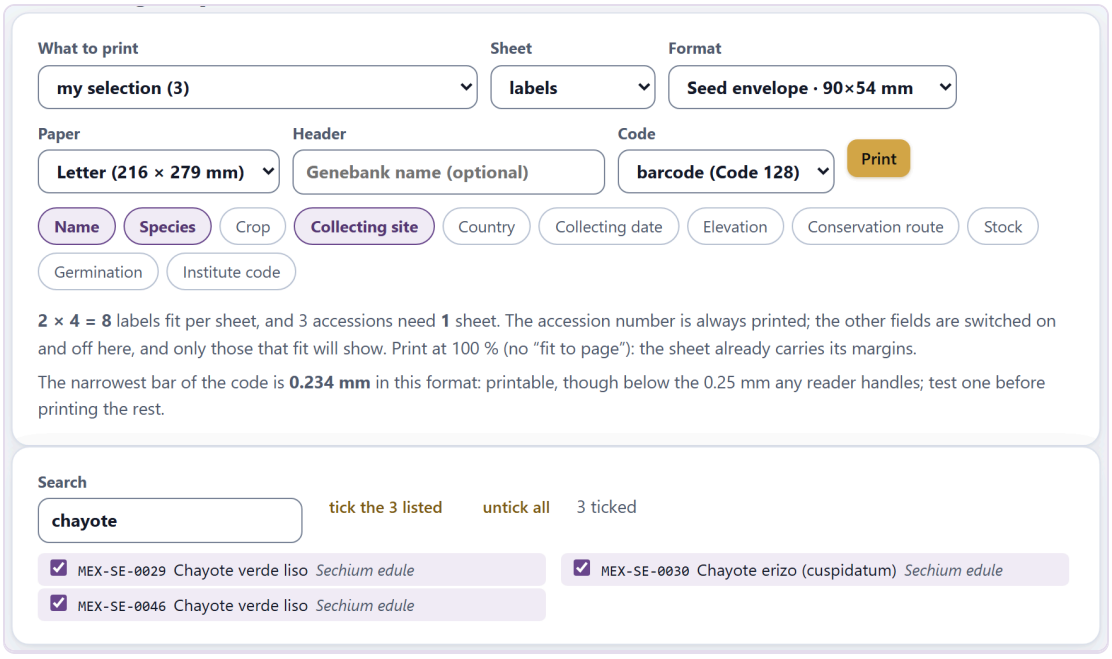


Figure 9.2. The three chayote accessions, picked with the search box for a shipment.

The selection stays ticked even if you change the search, and the selector option tells you how many you have: my selection (3). It lives only while the app is open: reloading the page loses it.

9.2 The five formats and how many fit on a sheet

The Format selector offers five label sizes, one per object labeled in a genebank; the name of each option carries its size. Paper chooses letter (216 × 279 mm, the default) or A4 (210 × 297 mm). The app draws each label with a gray border with rounded corners, which is the cutting line: it is designed to be printed on a full sheet of adhesive paper and cut out, not for pre-cut label sheets of a commercial size.

Table 9.2. The five formats, how many labels fit on each paper and how many sheets the example collection (46 accessions) needs.

Format	Size (mm)	Text (mm)	Letter	Sheets	A4	Sheets
Seed envelope	90 × 54	3.2	2 × 4 = 8	6	2 × 5 = 10	5
Cold-store jar	63 × 29	2.5	3 × 8 = 24	2	3 × 9 = 27	2
Field stake	95 × 35	4.2	2 × 7 = 14	4	2 × 7 = 14	4
Cryovial	33 × 13	1.9	5 × 17 = 85	1	5 × 18 = 90	1
Shipping label	100 × 62	3.4	1 × 4 = 4	12	1 × 4 = 4	12

The *Text* column is the height of the field text; the accession number is 1.25 times larger. On the envelope, 3.2 mm is about 9 typographic points; on the cryovial, 1.9 mm is about 5.4 points, which can be read with a magnifier or very good eyesight: that is what the code is for.

WORKED EXAMPLE · HOW MANY ENVELOPES FIT ON A LETTER SHEET

The app leaves an 8 mm margin at each edge of the paper and 2 mm between labels. In each direction, the labels that fit are those that go into the paper minus the two margins, counting that the last one has no gap after it:

```
columns = ⌊(216 − 2 × 8 + 2) / (90 + 2)⌋ = ⌊202 / 92⌋ = ⌊2.20⌋ = 2
rows = ⌊(279 − 2 × 8 + 2) / (54 + 2)⌋ = ⌊265 / 56⌋ = ⌊4.73⌋ = 4
per sheet = 2 × 4 = 8                    sheets = ⌈46 / 8⌉ = ⌈5.75⌉ = 6 (the last one with 6 envelopes)
on A4: ⌊(210 − 14) / 92⌋ = 2 columns; ⌊(297 − 14) / 56⌋ = 5 rows → 10 per sheet and 5 sheets
```

This is the computation summed up in the line below the buttons: 2 × 4 = 8 labels fit per sheet, and 46 accessions need 6 sheets. The shipping label, 100 mm wide, does not fit twice across (2 × 100 + 2 = 202 > 200 mm) and goes in a single column.

The preview shows all the sheets, one below the other, each with its footer sheet 1 of 6, which is not printed. With the whole collection and a QR on every label, building the sheets takes a moment: meanwhile, the app shows its waiting window.

9.3 What each label says

Each label carries, from top to bottom: the **header**, if you wrote one, in small gray type; the **accession number** (`ACCENUMB`), always, in bold monospaced type so that zero and O are not confused; the **fields** you switch on, one per line; and the **code**, at the bottom right if it is a barcode, and on the right, from top to bottom, if it is a QR.

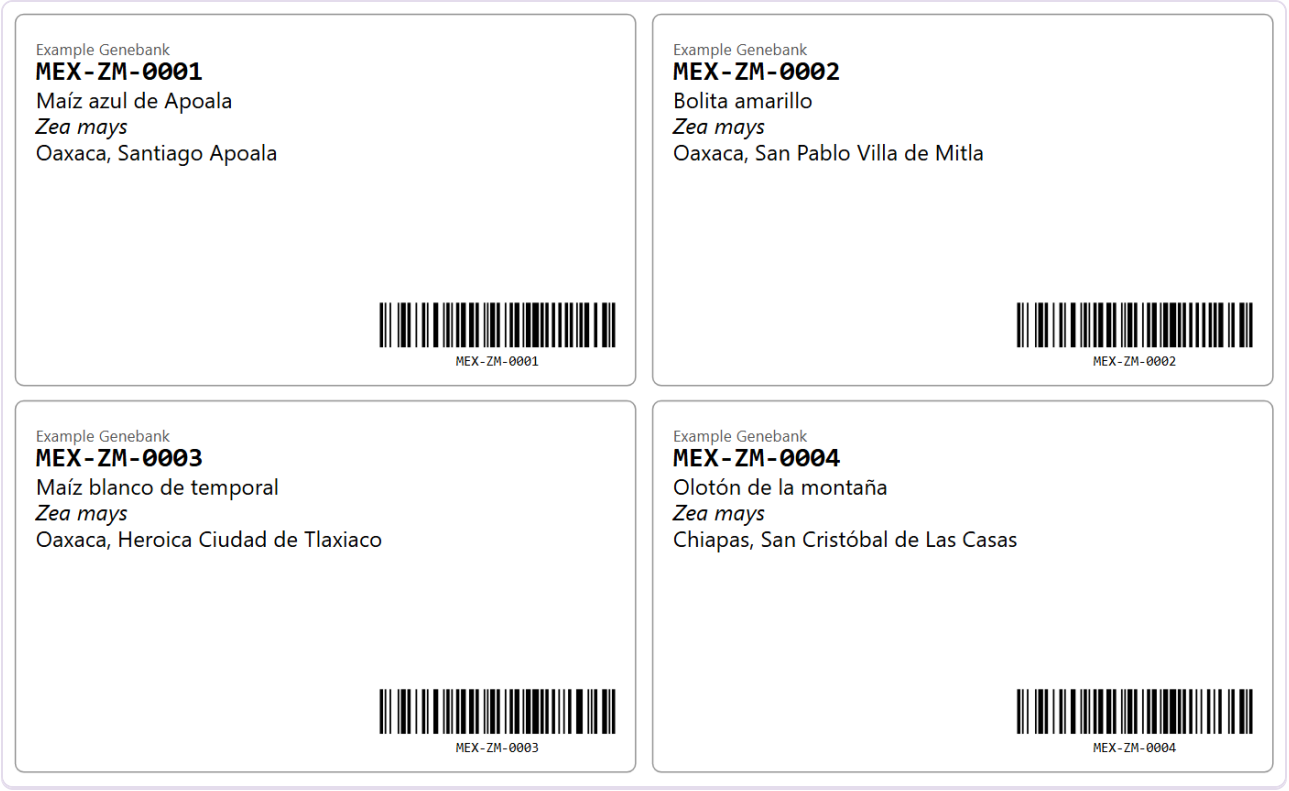


Figure 9.3. The first four envelope labels, with the three default fields, the header *Example Genebank* and the barcode.

The eleven field buttons are switched on and off with a click; the default ones are Name, Species and Collecting site. The app remembers from one session to the next the fields switched on, the header, the code and the chayote of the QR.

Table 9.3. The fields that can be printed and how they are written.

Button	Field	How it comes out (MEX-ZM-0001)
Name	`ACCENAME`	Maíz azul de Apoala
Species	`GENUS` `SPECIES` `SUBTAXA`	*Zea mays*, in italics; with the subspecies if there is one: *Zea mays subsp. parviglumis*
Crop	`CROPNAME`	Maíz
Collecting site	`COLLSITE`	Oaxaca, Santiago Apoala
Country	`ORIGCTY`	Mexico: the name of the country, in the language of the app, not the ISO code

Table 9.3. The fields that can be printed and how they are written. *(continued)*

Button	Field	How it comes out (MEX-ZM-0001)
Collecting date	`COLLDATE`	2019-08-12 (year-month-day, ISO 8601; with the interface in Spanish, day/month/year); an incomplete date such as `2021---` comes out as "2021"
Elevation	`ELEVATION`	followed by "m"
Conservation route	`GP_CONS`	Seed bank (or the one deduced from `STORAGE`)
Stock	`GP_STOCK`	1800, with no unit
Germination	`GP_GERMPCT`	88 %
Institute code	`INSTCODE`	the FAO code of the institute

An empty field leaves no blank line: it is skipped. A text that reaches the edge or the QR is cut and ends in "…" —the header, the accession number and each field—, with an approximate estimate of the width of the letters. And **only those that fit are printed**: if a line no longer fits before the barcode or the bottom edge, it is left out, and the app says so in an ochre warning below the controls, naming the fields that do not fit (Figure 9.4). Table 9.4 gives how many lines fit on each format; the ones that are dropped are the last ones you switched on.

Table 9.4. How many field lines fit on each label, besides the accession number.

Format	Code 128	QR or no code	Code 128 with header	QR with header
Seed envelope	9	11	8	10
Cold-store jar	4	7	3	6
Field stake	2	4	2	3
Cryovial	1	3	0	2
Shipping label	9	12	9	11

On the shipping label with QR or no code, 12 lines fit, more than the 11 fields there are. Look at the stake: with a barcode only two fit, so of the three default fields the collecting site is left out, and the app warns One field does not fit on "Field stake": 2 lines fit besides the number, so Collecting site is left out. On the cryovial with a barcode a single field comes out, and none if you also add a header. A long header is cut with "…" before the QR: keep it short, an acronym or the institute code.

9.4 The Code 128 barcode

The default option of the Code selector is barcode (Code 128). The code carries only the accession number, and below it the same number in readable type (except on the cryovial, where it does not fit). It is not a drawing: the app writes the code according to the standard, and the app's tests read it back to check it. With the 46 example accessions, the app's own reader recovers all 46 numbers without a single error.

THEORY IN BRIEF · HOW A CODE 128 IS WRITTEN

A Code 128 is a row of **symbols**. Each symbol has three bars and three spaces that measure from 1 to 4 **modules** and always add up to 11 modules; the module is the thinnest stripe of the code. There are 107 symbols, with values from 0 to 106. The code begins with a start symbol that sets the character set —**B** (value 104) for letters, digits and signs of printable ASCII, or **C** (105), which stores **two digits per symbol**—, continues with the data, a **check digit** and the stop symbol, which is 13 modules wide. Halfway through the code you can switch from one set to the other with symbols 99 (to C) and 100 (to B). In set B, the value of a character is its ASCII code minus 32.

$$\text{check} = (\text{start} + \Sigma_i\ i \times \text{value}_i) \bmod 103 \cdot \text{modules} = 11 \times (\text{symbols without the stop}) + 13$$

The app switches to set C when at least six digits come in a row, or four if they are the last ones and even in number: that way the code is shorter and, in the same space, has wider bars. The reader adds up the same products and discards the code if the check does not match: that is why a misread code almost never gives a wrong number, but no number at all.

WORKED EXAMPLE · THE CODE OF MEX-ZM-0001

The first seven characters go in set B; the four final digits are even in number and close the text, so the app switches to set C and stores them in two symbols.

start B = 104 · M = 77 − 32 = 45 · E = 37 · X = 56 · − = 13 · Z = 58 · M = 45 · − = 13 · to C = 99 · “00” = 0 · “01” = 1

104 + 1×45 + 2×37 + 3×56 + 4×13 + 5×58 + 6×45 + 7×13 + 8×99 + 9×0 + 10×1

= 104 + 45 + 74 + 168 + 52 + 290 + 270 + 91 + 792 + 0 + 10 = 1,896

1,896 mod 103 = 1,896 − 18 × 103 = **42** modules = 11 × 12 + 13 = **145**

widths: start 211214 · M 113123 · E 132113 · … · check 112133 · stop 2331112

Without the switch to set C, the four digits would take four symbols instead of three, and the code would be 156 modules wide. The 46 example accessions have numbers of the same form (three letters, hyphen, two letters, hyphen, four digits) and all of them are 145 modules wide.

The code takes a fixed width on each format, so the **module** comes from dividing that width by the modules of the code. Whether the code can be read depends on the module: each stripe must span several printer dots. A 300 dpi printer places a dot every 0.085 mm.

Table 9.5. The barcode on each format, for a 145-module number such as MEX-ZM-0001.

Format	Code width (mm)	Bar height (mm)	Module (mm)	Dots at 300 dpi
Seed envelope	34	6.4	0.234	2.8
Cold-store jar	26	5.0	0.179	2.1
Field stake	30	4.6	0.207	2.4

Table 9.5. The barcode on each format, for a 145-module number such as MEX-ZM-0001. *(continued)*

Format	Code width (mm)	Bar height (mm)	Module (mm)	Dots at 300 dpi
Cryovial	22	4.0	0.152	1.8
Shipping label	42	8.3	0.290	3.4

A longer accession number makes the module thinner, because the width does not change; that is why the app measures the module with the longest number of what you are about to print. To the right of the code lies the label margin, from 1.6 to 3.5 mm, which on every format is wider than the ten modules of quiet zone the standard requires; to the left there is nothing.

Below the controls, the app says how wide the narrowest bar is and whether it is enough. From 0.25 mm up anyone can read it: on the shipping label it says `The narrowest bar of the code is 0.290 mm in this format, plenty for any reader.` Between 0.19 and 0.25 mm —the envelope and the stake— the note adds that it is printable but that you should test one. Below 0.19 mm —the jar, with 0.179 mm, and the cryovial, with 0.152 mm— it shows an ochre warning: an office printer no longer draws the bars evenly, nor does a handheld reader tell them apart. On those labels, print with a thermal printer of 300 dpi or more and test the reader before going on, or use the QR.

> In this format the narrowest bar of the code is **0.152 mm** (with MEX-ZM-0001), below the 0.19 mm an office printer draws evenly and a handheld reader resolves; from 0.25 mm up any reader copes. Print one and test it, use a wider format or a shorter number.

> **2 fields do not fit** on "Cryovial": 1 line fits besides the number, so Species, Collecting site are left out (except on labels where an earlier field is empty). Remove fields, or use a larger format.

Figure 9.4. The two warnings for the cryovial with a barcode and the three default fields: the bar is too thin and two fields do not fit.

Code 128 only writes characters of printable ASCII. If an accession number carries ñ, accents or other characters outside it, that label comes out without a barcode: in the preview the app frames it with a dotted red line, which is not printed, and a warning says how many labels come out like that and suggests using the QR, which accepts any character, or fixing the number in Block 2.

9.5 The QR code and the chayote in the middle

With `Code` set to `QR`, each label carries on its right a square QR code with the accession number, and the `chayote in the middle` button appears, switched on by default, which draws a chayote on a white disc in the middle of the code. With `no code` the label carries text only.

THEORY IN BRIEF · HOW A QR IS WRITTEN

A QR is a square matrix of light and dark modules. The **version** sets its size: version v measures $17 + 4v$ modules per side. Three **finder patterns** in the corners, the timing lines and, from version 2 on, the alignment patterns orient the reader; the rest carries the message in **bytes** (codewords) plus **Reed–Solomon error-correction** bytes. There are four correction levels, which recover roughly 7% (L), 15% (M), 25% (Q) and 30% (H) of the bytes. Finally one of eight **masks** is applied, which inverts modules to avoid blotches and stripes that would confuse the reader; the app tries all eight, scores them with the four penalty rules of the standard and keeps the one with the lowest score. The level and the mask travel in 15 format bits, protected by their own code.

The app writes the text in byte mode (UTF-8), always at level H, in versions 1 to 10. At level H, 7 bytes fit in version 1, 14 in version 2, 24 in version 3 and 119 in version 10; an accession number of 8 to 14 characters goes in version 2.

WORKED EXAMPLE · THE QR OF MEX-ZM-0001

```
message = 4 mode bits (0100) + 8 count bits (11) + 11 × 8 bits = 100 bits
version 1, level H: 9 data bytes = 72 bits < 100 → does not fit · version 2, level H: 16 bytes = 128 bits → fits
version 2: 17 + 4 × 2 = 25 × 25 modules      44 bytes = 16 data + 28 correction
28 correction bytes → recovers up to 14 damaged bytes out of 44 (32%)
mask chosen: 1 · format bits (H, mask 1) = 001001110111110
```

The 46 example accessions all end up in version 2. The chayote takes a 7 × 7 module gap in the middle (26% of the side, rounded: 25 × 0.26 = 6.5 → 7), 49 of the 625 modules, 7.8% of the area, and the white disc stays inside the gap. For all 46, the app's own reader, which does not correct errors, still recovers the number with the gap left blank: the gap falls on padding and correction bytes, not on those of the number. Even if it fell on them, level H has correction to spare; that is why the drawing always goes with level H.

Figure 9.5. Jar labels with QR and the chayote in the middle. The collecting site is cut with "…" before the code.

The gap never covers the modules that orient the reader. In version 1 (numbers of up to 7 characters) it shrinks until it no longer touches the format information and is 3 × 3 modules; from version 7 on (texts longer than 58 characters) the alignment pattern falls right in the middle and no gap avoids it, so the QR comes out without the chayote even if the button is on.

The side of the QR is the height of the label minus its margins, never more than a third of the width; the text stops 1 mm before the code's square. Inside that square the app leaves, around the modules, the **4 blank modules** the standard requires.

Table 9.6. The QR on each format (version 2: 25 modules plus 4 blank on each side, 33 in total).

Format	QR side (mm)	Module (mm)	What the app says
Seed envelope	30.6	0.93	Each QR module is N mm in this format, plenty for any reader. Version 2, level H, with the 4-module quiet zone on each side the standard asks for.
Cold-store jar	21.4	0.65	
Field stake	28.4	0.86	
Shipping label	34.0	1.03	
Cryovial	9.8	0.30	An ochre warning (below).

Below 0.4 mm per module the app shows an ochre warning. With the cryovial it says:

> On this label each QR module is 0.30 mm, below the 0.4 mm that a laser printer and a phone camera resolve comfortably (already with the 4-module quiet zone on each side). Print one and test it before doing the rest; if it does not read, use the barcode, which copes better in so little space.

Only the cryovial falls below with numbers like those of the example. The warning is computed with the longest number of what you are about to print, which is the one that gives the densest QR.

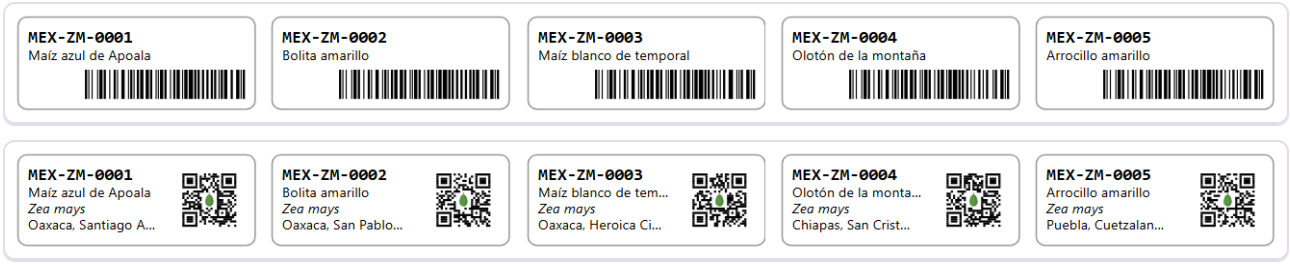


Figure 9.6. Cryovial labels (33 × 13 mm) with a barcode, above, and with a QR, below. With the barcode a single field fits; with the QR, three, but each module is 0.30 mm. Texts that do not fit end in "…".

BARCODE OR QR?

The barcode can be read by any handheld reader of the kind that connects to the computer as a keyboard: the number appears wherever the cursor is, in a spreadsheet or in the Block 2 search box. The QR can also be read by a phone camera, and in the same space it leaves more lines of text, but it needs larger modules. On envelopes, jars and shipping labels both work; on the cryovial, test both with your printer; on the field stake, the QR leaves twice as many fields.

9.6 Printing and checking

The Print button opens the browser's print dialog with the preview; the button is disabled if there are no accessions to print. With labels, the app prepares the printout at exact scale: each sheet measures the same as the chosen paper, carries its own 8 mm margins, and the page is declared at the size of the paper and with no browser margin. That way nothing shrinks and no blank sheet is left over at the end. The same happens if you print with Ctrl + P while you are in Block 9.

89.6

WHAT WAS MEASURED · MILLIMETERS

When the 46 envelope labels were printed to PDF, they came out as 6 sheets on letter and 5 on A4, with no blank sheet. The gray border of each envelope measured **89.6 mm** wide on both papers —the border runs 0.2 mm inside each edge of the 90 mm—, and the first column began 8 mm from the edge of the paper.

The worksheets are printed as a document: tables split across sheets and repeat their header row at the top of each sheet. With the whole example collection on letter paper and the app in English, the sowing list takes 2 sheets, the register book 2 and the delivery note 2. At the foot of the block, a note sums up what to take care of: Check the first sheet before printing a hundred.

1. In the dialog set the scale to **100%**, not "fit to page", as the line in section 1 also asks, and turn off the browser's headers and footers.
2. Print **one** sheet and measure a label with a ruler: the envelope must measure 90 × 54 mm. Some printer drivers shrink the sheet on their own; if the label comes out smaller, look in their options for "actual size" or "100%".
3. Scan the code with the reader. If the number it reads is the one on the label, print the rest.

For the field and the cold room, the paper matters as much as the ink: use polyester label stock or water-resistant paper, and laser toner or thermal transfer; inkjet ink runs with the condensation when a jar is taken out of the cold.

9.7 The sowing list

With Sheet set to sowing list, the preview turns into the sheet you take to the greenhouse or the field for a regeneration: one row per accession with how many seeds to sow, and three blank columns to write the plot, the date and the notes by hand. At the top go the title, a line with the number of accessions and of plants, and, on the right, the header and today's date. The natural thing is to combine it with the lots in trouble from Block 8.

Sowing list for regeneration

2026-10-31

10 accessions · 100 plants each · seeds to sow follow from each lot's germination

ACCESSION	NAME	SPECIES	GERM.	SEEDS	PLOT	DATE	NOTES
MEX-ZP-0011	Teocintle de Balsas	*Zea mays*	81 %	138			
MEX-SE-0013	Frijol vaquita	*Phaseolus vulgaris*	84 %	133			
PER-ST-0021	Papa nativa Huayro	*Solanum tuberosum*	85 %*	131			
MEX-VA-0025	Vainilla de Papantla	*Vanilla planifolia*	85 %*	131			
MEX-PA-0026	Aguacate criollo de Uruap…	*Persea americana*	85 %*	131			
MEX-TC-0027	Cacao criollo de Ocosingo	*Theobroma cacao*	85 %*	131			
COL-MA-0035	Yuca amarga regional	*Manihot esculenta*	85 %*	131			
MEX-PH-0041	Frijol tepari	*Phaseolus acutifolius*	79 %	141			
MEX-ZM-0043	Maíz ancho de riego	*Zea mays*	72 %	155			
MEX-SE-0046	Chayote verde liso	*Sechium edule*	85 %*	131			

* Assumed: these 6 accessions have no germination test recorded; the 85 % from the Block 8 calculator is used. Test germination before sowing if you can.

Plants per accession

100 those of one regeneration in the Block 8 assumptions Download as CSV

Figure 9.7. The sowing list of the 10 lots in trouble from Block 8, with 100 plants per accession. The six rows with an asterisk are not seed lots and have no germination test.

The seeds come from the same computation as in Block 8 (Section 8.7): the plants you want, divided by the lot's germination and by a fixed field establishment of 90%. The plants per accession are, to begin with, those of one regeneration in the Block 8 assumptions (the Plants per regeneration slider, 100 by default), and the remark next to the field says so. If you change them in the Plants per accession field, from 10 to 500 in steps of 10, they apply to every row and the remark disappears.

```
seeds = ⌈ plants / (germination/100 × 0.9) ⌉
```

WORKED EXAMPLE · THREE ROWS OF FIGURE 9.7

```
MEX-ZM-0043, 72%: 100 / (0.72 × 0.9) = 100 / 0.648 = 154.3 → 155 seeds
MEX-ZP-0011, 81%: 100 / 0.729 = 137.2 → 138 · MEX-PH-0041, 79%: 100 / 0.711 = 140.6 → 141
with 150 plants, MEX-ZM-0001 (88%): 150 / 0.792 = 189.4 → 190
```

THE ASSUMED GERMINATION AND THE MEASURED 0%

If an accession has no `GP_GERMPCT`, the list uses the assumption of the Block 8 calculator (Lot germination (%), 85% by default, Section 8.7), writes it with an asterisk — 85 %* — and explains it in a note below the table. With the whole example collection these are 17 accessions: all those that are not seed lots, because the field, in vitro, cryopreservation, in situ and on-farm routes have no germination test. Use the list with seed lots, and cross out the rows with an asterisk that are regenerated by replanting or subculture.

A measured 0%, on the other hand, is a datum: that lot has no viable seed. The row carries a dash under Seeds and an ochre note names it and suggests turning to the safety duplicate or collecting again.

The Download as CSV button saves sowing-list.csv, in UTF-8 with comma separator, with eleven columns: `ACCENUMB`, `name`, `taxon`, `plants`, `germination`, `assumed_germination` ("yes" when it is the assumption, empty when it was measured), `seeds` (empty with a measured 0%), `stock` (`GP_STOCK`), and the three empty ones `plot`, `sowing_date` and `notes`. With the interface in Spanish, the file is called lista-de-siembra.csv and its columns and the "yes" are in Spanish. Unlike on the sheet, the name is given in full.

Table 9.7. Three rows of sowing-list.csv with the whole collection (without the three empty columns).

ACCENUMB	name	taxon	plants	germination	assumed_germination	seeds	stock
MEX-ZM-0001	Maíz azul de Apoala	Zea mays	100	88		127	1800
MEX-ZM-0002	Bolita amarillo	Zea mays	100	89		125	1930
MEX-ZP-0012	Población silvestre de teocintle, reserva	Zea mays	100	85	yes	131	1200

Compare the `seeds` column with `stock` before sowing: if the lot does not have the seeds the list asks for, sow the ones there are and write it down in the notes.

9.8 The delivery note

The delivery note is the sheet that travels with a shipment. At the top it carries the title, a note about the International Treaty, the header and the date; then four lines to fill in by hand — Requester, Institution, Intended use and Shipping date —, the table of accessions and two signature lines: Released by (name and signature) and Received by (name and signature).

Germplasm delivery note

Example Genebank
2026-10-31

This material is supplied under the International Treaty terms when the accession is in the Multilateral System (MLSSTAT = 1).

Requester: _____ **Institution:** _____

Intended use: _____ **Shipping date:** _____

ACCESSION	NAME	SPECIES	ORIGIN	QUANTITY	SMTA
MEX-SE-0029	Chayote verde liso	*Sechium edule*	Mexico		—
MEX-SE-0030	Chayote erizo (cuspidat…	*Sechium edule*	Mexico		—
MEX-SE-0046	Chayote verde liso	*Sechium edule*	Mexico		—

Released by (name and signature) _____

Received by (name and signature) _____

Figure 9.8. The delivery note for the three chayote accessions of Figure 9.2.

The table has one row per accession: the number, the name (cut with "…" if it is longer than 24 characters), the species, the country of origin by its name, the Quantity left blank to write what is sent (seeds, grams, jars or cuttings) and the SMTA column, which comes from `MLSSTAT`: yes if the accession is in the Multilateral System (1), no if it is not (0) and a dash if the field is empty. With the whole example collection there are 21 "yes", 4 "no" and 21 dashes.

> **THE NOTE DOES NOT REPLACE THE AGREEMENT**
>
> The Standard Material Transfer Agreement (SMTA) of the International Treaty on Plant Genetic Resources for Food and Agriculture is a contract signed or accepted separately, for each shipment of material from the Multilateral System. The app's delivery note is the internal record of what left, how much and to whom; the SMTA column only reminds you which accessions need the agreement. A dash means you do not know: fill in `MLSSTAT` in Block 2 before shipping.

9.9 The register book

The register book is the printed inventory of the collection: the title, the line 46 accessions as of 2026-10-31 and one table per conservation route, each with its subtitle and its number of accessions. The routes come out in the order in which their first accession appears in the passport; with the example collection they are Seed bank · 28, Genetic reserve (in situ) · 2, In vitro culture · 3, Cryopreservation · 1, Field genebank or living collection · 8, On-farm conservation · 3 and Not declared · 1, which is the chile costeño (Section 8.9).

Table 9.8. The columns of the register book.

Column	What it shows
Accession	`ACCENUMB`.
Name	`ACCENAME`, cut with "…" if it is longer than 22 characters.
Species	Genus and species, in italics.
Site	`COLLSITE`, cut with "…" if it is longer than 26 characters.
Acquired	`ACQDATE`, the date the accession entered the genebank, as year-month-day (ISO 8601), like the date at the top of the sheet; with the interface in Spanish, both go as day/month/year.
Stock	`GP_STOCK`, in the units of the route.
Last germ.	`GP_GERMPCT` with "%", or empty.
Backup	`DUPLSITE`, the institute holding the safety duplicate, or a dash.

Collection register book

2026-10-31

46 accessions as of 2026-10-31

Seed bank · 28

ACCESSION	NAME	SPECIES	SITE	ACQUIRED	STOCK	LAST GERM.	BACKUP
MEX-ZM-0001	Maíz azul de Apoala	*Zea mays*	Oaxaca, Santiago Apoala	2019-11-04	1,800	88 %	—
MEX-ZM-0002	Bolita amarillo	*Zea mays*	Oaxaca, San Pablo Villa d…	2019-11-04	1,930	89 %	NOR051
MEX-ZM-0003	Maíz blanco de tempor…	*Zea mays*	Oaxaca, Heroica Ciudad de…	2019-11-04	2,060	90 %	NOR051
MEX-ZM-0004	Olotón de la montaña	*Zea mays*	Chiapas, San Cristóbal de…	2019-11-04	2,190	91 %	NOR051
MEX-ZM-0005	Arrocillo amarillo	*Zea mays*	Puebla, Cuetzalan del Pro…	2019-11-04	2,320	92 %	NOR051
MEX-ZM-0006	Cacahuacintle de Chol…	*Zea mays*	Puebla, San Andrés Cholula	2019-11-04	2,450	93 %	—

Figure 9.9. The beginning of the register book of the example collection.

Names and sites that are too long are cut and end in "…" ("Maíz blanco de tempor…"), and the stock carries a thousands separator, as in the rest of the app. The register keeps no history: it reflects the passport on the day you print it, with the tests and regenerations you have recorded in Block 8. Print it at the close of each season, sign it and file it with the passport export from Block 2: that way the collection has a paper record of every year.

When something goes wrong

What you see	Why it happens and what to do
Load your accessions in Block 2 first.	There is no passport. Load your collection or the example one (Chapter 2).
No accessions in this selection. and the Print button disabled	You chose my selection without ticking any, or the Block 7 core no longer exists. Tick them in the list or build the core again (Section 9.1).
The core option says (not built yet) and cannot be chosen	You have not built the core. Go to Block 7, load the characterization and come back.
Ochre warning that one or more fields do not fit	The label does not have lines for all of them (Table 9.4). Switch fields off, remove the header, use QR or a larger format.
A label with a dotted red border in the preview	Its number has characters that Code 128 does not accept (ñ, accents) and it comes out without a code. Switch to QR or fix the number in Block 2.
Ochre warning about the narrowest bar	The Code 128 module is below 0.19 mm (Section 9.4). Print a test one, use a wider format or the QR.
Ochre warning about the QR module	The format is too small for the QR (Section 9.5). Print a test one, or use the barcode.
The printed label is smaller than its format	The scale in the dialog is not 100%, or the printer driver shrinks the sheet. Set it to 100% or "actual size" (Section 9.6).
A row of the sowing list says 85 %*	The accession has no germination test and the list uses the Block 8 assumption (Section 9.7). If it is a seed lot, run the test; if it is not, cross out the row.
A row of the sowing list has a dash under Seeds	Its last test gave 0%: there is no viable seed. Turn to the safety duplicate or collect again.

DECISION RULE · WHAT TO PRINT IN BLOCK 9

If you are labeling envelopes, jars or shipments	**then** use Code 128: any handheld reader reads it and types the number wherever the cursor is.
If you are labeling field stakes or want to read with a phone	**then** use the QR: it leaves more lines of text and the camera reads it.
If you are labeling cryovials	**then** test both codes with your printer before printing the rest; the module is at the limit.
If you are about to print a batch	**then** print one sheet, measure a label with a ruler and scan it; only then print the rest.
If you are preparing a regeneration	**then** use the lots in trouble from Block 8 with the sowing list, and cross out the accessions that are not regenerated from seed.
If material is leaving the genebank	**then** print the delivery note, check the SMTA column and fill in `MLSSTAT` wherever there is a dash.
If you are closing a season	**then** print the register book and file it with the passport export.

Block 9 writes nothing into the passport: everything it prints comes from Block 2 and from the results of Blocks 7 and 8. If a label says something wrong, fix it in the passport, not on the paper.

10

CHAPTER TEN · BLOCK 10 · REPORT

Report

The last block computes nothing of its own: it gathers what Blocks 2 to 8 compute into a document that stands on its own, with its numbers, its figures, the methods written out in prose and only the references of the methods that were used, each one cited in the text. The preview is the file itself: what you see is what is printed and what is downloaded. It also packs everything into a ZIP — the report, the working tables and each figure in SVG— and exports the passport in the two formats used to share germplasm: MCPD, the one genebanks use with one another, and Darwin Core, the one used by GBIF. This chapter explains each control, what each section of the report on the example collection says, which block each number comes from and what you have to check before signing.

10.1 What the report gathers

10.2 Title, signature and language

10.3 The sections and the figures

10.4 The summary and the collection

10.5 Data quality

10.6 Geographic distribution

10.7 Diversity and collecting gaps

10.8 Characterization, core and management

10.9 The written methods and the references

10.10 Printing and downloading the report

10.11 The ZIP package

10.12 The passport in Darwin Core and in MCPD

10.1 What the report gathers

The block has two sections: 1 · What goes into the report, with the header data, the section buttons and the five output buttons, and 2 · The report, as it comes out, with the preview. The preview is not an approximate drawing: it is the same HTML document you download, placed in a frame on the page. If there are no accessions, the block says Load your accessions in Block 2 first.

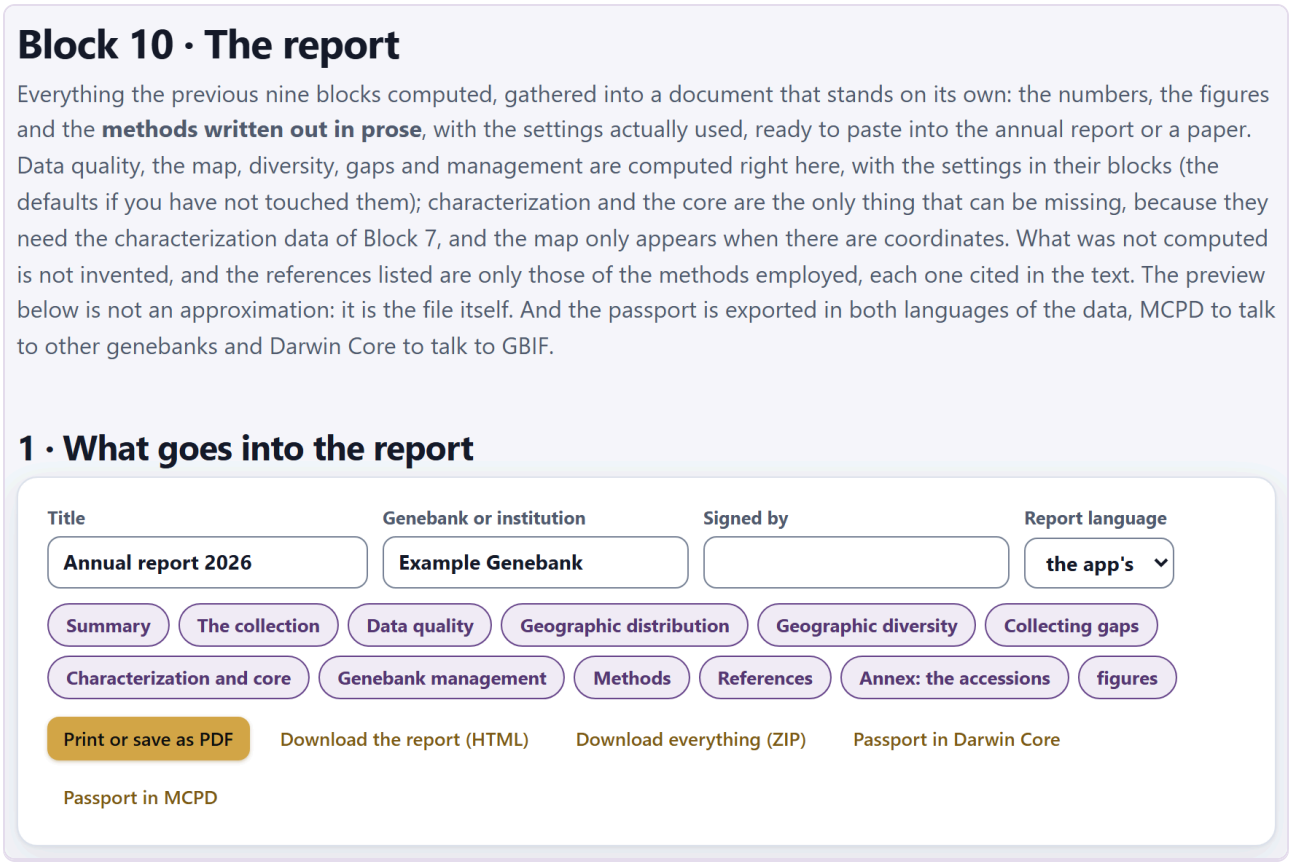


Figure 10.1. Section 1 with the example collection, the example characterization from Block 7 and the title and genebank of this chapter's example.

Every time you enter the block, the app **runs again** the analyses of Blocks 3 to 8 with whatever settings they have at that moment and builds the document; meanwhile, it shows its waiting window with the text Building the report. You do not need to have visited those blocks: if you never opened them, they come in with their default values. So, if you change a setting in another block —the grid size, the duplicate threshold, the core size— and come back here, the report already reflects it. The only thing that can be missing is the characterization and the core, which need the characterization data of Block 7; and the map, which only appears if there are coordinates.

Table 10.1. The eleven sections of the report, where each one comes from and when it appears.

Section	Comes from	When it appears
Summary	Blocks 2, 3, 6 and 8	Always; its button cannot be switched off.
The collection	Block 2	Always; its button cannot be switched off.
Data quality	Blocks 2 and 3	Always, unless you switch it off. The review is computed right here, with the duplicate threshold and the dismissed pairs you left in Block 3.
Geographic distribution	Block 4	When there are accessions with coordinates.
Geographic diversity	Block 5	With the units and the kind of diversity set in Block 5.
Collecting gaps	Block 6	With the territory, the grid and the weights of Block 6.
Characterization and core	Block 7	Only if you loaded a characterization in Block 7. Otherwise, the button is gray.
Genebank management	Block 8	With the genebank assumptions and the constants of Block 8.
Methods	all	Always; one paragraph per analysis present. Its button cannot be switched off.
References	all	Always; only those cited in the methods. Its button cannot be switched off.
Annex: the accessions	Block 2	Always, unless you switch it off.

Block 9 contributes nothing to the report: its labels and worksheets are printed there.

All the numbers in this chapter come from the example collection of Block 2 (button Open the example collection), with the example characterization of Block 7 and its default 10% core, and with the other blocks at their default values. The date of the report is that of the day it is built; here, 31 October 2026.

10.2 Title, signature and language

The three fields at the top fill in the header of the document. The app remembers them from one session to the next.

Table 10.2. The header data.

Control	What it does
Title	The large title of the report and the name of the browser tab when the file is opened. If empty, the report is called Collection report (in Spanish, *Informe de la colección*).
Genebank or institution	Appears in the gray line below the title.
Signed by	Also in that line, after the genebank. Write the name of the person responsible for the collection.
Report language	the app's (default), español or English. It changes the whole document —titles, tables, texts, methods, references and the labels inside the figures— without changing the language of the app.

The header line joins what you wrote with the date and the number of accessions, separated by middle dots. With the example it says Example Genebank · 31 October 2026 · 46 accessions; in Spanish the date comes out as *31 de octubre de 2026*. While you type in the three fields, the preview is rebuilt without running the analyses again and the cursor stays in place.

With English, the figures are drawn in English —the accumulation curve says "accessions examined" and "expected types", the principal coordinates plot "axis 1" and "axis 2"— and so are the separate figures of the ZIP package. The references change their "y" into "and" and FAO's city into "Rome". Texts that come from your data (names, sites, remarks) stay as you wrote them.

10.3 The sections and the figures

Below the fields there is one button per section, in the order they appear in the document, and at the end the figures button. When it is switched on (with a colored background), the section goes in; a click switches it off and the preview is rebuilt. By default all of them go in. Four buttons cannot be switched off: Summary, The collection, Methods and References; when you hover over them they say this section is always in the report. A gray button is a section without data: with the example collection this only happens to Characterization and core, until you load the characterization in Block 7. For screen-reader users, each button announces whether it is on or off.

What is useful is to remove the sections that are not relevant: for the annual report to management, perhaps the summary, the collection, management and the annex are enough; for a paper on diversity, switch off management and the annex. When you switch off an analysis section, its methods paragraph and its references do **not** disappear: the methods are written with everything that was computed, whether its section is included or not (Section 10.9).

The figures button removes or restores the report's five figures: the collecting map, the accumulation curve, the gap map, the principal coordinates and the workload. Switching it off makes the file lighter; the sections keep their boxes, their tables and their notes, and the ZIP package also comes out without the figures folder (Section 10.11). Switching a section or the figures on or off, or changing the language, runs the analyses again and takes a moment.

10.4 The summary and the collection

The document begins with the title, the header line and the **summary**: six boxes with the basic passport numbers and, below, a list of findings, each one counted with its number and in singular or plural as needed.

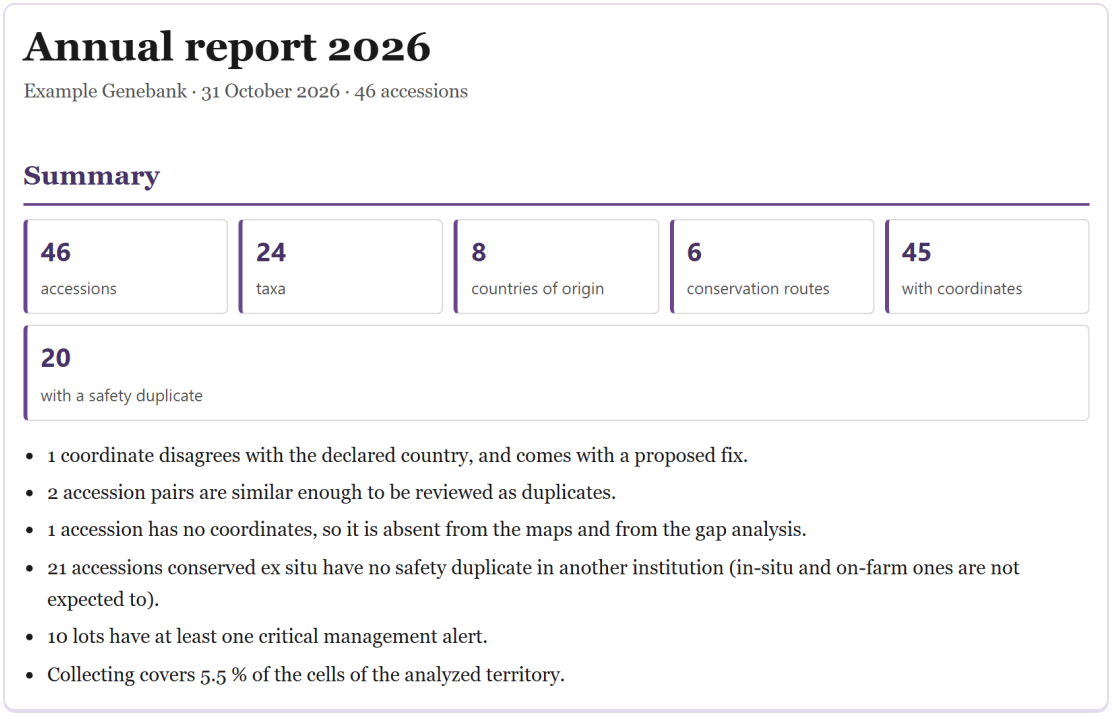


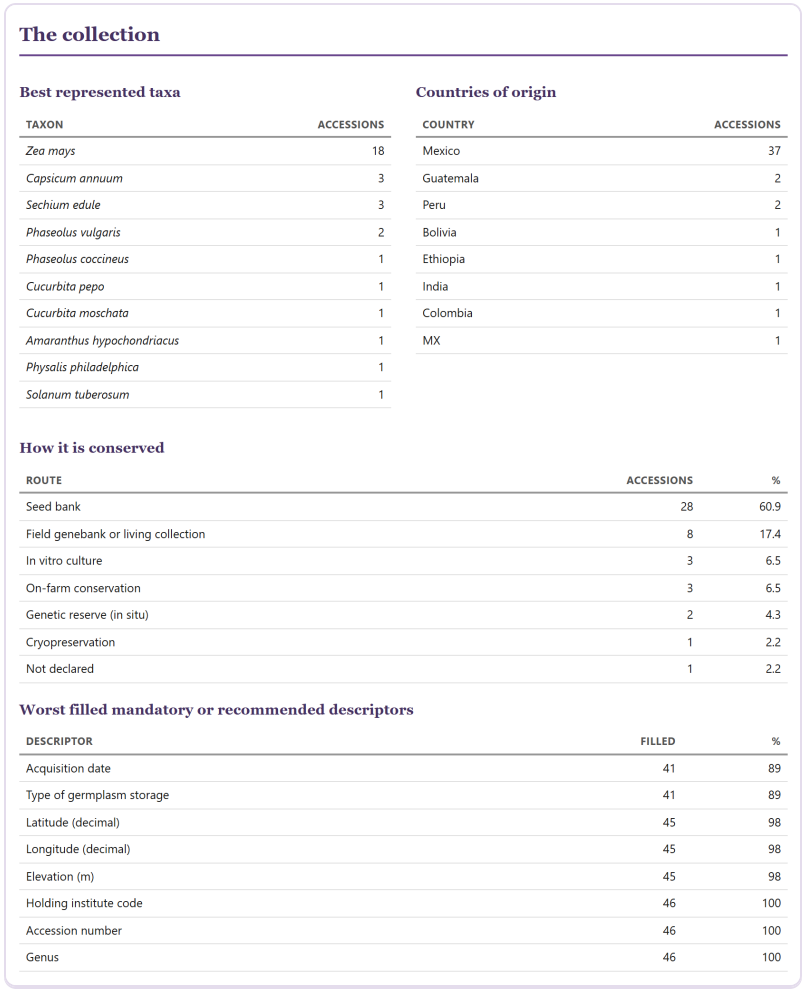
Figure 10.2. The beginning of the report on the example collection.

Table 10.3. The summary boxes and how they are counted.

Box	Ex.	What it counts
accessions	46	The records of the passport.
taxa	24	The distinct combinations of genus and species (`GENUS` + `SPECIES`), without the subspecies.
countries of origin	8	The distinct values of `ORIGCTY`, as written: the "MX" of the tepary bean, one of the planted errors (Section 3.4), counts as one more country.
conservation routes	6	The routes with at least one accession, declared in `GP_CONS` or deduced from `STORAGE`; "Not declared" is not counted.
with coordinates	45	Those with numeric latitude and longitude.
with a safety duplicate	20	Those with a safety duplicate in another institution (`DUPLSITE` different from `INSTCODE`).

The findings only appear when there is something to count: coordinates that disagree with the country and come with a proposed fix, coordinates with a clear error that the app cannot fix (on a separate line), candidate duplicate pairs, accessions without coordinates, ex-situ accessions without a safety duplicate, lots with a critical alert (from Block 8) and the grid coverage (from Block 6). With the example there are six. The safety-duplicate finding counts 21 and not 26 (the 46 minus the 20 that do have one) because the 2 in-situ populations and the 3 on-farm plots are not expected to have a duplicate in a cold room; it is the same number as in Block 8.

The section **The collection** describes the passport in four tables: the ten taxa with the most accessions, the ten countries of origin with the most accessions, how many accessions there are per conservation route and with what percentage, and the eight worst-filled mandatory or recommended descriptors.



The collection

Best represented taxa

TAXON	ACCESSIONS
Zea mays	18
Capsicum annuum	3
Sechium edule	3
Phaseolus vulgaris	2
Phaseolus coccineus	1
Cucurbita pepo	1
Cucurbita moschata	1
Amaranthus hypochondriacus	1
Physalis philadelphica	1
Solanum tuberosum	1

Countries of origin

COUNTRY	ACCESSIONS
Mexico	37
Guatemala	2
Peru	2
Bolivia	1
Ethiopia	1
India	1
Colombia	1
MX	1

How it is conserved

ROUTE	ACCESSIONS	%
Seed bank	28	60.9
Field genebank or living collection	8	17.4
In vitro culture	3	6.5
On-farm conservation	3	6.5
Genetic reserve (in situ)	2	4.3
Cryopreservation	1	2.2
Not declared	1	2.2

Worst filled mandatory or recommended descriptors

DESCRIPTOR	FILLED	%
Acquisition date	41	89
Type of germplasm storage	41	89
Latitude (decimal)	45	98
Longitude (decimal)	45	98
Elevation (m)	45	98
Holding institute code	46	100
Accession number	46	100
Genus	46	100

Figure 10.3. The section *The collection*.

The routes table adds up to the 46 accessions and 100%: the chile costeño (MEX-CH-0042), which declares neither route nor storage, goes in the `Not declared` row; in the annex it appears with a dash in the route column. The descriptors table serves as a to-do list for Block 2: here, five accessions without an acquisition date (`ACQDATE`) and five without a storage type (`STORAGE`). If all descriptors are complete, the table shows the first eight with 100%.

TAXA, SPECIES AND TYPES

The report gives three different counts of species diversity, and all three are correct: 24 **taxa** in the summary, which are all the species of the passport; 18 **types in total** in geographic diversity, which are the species of the accessions that fall inside a Mexican state, because the default units of Block 5 are the states; and 20 types with **a single accession** among the risks of the gap analysis, which Block 6 counts over the whole collection. When you quote a number, say which set it comes from.

10.5 Data quality

The section sums up the Block 3 review in six boxes, shows the ten candidate duplicate pairs with the highest scores and closes with a note: a high-scoring pair is not a proven duplicate, and merging is done by hand in Block 3.

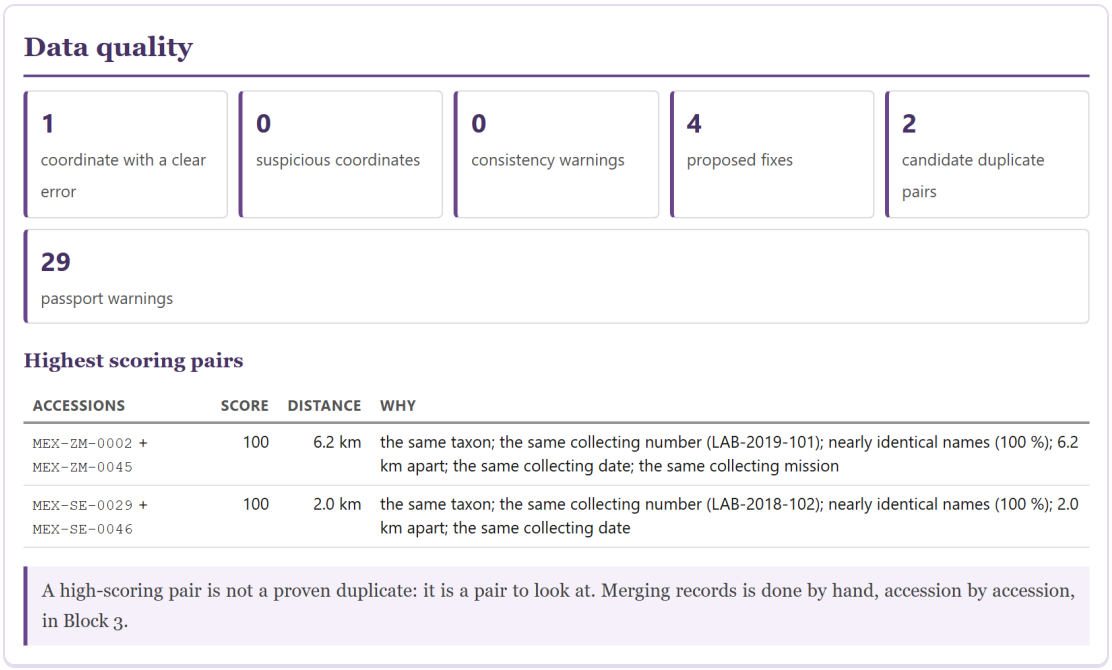


Figure 10.4. The section *Data quality*.

Table 10.4. The quality boxes, with the example collection; with a single item, the label is in the singular ("1 coordinate with a clear error").

Box	Ex.	Where it comes from
coordinates with a clear error	1	Those that do not fall in the declared country and have a diagnosis: here, MEX-ZM-0039, with the unsigned longitude (Section 3.2). It is the Location problems of Block 3.
suspicious coordinates	0	The doubtful ones: latitude equal to longitude, round values, outside the country without a diagnosis.
consistency warnings	0	Names written in several ways and inconsistent catalogs.

Table 10.4. The quality boxes, with the example collection; with a single item, the label is in the singular ("1 coordinate with a clear error"). *(continued)*

Box	Ex.	Where it comes from
proposed fixes	4	Those of Block 3: the sign of the longitude, "MX" for "MEX", "BOLITA AMARILLO" in capitals and the extra spaces of the chayote.
candidate duplicate pairs	2	The pairs above the Block 3 threshold, minus those you dismissed.
passport warnings	29	The problems of the Block 2 validation (Section 2.6).

The coordinate with a clear error is the same as in the first finding of the summary. If you apply the fix in Block 3, the box drops to 0, the finding disappears and the point of MEX-ZM-0039 returns to Veracruz on the map.

The pairs table shows the numbers of the two accessions, the rounded score, the distance between the collecting points and the reasons for the similarity, in the same words as Block 3. With the two pairs planted in the example collection —the Bolita amarillo maize and the chayote verde liso, each entered twice— the table has two rows, both with a score of 100. If you merge the pairs in Block 3, they disappear from the report.

10.6 Geographic distribution

The section shows the Block 4 collecting map with the framing set in that block (with the example, the whole world, because there are accessions from other continents) and its legend, and below it two tables: the twelve Mexican states and the twelve countries with the most accessions, counted **by the coordinate**, not by the text of the passport.

The points are colored by conservation route, as in Block 4, and the legend says how many points there are of each: 27 from the seed bank (the seed accession without coordinates does not count), 8 from the field genebank, 3 in vitro, 3 on farm, 2 in situ, 1 cryopreserved and 1 not declared; in total, the 45 of the caption.

Here the difference between what is declared and what is measured shows: the section The collection counts 37 accessions from Mexico by `ORIGCTY`, and this one, 36 points inside Mexico, because the tuxpeño de la costa (MEX-ZM-0039), which claims to be from Mexico, has the unsigned longitude and falls in Myanmar. The tepary bean, declared as "MX", does fall in Sonora. The note below says how many points fall outside Mexico: 9 of the points fall outside Mexico, so they only appear in the country count. The states that add up to 36 are the ten in the table; with more than twelve, the rest are not shown.

Geographic distribution

Collecting sites of the 45 georeferenced accessions.

By Mexican state

STATE	ACCESSIONS
Oaxaca	7
Puebla	7
Veracruz	6
Chiapas	4
Michoacán	3
Chihuahua	2
Yucatán	2
México	2
Jalisco	2
Sonora	1

By country

COUNTRY	ACCESSIONS
Mexico	36
Guatemala	2
Peru	2
Bolivia	1
Ethiopia	1
India	1
Colombia	1
Myanmar	1

9 of the points fall outside Mexico, so they only appear in the country count. State and country are assigned from the coordinate, not from the text of the field: if a point does not fall where its passport says, it shows here.

Figure 10.5. The section *Geographic distribution*. The point in Myanmar is MEX-ZM-0039, with the unsigned longitude.

10.7 Diversity and collecting gaps

The section **Geographic diversity** takes the Block 5 analysis: five boxes with the whole collection, a table with the twelve richest units, the accumulation curve with its estimators, a note when Chao1 is not reliable and the Mantel test in one sentence.

The boxes are those of Block 5 (Chapter 5): 18 species in the 36 accessions that fall in a state, a Shannon of 2.34 for the whole, a Whittaker beta of 5.67 and a Good-Turing coverage of 61.1%. The Chao1 of 116 types, six times what was observed, comes with the note `The Chao1 estimator leans on the number of types seen twice, and here there are very few: treat it as a rough bound and give more weight to Chao2 and the jackknife.` With 36 accessions, no estimate of total richness is firm; the curve, which keeps rising, says the same without numbers.

The `Exclusive` column counts the species that are found only in that unit, what would be lost if the unit were lost: 4 in Puebla, 4 in Veracruz, 3 in Chiapas and 1 in Michoacán, the State of Mexico, Jalisco and Sonora. Oaxaca, with 7 accessions, has none: its three species are also found in other states.

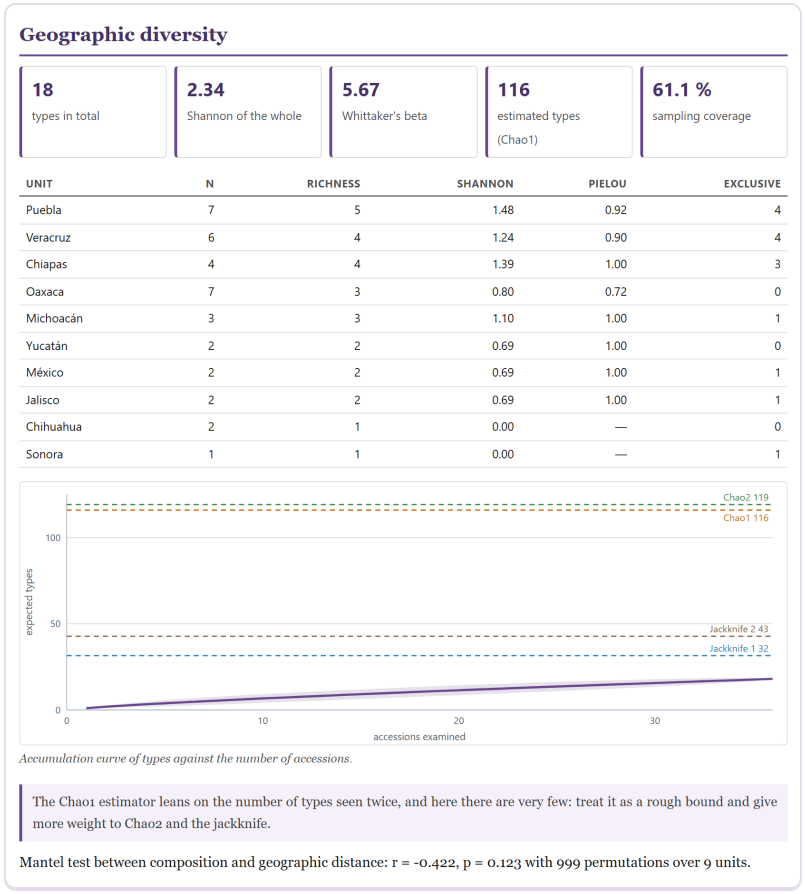


Figure 10.6. The section *Geographic diversity*, with the Mexican states as units and species as the type, the default values of Block 5.

The section **Collecting gaps** takes the Block 6 analysis (Chapter 6): four boxes —grid coverage, collected cells, empty cells and sites in the minimum set—, the map of cells with the ten priorities numbered, the table of those ten priorities, the table of types at risk and a note recalling that the analysis is geographic and compositional, not ecogeographic.

The priorities table gives, for each cell, its center, the state, the distance to the nearest collected point and the score: the first is the cell at 21.50, −99.50 in San Luis Potosí, 228 km from the nearest collection, with 86.9 points; the tenth, the one at 20.50, −99.50 in Hidalgo, with 70.0. The risks table counts how many types have each Block 6 signal: 20 with `a single accession`, 14 with `no safety duplicate`, 6 `only in a field collection`, 1 `only in situ, no ex-situ backup` and 1 `all from one state or country`. The same type can have several signals, so the column does not add up.

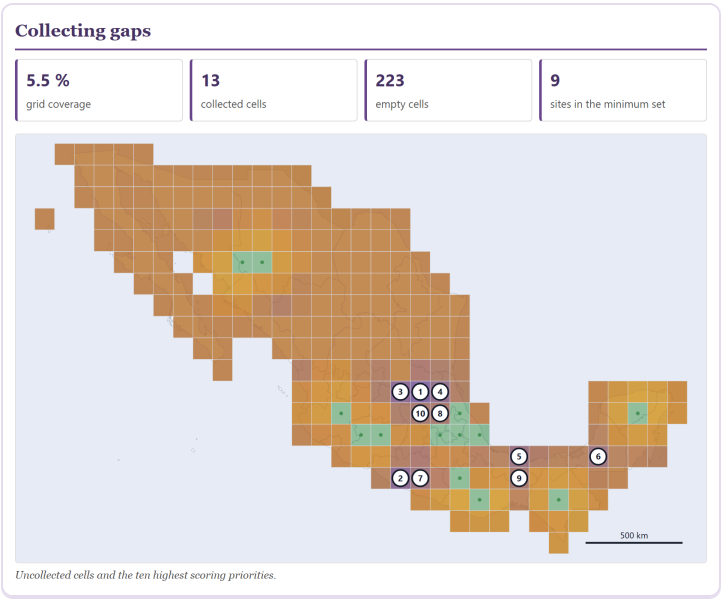


Figure 10.7. The beginning of the section *Collecting gaps*, with Mexico in 1° cells.

10.8 Characterization, core and management

The section **Characterization and core collection** only appears if you loaded a characterization in Block 7. It carries four boxes, the principal coordinates plot with its legend, the table of the four validation indicators of the core with what each one says, and a note when the core has fewer than ten accessions.

The numbers are those of Block 7 (Chapter 7): 46 accessions characterized with 12 descriptors, 4 groups with a mean silhouette of 0.14, and a core of 5 accessions —MEX-AG-0024, MEX-TC-0027, MEX-CU-0040, MEX-CH-0042 and MEX-ZM-0043—, 10.9% of those characterized. In the plot, the color is the cluster and the large outlined symbol marks the core accessions. The validation gives MD% = 0.0, VD% = 62.5, CR% = 75.2 and VR% = 91.4, and since the core has fewer than ten accessions, the report adds: The core has fewer than ten accessions: the validating t and F tests lack power and their percentages should be read with caution.

The section **Genebank management** takes the Block 8 dashboard (Chapter 8): four boxes, a table with one row per alert type and how many lots have it, the workload chart for the next 25 years and a note: a viability projection does not replace a germination test.

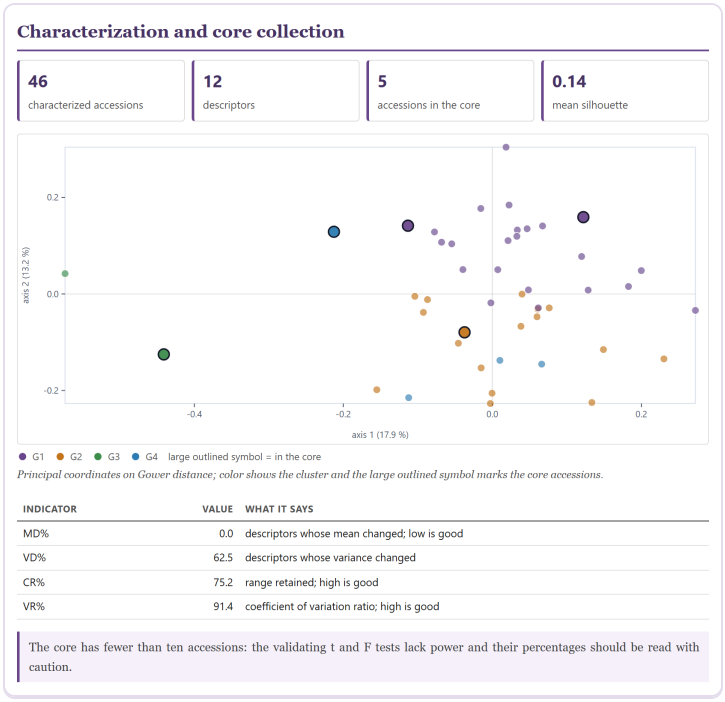


Figure 10.8. The section *Characterization and core collection* with the example characterization and the default 10% core.

The boxes match the Block 8 dashboard: 10 lots in trouble, 21 that need attention and 15 maize lots with projected viability; the 14 lots without published constants are the 13 seed lots of the other species and the chile costeño, which declares no route.

Table 10.5. The alert types of the example, in the order of the report: critical first (red dot), then warnings (ochre) and information last (blue).

Alert	Lots	Severity
Last germination test below the threshold	4	critical
In-vitro subculture overdue	3	critical
Fewer field plants than the minimum	3	critical
Field replanting overdue	2	critical
No safety duplicate in another institution	21	warning
Site visit overdue	5	warning
No date for the last replanting	2	warning
Stock close to the minimum	2	warning
Cryovial check overdue	1	warning
No viability constants: not projected	14	information

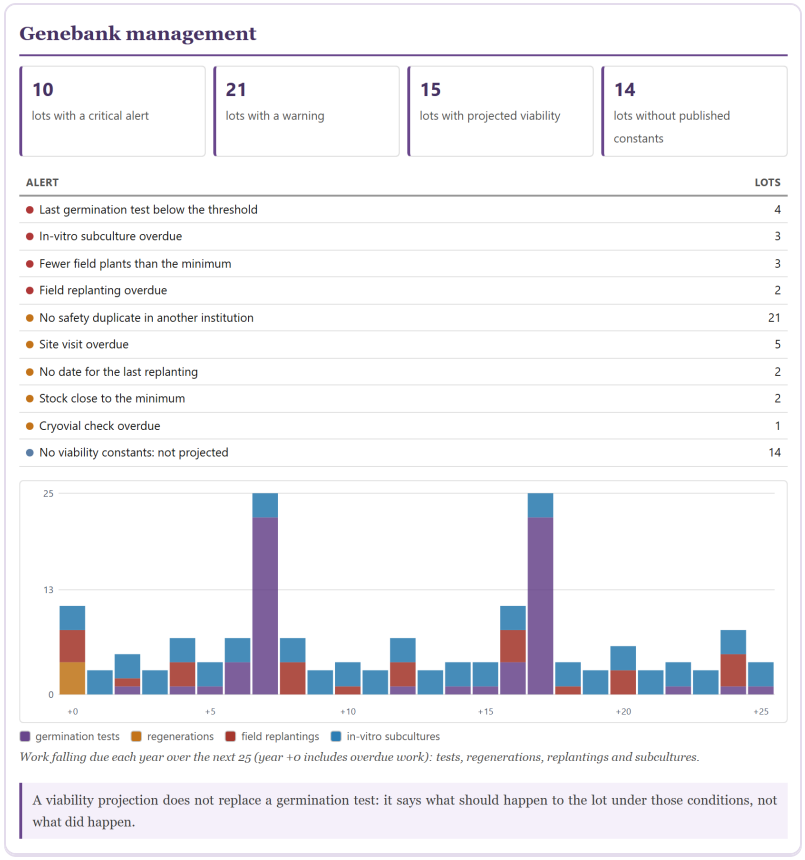


Figure 10.9. The section *Genebank management* with the default configuration of Block 8.

The names are generic: the report says what is happening and to how many lots, not the numbers of each one. To find out which are the 4 lots with germination below the threshold —the Balsas teosinte with 81%, the vaquita bean with 84%, the tepary bean with 79% and the maíz ancho de riego with 72%— look at the Block 8 table or at data/work-plan.csv in the package (Section 10.11). The workload chart carries its legend —germination tests, regenerations, field replantings and in-vitro subcultures— and an axis with 0, 13 and 25 tasks; the peaks in years +7 and +17 are the germination tests of 22 seed lots that fall due together.

At the end of the document comes the **annex**: one row per accession with its number, name, species (without subspecies), country of origin by its name, collecting site, conservation route and stock. A country code the app does not recognize, such as "MX", comes out as it is. It is the list that is filed with the report; if you do not need it, switch off its button.

10.9 The written methods and the references

The **Methods** section is the most useful thing the block has and the one that demands the most care. The app writes in prose one paragraph for each analysis present, with the numbers and the settings it took from each block, each method with its citation, ready to be pasted into the materials and methods section of a report or a paper. What was not computed does not appear.

Methods

Passport data

The collection was described with the FAO/Bioversity multi-crop passport descriptors MCPD v2.1 (Alercia et al., 2015), extended with seven local fields for what the standard, written for ex-situ banks, does not cover: the active conservation route, the in-situ site, the keeper, the stock, the germination percentage of the last test and the dates of the last germination test and the last regeneration. 46 accessions were recorded. Every value was validated against its descriptor's code list and against the rest of the record (dates consistent with one another, coordinates inside the declared country, storage compatible with the route). To share it with GBIF, the passport is also exported in Darwin Core (Wieczorek et al., 2012); descriptors without a term of their own in that standard travel in dynamicProperties under their original name.

Quality control and duplicates

Coordinates were checked against the bounding box of the declared country to detect the three classic mistakes — unsigned longitude, flipped latitude and latitude and longitude swapped—, each with a proposed fix that was applied only after review. Scientific names and site names were screened with Levenshtein distance to group spelling variants. Duplicates were searched with a blocking scheme (collecting number, donor numbers, name prefix, half-degree cell and taxon with date) and scored with a weighted combination of Jaro-Winkler name similarity (Winkler, 1990), haversine geographic distance and agreement of numbers and dates; pairs above the threshold were reviewed one by one. 2 candidate pairs were examined.

Cartography

Maps were drawn offline from the public-domain Natural Earth outlines bundled with the application (Mexican admin-1 at 1:10 million and world admin-0 at 1:110 million), in an equirectangular projection corrected by the cosine of the latitude. The state and country of each accession were assigned by point-in-polygon, not from the text of the corresponding field, so that the map exposes coordinates that disagree with what was declared.

Figure 10.10. The first three paragraphs of the methods in the example.

Table 10.6. The methods paragraphs and what each one takes from the app, with the numbers of the example.

Paragraph	What it says
Passport data	MCPD v2.1 with seven local fields (the route, the in-situ site, the keeper, the stock, the percentage and date of the last germination test and the date of the last regeneration); "46 accessions were recorded"; validation by descriptor and across fields; the export in Darwin Core.
Quality control and duplicates	The three coordinate mistakes, Levenshtein, the blocking scheme and Jaro-Winkler scoring; "2 candidate pairs were examined".
Cartography	Natural Earth offline and point-in-polygon assignment.
Geographic diversity	The unit and type of Block 5 ("grouped by state", "the diversity of species"), the indices, rarefaction, Chao1, Chao2, the jackknife, Good-Turing, Jaccard dissimilarity, UPGMA and the Mantel test "using 999 permutations".
Collecting gaps	The "1° grid", the four weights of the priority index (40, 25, 20 and 15), greedy complementarity and what the analysis is not.
Characterization and core collection	All the options of Block 7 (below).

Table 10.6. The methods paragraphs and what each one takes from the app, with the numbers of the example. *(continued)*

Paragraph	What it says
Viability and management	The Ellis and Roberts equation with the store temperature (–18 °C) and seed moisture content (6%) of the genebank configuration, where the constants come from, the regeneration threshold (85%) and the interval between tests (10 years), and the number of plants needed to retain an allele.
Software	That everything was computed with GermplasmPro, version 1.0.0 (Barrera-Guzmán, 2026), in the browser and without sending any data.

The characterization paragraph follows every setting of Block 7. With the default values it says, among other things:

> *"Hierarchical clustering used Ward's method in its D2 variant (Murtagh and Legendre, 2014), and the tree was cut into 4 groups, a number chosen by hand; the mean silhouette of that cut was 0.14 (Rousseeuw, 1987). The core collection, in the sense of Brown (1989), was requested at 10 % of the characterized accessions (5 of 46) and ended up with 5. It was built by strata —the 4 clusters—, with logarithmic allocation of the size among strata, picking within each stratum the accessions most distinct from one another by Gower distance. Every stratum was represented."*

If in Block 7 you choose UPGMA, 3 groups and a 20% core, the paragraph changes to "used UPGMA, the unweighted pair-group average (Sokal and Michener, 1958)", "cut into 3 groups", "the mean silhouette of that cut was 0.27" and "requested at 20 % … (9 of 46) and ended up with 9". It also says whether you required at least one accession per stratum, how many strata were left without a representative and, with the M method, that the core was built by maximizing classes (Schoen and Brown, 1993), without strata.

! THE METHODS ARE A DRAFT, AND THE RESPONSIBILITY IS YOURS

The app writes with what it knows; you sign. Read each paragraph before copying it, check that it says what you did and that every number matches the blocks, and adapt the style to that of your journal or your institution. Add what the app cannot know: where the passport and characterization data come from, who collected and with what permit, what conditions your cold room really has, why you chose each setting. If you cite a method, consult the original source: the app supplies the reference, but what the text claims about it, you claim. And if after downloading the report you change something in a block, the old text is out of date: go back into Block 10 and download it again.

The **references** are built from what is cited in the methods, in alphabetical order and numbered, in the language of the report. With the full example there are 24. Among them is that of the program itself, cited by the *Software* paragraph; without characterization, Brown, Gower, Hu et al., Murtagh and Legendre and Rousseeuw, cited only by that paragraph, disappear; and Darwin Core (Wieczorek et al., 2012) is always there, because the passport paragraph mentions the export.

References

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Figure 10.11. The 24 references of the example report.

10.10 Printing and downloading the report

The Print or save as PDF button opens the browser's print dialog with the report only, without the app around it. For a PDF, choose *Save as PDF* as the printer. If the browser does not allow printing the frame, the app opens the report in a new tab and prints it from there; if the browser also blocks new windows, it says The browser blocked the window; download the report and open it with a double click.

The Download the report (HTML) button saves the document as report-2026-10-31.html (with the day's date; with the interface in Spanish, informe-2026-10-31.html) and says Report downloaded. for four seconds. It is a single file of about 315 kB, with the figures inside, without fonts or external files: it opens with a double click in any browser, without internet, and can be printed from there, or opened in a word processor to edit it. The messages of the buttons are also announced to screen readers.

EXAMPLE · THE REPORT ON THE EXAMPLE COLLECTION ON PAPER

Printed to PDF on letter paper, with the default margins and without browser headers, the full report takes 13 sheets: the summary and the collection on the first two, quality, the map and diversity from the second to the fifth, the gaps on the fifth and sixth, characterization and management from the sixth to the eighth, the methods on the eighth and ninth, the references on the tenth and eleventh, and the annex of 46 accessions on the last three. The text is set in a serif face at 10.5 points and the tables in a sans-serif face, with the numbers aligned to the right.

When printing, titles and subtitles are not left alone at the foot of a sheet: they move to the next one with what they head. Figures are printed at 82% of the width and are not split; long tables are split between rows, and the table header is repeated on the next sheet. Accession numbers are not broken across two lines.

10.11 The ZIP package

The Download everything (ZIP) button builds a package with the report, the working tables and the separate figures, and saves it as germplasmpro-2026-10-31.zip. While building it, it says Building the package... and when it finishes it counts the files: with the full example, 15 files in the package. The ZIP is written by the app itself, without outside libraries. The names of the files and folders inside follow the language of the report: those given here are the English ones; with the report in Spanish they are informe.html, datos/, figuras/, LEEME.txt and so on.

Table 10.7. The files of the package with the full example collection.

File	What it is
report.html	The report, identical to the preview: with the sections, the figures and the language you chose.
data/passport-mcpd.csv	The passport with the 42 MCPD v2.1 descriptors (Section 10.12).
data/passport-full.csv	The 42 descriptors and the seven local `GP_` fields: 49 columns.
data/darwin-core.csv	The passport in Darwin Core (Section 10.12).
data/quality.csv	The same table as Download the quality report in Block 3 (quality-report.csv), produced by the same function: one row per finding, with `type` (geography, consistency, correction, duplicate or passport), `level`, `accession`, `field` (for consistency findings, the name of the warning) and `detail`. Only if there are findings.
data/work-plan.csv	One row per alert of each Block 8 lot: `accession`, `taxon`, `route`, `level`, `alert` and `detail`, with the numbers of each lot. Only if there are alerts.

Table 10.7. The files of the package with the full example collection. *(continued)*

File	What it is
figures/*.svg	Eight SVG figures 900 pixels wide, in the language of the report and with the colors of the light theme: collecting-map, accumulation-curve, diversity-map, collecting-gaps, complementarity-curve, principal-coordinates, dendrogram and workload. Only with the figures button on; those of Block 7, only with characterization.
README.txt	What the package contains, the date and time it was generated, the number of accessions and of files, and the difference between the three passport formats.

The package carries three figures that the report does not —the diversity map, the complementarity curve and the dendrogram—. Each separate figure is an SVG file in its own right, without the HTML legend or the caption of the report: write your own caption when you put it into another document. Since SVG is vector-based, it can be enlarged without losing sharpness and edited in any drawing program; if the journal asks for an image format, convert it with that program.

> **WHAT TO FILE EACH YEAR**
>
> The package is a snapshot of the collection on a date: the passport as it stood, the tables behind every number in the report and the figures. Keep it with the signed report and the Block 9 register book (Section 9.9). To reopen the collection in the app, use passport-full.csv in Block 2: it is the only one of the three that keeps the route, the stock and the germination and regeneration data in their own columns.

10.12 The passport in Darwin Core and in MCPD

The last two buttons export only the passport, without the report, as CSV with UTF-8 encoding and comma separator:

- Passport in MCPD saves passport-mcpd-2026-10-31.csv with the 42 MCPD v2.1 descriptors, in the order of the standard and with its names (`PUID`, `INSTCODE`, `ACCENUMB`...), and says 46 accessions in MCPD v2.1. It is the format for sending the list to another genebank or to a germplasm information system. The `GP_` fields are left out.
- Passport in Darwin Core saves darwin-core-2026-10-31.csv with 33 Darwin Core columns, the GBIF standard, and says 46 records in Darwin Core.

Nineteen MCPD descriptors have a term of their own in Darwin Core and are translated; the other thirty —among them `SAMPSTAT`, `STORAGE`, `MLSSTAT`, `DUPLSITE` and the seven `GP_`— travel in the `dynamicProperties` column as a JSON object, with their original name preceded by `mcpd:`. Between the translated columns and that one, the file keeps the complete passport.

Table 10.8. Some Darwin Core columns and where they come from, with the row of MEX-ZM-0001 and the report in English.

Darwin Core term	Comes from	MEX-ZM-0001
`occurrenceID`	`INSTCODE` : `ACCENUMB`	MEX006:MEX-ZM-0001
`basisOfRecord`	the route	LivingSpecimen (in situ and on farm: HumanObservation)
`collectionCode`	fixed	germplasm
`recordedBy`	`COLLNAME`	empty: the example does not include the name of the collector
`eventDate` , `year` , `month` , `day`	`COLLDATE`	2019-08-12 · 2019 · 8 · 12
`scientificName`	`GENUS` `SPECIES` `SUBTAXA`	Zea mays
`country` , `countryCode`	`ORIGCTY`	Mexico · MX (two-letter ISO code)
`decimalLatitude` , `decimalLongitude` , `geodeticDatum`	`DECLATITUDE` , `DECLONGITUDE`	17.6871 · −97.2788 · EPSG:4326
`minimum–` and `maximumElevationInMeters`	`ELEVATION`	1993 · 1993
`samplingProtocol`	`COLLSRC`	Farm store
`degreeOfEstablishment`	`SAMPSTAT` from 300 to 599	cultivated
`preparations`	the route	Seed bank
`dynamicProperties`	the remaining 30	{"mcpd:ACCENAME":"Maíz azul de Apoala", "mcpd:SAMPSTAT":"300", "mcpd:STORAGE":"13;40", … "mcpd:GP_GERMPCT":"88", …}

The texts of `country` , `samplingProtocol` , `georeferenceProtocol` and `preparations` come out in the language of the report: if you are going to upload the file to GBIF, set the report to English before exporting. `recordedBy` is the person who collected, and is only filled from `COLLNAME` ; the code of the collecting institute travels in `dynamicProperties` . Wild material has no term in `degreeOfEstablishment` : it is left empty and the full `SAMPSTAT` code goes in `dynamicProperties` . If the country is miswritten in the passport, like the "MX" of the tepary bean, `country` and `countryCode` are left empty: fix the passport in Block 3 first.

When something goes wrong

What you see	Why it happens and what to do
Load your accessions in Block 2 first.	There is no passport. Load your collection or the example one (Chapter 2).
The Characterization and core button is gray	No characterization is loaded. Load it in Block 7 (Chapter 7) and come back.
The Summary or the Methods cannot be switched off	They are among the four sections that always go into the report (Section 10.3).

What you see	Why it happens and what to do
The geographic distribution does not appear	No accession has coordinates. Fill them in in Block 2.
The summary says a coordinate disagrees with its country	Check it in Block 3 and apply the proposed fix, or explain in the text why you keep it.
I want the numbers of each lot with an alert	The report only counts lots by alert type. Look at the Block 8 table or at the work plan in the ZIP.
I changed a setting in another block and the downloaded report does not have it	The report is built when you enter the block. Go back to Block 10 and download again.
The browser blocked the window; download the report and open it with a double click.	The browser did not let the tab for printing open. Download the HTML and print it from there, or allow pop-up windows for the app.
The Darwin Core comes out with countries and protocols in Spanish	They come out in the language of the report. Choose English in Report language and export again.
The ZIP carries fewer than 15 files	There are no quality findings, or no alerts, or no characterization, or the figures are switched off: each removes its own files.

DECISION RULE · BEFORE HANDING IN THE REPORT

If the report goes to management or into the annual report	**then** keep the summary, the collection, management and the annex, and fill in the genebank and the signatory.
If it goes into a paper	**then** use the methods as a draft: check them against the blocks, consult the sources they cite and add what the app does not know.
If the summary reports coordinates that disagree or pairs to review	**then** resolve them in Block 3 before building the final report.
If you changed a setting in another block	**then** enter Block 10 again and download again.
If the report is to be in English	**then** choose English: text, figures, references and Darwin Core change together.
If you need the figures in another document	**then** download the ZIP and use the SVGs in figures/, with your own caption.
If you are going to share the list with another genebank	**then** send the passport in MCPD.
If you are going to publish the data in GBIF	**then** use Darwin Core in English and check the empty countries.
If you are closing the year	**then** file the ZIP with the signed report and the Block 9 register book.

This closes the ten blocks. The report says what the collection and the analyses say, but the interpretation, the sources and the signature are yours. What follows in the manual are the appendices: the files the app reads and writes, the decision rules gathered together, the glossary, what to do when something goes wrong, the references and the license.

A

APPENDICES

Appendices

What you need to have at hand and did not fit in any chapter: which files the app reads and writes, with the formats and the code lists of the passport; the decision rules of the ten blocks gathered in a single list; the vocabulary of genebanks; what to do when something goes wrong; the references behind each computation, and what you need to know about the tool itself: the example data, what it stores and where, the keyboard, the license and how to cite it.

A The files the app reads and writes

B The decision rules, gathered together

C Glossary

D When something goes wrong

E References

F The tool: data, privacy, license and citation

A The files the app reads and writes

GermplasmPro runs in the browser and sends nothing to any server: every file comes in from and goes out to a folder on your computer. This appendix says which they are, what format the passport data require and which columns each downloaded table carries.

A.1 · What the app reads

Format	Where and what for
.csv, .tsv, .txt	Block 2, Open a file (CSV, TSV or project): the passport, one row per accession. Block 7, Load my characterization (CSV): the characterization, which is joined to the passport by accession number.
pasted	Blocks 2 and 7, Paste from a spreadsheet: a block of cells copied with its header row. In Block 2 the same window carries the guided template for writing the table from scratch (Section 2.4).
project **.json**	Block 2, the same open button: the collection, the characterization and the settings of Blocks 3, 8 and 9 saved earlier with Save the project (.json).
on screen	Block 8: the viability constants of other species (Add constants for a species) and the records of tests, regenerations, subcultures, checks and visits, which are typed into the app itself and go straight into the passport.

When reading a text, the app picks the separator on its own —tab, semicolon, comma or vertical bar— by testing which one splits every line into the same number of fields, not which one appears most often. It respects quotes, commas and line breaks inside a quoted cell, and the byte order mark at the beginning; if replacement characters appear when decoding as UTF-8, it reads the file again as Windows-1252, which is what a spreadsheet on Windows usually saves. Completely empty rows are discarded, a repeated header is numbered so as not to lose the column, and a column without a header is named as in the spreadsheet: Column A, Column B… (Columna A… with the interface in Spanish).

> **A SPREADSHEET WORKBOOK IS NOT READ, AND THAT IS ON PURPOSE**
>
> An `.xlsx`, `.xls` or `.ods` file does not open; save it as .csv or copy and paste the cells. Reading it would mean bringing in someone else's library of hundreds of thousands of lines for a problem that is solved in two clicks. The choice was for the app not to load a single line that is not its own (Appendix F): that is what lets it open with a double click, without installing anything, and work without internet.

A.2 · The passport formats

Chapter 2 gives the full list of the 49 descriptors —the 42 of MCPD v2.1 and the seven of the GP_ extension— with their name in the app and an example (Tables 2.2 and 2.3). Here is what the validation requires of each class of descriptor. An **error** is a value that cannot be; a **warning**, one worth checking.

Class and descriptors	Required format
Dates · `COLLDATE`, `ACQDATE`, `GP_GERMDATE`, `GP_REGENDATE`	`YYYYMMDD`, with hyphens where unknown: `201907--`, `2019----`. Error if it does not follow that pattern, if the year is before 1700 or after the current one, if the month is not from 01 to 12, if the day is not from 01 to 31 or if that day does not exist in that month (30 February).
Decimal coordinates · `DECLATITUDE`, `DECLONGITUDE`	Signed decimal degrees: negative latitude south of the equator and negative longitude west of Greenwich. A decimal comma is accepted and becomes a point on import. Error if it is not a number, if it falls outside −90 to 90 or −180 to 180, or if only one of the two is given; warning if it is exactly zero.

Class and descriptors	Required format
Degrees, minutes and seconds · `LATITUDE`, `LONGITUDE`	`DDMMSSH` for latitude (`173832N`) and `DDDMMSSH` for longitude (`0971952W`); unknown minutes or seconds, with hyphens. If only these are given, the app computes the decimal ones on import; if both are given and differ by more than 0.02°, warning.
Numbers · `ELEVATION`, `COORDUNCERT`, `GP_STOCK`, `GP_GERMPCT`	A number, with a decimal point or comma. Error if germination falls outside 0 to 100 or if uncertainty or stock is negative; warning if elevation falls outside −430 to 6,500 m.
Country · `ORIGCTY`	Three-letter ISO 3166-1 alpha-3 code (`MEX`, `GTM`, `PER`), converted to capitals on import. Error if it is not three letters; warning if it is well formed but not in the list the app carries.
Institutes · `INSTCODE`, `COLLCODE`, `BREDCODE`, `DONORCODE`, `DUPLSITE`	WIEWS code: three country letters and three digits (`MEX006`). `DUPLSITE` accepts several, separated by semicolons. A malformed code is only a warning.
Closed lists · `SAMPSTAT`, `COLLSRC`, `MLSSTAT`, `GEOREFMETH`, `GP_CONS`	One of the codes in the table below, regardless of case. Any other value is an error.
Multi-valued list · `STORAGE`	One or more codes from the list, separated by semicolons (`13;40`). Error if any of them does not exist.
Free text · the rest	No format. `ACCENAME` and `OTHERNUMB` separate several values with semicolons.

Empty mandatory descriptors —`INSTCODE`, `ACCENUMB` and `GENUS`— are errors, and the twelve highly recommended ones, when empty, warnings. An in-situ or on-farm record is not asked for `STORAGE`, `ACQDATE`, `DUPLSITE` or `INSTCODE`. The rules that cross two fields —status against collecting source, order of the dates, in-situ route against storage, safety duplicate in the same institution— are in Table 2.6 of Section 2.6.

A.3 · The code lists

They are those of MCPD v2.1 (Alercia, Diulgheroff and Mackay, 2015), with the meaning given in the standard; `GP_CONS` belongs to the extension. There is no need to learn them: when you type in those cells, the table and the record card of Block 2 drop down the list with its meaning.

Descriptor	Codes
SAMPSTAT biological status · 21	`100` wild · `110` natural · `120` semi-natural/wild · `130` semi-natural/sown · `200` weedy · `300` traditional cultivar/landrace · `400` breeding or research material · `410` breeder's line · `411` synthetic population · `412` hybrid · `413` founder stock or base population · `414` inbred line · `415` segregating population · `416` clonal selection · `420` genetic stock · `421` mutant · `422` cytogenetic stocks · `423` other genetic stocks · `500` advanced or improved cultivar · `600` GMO · `999` other
COLLSRC collecting source · 22	`10` wild habitat · `11` forest or woodland · `12` shrubland · `13` grassland · `14` desert or tundra · `15` aquatic habitat · `20` farm or cultivated habitat · `21` field · `22` orchard · `23` backyard, kitchen or home garden · `24` fallow land · `25` pasture · `26` farm store · `27` threshing floor · `28` park · `30` market or shop · `40` institute, research organization, genebank · `50` seed company · `60` weedy, disturbed or ruderal habitat · `61` roadside · `62` field margin · `99` other
STORAGE storage · 9	`10` seed collection · `11` short-term seed · `12` medium-term · `13` long-term (all four give the seed route) · `20` field collection · `30` in vitro collection · `40` cryopreserved collection · `50` DNA collection · `99` other (no route)

Descriptor	Codes
MLSSTAT Multilateral System · 3	`0` not in the Multilateral System · `1` included · `99` other
GEOREFMETH georeferencing · 4	`GPS` · `determined from map` · `gazetteer` · `determined by locality description`
GP_CONS active route · 7	`seed` seed bank · `field` field genebank or living collection · `invitro` in vitro culture · `cryo` cryopreservation · `dna` DNA bank (these five, ex situ) · `insitu` genetic reserve · `onfarm` on-farm conservation (these two, in situ). If empty, it is deduced from the first `STORAGE` code that has a route.

A.4 · What the app writes

Every CSV the app writes uses a comma as separator and carries the UTF-8 byte order mark at the beginning: without it, a spreadsheet on Windows would open it as Windows-1252 and "Pérez" would come out as "PÃ©rez". The app removes it when reading, so its own CSV files are re-imported unchanged. The Block 10 files carry the date of the report in their name. The names below are those given with the interface in English; with the interface in Spanish the files are named in Spanish (pasaporte-mcpd.csv, informe-calidad.csv…), and in the ZIP the names follow the language of the report.

Block	Button	File and content
2	Download the template	passport-template.csv: the 49 descriptors with an example row. From the guided template, Download this template as CSV gives germplasmpro-template.csv, with the ticked columns only.
2	Export MCPD (standard only) · Export everything (with the extension)	passport-mcpd.csv, with the 42 descriptors of the standard; passport-full.csv, with all 49.
2	Save the project (.json)	germplasmpro-collection.json (version 2 of the format): the passport, the Block 7 characterization, the threshold and the dismissed pairs of Block 3, the genebank assumptions, the custom viability constants and the calculators of Block 8, and the label preferences of Block 9.
3	Download the quality report	quality-report.csv: one row per finding of geography, consistency, correction, duplicate or passport validation.
4	Export SVG · Export PNG (2×) · Points as CSV (with state and cell)	collecting-map.svg and .png; map-points.csv, with the accessions on view.
5	Table as CSV · Curve as SVG · Heat map as SVG	geographic-diversity.csv; accumulation-curve.svg; composition-similarity.svg.
6	Collecting route as CSV · Types at risk as CSV · Map as SVG	collecting-priorities.csv, with the first 200 cells; types-at-risk.csv; collecting-gaps.svg.
7	Download the template	characterization-template.csv: your accession numbers and empty columns for six descriptors.
7	Export the core as CSV · Characterization + group + core · Distance matrix	core-collection.csv; characterization-full.csv; gower-distances.csv. Also PCoA as SVG and Tree as SVG: pcoa-gower.svg and gower-tree.svg.
8	Work plan as CSV	work-plan.csv: one row per accession with some alert. If there are none, the app says so and downloads nothing.

Ap

Block	Button	File and content
9	Print · Download as CSV	The labels, the delivery note and the register book are printed (or saved as PDF from the print dialog); the sowing list is also available as sowing-list.csv.
10	Print or save as PDF · Download the report (HTML)	report-YYYY-MM-DD.html: the report in a single file, which opens in any browser.
10	Passport in Darwin Core · Passport in MCPD	darwin-core-YYYY-MM-DD.csv, with 33 terms; passport-mcpd-YYYY-MM-DD.csv, with the 42 descriptors.
10	Download everything (ZIP)	germplasmpro-YYYY-MM-DD.zip: report.html, README.txt, the data/ folder with the three passports (MCPD, complete and Darwin Core) and, if they exist, quality.csv and work-plan.csv, and the figures/ folder with the figures in SVG (Section 10.11).

Figures opened in the figure studio are exported from there in the format and size you choose.

A.5 · The columns of each table

File	Columns, in order
passport-mcpd.csv	the MCPD keys in the order of Table 2.2: PUID, INSTCODE, ACCENUMB … STORAGE, MLSSTAT, REMARKS
passport-full.csv	the same 42 and, at the end, GP_CONS, GP_SITEID, GP_KEEPER, GP_STOCK, GP_GERMPCT, GP_GERMDATE, GP_REGENDATE
quality-report.csv	type, level, accession, field, detail
map-points.csv	ACCENUMB, ACCENAME, TAXON, CROPNAME, DECLATITUDE, DECLONGITUDE, ELEVATION, STATE, POINT_COUNTRY, PASSPORT_COUNTRY, CELL_1DEG, ROUTE
geographic-diversity.csv	unit, key, accessions, richness, rarefied, shannon, gini_simpson, pielou, hill_q1, exclusive, mean_elevation, lat, lon
collecting-priorities.csv	rank, latitude, longitude, state, country, km_to_nearest, neighborhood_richness, neighborhood_accessions, exclusive_nearby, score, cell_degrees
types-at-risk.csv	type, accessions, states_or_countries, routes, duplicates, with_coordinates, risks, weight
characterization-template.csv	ACCENUMB, DIAS_FLORACION, ALTURA_PLANTA, LARGO_FRUTO, COLOR_EXTERNO, FORMA_FRUTO, HABITO
core-collection.csv	ACCENUMB, ACCENAME, TAXON, CROPNAME, group, stratum, core, pcoa1, pcoa2; characterization-full.csv carries every accession and adds the characters in use
gower-distances.csv	accession and one column per accession: the full matrix
work-plan.csv	ACCENUMB, ACCENAME, TAXON, route, urgency, stock, unit, last_germination, projected_germination, years_since_test, years_to_threshold, actions

File	Columns, in order
sowing-list.csv	`ACCENUMB, name, taxon, plants, germination, assumed_germination, seeds, stock, plot, sowing_date, notes`
darwin-core-….csv	`occurrenceID, basisOfRecord, institutionCode, collectionCode, catalogNumber, otherCatalogNumbers, fieldNumber, recordedBy, eventDate, year, month, day, scientificName, genus, specificEpithet, infraspecificEpithet, scientificNameAuthorship, vernacularName, country, countryCode, locality, decimalLatitude, decimalLongitude, geodeticDatum, coordinateUncertaintyInMeters, georeferenceProtocol, minimumElevationInMeters, maximumElevationInMeters, samplingProtocol, degreeOfEstablishment, preparations, occurrenceRemarks, dynamicProperties`

These are the column names with the interface in English; with the interface in Spanish they come out in Spanish. The MCPD passport columns and the keys of the characterization template are the same in both languages (the characterization import accepts `accession` or `accesion`). In the ZIP, data/quality.csv has the same columns and rows as quality-report.csv from Block 3 (geography, consistency, correction, duplicate and passport validation), and data/work-plan.csv has one row per alert: accession, taxon, route, level, alert, detail. How each descriptor is translated into Darwin Core is in Section 10.12.

B The decision rules, gathered together

The scales and thresholds the app places next to each number, in a single list; when they are adjustable, the default value is given. They are guides for reading, not laws: what you know about your species, your genebank and your sites always prevails. What is worth *doing* in each case is in the decision rule box that closes each chapter.

Block	Quantity	Scale or threshold
2	Completeness of a descriptor	≥ 90% green bar · 50–90% ochre · < 50% red
2	Elevation of the collecting site	−430 to 6,500 m plausible; outside, warning in Block 2 · outside −500 to 7,000 m, impossible: left out of the elevation coloring, the belts and the gradient (Blocks 4 to 6)
2	Coordinates	exactly zero: warning · decimal and degrees-minutes-seconds values differing by more than 0.02°: warning · only one of the two decimal values: error
3	Point against its country	inside the country's rectangle with a one-degree margin: no problem · outside: sign change and swap are tried · (0, 0): error without a fix
3	Score of a possible duplicate	default threshold 50 points (slider from 25 to 95) · no single clue is enough; a matching collecting number is almost a signature · lowering the threshold proposes more pairs and gets fewer right
5	Mantel test (999 permutations)	p < 0.05 with positive r: diversity has geographic structure · p ≥ 0.05: no evidence at that scale
5	Richness estimators	fewer than three doubletons: warning `Careful with Chao1 in this collection.`; read jackknife 1, the most stable
5	Good-Turing coverage	probability that the next accession is of a type you already have; its complement, that it brings a new one
6	Coverage of the territory	< 25% of cells with collections: the number is shown in red

Block	Quantity	Scale or threshold
6	Priority index (0 to 100)	default weights: remoteness 40 · neighborhood richness 25 · state deficit 20 · nearby exclusive types 15 · remoteness capped at 250 km · neighborhood of about 220 km · robust priority: the cell that stays at the top when each weight is raised
6	Minimum set	the cells that, by complementarity, capture at least 90% of the types
7	Mean silhouette	> 0.25 the groups are real: stratify by group · close to 0, an almost arbitrary cut · negative, the accession would be better in another group
7	Core validation (Hu et al., 2000)	MD% < 20 · VD% < 20 · CR% > 80 · VR% ≥ 100 · classes captured > 80% · the two thresholds of the paper are those of MD% and CR%
7	Core size	10% by default · fewer than 10 accessions: warning that the core is too small to judge
8	Germination of a seed lot	regeneration threshold 85% · test or projection below the threshold: in trouble · less than 3 years to cross it: needs attention · test every 10 years
8	Seed stock	< 400 seeds: in trouble · < 600 (1.5 times the minimum): needs attention
8	The other routes, by default	field: replanting every 8 years, minimum 25 plants · in vitro: subculture every 12 months, minimum 10 jars · cryopreservation: check every 5 years, minimum 20 cryovials · genetic reserve: visit every 2 years · on farm: every year
8	Plants to regenerate	$P(\text{retaining an allele of frequency } p) = 1 - (1 - p)^{k \cdot n}$, k = 2 in outcrossing and 1 in selfing species · with p = 5% and 95% confidence: 30 plants for an outcrossing species, 59 for a selfing one
9	Narrowest bar of the Code 128	≥ 0.25 mm any reader reads it · 0.19–0.25 mm printable, but test one · < 0.19 mm warning: thermal printer
9	QR module	< 0.4 mm: ochre warning · with the chayote in the middle, always correction level H

C Glossary

Term	What it is
Accession (*accesión*)	A distinct sample of a species that the genebank conserves and manages as a unit, with its unique number (`ACCENUMB`), which is never repeated or reused.
Accidental duplicate (*duplicado accidental*)	The same collection entered twice into the catalog, almost always with the name written differently. It protects nothing: it inflates the collection and disorders the analyses. Block 3 looks for it.
Blocking (*bloqueo*)	Comparing only the pairs of accessions that share some key —the collecting or donor number, the start of the name, the genus with the cell, the taxon with the year and month of collection— instead of all against all. It is what makes searching for duplicates in a large collection feasible.
Code 128	A linear barcode that accepts letters, digits and signs; the one for envelopes, jars and shipments. Any handheld reader reads it and types the number wherever the cursor is.

Term	What it is
Collecting gap (*vacío de colecta*)	A cell of the territory without any accession. In GermplasmPro the analysis is geographic and based on the collection, not ecogeographic: it uses no climate or soil layers.
Complementarity (*complementariedad*)	Choosing sites one by one, each time the one that adds the most types not yet held. It gives the minimum set of cells that gathers the diversity of the collection.
Conservation route (*ruta de conservación*)	The way an accession is managed today: seeds, field, in vitro, cryopreservation, DNA, genetic reserve or on farm. It decides the alerts of Block 8.
Core collection (*colección núcleo*)	A small subset that keeps, with a minimum of repetition, most of the diversity of the whole collection (Frankel and Brown, 1984); it is characterized and evaluated in depth instead of the whole collection.
Core allocation (C, P, L, S) (*reparto del núcleo*)	How many places in the core each stratum gets (Brown, 1989): constant, which favors small strata; proportional, which reproduces the imbalance of the collection; logarithmic or square root, which lie in between.
Cryopreservation (*criopreservación*)	Shoot tips, embryos or pollen at –196 °C, in liquid nitrogen. The long-term backup of clonal material; it is checked by thawing a cryovial.
Darwin Core	The standard set of terms for publishing biodiversity records, the one required by GBIF and other portals (Wieczorek et al., 2012).
Ex situ / in situ	Away from its place —in a seed bank, a field genebank, in vitro or cryopreserved — / in its place, where the population keeps evolving: the genetic reserve and the farm.
Exclusive type (*tipo exclusivo*)	A type (species, crop, taxon...) found in only one geographic unit of the collection. If that unit is lost, the type is lost with it.
Field genebank or living collection (*banco de campo o colección viva*)	Living plants in a plot, an orchard or an arboretum. The route for trees, recalcitrant species and vegetatively propagated crops; expensive and exposed.
Genetic reserve (*reserva genética*)	A protected site where a wild population is conserved in situ, almost always of crop wild relatives. `GP_CONS` = `insitu`; its key goes in `GP_SITEID`.
Good-Turing coverage (*cobertura de Good-Turing*)	1 – f_1/N: the probability that the next accession you collect is of a type you already have.
Gower distance (*distancia de Gower*)	A distance between accessions for mixed characters —numeric, categorical and ordinal— with empty cells: each character contributes its relative difference and the average is taken only over what both have measured (Gower, 1971).
GP_ extension (*extensión GP_*)	The seven fields GermplasmPro adds to MCPD for management and in-situ conservation: active route, in-situ site and keeper, stock, last germination with its date and last regeneration. They are never confused with the standard and are exported separately.
Hill numbers (*números de Hill*)	Diversity expressed as "effective types": q0 is richness, q1 the exponential of Shannon and q2 the inverse of Simpson (Jost, 2006). Always q0 ≥ q1 ≥ q2.
Jaro-Winkler, Levenshtein	Two measures of similarity between texts. Jaro-Winkler (Winkler, 1990) compares accession names and rewards a common beginning; Levenshtein counts the letters that must be changed, and flags misspelt genera.
M method (*método M*)	Building the core by pursuing, one by one, the classes of the qualitative and ordinal characters it does not yet have (Schoen and Brown, 1993).

Term	What it is
Mantel test (*prueba de Mantel*)	Correlation between two distance matrices —here, composition and geography— with its p-value by permutation (Mantel, 1967).
MCPD v2.1	The FAO/Bioversity multi-crop passport descriptors, version 2.1 (Alercia, Diulgheroff and Mackay, 2015): 42 descriptors with a fixed key, the common way of writing the passport.
MD%, VD%, CR%, VR%	The criteria of Hu et al. (2000) for judging a core: means that differ, variances that differ, range retained and coefficient of variation ratio (the relative variation retained) compared with the collection.
Multilateral System (*Sistema Multilateral*)	That of the International Treaty on Plant Genetic Resources for Food and Agriculture; its material is supplied with the Standard Material Transfer Agreement (SMTA; in Spanish, ANTM). It is recorded in `MLSSTAT`.
On-farm conservation (*conservación en la parcela*)	In-situ conservation on the land of whoever sows it, where the variety remains under farmer selection. `GP_CONS` = `onfarm`; the keeping person or community goes in `GP_KEEPER`.
Outcrossing / selfing (*alógama / autógama*)	A species that crosses with other plants / that self-fertilizes. In a regeneration, each outcrossing plant contributes two independent gametes, and a selfing one, in practice, only one; that is why a selfing species needs almost twice as many plants.
Passport (*pasaporte*)	What is known about an accession: what it is, where it came from, who collected it and when, what material it is and how it is conserved. The input of every block.
Principal coordinates (PCoA) (*coordenadas principales*)	A two-axis map whose distances resemble as closely as possible those of a distance matrix, here the Gower one (Gower, 1966).
Priority index (*índice de prioridad*)	The 0 to 100 score of an empty cell in Block 6: the sum of remoteness, neighborhood richness, state deficit and nearby exclusive types, each scaled from 0 to 1 by its weight.
QR code (*código QR*)	A two-dimensional code with Reed–Solomon error correction. At level H it can be recovered even with 30% of the symbol covered, which allows the chayote to be drawn in the middle.
Rarefaction (*rarefacción*)	The richness a unit would have with a fixed number of accessions; it compares units with different collecting effort (Hurlbert, 1971; Heck, van Belle and Simberloff, 1975).
Regeneration (*regeneración*)	Sowing an accession to obtain new seed when germination or stock falls below the minimum. Each one is a bottleneck: regenerate seldom and with many plants.
Richness estimators (*estimadores de riqueza*)	Chao1, Chao2, jackknife 1 and jackknife 2: how many types the collection would have if collecting continued, computed from the rare types (with one or two accessions, or in one or two units).
Safety duplicate (*duplicado de seguridad*)	A sample of the same accession kept in another institution, so that a disaster does not take away the only copy. It is recorded in `DUPLSITE`; one in the same institution does not count.
Seed storage behavior (*comportamiento de la semilla*)	*Orthodox*: tolerates drying to 3–7% moisture and freezing. *Intermediate*: tolerates drying, but not deep cold. *Recalcitrant*: tolerates neither drying nor freezing, and cannot be kept in a seed bank.
Silhouette (*silueta*)	For each accession, how close it is to its own group compared with the neighboring group, from −1 to 1 (Rousseeuw, 1987). Its mean says whether the groups are real.

Term	What it is
Slow-growth in vitro culture (*cultivo in vitro de crecimiento lento*)	Plantlets or shoot tips in jars, with growth slowed by cold or by the medium, subcultured every so many months.
UPGMA, Ward.D2	Two ways of building a clustering tree: UPGMA joins the groups with the smallest average distance (Sokal and Michener, 1958); Ward.D2, those that least increase the within-group sum of squares (Murtagh and Legendre, 2014).
Viability constants (*constantes de viabilidad*)	K_E, C_W, C_H and C_Q: those of the Ellis and Roberts equation for a species. The app carries the published ones for maize; those for other species are taken from Kew's Seed Information Database or from the literature.
Viability equation (*ecuación de viabilidad*)	That of Ellis and Roberts (1980): with the temperature and the seed moisture content it gives the deviation σ of the life span, and with it how much the germination of a lot drops over the years.
WIEWS code (*código WIEWS*)	The code of an institution in FAO's world information system: three country letters and three digits (`MEX006`). It is used by `INSTCODE`, `DUPLSITE` and the other institute descriptors.

D When something goes wrong

Each chapter closes with its own table of problems for the block. These are the general ones, those that affect the whole app.

What happens	What to do
A block says `Load your accessions in Block 2 first.`	There is no passport in memory. Load your collection, open your .json project or use the example collection (Chapter 2).
When you open the app, yesterday's collection is not there	It is saved in the browser of that computer: another browser, another system account, a private window or having cleared the browsing data leave it behind. Open the .json project you saved.
The app does not open with a double click, or opens blank	Your institution blocks local pages. Use Open GermplasmPro.bat (or server.ps1), which opens the app at http://localhost:9600, or the online version.
The file is an .xlsx and does not open	Save it as .csv or copy the cells and use `Paste from a spreadsheet` (Appendix A.1).
Accents come out as "Ã©" or "�"	The file is neither UTF-8 nor Windows-1252. Save it from your spreadsheet as "CSV UTF-8" and open it again.
The paste window says `Everything landed in a single column.`	The separator is not the right one: pick it in the paste window. If you really copied a single column, you can go on.
The first data row was taken as headers, or the other way round	Fix it with the `The first row holds the headers` box of the paste window; if there is only one row and it is a header, the app warns that there is no data left to import.
The coordinates carry a decimal comma (`19,149`)	With tab or semicolon as separator, the app understands them, warns when pasting how many values use it and converts them to a point on import. In a comma-separated file, a decimal comma not enclosed in quotes splits the number into two columns: save it with semicolons or copy and paste from the spreadsheet.

What happens	What to do
Opening a .json says `That file is not a GermplasmPro project.`	It was not written by `Save the project (.json)`. If it is a table, save it as .csv.
I opened a project on another computer and my constants, my genebank assumptions or the pairs I dismissed are missing	The project was saved with an earlier version of the app, which only carried the passport and the characterization. Set them again on the original computer and save the project again: from version 2 of the format they travel with it (Appendix F.3).
Next to the Block 2 counter it says *Could not save in this browser…*	The browser storage is full or the window is private: what you do only lives in that tab. Save the .json project now and before closing.
I downloaded several tables in a row and only some came down	The browser throttles consecutive downloads: allow them for this site, or use `Download everything (ZIP)` in Block 10.
The report does not open for printing	The browser blocked the window, and the app says so: download the report with `Download the report (HTML)` and open it with a double click.
The printed labels do not measure what their format says	The scale in the print dialog is not 100% or the printer shrinks the sheet. Set it to 100% or "actual size" (Section 9.6).
The numbers do not match those of the manual	The manual uses the uncorrected example collection and the default values; a setting saved earlier (threshold, weights, assumptions) changes the result. Go back to the defaults or check the state the chapter starts from.
I want to check that the computations come out right in this browser	Open `tests/index.html`: these are the program's tests; all must come out green (with `?all` at the end of the address the passing ones are listed too).

E References

The references on the app's home page, those the Block 10 report builds and those cited by the chapters, by family. The report cites only those of the methods the study used (Section 10.9).

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The program

Barrera-Guzmán, L.Á. (2026). *GermplasmPro: the genebank, accession by accession* (version 1.0.0) [software]. https://github.com/luisangelbg/GermplasmPro

F The tool: data, privacy, license and citation

F.1 · What the app contains

GermplasmPro is written in HTML, CSS and JavaScript without third-party libraries: not for reading files, not for the charts and maps, not for writing the .zip, the QR code or the Code 128, which are generated in the browser itself according to their standards. It carries no fonts and requests none from the internet: it uses the system fonts. The only outside items are two, declared in LICENSES-TERCEROS.md:

- **The Lucide icons** (version 1.49.0) of the top bar, the figure studio and the Animations button, drawn with a thinner stroke, under the ISC license; those Lucide took from the Feather project also carry the MIT license. Both are permissive and compatible with GPL-3.0.
- **The Natural Earth outlines** of the countries (1:110 million scale) and of the 32 Mexican states (1:10 million), simplified for the app, with which the maps are drawn and each accession is assigned to its country and state. They are in the public domain: they require neither license nor attribution, and are declared only so that it is known where they come from.

The formulas are those of the original papers of each method (Appendix E), and the code lists, those of MCPD v2.1. The viability constants included are those published by Ellis and Roberts for *Zea mays*; for the other species you supply them, with their source.

F.2 · The example data

The example collection of Block 2 is **fictitious**. Its collecting sites are real localities with approximate coordinates and elevations, to which the program adds a small offset with a fixed seed; the accession numbers, the names of people, the stocks and the dates are made up. It holds 46 accessions of 24 taxa and 18 genera, from seven countries, across the five routes (Section I.7). The example characterization of Block 7 is **synthetic**: the program generates it with a fixed seed for the loaded accessions —six quantitative characters drawn around a profile per genus, four qualitative and two ordinal, with about 8% empty cells—, so its groups reflect those profiles and not measured plants. Errors were planted on purpose in the last eight accessions of the collection, from MEX-ZM-0039 to MEX-SE-0046:

Accession	Planted error	Who finds it
MEX-ZM-0039 · Tuxpeño de la costa	The longitude lost its sign: 96.06 instead of −96.06, and the point falls outside Mexico.	Block 3, with a fix
MEX-CU-0040 · Calabaza pipiana	Collecting date 20190230, a 30 February.	Block 2, error
MEX-PH-0041 · Frijol tepari	Country written with two letters (`MX`) and an elevation of 25,000 m.	Block 2, error and warning; Block 3 proposes `MEX`
MEX-CH-0042 · Chile costeño	Storage code 15, which does not exist; the accession is left without a route.	Block 2, error
MEX-ZM-0043 · Maíz ancho de riego	No coordinates; the site, "Por el rumbo de Guerrero" (somewhere around Guerrero), was written from memory.	Block 2, warnings
MEX-ZM-0044 · Maíz reventador	Wild status (100) with farm store as source (26), and an acquisition date earlier than the collecting date.	Block 2, warnings
MEX-ZM-0045 · BOLITA AMARILLO	The same collection as MEX-ZM-0002 (LAB-2019-101), in capitals and with an extra space.	Block 3, duplicate
MEX-SE-0046 · Chayote verde liso	The same collection as MEX-SE-0029 (LAB-2018-102), with a space at the end of the name.	Block 3, duplicate

F.3 · Privacy: what is stored and where

Everything is computed in your browser. The app makes no request to the internet: it does not send the passport, it does not query catalogs or portals and it loads nothing from outside; the maps use the outlines it carries inside. Only two links in the top bar go out to the network, and only if you click them: the one to the suite portal and Report a problem, which opens GitHub in another tab.

What the app remembers stays in the local storage of that browser, on that computer, under keys beginning with germplasmpro:: the collection and the time of the last save; the characterization; the genebank assumptions and the custom constants of Block 8; the duplicate threshold and the dismissed pairs of Block 3; the fields, header and code of the labels; the title, genebank and signatory of the report; the language and the theme. The top bar separately stores whether the sidebar is collapsed and whether animations are on. None of that leaves the computer, and it is erased when you clear the browsing data for that site; Empty, in the Block 2 table, erases the collection after asking for confirmation.

> **THE BACKUP IS YOURS**
>
> Browser storage is a convenience, not a backup, and sometimes it cannot be used: with the space full or in a private window, Block 2 warns that it could not save. The .json project carries the passport, the characterization, the genebank assumptions, the custom constants, the dismissed pairs and the threshold, and the label preferences; the report title, the language and the theme stay in the browser. Save the project at the end of each session, export the passport every season and write down the constants of your species with their source.

F.4 · Keyboard and accessibility

The ? button in the top bar, or the ? key outside a text field, opens the list of shortcuts with a note on where you are:

Keys	What they do
Alt + ← · Alt + →	Previous or next block. In focused mode, previous or next section, and at the ends, the block.
← / → · Home / End	With the focus on the sidebar, move along it; Home and End go to the first and the last.
Ctrl + K	Search blocks, sections, controls and actions, also from a text field.
[	Collapse or show the sidebar.
?	Show the list of shortcuts.
Esc	Close windows and panels.
Ctrl + P	In Block 9, print the label sheet at exact scale, like the Print button.

The first Tab offers Skip to content; buttons and icons have names for screen readers, and status messages and block changes are announced to the screen reader. The traffic lights do not depend on color alone: they carry a word (good, check) or their written threshold. If your system asks for reduced motion, animations stay still, and the Animations button switches them off in any case.

F.5 · License

The program is distributed under the **GNU General Public License, version 3 or later** (GPL-3.0-or-later): it may be freely used, copied, studied and modified, on condition that modified versions are distributed under the same license and keep the attribution. The full text is in the file LICENSE. This manual is © 2026 Luis Ángel Barrera-Guzmán, all rights reserved.

F.6 · How to cite it

Barrera-Guzmán, L.Á. (2026). GermplasmPro: the genebank, accession by accession (version 1.0.0) [software]. https://github.com/luisangelbg/GermplasmPro

Barrera-Guzmán, L.Á. (2026). *GermplasmPro: User's Manual* (1st English ed.). Distributed with GermplasmPro, version 1.0.0. https://github.com/luisangelbg/GermplasmPro

The citation of the program is the one given by the How to cite button on the app's home page and the one that closes every Block 10 report in English. GermplasmPro has no DOI: the citation carries the address of the repository, where the source code is; the app is used online at https://luisangelbg.github.io/GermplasmPro/. The CITATION.cff file in the program folder gives the same data —author, with his ORCID 0000-0001-8057-2583; version 1.0.0, of 1 October 2026; license GPL-3.0-or-later— in the format read by reference managers and GitHub.